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Homological and Computational Methods in Commutative Algebra

Dedicated to Winfried Bruns on the
Occasion of his 70th Birthday



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Dedicated to Winfried Bruns on the Occasion
of his 70th Birthday

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Foreword

Winfried Bruns was born on May 5, 1946, at Oker near Goslar in Germany. From 1956 to 1965, he attended the classical gymnasium at Goslar whose focus is on old languages—Latin and Greek. After his graduation from school, he served in the army for two years, and then started studying mathematics at TU Hannover in 1967. During his studies and until his diploma, he was supported by a distinguished grant for excellent students from the Studienstiftung des Deutschen Volkes.

After his graduation, he became scientific assistant at TU Clausthal where, in 1972 he received his PhD for his dissertation *Beispiele reflexiver Differentialmoduln* with Udo Vetter as his advisor. Large parts of his dissertation were later published in *Journal für reine und angewandte Mathematik*.

At that time basic element theory had become very popular, as it turned out to be an important tool to study generating sets of modules. By extending techniques on basic elements, as developed by Eisenbud and Evans in their influential paper “Generating modules efficiently: Theorems from algebraic K -theory” Winfried proved a beautiful and surprising theorem in his paper “*Jede endliche freie Auflösung ist freie Auflösung eines von drei Elementen erzeugten Ideals*”. This was the beginning of an outstanding scientific career. The methods used in this paper, namely, those of basic element theory, were also the subject of his habilitation dissertation from 1977.

In 1979, Winfried became a full professor at the University of Osnabrück, Abteilung Vechta, and in 1995 moved to the Mathematical Department of the same university in 1995. Before his time in Vechta, however, he spent 1 year as Visiting Lecturer at The University of Illinois at Urbana-Champaign, mainly to work with Graham Evans. This cooperation had a formative effect on his work and resulted in a lifelong friendship with Graham. As one of the achievements of his visit to Urbana Champaign, in 1980 Winfried, jointly with Evans and Griffith, published the paper *Syzygies, ideals of height two and vector bundles*. In this paper, among other results, the so-called syzygy problem, in some special cases was answered in the affirmative. One year later, Evans and Griffith succeeded in proving the syzygy theorem in general for any Noetherian local domain containing a field. In the following this theorem has been generalized in various directions, also by Winfried.

During his time in Vechta, Winfried published numerous influential papers, covering many key aspects of modern Commutative Algebra, including generic maps, generic resolutions, divisors on varieties of complexes, straightening laws on modules, straightening closed ideals. The theory of algebras with straightening laws had been developed by De Concini, Eisenbud and Procesi at the beginning of the 1980s, for which determinantal rings are prominent examples. Winfried extended the scope of this theory by showing that it provides upper bounds for the arithmetical rank of ideals generated by a poset ideal of the poset underlying the algebra with straightening law, and further introduced modules with straightening laws in order to study the symmetric algebra of generic modules.

A few years later, in 1988, Winfried and Vetter published their famous Springer Lecture Notes *Determinantal Rings*, Vol. 1327. These lecture notes are to date the standard reference on this subject. Besides the classical theory developed by Hochster and Eagon, powers and symbolic powers of determinantal ideals are considered, ASL theory is employed, the canonical class of determinantal rings is determined, and much more is presented.

Gröbner basis entered the theory of determinantal ideals with the work of Sturmfels in 1990. This aspect of the theory had a high impact on Winfried's later work and his future scientific cooperations. In 2003, jointly with Aldo Conca, he wrote the article *Gröbner bases and determinantal ideals*, in which they follow the line of investigations started by Sturmfels. Ever since their first joint paper *KRS and powers of determinantal ideal*, in 1998, Aldo and Winfried have enjoyed an extremely fruitful cooperation, which continues today. They wrote a series of brilliant papers on the theory of determinantal rings in which they studied Gröbner bases and powers of determinantal ideals. In other papers jointly with Tim Römer they considered Koszul cycles, and Koszul homology and syzygies of Veronese subalgebras. More recently, Matteo Varbaro joined the team, and in 2013 Winfried, together with Aldo and Matteo, published in *Advances in Mathematics* a fundamental paper dealing with the very difficult problem of better understanding the relations of the algebra generated by the t -minors of a matrix of indeterminates.

Winfried has written several monographs. Besides his Springer Lecture Notes with Vetter, his book *Cohen–Macaulay rings* (with Jürgen Herzog as coauthor) became extremely popular. Its first edition was published in 1993, with a revised edition in 1998. Together with the book *Commutative algebra with a view towards algebraic geometry* by David Eisenbud, this monograph is a must-read for any advanced student in Commutative Algebra.

His most recent monograph is his book *Polytopes, Rings and K -theory*, with Joseph Gubeladze. This book treats the interaction between discrete convex geometry, commutative ring theory, algebraic K -theory, and algebraic geometry, and is the culmination of years of scientific cooperation with Joseph which began 1996. So far Winfried and Joseph have published 20 joint papers containing some spectacular results. Among these, one should mention first, the paper *Normality and covering properties of affine semigroups* which appeared 1999 in *Journal für die reine and angewandte Mathematik*. In that paper they succeeded in finding a six-dimensional counterexample to a conjecture of Sebö, who conjectured that a

finite rational cone admits a unimodular covering by simplicial cones spanned by elements of the Hilbert basis. It was known previously by Sebö and others that three-dimensional rational cones even admit unimodular triangulations, and that such triangulations in general no longer exist in dimension 4. In their paper, Winfried and Joseph also presented an algorithm to decide whether a finite rational cone admits a unimodular covering. The computational and algorithmic aspects of the theory of polytopes and affine monoids have always been one of Winfried's especial concerns. The same year, in the same journal, Winfried together with Gubeladze, Henk, Martin and Weismantel showed that the Bruns-Gubeladze counterexample to unimodular coverings turned out also to be a counterexample to the integral Carathéodory property of cones which requires that any integral vector of the n -dimensional integral polyhedral pointed cone can be written as a nonnegative integral combination of at most n elements of the Hilbert basis of the cone. This result showed that Carathéodory's theorem for convex cones does not have an integer analogue.

The discovery of the counterexamples certainly promoted experimental mathematics a lot. Progress has been made possible by the huge capacity of modern computers and powerful computer algebra programs, such as Normaliz. It is not going too far to say that the software *Normaliz*, which was created by Winfried together with his former PhD student, Koch, is one of his passions. *Normaliz* is regularly updated by Winfried and his team consisting at the moment of Ichim, Sieg, Römer and Söger. The program is freely downloadable and also integrated in other algebraic software, such as Macaulay 2, Singular and polymake. It has become an indispensable tool in the study of polytopes and affine monoids.

Winfried has six PhD students, but has supported many more young students, helping them to find their way into mathematics. In 2002 he was awarded the Osnabrück University prize for excellence in teaching. Together with Holger Brenner, Tim Römer and other colleagues in the department, Winfried successfully applied for the Graduiertenkolleg at DFG (the German national science foundation). The Graduiertenkolleg was established 2013 and has since been running since then extremely successfully providing PhD positions and Postdocs for many young researchers. It is his merit to have made the mathematics department of Osnabrück one of the most important centers of Commutative Algebra in Europe.

Winfried's international contacts are countless. So far he has published more than 100 papers, and since 1975 he has presented invited lectures at almost all major international conferences on Commutative Algebra. He has also organized several important international conferences at Oberwolfach, Luminy, Cortona, and Berkeley, thereby promoting the exchange of ideas and giving young mathematicians the chance to present their results to large audiences of experts. Over the years he has maintained a particularly close scientific relationship with the Commutative Algebra research team in Genoa, and also the algebraists in Messina. This was partly made possible by DAAD funded programs. In recognition of his scientific merits he was elected as a corresponding member of the *Accademia Peloritana dei Pericolanti di Messina*. He also served on the editorial board of *Communications in Algebra*, and he is still a member of the editorial board of *Homology, Homotopy*

and Applications. Furthermore, he has refereed research projects for the DFG, the DAAD, the Alexander von Humboldt-Stiftung, the NSF and others.

The authors and editors of this volume wish Winfried good health and joy and happiness with his family. Commutative Algebra as we know it today would not be the same without his work, which has so greatly influenced and shaped our field. So we wish him happy research and expect many more great theorems.

Essen, Germany
3 January 2017

Prof. Jürgen Herzog

Preface

The present volume is an outcome of the INdAM Conference “Homological and Computational Methods in Commutative Algebra”, which took place at the Palazzone in Cortona, Italy, from May 30 to June 3, 2016.

The volume reflects much of the spirit of the INdAM Conference. It provides a snapshot of the current developments in Commutative Algebra, with special emphasis on homological notions, combinatorics, and symbolic computations. The fifteen chapters address a wide range of topics, from themes that are at the heart of the discipline to themes that establish connections with related areas.

Both the conference and this volume are dedicated to Prof. Winfried Bruns on the occasion of his 70th birthday.

We thank the participants and the speakers at the meeting as well as the authors of the chapters in this volume for their wonderful collaboration.

In particular, we wish to thank Prof. Jürgen Herzog for writing the Foreword, which provides a very nice overview of Professor Bruns’ contribution to the subject.

Genova, Italy
San Francisco, CA, USA
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3 January 2017

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Betti Sequences over Standard Graded Commutative Algebras with Two Relations

Luchezar L. Avramov and Zheng Yang

Abstract It is well known that if Q is a standard graded polynomial ring in d variables over a field, and R is a quotient of Q with $\text{proj dim}_Q R \leq 1$, then the Betti sequence of every R -module N is constant after d steps. We prove that if $\text{proj dim}_Q R = 2$, then the Betti sequence of N is determined explicitly by some initial segment of known length, which can be read off a presentation of N . There are three cases, depending on the degrees of the defining relations of R and on whether these have a common divisor.

1 Introduction

In this paper we work with *standard graded* commutative algebras R over some field. Such an R is canonically isomorphic to P/I , where P is the symmetric algebra of R_1 , and I is the kernel of the natural homomorphism $P \rightarrow R$.

We are concerned with the *sequences of Betti numbers* $(\beta_i^R(N))_{i \geq 0}$ of finitely generated graded R -modules N . When I is principal, such a sequence is determined by $\beta_i^R(N)$ with $i \leq d$, where $d = \text{depth } R - \text{depth}_R N$: For $i > d$, the Hilbert Syzygy Theorem and the Auslander-Buchsbaum Equality yield $\beta_i^R(N) = 0$ if $I = 0$, and Eisenbud [10] proves that $\beta_i^R(N) = \beta_d^R(N)$ holds if $I = (f) \neq 0$. More generally, $(\beta_i^R(N))_{i \geq 0}$ has the same properties when $R = Q/J$ for some standard graded algebra Q , if the projective dimension $\text{proj dim}_Q N$ is finite and $\text{proj dim}_Q R \leq 1$.

In Sect. 8 we establish an equally explicit statement when $\text{proj dim}_Q R = 2$.

Theorem 1.1 *Let Q be a standard graded algebra with $\text{rank}_k Q_2 = \binom{\text{rank}_k Q_1}{2}$, and let J be a homogeneous ideal minimally generated by r elements in $Q_{\geq 2}$.*

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Set $R = Q/J$. Let N be a finitely generated graded R -module, $\beta_i = \beta_i^R(N)$,

$$d = \text{depth } R - \text{depth}_R N, \quad \text{and} \quad l = d + \max\{2\beta_d, 2\beta_{d+1} - 2\beta_d - 1\}.$$

When $\text{proj dim}_Q R = 2$ and $\text{proj dim}_Q N$ is finite the following cases occur:

(a) $r = 2 = \text{grade } J$ and $J_2 \neq 0$. In this case

$$\beta_i = (\beta_{l+1} - \beta_l)(i - l) + \beta_l \quad \text{for } i \geq l.$$

(b) $r = 2 = \text{grade } J$ and $J_2 = 0$. In this case

$$\beta_i = \frac{\beta_{l+2} - \beta_l}{2}(i - l) + \beta_l \quad \text{for even } i - l \geq 0.$$

$$\beta_i = \frac{\beta_{l+2} - \beta_l}{2}(i - l - 1) + \beta_{l+1} \quad \text{for odd } i - l \geq 1.$$

(c) $r = 2 = \text{grade } J + 1$ or $r \geq 3$. In this case

$$\beta_i = \beta_{d+3}\varphi_{i-d-3} + (r-1)\beta_{d+2}\varphi_{i-d-4} \quad \text{for } i \geq d+4,$$

where $(\varphi_i)_{i \geq 0}$ is defined recursively by

$$\varphi_{i+2} = \varphi_{i+1} + (r-1)\varphi_i \quad \text{and} \quad \varphi_0 = \varphi_1 = 1.$$

Much of this information is already available. The possible types of ideals J are described by the (graded) Hilbert-Burch Theorem; see [8]. The result in case (c) is due to Iyengar [14]. In case (b), a slightly weaker conclusion is known from work of Gulliksen [12], Avramov [2], and Avramov and Buchweitz [5]. When Q is a polynomial ring, in case (b) there are Betti sequences that are not eventually arithmetic, but there are no such sequences in case (a) if $\text{rank}_k J_2 = 2$; see [3].

This paper started as an exploration of the only open case: (a) with $\text{rank}_k J_2 = 1$. As in [2, 10, 12], it utilizes the action of a commutative ring $\mathcal{R}$ of cohomology operators on $\mathcal{L} = \text{Ext}_R^*(L, k)$, where $L = \Omega_R^j(N)$ and $k = R/R_{\geq 1}$. Homology products turn $L = \text{Tor}_*^Q(L, k)$ into a graded module over $\Lambda = \text{Tor}_*^Q(R, k)$, and $\mathcal{R}$ acts also on $\text{Ext}_\Lambda^*(L, k)$. It is proved in [5] that for $j \gg 0$ the $\mathcal{R}$ -modules $\mathcal{L}$ and $\text{Ext}_\Lambda^*(L, k)$ are naturally isomorphic. The upshot is that we can draw on Kronecker's description of the indecomposable Λ -modules over the exterior algebra Λ on two generators. The relevant material from [5] is summarized in Sects. 2 and 4.

The plan for the proof of (a) is to write L as a direct sum of shifts $\Sigma^p X$ of indecomposable Λ -modules X from Kronecker's list and express $(\beta_i^\Lambda(L))_i$ in terms of the explicitly known sequences $(\beta_i^\Lambda(\Sigma^p X))_i$. From the resulting expression one can, in theory, extract a criterion for $(\beta_i^\Lambda(L))_i$ to be arithmetic; it takes the form of a system of linear equations for the multiplicities of the $\Sigma^p X$ in the decomposition of L . Since infinitely many equations and variables are involved, we first look for restrictions imposed on those multiplicities by the homological origin of L .

The desired information on the Λ -module structure of L is gathered by analyzing the $\mathcal{R}$ -module structure of $\mathcal{L}$; this reverses the approach in [5]. In Sect. 3 we prove that quadratic forms in J give rise to operators in $\mathcal{R}$, which are *squares* of (non-central) operators of degree 1. This result is of general cohomological interest.

Up through the end of Sect. 5 we work entirely in the more general context of modules over local rings. Starting with Sect. 6, we exploit the rigidity imposed by the *internal gradings* on Λ , L , $\mathcal{R}$, and $\mathcal{L}$ (they are induced by the gradings on Q , R , L , and k). In particular, most bigradings of Λ drastically simplify its representation theory, as they eliminate the continuous families of indecomposables from Kronecker's list. In Sect. 7 we use that information to prove that the multiplicities of interest, satisfy restrictions that explicitly involve internal degrees.

In Sect. 8, we assemble all the data obtained earlier in the paper to prove case (a) of the Main Theorem; a small subset of these data goes into the proof of (b).

In recent joint work [7] with Alexandra Seceleanu, we answer a question in [3] and partly generalize (a): If Q is a polynomial ring and J is generated by a homogeneous regular sequence of length c containing $c - 1$ quadrics, then the Betti sequence of every (Q/J) -module N is eventually given by some polynomial in i of degree at most $c - 1$. The approach is completely different and gives no estimate for the start of the polynomial behavior. For $c \geq 3$, we do not know if such behavior occurs when the hypothesis on Q is replaced by a finiteness assumption on $\text{proj dim}_Q N$.

2 Background

In this section $(Q, \mathfrak{q}, k)$ is a local ring and $\mathbf{g} = g_1, \dots, g_c$ a Q -regular sequence. We set $J = (\mathbf{g})$ and form the local ring $(R, \mathfrak{m}, k) = (Q/J, \mathfrak{q}/J, k)$.

Throughout the section, N denotes a finitely generated R -module. The graded k -space $\text{Ext}_R^*(N, k)$ has a natural structure of left graded module over a k -algebra of cohomology operators $\mathcal{R}$, which is a polynomial ring in c indeterminates of cohomological degree 2. That algebra can be introduced in a variety of ways, discussed and compared in [6]. Here we recall the construction given in [5].

We start by enouncing some, mostly standard, conventions.

2.1 Differential Graded Algebras and Modules

In this section all gradings arise from (co)homological constructions.

Let $U = \cdots \rightarrow U_n \xrightarrow{\partial_n} U_{n-1} \rightarrow \cdots$ be a complex of Q -modules. The same complex may also be written as $U = \cdots \rightarrow U^{-n} \xrightarrow{\partial^{-n}} U^{-n+1} \rightarrow \cdots$ with $U^{-n} = U_n$ and $\partial^{-n} = \partial_n$ for all n . When p is an integer, $\Sigma^p U$ denotes the complex with $(\Sigma^p U)_n = U_{n-p}$ for $n \in \mathbb{Z}$ and $\partial_n^{\Sigma^p U} = (-1)^p \partial_{n-p}^U$; in particular, $(\Sigma^p U)^n = U^{n+p}$.

All DG algebras are assumed to be graded-commutative. When U is a left DG A -module, $\mathrm{Hom}_Q(U, k)$ is the left DG A -module with $\mathrm{Hom}_Q(U, k)^j = \mathrm{Hom}_Q(U_j, k)$, on $v: U_j \rightarrow k$ the differential ∂' acts by $\partial'(v) = (-1)^{j+1} v \partial^U: U_{j+1} \rightarrow k$, and an element $a \in A_i$ acts on v by $(av)(u) = (-1)^{ij} v(au): U_{j-i} \rightarrow k$.

Graded algebras are viewed as DG algebras with zero differential, and graded modules over graded algebras as DG modules with zero differential. The graded algebra (or module) underlying a DG algebra (or module) U is denoted by $U^\natural$.

Rings are viewed as graded algebras concentrated in degree zero.

2.2 Cohomology Operators

The notation introduced at the beginning of this section is in force.

The Koszul complex E on $\mathbf{g}$ is a DG Q -algebra. We use the quasi-isomorphism of DG algebras $E \rightarrow H_0(E) = R$ to turn R -modules into DG E -modules. Any subset $\mathbf{x} = \{x_1, \dots, x_c\}$ of E_1 such that $\partial^E(x_h) = g_h$ for $h = 1, \dots, c$ is a basis of E_1 .

Let $\mathcal{R}$ be the symmetric algebra of the graded k -vector space $\Sigma^{-2} \mathrm{Hom}_Q(J, k)$.

Let $\mathcal{S}$ be the symmetric algebra of the graded Q -module $\Sigma^{-2} \mathrm{Hom}_Q(E_1, Q)$. When $\mathbf{x}$ is chosen as above, we let $\chi = \{\chi_1, \dots, \chi_c\}$ be the dual basis of $\mathcal{S}^2$; then $\mathcal{S}$ becomes the graded polynomial ring $Q[\chi_1, \dots, \chi_c]$. The canonical isomorphism

$$\mathrm{Hom}_Q(E_1, Q) \otimes_Q k \xrightarrow{\cong} \mathrm{Hom}_Q(J, k)$$

extends to an isomorphism $\mathcal{S} \otimes_Q k \xrightarrow{\cong} \mathcal{R}$ of graded k -algebras. Identifying them, and letting χ_h also denote the element $\chi_h \otimes 1$ of $\mathcal{R}^2$, we write $\mathcal{R} = k[\chi_1, \dots, \chi_c]$.

Choose a free resolution G of N over Q , that is also a DG module over E ; see [4, 2.2.7]. Let G' denote the DG E -module $\mathrm{Hom}_Q(G, k)$, as in Sect. 2.1.

Let $\mathcal{S} \widetilde{\otimes}_Q G'$ be the graded vector space with

$$(\mathcal{S} \widetilde{\otimes}_Q G')^n = \bigoplus_{2l+j=n} \mathcal{S}^{2l} \otimes_Q G'^j \quad \text{for } n \in \mathbb{Z}$$

and let $\partial^n: (\mathcal{S} \widetilde{\otimes}_Q G')^n \rightarrow (\mathcal{S} \widetilde{\otimes}_Q G')^{n+1}$ be the k -linear map satisfying

$$\partial(\chi \otimes \gamma) = \chi \otimes \partial'(\gamma) + \sum_{h=1}^c \chi_h \chi \otimes x_h \gamma \quad \text{for all } \chi \in \mathcal{S} \text{ and } \gamma \in G'.$$

The evident actions of $\mathcal{S}$ and k turn $\mathcal{S} \widetilde{\otimes}_Q \mathrm{Hom}_Q(G, k)$ into a DG $\mathcal{R}$ -module. By [5, 3.1], any two choices of a minimal generating set $\mathbf{g}$ of J , of a subset $\mathbf{x} \subset E_1$ with $\partial(\mathbf{x}) = \mathbf{g}$, and of a DG E -module G resolving N produce DG $\mathcal{S}$ -modules linked by quasi-isomorphisms that are unique up to homotopy. As a consequence, a graded $\mathcal{R}$ -module $H^*(\mathcal{S} \widetilde{\otimes}_Q \mathrm{Hom}_Q(G, k))$ is defined uniquely up to a unique isomorphism.

It is proved in [5, 3.1] that there exists a unique isomorphism

$$\mathrm{Ext}_R^*(N, k) \cong H^*(\mathcal{S} \widetilde{\otimes}_Q \mathrm{Hom}_Q(G, k)).$$

This gives $\mathrm{Ext}_R^*(N, k)$ a well-defined structure of graded $\mathcal{R}$ -module.

The rest of this section will not be used before Sect. 5.

2.3 Exterior Algebras

Let Λ be the exterior algebra of the graded k -vector space $\Sigma^1(J \otimes_Q k)$.

The natural isomorphism $E_1 \otimes_Q k \xrightarrow{\cong} J \otimes_Q k$ extends to an isomorphism of graded k -algebras $E \otimes_Q k \xrightarrow{\cong} \Lambda$. We identify these algebras, let x_h also stand for the element $x_h \otimes 1$ of Λ_1 , and write $\Lambda = k\langle x_1, \dots, x_c \rangle$. The bases $\mathbf{x}$ of Λ_1 and χ of $\mathcal{R}^2$ are dual to each other under the natural isomorphism $\mathrm{Hom}_k(J \otimes_Q k, k) \xrightarrow{\cong} \mathrm{Hom}_Q(J, k)$.

Let $\mathbf{G}$ be a DG Λ -module with $\mathrm{rank}_k \mathbf{G}_j$ finite for every j and $\mathbf{G}_j = 0$ for $j \ll 0$, and let $\mathbf{G}'$ denote the DG Λ -module $\mathrm{Hom}_k(\mathbf{G}, k)$; see Sect. 2.1.

Let $\mathcal{R} \widetilde{\otimes}_k \mathrm{Hom}_k(\mathbf{G}, k)$ be the vector space with $(\mathcal{R} \widetilde{\otimes}_k \mathbf{G}')^n = \bigoplus_{2l+j=n} \mathcal{R}^{2l} \otimes_k \mathbf{G}'^j$. For each $n \in \mathbb{Z}$, let $\partial^n: (\mathcal{R} \widetilde{\otimes}_k \mathbf{G}')^n \rightarrow (\mathcal{R} \widetilde{\otimes}_k \mathbf{G}')^{n+1}$ be the k -linear map satisfying

$$\partial(\chi \otimes \gamma) = \chi \otimes \partial'(\gamma) + \sum_{h=1}^c \chi_h \chi \otimes x_h \gamma \quad \text{for } \chi \in \mathcal{R} \text{ and } \gamma \in \mathbf{G}'.$$

The evident action on $\mathcal{R}$ turns $\mathcal{R} \widetilde{\otimes}_k \mathrm{Hom}_k(\mathbf{G}, k)$ into a left DG $\mathcal{R}$ -module.

We define a graded $\mathcal{R}$ -module $\mathrm{Ext}_\Lambda^*(\mathbf{G}, k)$ by setting

$$\mathrm{Ext}_\Lambda^*(\mathbf{G}, k) = H^*(\mathcal{R} \widetilde{\otimes}_k \mathrm{Hom}_k(\mathbf{G}, k)).$$

For completeness, we note that the left hand side of this formula can be defined independently, as Ext in the category of DG Λ -modules; the equality then is replaced by an isomorphism, obtained from a computation with a semifree resolution that goes back to Henri Cartan.

2.4 Homology Products

Let G be the DG E -module chosen in Sect. 2.2; set $\mathbf{G} = G \otimes_Q k$ and $\mathbf{N} = H_*(\mathbf{G})$.

As E and G are free resolutions over Q , there are unique isomorphisms

$$\Lambda \cong \mathrm{Tor}_*^Q(R, k) \quad \text{and} \quad \mathbf{N} \cong \mathrm{Tor}_*^Q(N, k).$$

These maps carry products in Λ and $\mathbf{N}$ to classical *homology products* in $\mathrm{Tor}_*^Q(?, k)$, which can be defined from arbitrary Q -free resolutions; see [4, 2.3.2 and 2.3.3].

On the other hand, we obtain an isomorphism of graded $\mathcal{R}$ -modules

$$\mathrm{Ext}_R^*(N, k) \cong \mathrm{Ext}_\Lambda^*(\mathbf{G}, k),$$

because the formulas in Sects. 2.2 and 2.3 yield

$$\mathcal{S} \widetilde{\otimes}_Q \mathrm{Hom}_Q(G, k) = \mathcal{R} \widetilde{\otimes}_k \mathrm{Hom}_k(\mathbf{G}, k).$$

Assume that the DG E -module G , chosen in Sect. 2.2, satisfies $\partial^G(G) \subseteq qG$.

The DG Λ -modules $\mathbf{G}$ and $\mathrm{Hom}_k(\mathbf{G}, k)$ then have zero differentials, so the isomorphisms in Sects. 2.4 and 2.3 yield an isomorphism of graded $\mathcal{R}$ -modules

$$\mathrm{Ext}_R^*(N, k) \cong \mathrm{Ext}_\Lambda^*(\mathbf{N}, k).$$

The right-hand side of the last formula can also be described in classical terms:

$$\mathrm{Ext}_R^n(N, k) = \bigoplus_{p+q=n} {}^{\mathrm{gr}}\mathrm{Ext}_\Lambda^p(\mathbf{N}, k)^q \quad \text{for } n \in \mathbb{Z},$$

where ${}^{\mathrm{gr}}\mathrm{Ext}_\Lambda(\mathbf{N}, k)$ is the bigraded k -vector space obtained from a resolution of $\mathbf{N}$ by means of graded free Λ -modules, with differentials preserving the gradings defined by the gradings of Λ and $\mathbf{N}$.

3 Square Roots of Cohomology Operators

Let (Q, q, k) be a complete local ring and J an ideal contained in q^2 , generated by some Q -regular sequence of length c . Set $R = Q/J$, and then

$$e = \mathrm{rank}_k(q/q^2) \quad \text{and} \quad b = \mathrm{rank}_k J/(q^3 \cap J).$$

By a theorem of Gulliksen and Schoeller, k has a minimal Q -free resolution K that has a DG Q -algebra structure; see [4, 2.2.8]. Let $\{t_1, \dots, t_e\}$ and $\{g_1, \dots, g_c\}$ be minimal generating sets of q and J , respectively. Each g_i is in q^2 , so we have

$$g_i = \sum_{h=1}^e a_{hi} t_h \quad \text{for } i = 1, \dots, c, \quad \text{with } a_{hi} \in q. \quad (1)$$

Since $H_0(K) = k$, there exist $u_1, \dots, u_e \in K_1$ satisfying $\partial^K(u_i) = t_i$ for $i = 1, \dots, e$.

Let E be the Koszul complex from Sect. 2.2. By the universal property of exterior algebras, there is a homomorphism $\varphi: E^{\natural} \rightarrow K^{\natural}$ of the graded Q -algebras satisfying $\varphi(x_i) = \sum_{h=1}^e a_{hi} u_h$ for each i . As $\partial^K \varphi(x_i) = \varphi \partial^E(x_i)$ holds by design, φ is a morphism of DG algebras by the Leibniz rule. It turns K into a DG E -module.

Next we specialize the discussion to the case $N = k$.

3.1 Composition Products

Let $\mathcal{R}$ be the symmetric algebra of the graded k -vector space $\Sigma^{-2} \text{Hom}_Q(J, k)$.

The inclusions $\partial^K(K) \subseteq \mathfrak{q}K$ and $\varphi(x_i) \in \mathfrak{q}K$ yield, respectively, $\partial' = 0$ and $x_h \text{Hom}_Q(K, k) = 0$ in the formula in Sect. 2.2. It follows that $\mathcal{S} \widetilde{\otimes}_Q \text{Hom}_Q(K, k)$ has zero differential, so we get a canonical isomorphism of graded $\mathcal{R}$ -modules

$$\mathcal{R} \otimes_k \text{Ext}_Q^*(k, k) \cong \text{Ext}_R^*(k, k). \quad (2)$$

We use this map to identify the modules involved; by extension, we identify $\chi \in \mathcal{R}$ and $\chi \text{id}^k \in \text{Ext}_R^*(k, k)$, where id^k is the identity map in $\text{Hom}_R(k, k) = \text{Ext}_R^0(k, k)$.

Set $\mathcal{E} = \text{Ext}_R^*(k, k)$. *Composition products* turn $\mathcal{E}$ into a graded k -algebra with unit element $\text{id}^k \in \text{Hom}_R(k, k)$. It is proved in [6, Theorem 2.7 on p. 700] that the subset $\mathcal{R} \subseteq \mathcal{E}$, described above, is a subalgebra of the center of $\mathcal{E}$.

Let N be an R -module and set $\mathcal{N} = \text{Ext}_R^*(N, k)$. Composition products also turn $\mathcal{N}$ into a graded left $\mathcal{E}$ -module. The vector space underlying $\mathcal{N}$ decomposes as $\mathcal{N}^{\text{even}} \oplus \mathcal{N}^{\text{odd}}$, where $(\mathcal{N}^{\text{even}})^i = \mathcal{N}^i$ for even i and $(\mathcal{N}^{\text{even}})^i = 0$ for odd i , and $\mathcal{N}^{\text{odd}}$ is defined accordingly. These subspaces are $\mathcal{R}$ -submodules, since $\mathcal{R}^{\text{odd}} = 0$.

Theorem 3.1 *With notation as above, assume $\text{rank}_k(\mathfrak{q}^2/\mathfrak{q}^3) = \binom{e+1}{2}$ and $b \geq 1$.*

There exist a subset $\{\tau_1, \dots, \tau_b\}$ of $\mathcal{E}^1$ and a basis $\{\chi_1, \dots, \chi_c\}$ of $\mathcal{R}^2$ satisfying

$$\chi_h = \tau_h^2 \quad \text{for } 1 \leq h \leq b. \quad (3)$$

When N is an R -module and $\mathcal{N} = \text{Ext}_R^(N, k)$, the maps of upper degree one*

$$\alpha_h: \mathcal{N}^{\text{even}} \rightarrow \mathcal{N}^{\text{odd}} \quad \text{and} \quad \beta_h: \mathcal{N}^{\text{odd}} \rightarrow \mathcal{N}^{\text{even}} \quad (4)$$

induced by left multiplication by τ_h are $\mathcal{R}$ -linear, and the composed maps

$$\beta_h \alpha_h: \mathcal{N}^{\text{even}} \rightarrow \mathcal{N}^{\text{even}} \quad \text{and} \quad \alpha_h \beta_h: \mathcal{N}^{\text{odd}} \rightarrow \mathcal{N}^{\text{odd}} \quad (5)$$

are given by left multiplication by χ_h .

In the proof of Theorem 3.1 we use the structure of the algebra $\mathcal{E}$.

3.2 Homotopy Lie Algebras

The graded vector space $\text{Tor}_*^R(k, k)$ is canonically a graded k -algebra with divided powers defined for the elements of positive even degree; see [13, 2.3.4].

For every $i \in \mathbb{Z}$, let $\pi_i(R)$ denote the quotient of $\text{Tor}_i^R(k, k)$ by its k -subspace spanned by all the products of elements of positive degree and by the divided powers of degree two or higher of all the elements of positive even degree.

Let $\pi^*(R)$ be the graded k -space with $\pi^i(R) = \text{Hom}_k(\pi_i(R), k)$, viewed as a subspace of $\mathcal{E}$ through the canonical identifications $\mathcal{E}^i = \text{Hom}_k(\text{Tor}_i^R(k, k), k)$.

The subspace $\pi^*(R)$ contains the commutators $[\vartheta, \theta] = \vartheta\theta - (-1)^{ij}\theta\vartheta$ of all $\vartheta \in \pi^i(R)$ and $\theta \in \pi^j(R)$, and the squares ϑ^2 of all $\vartheta \in \pi^{\text{odd}}(R)$ (the products are in $\mathcal{E}$). Thus $\pi^*(R)$ is a graded Lie algebra, called the *homotopy Lie algebra* of R . In fact, $\mathcal{E}$ is the universal enveloping algebra of $\pi^*(R)$; see [17, § 1] or [4, Ch. 10].

The components of $\pi^*(R)$ of low degree have useful descriptions; see [4, 7.1.5].

There are canonical isomorphisms of vector spaces

$$\text{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k) \cong \text{Hom}_R(\mathfrak{m}, k) \xrightarrow{\cong} \mathcal{E}^1 = \pi^1(R). \quad (6)$$

Since R is complete, Cohen's Structure Theorem yields an isomorphism $P/I \cong R$ with a regular local ring $(P, \mathfrak{n}, k)$ and an ideal I contained in $\mathfrak{n}^2$. We call any such isomorphism a *minimal Cohen presentation* of R .

For any minimal Cohen presentation there are canonical isomorphisms

$$\text{Hom}_k(I/\mathfrak{n}I, k) \cong \text{Hom}_P(I, k) \xrightarrow{\cong} \pi^2(R). \quad (7)$$

The assignment $R \mapsto \pi^*(R)$ defines a contravariant functor from the category of local rings with residue field k and ring homomorphisms inducing the identity on k to the category of graded Lie algebras over k and their homomorphisms.

A minimal Cohen presentation $R \cong P/I$ with regular local ring $(P, \mathfrak{n}, k)$ induces an exact sequence of graded Lie algebras.

$$0 \rightarrow \pi^{\geq 2}(R) \rightarrow \pi^*(R) \rightarrow \pi^*(P) \rightarrow 0$$

Indeed, $\text{Tor}_*^P(k, k)$ is the exterior algebra on $\Sigma(\mathfrak{n}/\mathfrak{n}^2)$, whence $\pi_*(P) = \Sigma(\mathfrak{n}/\mathfrak{n}^2)$, giving $\pi^{\geq 2}(P) = 0$; exactness in degrees different from 1 follows. By (6), exactness in degree 1 means that the induced map $\text{Hom}_R(\mathfrak{m}, k) \rightarrow \text{Hom}_P(\mathfrak{n}, k)$ is bijective; this is indeed the case, because I is contained in $\mathfrak{n}^2$.

When Q and J are as in Theorem 3.1, and $\psi: Q \rightarrow R$ is the canonical map, there is a natural exact sequence of graded Lie algebras

$$0 \rightarrow \Sigma^{-2} \text{Hom}_Q(J, k) \xrightarrow{\zeta} \pi_*(R) \xrightarrow{\pi_\psi} \pi_*(Q) \rightarrow 0$$

Indeed, it is shown in the proof of [13, 3.4.1] that π_i^ψ is bijective for $i \geq 3$, and that there is a natural exact sequence of k -vector spaces

$$0 \rightarrow \pi_2(Q) \xrightarrow{\pi_2(\psi)} \pi_2(R) \rightarrow J/\mathfrak{q}J \rightarrow \pi_1(Q) \xrightarrow{\pi_1(\psi)} \pi_1(R) \rightarrow 0$$

The desired properties of $\pi^i(\psi)$ follow, because $\pi^i(\psi) = \text{Hom}_k(\pi_i(\psi), k)$ holds for each i , by definition, and $J \subseteq \mathfrak{q}^2$ means that $\pi_1(\psi)$ is bijective; see (6).

Proof (Proof of Theorem 3.1) From the exact sequence of k -vector spaces

$$0 \rightarrow (q^3 \cap J)/qJ \rightarrow J/qJ \rightarrow J/(q^3 \cap J) \rightarrow 0 \quad (8)$$

we get $\text{rank}_k(q^3 \cap J)/qJ = c - b$. Choose $g_1, \dots, g_b$ in J so that their residue classes form a basis of $J/(q^3 \cap J)$, then choose $g_{b+1}, \dots, g_c$ in $q^3 \cap J$ whose classes form a basis of $(q^3 \cap J)/qJ$. Now (8) shows that the images of $g_1, \dots, g_c$ form a basis of J/qJ over k , and so minimally generate J ; thus, $g_1, \dots, g_c$ is a Q -regular sequence.

The ring $Q' = Q/(g_{b+1}, \dots, g_c)$ is local with maximal ideal $q' = Q'q$, and the images of $g_1, \dots, g_b$ form a Q' -regular sequence that generates the ideal $J' = Q'J$.

Let $P \twoheadrightarrow Q$ be the surjective ring homomorphism with regular local ring (P, n, k) , obtained from some arbitrarily chosen minimal Cohen presentation of Q . Due to the inclusion $J \subseteq q^2$ we have $J' \subseteq q'^2$, so the surjective ring homomorphisms

$$P \twoheadrightarrow Q \twoheadrightarrow Q' \twoheadrightarrow R \quad (9)$$

induce isomorphisms of k -vector spaces

$$n/n^2 \cong q/q^2 \cong q'/q'^2 \cong m/m^2. \quad (10)$$

Thus, setting $I = \text{Ker}(P \rightarrow R)$ we get a minimal Cohen presentation $R \cong P/I$.

Since $J = QI$, the homomorphisms in (9) induce surjective k -linear maps

$$I/nI \rightarrow J/qJ \rightarrow J'/q'J'. \quad (11)$$

Thus, we may choose $f_1, \dots, f_b$ in I so that f_h maps to g_h for $h = 1, \dots, b$, and $f_{b+1}, \dots, f_c$ in $n^3 \cap I$ so that f_h maps to g_h for $h = b+1, \dots, c$. Since the classes of $g_1, \dots, g_c$ form a basis of J/qJ , it follows from (11) that we can choose $f_{c+1}, \dots, f_r$ in I so that $\{f_1, \dots, f_r\}$ is a minimal generating set of I .

We claim that the images of $f_1, \dots, f_b$ form a k -basis of $I/(n^3 \cap I)$. They are linearly independent because their images in $J/(q^3 \cap J)$ form a basis, so it suffices to show that f_h is in n^3 for $h \geq b+1$. This is how f_h was chosen for $b+1 \leq h \leq c$. On the other hand, (10) gives $\text{rank}_k(n/n^2) = e$, whence $\text{rank}_k(n^2/n^3) = \binom{e+1}{2}$ because P is regular. Since $\text{rank}_k(q^2/q^3) = \binom{e+1}{2}$ holds by hypothesis, the surjection $n^2/n^3 \twoheadrightarrow q^2/q^3$ is bijective. This gives $f_h \in n^3$ for $c+1 \leq h \leq r$, as desired.

Choose $\{t_1, \dots, t_e\}$ to minimally generate n . As $f_1, \dots, f_r$ are in n^2 , we have

$$f_h = \sum_{1 \leq i \leq j \leq e} a_{ij}^h t_i t_j \quad \text{for } h = 1, \dots, r \quad (12)$$

with $a_{ij}^h \in P$. Let $\bar{a}$ denote the image in k of $a \in R$. We then have

$$\bar{a}_{ij}^h = 0 \quad \text{for } b+1 \leq h \leq r. \quad (13)$$

$$\text{rank}_k(\bar{a}_{ij}^h)_{1 \leq i \leq j \leq e}^{1 \leq h \leq b} = b. \quad (14)$$

Indeed, the set $\{t_i t_j\}_{1 \leq i \leq j \leq e}$ maps to a basis of $\mathfrak{n}^2/\mathfrak{n}^3$, so $f_{b+1}, \dots, f_r \in \mathfrak{n}^3$ implies (13). The linear independence of the images of $f_1, \dots, f_b$ in $I/(\mathfrak{n}^3 \cap I)$ and the linear maps $I/(\mathfrak{n}^3 \cap I) \cong (I + \mathfrak{n}^3)/\mathfrak{n}^3 \subseteq \mathfrak{n}^2/\mathfrak{n}^3$ imply (14).

The ring homomorphisms in (9) induce a commutative diagram

$$\begin{array}{ccccccc}
 0 & \rightarrow & \Sigma^{-2} \operatorname{Hom}_{Q'}(J', k) & \longrightarrow & \pi^*(R) & \longrightarrow & \pi^*(Q') \rightarrow 0 \\
 & & \downarrow \gamma' & & \parallel & & \downarrow \\
 0 & \rightarrow & \Sigma^{-2} \operatorname{Hom}_Q(J, k) & \xrightarrow{\zeta} & \pi^*(R) & \longrightarrow & \pi^*(Q) \rightarrow 0 \\
 & & \downarrow \gamma & & \parallel & & \downarrow \\
 0 & \longrightarrow & \pi^{\geq 2}(R) & \longrightarrow & \pi^*(R) & \longrightarrow & \pi^*(P) \rightarrow 0
 \end{array}$$

of graded Lie algebras; see Sect. 3.2. The rows are exact, by Sect. 3.2. Due to the commutativity of the diagram, the maps in the right-hand column are surjective and those in the left-hand one are injective.

Let $\{\phi_1, \dots, \phi_r\}$ be the basis of $\pi^2(R)$, dual to the basis of $I/\mathfrak{n}I$ defined by $\{f_1, \dots, f_r\}$; see (7). The maps γ and γ' are the k -duals of those in (11). The diagram then shows that $\{\phi_1, \dots, \phi_b\}$ is a basis of $\mathfrak{Z}(\gamma\gamma')$, that $\{\phi_1, \dots, \phi_c\}$ is one for (γ) , and that (γ) is equal to $\mathcal{R}^2$.

Let $\{\vartheta_1, \dots, \vartheta_e\}$ be the basis $\pi^1(R)$, dual to the basis of $\mathfrak{n}/\mathfrak{n}^2$ defined by $\{t_1, \dots, t_e\}$; see (6). Sjödin [16, Theorem 4] (see also [4, 10.2.2]) yields equalities

$$\begin{aligned}
 [\vartheta_i, \vartheta_j] &= \sum_{h=1}^b \bar{a}_{ij}^h \phi_h \quad \text{for } 1 \leq i < j \leq e, \\
 \vartheta_i^2 &= \sum_{h=1}^b \bar{a}_{ii}^h \phi_h \quad \text{for } 1 \leq i \leq e,
 \end{aligned} \tag{15}$$

where the summations go up only to b , in view of (13).

Let $[\pi^1(R), \pi^1(R)]$ denote the subspace of $\pi^2(R)$, spanned by the commutators and the squares of the elements in $\pi^1(R)$. Formulas (15) give

$$[\pi^1(R), \pi^1(R)] \subseteq (\phi_1, \dots, \phi_b).$$

Equality holds because the system of Eq. (15) can be solved for $\phi_1, \dots, \phi_b$, due to the condition in (14). All ϑ and ξ in $\pi^1(R)$ satisfy $[\vartheta, \xi] = (\vartheta + \xi)^2 - \vartheta^2 - \xi^2$, so the squares of the elements in $\pi^1(R)$ span $[\pi^1(R), \pi^1(R)]$ over k . Now pick $\tau_1, \dots, \tau_b$ in $\pi^1(R)$ so that $\{\tau_1^2, \dots, \tau_b^2\}$ is a basis of $\mathfrak{Z}(\gamma\gamma')$.

By setting $\chi_h = \tau_h^2$ for $h = 1, \dots, b$, then completing $\{\chi_1, \dots, \chi_b\}$ to a basis $\{\chi_1, \dots, \chi_c\}$ of $\mathcal{R}^2$, we obtain a basis of $\mathcal{R}^2$ that satisfies condition (3).

The maps $\alpha_h: \mathcal{N}^{\text{even}} \rightarrow \mathcal{N}^{\text{odd}}$ and $\beta_h: \mathcal{N}^{\text{odd}} \rightarrow \mathcal{N}^{\text{even}}$, induced by left multiplication by τ_h , are $\mathcal{R}$ -linear because $\mathcal{R}$ lies in the center of $\mathcal{E}$; see Sect. 3.1.

As $\tau_h^2 = \chi_h$, the composed maps $\beta_h \alpha_h$ and $\alpha_h \beta_h$ are induced by left multiplication by χ_h .

4 Graded Exterior Algebras with Two Generators

Let Λ be a graded exterior algebra on a two-dimensional vector space concentrated in degree 1. When $\{x_1, x_2\}$ is a basis of Λ_1 , we write $k\langle x_1, x_2 \rangle$ for Λ .

The representation theory of the graded algebra Λ can be deduced from Kronecker's classification of pairs of matrices, up to conjugacy. We start by recalling some general facts about the representation theory of graded modules.

4.1 Direct Sum Decompositions of Graded Modules

Let A be a graded k -algebra with $A_0 = k$ and $A_i = 0$ for $i < 0$ or $i > 0$.

We say that A has the *graded Krull-Remak-Schmidt property* if every finitely generated graded A -module is a direct sum of indecomposable graded A -modules, and the summands in any such sum are defined uniquely up to isomorphism. When this holds, the *multiplicity* of an indecomposable module X in a graded module N is the largest $s \in \mathbb{N}$ for which N contains a direct summand isomorphic to X^s .

If $\text{rank}_k A$ is finite, or if A is finitely generated and commutative, then A has the graded Krull-Remak-Schmidt property. Let γ^b denote the functor that forgets gradings. A graded A -module L is indecomposable if and only if the A^b -module L^b is indecomposable. If L and M are indecomposable graded modules with $M^b \cong L^b$, then $M \cong \Sigma^p L$ for some $p \in \mathbb{Z}$. These statements are proved by Gordon and Green [11, 3.2, 4.1] and by Auslander and Reiten [1, Propositions 8, 9], respectively.

4.2 Indecomposable Graded Λ -Modules

For $n \in \mathbb{N}$, let $X^{[n]}$ denote any one of the graded Λ -modules described below.

The vector space $A^{[n]}$ with basis $\{u_1, \dots, u_n, v_1, \dots, v_{n+1}\}$, where $u_i \in A^{[n]}_0$ and $v_j \in A^{[n]}_1$, and Λ acts by the formulas $x_1 v_i = 0 = x_2 v_i$ and

$$x_1 u_i = v_i \text{ for } 1 \leq i \leq n \quad \text{and} \quad x_2 u_i = -v_{i+1} \text{ for } 1 \leq i \leq n.$$

The vector space $B^{[n]}$ with basis $\{u_1, \dots, u_n, v_1, \dots, v_{n-1}\}$, where $u_i \in B^{[n]}_0$ and $v_j \in B^{[n]}_1$, and Λ acts by the formulas $x_1 v_i = 0 = x_2 v_i$ and

$$x_1 u_i = \begin{cases} v_i & \text{for } 1 \leq i \leq n-1, \\ 0 & \text{for } i = n, \end{cases} \quad \text{and} \quad x_2 u_i = \begin{cases} 0 & \text{for } i = 1, \\ -v_{i-1} & \text{for } 2 \leq i \leq n. \end{cases}$$

The vector space $C^{[n]}$ with basis $\{u_1, \dots, u_n, v_1, \dots, v_n\}$, where $u_i \in C^{[n]}_0$ and $v_j \in C^{[n]}_1$, and Λ acts by the formulas $x_1 v_i = 0 = x_2 v_i$ and

$$x_1 u_i = \begin{cases} v_{i+1} & \text{for } 1 \leq i \leq n-1, \\ 0 & \text{for } i = n, \end{cases} \quad \text{and} \quad x_2 u_i = -v_i \text{ for } 1 \leq i \leq n.$$

The vector spaces ${}_\lambda D^{[n]}$ for $\lambda \in k$, with bases $\{u_1, \dots, u_n, v_1, \dots, v_n\}$, where $u_i \in {}_\lambda D^{[n]}_0$ and $v_j \in {}_\lambda D^{[n]}_1$, and Λ acts by the formulas $x_1 v_i = 0 = x_2 v_i$ and

$$x_1 u_i = v_i \text{ for } 1 \leq i \leq n \quad \text{and} \quad x_2 u_i = \begin{cases} \lambda v_i - v_{i+1} & \text{for } 1 \leq i \leq n-1, \\ \lambda v_n & \text{for } i = n. \end{cases}$$

Notice that the subset $\{u_1, \dots, u_n\} \subset X_0^{[n]}$ minimally generates $X^{[n]}$, and that

$$B^{[1]} = k.$$

When k is algebraically closed, each indecomposable graded Λ -module is isomorphic to $\Sigma^p X^{[n]}$ for some uniquely determined integers p and n and some module X in $\{A, B, C, \{{}_\lambda D\}_{\lambda \in k}, \Lambda\}$. In view of Sect. 4.1, this follows from Kronecker's classification, for which the first modern proof may be the one by Dieudonné [9].

4.3 Indecomposable Graded $\mathcal{R}$ -Modules

Let $\mathcal{R}$ be a graded polynomial ring $k[\chi_1, \chi_2]$ with $\chi_i \in \mathcal{R}^2$.

For each $n \in \mathbb{N}$, let $\mathcal{X}_{[n]}$ denote one of the $\mathcal{R}$ -modules described below:

$$\begin{aligned} \mathcal{A}_{[n]} &= \Sigma^{2-2n} \operatorname{Hom}_k \left(\frac{\mathcal{R}}{(\chi_1, \chi_2)^n}, k \right), & \mathcal{B}_{[n]} &= \Sigma^{2n-2} (\chi_1, \chi_2)^{n-1}, \\ \mathcal{C}_{[n]} &= \Sigma^{2n} \frac{(\chi_1, \chi_2)^n}{(\chi_2)^n}, & {}^\lambda \mathcal{D}_{[n]} &= \Sigma^{2n} \frac{(\chi_1, \chi_2)^n}{(\chi_1 + \lambda \chi_2)^n} \quad \text{for } \lambda \in k. \end{aligned}$$

Notice that $\mathcal{X}_{[n]}$ is minimally generated by n elements in $\mathcal{X}_{[n]}^0$, and that

$$\mathcal{A}_{[1]} = k \quad \text{and} \quad \mathcal{B}_{[1]} = \mathcal{R}.$$

Now we can describe the cohomology of the indecomposable graded Λ -modules.

4.4 Cohomology of Graded Λ -Modules

Let $\{x_1, x_2\}$ be a k -basis of Λ_1 and $\{\chi_1, \chi_2\}$ the dual basis of $\mathcal{R}^2$. If N is a graded Λ -module, then $\text{Ext}_\Lambda^*(N, k)$ is a graded $\mathcal{R}$ -module, as explained in Sect. 2.3.

For every $n \geq 1$ there are isomorphisms of graded $\mathcal{R}$ -modules

$$\begin{aligned} \text{Ext}_\Lambda^*(A^{[n]}, k) &\cong \mathcal{A}_{[n]} \oplus \Sigma^{-(2n+1)} \mathcal{R}, & \text{Ext}_\Lambda^*(B^{[n]}, k) &\cong \mathcal{B}_{[n]}, \\ \text{Ext}_\Lambda^*(C^{[n]}, k) &\cong \mathcal{C}_{[n]}, & \text{Ext}_\Lambda^*(\lambda D^{[n]}, k) &\cong {}^\lambda \mathcal{D}_{[n]}, \\ \text{Ext}_\Lambda^*(\Lambda, k) &\cong k. \end{aligned}$$

These isomorphisms are proved in [5, §4], where cohomological notation is used and the names $L(n)$, $L(1-n)$, $M(n, \infty)$, and ${}_\lambda D^{[n]}$ for $n \geq 1$ and $\lambda \in k$ are used for the indecomposable non-free Λ -modules. The translation runs as follows: $L(n)$ stands for $\Sigma^{-1} A^{[n]}$, $L(1-n)$ for $B^{[n]}$, $M(n, \infty)$ for $C^{[n]}$, and $M(n, \lambda)$ for ${}_\lambda D^{[n]}$. Furthermore, by [5, 3.12] the graded $\mathcal{R}$ -module $\text{Ext}_\Lambda^*(X^{[n]}, k)$ is equal to the total cohomology of a complex of free graded $\mathcal{R}$ -modules constructed from $X^{[n]}$. The one associated to $L(n)$ is denoted by $\mathcal{L}^\bullet(-n)$ for $n \in \mathbb{Z}$, and the one associated to $M(n, \lambda)$ is denoted by $\mathcal{M}^\bullet(n, \lambda)$ for $n \in \mathbb{N}$ and $\lambda \in k \cup \{\infty\}$. The cohomology of these complexes is computed in [5, 4.5.2, 4.5.3], giving the results recorded above.

Recall that when $\mathcal{N}$ is a graded $\mathcal{R}$ -module, its associated prime ideals are homogeneous; $\text{Ass}_{\mathcal{R}}(\mathcal{N})$ denotes the set of associated primes of $\mathcal{R}$ associated to $\mathcal{N}$.

Lemma 4.1 *The $\mathcal{R}$ -modules $\mathcal{X}_{[n]}$ in 4.3 are indecomposable and satisfy*

$$\begin{aligned} \text{Ass}_{\mathcal{R}}(\mathcal{A}_{[n]}) &= \{\mathfrak{p}\} \text{ with } \mathfrak{p} = (\chi_1, \chi_2) & \text{and } (\mathcal{A}_{[n]})_{\mathfrak{p}} &\cong \text{Hom}_k(\mathcal{R}_{\mathfrak{p}}/\mathfrak{p}^n \mathcal{R}_{\mathfrak{p}}, k). \\ \text{Ass}_{\mathcal{R}}(\mathcal{B}_{[n]}) &= \{\mathfrak{p}\} \text{ with } \mathfrak{p} = (0) & \text{and } (\mathcal{B}_{[n]})_{\mathfrak{p}} &\cong \mathcal{R}_{\mathfrak{p}}. \\ \text{Ass}_{\mathcal{R}}(\mathcal{C}_{[n]}) &= \{\mathfrak{p}\} \text{ with } \mathfrak{p} = (\chi_2) & \text{and } (\mathcal{C}_{[n]})_{\mathfrak{p}} &\cong \mathcal{R}_{\mathfrak{p}}/\mathfrak{p}^n \mathcal{R}_{\mathfrak{p}}. \\ \text{Ass}_{\mathcal{R}}({}^\lambda \mathcal{D}_{[n]}) &= \{\mathfrak{p}\} \text{ with } \mathfrak{p} = (\chi_1 + \lambda \chi_2) & \text{and } ({}^\lambda \mathcal{D}_{[n]})_{\mathfrak{p}} &\cong \mathcal{R}_{\mathfrak{p}}/\mathfrak{p}^n \mathcal{R}_{\mathfrak{p}}. \end{aligned}$$

Proof The formulas for $\mathcal{A}_{[n]}$ hold because $(\chi_1, \chi_2)^n \mathcal{A}_{[n]} = 0$, and for $\mathcal{B}_{[n]}$ because its shift is an ideal. Setting $\mathcal{J}_i = (\chi_1^i \chi_2^{n-i}, \dots, \chi_2^n)$ we get inclusions of $\mathcal{R}$ -modules

$$0 \subset \mathcal{J}_1/(\chi_2^n) \subset \dots \subset \mathcal{J}_n/(\chi_2^n) = \mathcal{C}_{[n]}$$

with subquotients isomorphic to $\mathcal{R}/(\chi_2)$; the formulas for $\mathcal{C}_{[n]}$ follows. A similar argument, where χ_2 is replaced with $\chi_1 + \lambda \chi_2$, proves the formulas for ${}^\lambda \mathcal{D}_{[n]}$.

As $\text{Ass}_{\mathcal{R}}(\mathcal{X}_{[n]}) = \{\mathfrak{p}\}$, every non-zero submodule $\mathcal{Y}$ of $\mathcal{X}_{[n]}$ has $\text{Ass}_{\mathcal{R}}(\mathcal{Y}) = \{\mathfrak{p}\}$; in particular, $\mathcal{Y}_{\mathfrak{p}}$ is non-zero. Thus, a non-trivial decomposition of $\mathcal{X}_{[n]}$ over $\mathcal{R}$ would yield a non-trivial decomposition for $(\mathcal{X}_{[n]})_{\mathfrak{p}}$ over the local ring $\mathcal{R}_{\mathfrak{p}}$ with maximal ideal $\mathfrak{p} \mathcal{R}_{\mathfrak{p}}$. However, $(\mathcal{X}_{[n]})_{\mathfrak{p}}$ is indecomposable: For $\mathcal{X} \neq \mathcal{A}$ this holds because $(\mathcal{X}_{[n]})_{\mathfrak{p}}$ is cyclic. The module $(\mathcal{A}_{[n]})_{\mathfrak{p}}$ has finite k -rank, so it is isomorphic

to its double dual over k . Thus, it is enough to show that $\text{Hom}_k((\mathcal{A}_{[n]})_{\mathfrak{p}}, k)$ is indecomposable over $\mathcal{R}_{\mathfrak{p}}$; to see this, just note that it is isomorphic to $\mathcal{R}_{\mathfrak{p}}/\mathfrak{p}^n \mathcal{R}_{\mathfrak{p}}$.

5 Regular Sequences of Length Two

In this section $(Q, \mathfrak{q}, k)$ denotes a local ring with *algebraically closed residue* field k , and J an ideal generated by a Q -regular sequence g_1, g_2 contained in $\mathfrak{q}^2$.

Let R be the local ring Q/J and N be a finitely generated R -module.

5.1 Direct Sum Decompositions

Let $\mathbf{N}$ denote the graded module $\text{Tor}_*^Q(N, k)$ over the graded exterior algebra $\Lambda = \text{Tor}_*^Q(R, k) = k\langle x_1, x_2 \rangle$; see Sect. 2.4. When $\text{proj dim}_Q N$ is finite, we have

$$\mathbf{N} \cong \left(\bigoplus_{X \in \{A, B, C, \lambda D\}} \bigoplus_{(n, p) \in \mathbb{N} \times \mathbb{Z}} (\Sigma^p X^{[n]})^{\mu_X^{\mathbf{N}}(n, p)} \right) \oplus \bigoplus_{p \in \mathbb{Z}} (\Sigma^p \Lambda)^{\mu_{\Lambda}^{\mathbf{N}}(p)} \quad (16)$$

as graded Λ -modules with non-negative integers $\mu_X^{\mathbf{N}}(n, p)$ and $\mu_{\Lambda}^{\mathbf{N}}(p)$, and these integers do not depend on the choice of the decomposition; see Sects. 4.2 and 4.1. Set

$$\mu_X^{\mathbf{N}}(p) = \sum_n \mu_X^{\mathbf{N}}(n, p) \quad \text{and} \quad v_X^{\mathbf{N}}(p) = \sum_n \mu_X^{\mathbf{N}}(n, p)n. \quad (17)$$

$$\mu_X^{\mathbf{N}} = \sum_p (-1)^p \mu_X^{\mathbf{N}}(p) \quad \text{and} \quad v_{\Lambda}^{\mathbf{N}} = \sum_p (-1)^p v_{\Lambda}^{\mathbf{N}}(p)n. \quad (18)$$

Thus $\mu_X^{\mathbf{N}}(p)$ is the number of the indecomposable components of $\mathbf{N}$ isomorphic to $\Sigma^p X^{[n]}$, and $v_X^{\mathbf{N}}(p)$ is the minimal number of generators of their direct sum.

Our goal is to find restrictions on the numbers in (17) and (18), for various values of X and p . The equality in the next proposition was obtained in [5, 4.9].

Proposition 5.1 *The graded Λ -module $\mathbf{N} = \text{Tor}_*^Q(N, k)$ satisfies the equality*

$$\mu_A^{\mathbf{N}} = \mu_B^{\mathbf{N}}. \quad (19)$$

Proof For each Λ -module $\mathbf{N}$, set $\varepsilon(\mathbf{N}) = \sum_j (-1)^j \text{rank}_k \mathbf{N}_j$. We obtain

$$\begin{aligned} \varepsilon(\Sigma^p A^{[n]}) &= (-1)^{p+1}, \quad \varepsilon(\Sigma^p B^{[n]}) = (-1)^p, \quad \text{and} \quad \varepsilon(\Sigma^p \Lambda) = 0 \quad \text{for} \quad p \in \mathbb{Z}, \\ \varepsilon(\Sigma^p C^{[n]}) &= 0 = \varepsilon(\Sigma^p \lambda D^{[n]}) \quad \text{for} \quad (n, p) \in \mathbb{N} \times \mathbb{Z}, \end{aligned}$$

by counting degrees in the defining formulas in Sect. 4.2. Thus, we get

$$\varepsilon(\mathbf{N}) = \sum_{X,n,p} \mu_X^{\mathbf{N}}(n,p) \varepsilon(\Sigma^p X^{[n]}) = \sum_{n,p} (-1)^{p+1} \mu_A^{\mathbf{N}}(n,p) + \sum_{n,p} (-1)^p \mu_B^{\mathbf{N}}(n,p).$$

Recall now that $\varepsilon(\mathbf{N}) = 0$ holds, as the Q -module N has a non-zero annihilator.

5.2 Cohomology of Syzygy Modules

For each integer $l \geq 0$, we let $\Omega_R^l(N)$ denote the cokernel of the differential ∂_{l+1} in some minimal free resolution of N over R ; in particular, $\Omega_R^0(N) = N$. Tracking projective dimensions in a minimal R -free resolution of N gives

$$\text{proj dim}_Q \Omega_R^l(N) = \sup\{2, \text{proj dim}_Q N - l\}. \quad (20)$$

Assume $\text{proj dim}_Q N = 2$ holds. Iyengar [14, 2.1] shows that any minimal free resolution of N over Q has a structure of DG module over the Koszul complex E on g_1, g_2 ; see also [4, 2.2.5]. There are then isomorphisms of graded $\mathcal{R}$ -modules

$$\text{Ext}_R^*(\Omega_R^l(N), k) \cong \Sigma^l \text{Ext}_R^{\geq l}(N, k) \cong \Sigma^l \text{Ext}_\Lambda^{\geq l}(\mathbf{N}, k), \quad (21)$$

which come from the $\mathcal{R}$ -linearity of connecting maps and Sect. 2.4, respectively.

The next theorem is a sharper version of some results in [5, 4.1 and 4.7].

Theorem 5.2 *Let $L = \Omega_R^l(N)$ for some R -module N with $\text{proj dim}_Q N = 2$ and*

$$l \geq \max\{2\beta_0^Q(N), 2\beta_1^Q(N) - 2\beta_0^Q(N) - 1\}. \quad (22)$$

For all $n \geq 1$ and $\lambda \in k$ the following equalities hold:

$$\mu_A^{\mathbf{L}}(n,p) = \mu_\Lambda^{\mathbf{L}}(p) = 0 \quad \text{for } p \in \mathbb{Z}. \quad (23)$$

$$\mu_B^{\mathbf{L}}(n,p) = \mu_C^{\mathbf{L}}(n,p) = \mu_{\lambda D}^{\mathbf{L}}(n,p) = 0 \quad \text{for } p \neq 0, 1. \quad (24)$$

$$\mu_B^{\mathbf{L}}(0) = \mu_B^{\mathbf{L}}(1). \quad (25)$$

When $p = 0, 1$, there are equalities

$$\beta_{2u+p}^{\mathbf{L}}(\mathbf{L}) = \mu_B^{\mathbf{L}}(0)u + \left(v_B^{\mathbf{L}}(p) + v_C^{\mathbf{L}}(p) + \sum_{\lambda} v_{\lambda D}^{\mathbf{L}}(p) \right) \quad \text{for } u \geq 0. \quad (26)$$

Theorem 5.2 is proved after Lemma 5.3, for which we introduce a tracking device. Assume $\text{proj dim}_Q N = 2$ holds, and for $p = 0, 1$ set $\mathbf{N} = \text{Tor}_*^Q(N, k)$ and

$$a_p^{\mathbf{N}} = \begin{cases} \max\{n \in \mathbb{N} \mid \mu_{\mathbf{A}}^{\mathbf{N}}(n, p) \neq 0\} & \text{if } \mu_{\mathbf{A}}^{\mathbf{N}}(p) > 0, \\ 0 & \text{if } \mu_{\mathbf{A}}^{\mathbf{N}}(p) = 0. \end{cases} \quad (27)$$

These numbers satisfy the following bounds:

$$\begin{aligned} a_0^{\mathbf{N}} &\leq \text{rank}_k \mathbf{N}_0 = \beta_0^Q(N). \\ 1 + a_1^{\mathbf{N}} &\leq \text{rank}_k \mathbf{N}_2 = \beta_2^Q(N) = \beta_1^Q(N) - \beta_0^Q(N). \end{aligned} \quad (28)$$

Lemma 5.3 *Under the hypotheses of Theorem 5.2, the following hold:*

$$\mu_{\mathbf{A}}^{\mathbf{L}}(n, p) = 0 \quad \text{for } n \geq 1 \quad \text{if } p \neq 0, 1 \quad \text{or } l \geq \max\{2a_p^{\mathbf{N}} + p - 1\}_{p=0,1}. \quad (29)$$

$$\mu_{\mathbf{B}}^{\mathbf{L}}(1, p) = 0 \quad \text{if } p \neq 0, 1, 2. \quad (30)$$

$$\mu_{\mathbf{B}}^{\mathbf{L}}(1, 2) = 0 \quad \text{if } l \geq \max\{2a_p^{\mathbf{N}} + p\}_{p=0,1}. \quad (31)$$

$$\mu_{\mathbf{B}}^{\mathbf{L}}(n, p) = 0 \quad \text{for } n \geq 2 \quad \text{if } p \neq 0, 1. \quad (32)$$

$$\mu_{\mathbf{C}}^{\mathbf{L}}(n, p) = 0 \quad \text{for } n \geq 1 \quad \text{if } p \neq 0, 1. \quad (33)$$

$$\mu_{\lambda \mathbf{D}}^{\mathbf{L}}(n, p) = 0 \quad \text{for } n \geq 1 \quad \text{if } p \neq 0, 1. \quad (34)$$

$$\mu_{\lambda}^{\mathbf{L}}(p) = 0 \quad \text{if } p \neq 0. \quad (35)$$

Proof Since $\text{proj dim}_Q L \leq 2$, see (20), every indecomposable direct summand $\mathbf{X}$ of $\mathbf{L}$ has $\mathbf{X}_i = 0$ for $i \neq 0, 1, 2$. By consulting the list in Sect. 4.2, we see that $\mathbf{X}$ must be isomorphic to Λ , to $\Sigma^p \mathbf{X}^{[n]}$ for some $\mathbf{X} \in \{\mathbf{A}, \mathbf{B}, \mathbf{C}, {}_{\lambda} \mathbf{D}\}$ with $n \geq 1$ and $p \in \{0, 1\}$, or to $\Sigma^2 \mathbf{B}^{[1]}$. This proves the assertions that do not depend on l .

Next we prove (35). Assume, by way of contradiction, that $\mathbf{L}$ has a direct summand $\mathbf{K}$ isomorphic to Λ . As $N \neq 0$ implies $l \geq 2$, we have an exact sequence

$$0 \rightarrow L \rightarrow F \rightarrow M \rightarrow 0 \quad (36)$$

of R -modules, where $M = \Omega_R^{l-1}(N)$ and $F \rightarrow M$ is a projective cover. It induces a homomorphism of graded Λ -modules $\iota: \mathbf{L} \rightarrow \mathbf{F}$, where $\mathbf{F} = \text{Tor}_*^Q(F, k)$. Since the map $F \otimes_Q k \rightarrow M \otimes_Q k$ is bijective, the homology exact sequence of $\text{Tor}_*^Q(?, k)$ defined by (36) implies $\iota_0(\mathbf{L}_0) = 0$, hence $\iota(\mathbf{K}) = 0$, and thus $\text{Ker}(\iota_2) \supseteq \mathbf{K}_2 \neq 0$. On the other hand, we have $\text{Tor}_3^Q(M, k) = 0$, so the same homology exact sequence shows that ι_2 is injective. This contradiction proves that $\mu_{\lambda}^{\mathbf{L}}(p) = 0$ holds.

Assume that we have $\mu_{\Lambda}^{\mathbf{L}}(n, p) \neq 0$ for some $p \notin \{0, 1\}$, or $\mu_{\mathbf{B}}^{\mathbf{L}}(1, 2) \neq 0$. From Sect. 4.4, we see that the $\mathcal{R}$ -module $\text{Ext}_{\mathcal{R}}^*(L, k)$ contains a direct summand isomorphic to $\Sigma^{-r}\mathcal{R}$ with $r = 2n + 1 + p$ or $r = 2$, respectively. This means that $\text{Ext}_{\Lambda}^*(\mathbf{N}, k)$ contains a direct summand isomorphic to $\Sigma^{-r-l}\mathcal{R}$. We already know that $\mathbf{N}$ is a direct sum of indecomposable Λ -modules $\mathbf{X}$ with $\mathbf{X}_i = 0$ for $i \neq 0, 1, 2$. In view of Sect. 4.4, those $\mathbf{X}$ that contribute to $\text{Ext}_{\Lambda}^*(\mathbf{N}, k)$ a direct summand of the desired form are isomorphic to $\Sigma^q A^{[m]}$ for some $q \in \{0, 1\}$ with $2m + 1 + q = r + l$. Both Λ and $\mathcal{R}$ have the graded Krull-Remak-Schmidt property (see Sect. 4.1), so $\mathbf{N}$ has to contain a summand of one of those types. Accordingly, one of the following inequalities hold:

$$\begin{aligned} l &= 2m + 1 + q - r \leq 2m + q - 2 \leq 2a_q^{\mathbf{N}} + q - 2 \quad \text{or} \\ l &= 2m + 1 + q - 2 = 2m + q - 1 \leq 2a_q^{\mathbf{N}} + q - 1. \end{aligned}$$

Since $l \geq \max\{2a_q^{\mathbf{N}} + q - 1\}_{q=0,1}$ and $l \geq \max\{2a_q^{\mathbf{N}} + q\}_{q=0,1}$ hold by hypothesis, we have a contradiction that completes the proofs of (29) and (31).

Proof (Proof of Theorem 5.2) The hypothesis on l and the inequalities (28) yield

$$l \geq \max\{2\beta_0^Q(N), 2\beta_1^Q(N) - 2\beta_0^Q(N) - 1\} \geq \max\{2a_0^{\mathbf{N}}, 2a_1^{\mathbf{N}} + 1\}.$$

Lemma 5.3 shows that (23) and (24) hold, so $\mathbf{N}$ is a direct sum of modules isomorphic to $\Sigma^p \mathbf{X}^{[n]}$ with $\mathbf{X} \in \{\mathbf{B}, \mathbf{C}, {}_{\lambda}\mathbf{D}\}$, $n \geq 1$, and $p = 0, 1$. For $u \geq 0$ we get

$$\begin{aligned} \beta_{2u}^{\Lambda}(\mathbf{B}^{[n]}) &= u + n & \beta_{2u}^{\Lambda}(\mathbf{C}^{[n]}) &= n & \beta_{2u}^{\Lambda}({}_{\lambda}\mathbf{D}^{[n]}) &= n, \\ \beta_{2u+1}^{\Lambda}(\mathbf{B}^{[n]}) &= 0 & \beta_{2u+1}^{\Lambda}(\mathbf{C}^{[n]}) &= 0 & \beta_{2u+1}^{\Lambda}({}_{\lambda}\mathbf{D}^{[n]}) &= 0, \end{aligned}$$

from the isomorphisms in Sect. 4.4. Thus, we obtain

$$\begin{aligned} \beta_{2u+p}^{\Lambda}(\mathbf{L}) &= \sum_n \mu_{\mathbf{B}}^{\mathbf{L}}(n, p)(u + n) + \sum_n \mu_{\mathbf{C}}^{\mathbf{L}}(n, p)n + \sum_n \sum_{\lambda} \mu_{{}_{\lambda}\mathbf{D}}^{\mathbf{L}}(n, p)n \\ &= \mu_{\mathbf{B}}^{\mathbf{L}}(p)u + \left(v_{\mathbf{B}}^{\mathbf{L}}(p) + v_{\mathbf{C}}^{\mathbf{L}}(p) + \sum_{\lambda} v_{{}_{\lambda}\mathbf{D}}^{\mathbf{L}}(p) \right) \quad \text{for } p = 0, 1. \end{aligned}$$

In order to finish the proof of (26), it suffices to show that (25) holds. It does, because (24), (19), and (23) yield $\mu_{\mathbf{B}}^{\mathbf{L}}(0) - \mu_{\mathbf{B}}^{\mathbf{L}}(1) = \mu_{\mathbf{B}}^{\mathbf{L}} = \mu_{\Lambda}^{\mathbf{L}} = 0$.

Now we turn to results that depend on the existence of quadratic relations.

Recall that when $\mathcal{J}$ is a homogeneous ideal of $\mathcal{R}$, the elements of $\mathcal{N}$ annihilated by some power of $\mathcal{J}$ form a graded submodule, denoted by $H_{\mathcal{J}}^0(\mathcal{N})$.

Theorem 5.4 Assume that $J \not\subseteq \mathfrak{q}^3$ holds, and choose a basis $\{\chi_1, \chi_2\}$ of $\mathcal{R}^2$ such that $\chi_1 = \tau_1^2$ for some element τ_1 in $\mathcal{E}^1$; see Theorem 3.1.

Let N be an R -module with $\text{proj dim}_Q N = 2$.

The numbers a_p^N from (27) satisfy the inequalities

$$a_1^N - 1 \leq a_0^N \leq a_1^N + 1. \quad (37)$$

If $L = \Omega_R^l(N)$ for some integer l satisfying (22), then the following equalities hold for the graded Λ -module $\mathbf{L} = \text{Tor}_*^Q(L, k)$:

$$\mu_{\mathbf{C}}^{\mathbf{L}}(n, 0) = \mu_{\mathbf{C}}^{\mathbf{L}}(n, 1) \quad \text{for } n \geq 1. \quad (38)$$

$$\mu_{\lambda \mathbf{D}}^{\mathbf{L}}(n, 0) = \mu_{\lambda \mathbf{D}}^{\mathbf{L}}(n, 1) \quad \text{for } n \geq 1 \quad \text{when } \lambda \neq 0. \quad (39)$$

$$\mu_{0\mathbf{D}}^{\mathbf{L}}(n, 0) = \mu_{0\mathbf{D}}^{\mathbf{L}}(n, 1) \quad \text{for } n \geq 1 \quad \text{if } \text{rank}_k(J/\mathfrak{q}^3 \cap J) = 2. \quad (40)$$

When $p = 0, 1$, there are equalities

$$\beta_{2u+p}^{\Lambda}(\mathbf{L}) = \mu_{\mathbf{B}}^{\mathbf{L}}(0)u + (v_{\mathbf{B}}^{\mathbf{L}}(p) + v_{0\mathbf{D}}^{\mathbf{L}}(p)) \quad \text{for } u \geq 0. \quad (41)$$

Proof Set $\mathcal{N} = \text{Ext}_R^*(N, k)$ and $\mathcal{L} = \text{Ext}_R^*(L, k)$.

Let first $\mathfrak{p}$ denote the maximal ideal (χ_1, χ_2) of $\mathcal{R}$. Due to the isomorphisms in Sect. 4.4, Lemmas 4.1 and 5.3 yield isomorphisms of graded $\mathcal{R}$ -modules

$$H_{\mathfrak{p}}^0(\mathcal{N}^{\text{even}}) \cong \bigoplus_{n \geq 1} (\mathcal{A}_{[n]})^{\mu_{\mathbf{A}}^N(n, 0)} \quad \text{and} \quad H_{\mathfrak{p}}^0(\mathcal{N}^{\text{odd}}) \cong \bigoplus_{n \geq 1} (\Sigma^{-1} \mathcal{A}_{[n]})^{\mu_{\mathbf{A}}^N(n, 1)}.$$

We use them to identify the parties involved. In particular, we get

$$\max\{i \mid (H_{\mathfrak{p}}^0(\mathcal{N}^{\text{even}}))^i \neq 0\} = \max\{i \mid \mathcal{A}_{[n]}^i \neq 0\}_{\mu_{\mathbf{A}}^N(n, 0) \neq 0} = 2a_0^N - 2.$$

$$\max\{i \mid (H_{\mathfrak{p}}^0(\mathcal{N}^{\text{odd}}))^i \neq 0\} = \max\{i \mid \mathcal{A}_{[n]}^i \neq 0\}_{\mu_{\mathbf{A}}^N(n, 1) \neq 0} = 2a_1^N - 1.$$

From Theorem 3.1 we get, by functoriality, degree one homomorphisms

$$\alpha': H_{\mathfrak{p}}^0(\mathcal{N}^{\text{even}}) \rightarrow H_{\mathfrak{p}}^0(\mathcal{N}^{\text{odd}}) \quad \text{and} \quad \beta': H_{\mathfrak{p}}^0(\mathcal{N}^{\text{odd}}) \rightarrow H_{\mathfrak{p}}^0(\mathcal{N}^{\text{even}})$$

of graded $\mathcal{R}$ -modules, such that $\alpha' \beta'$ and $\beta' \alpha'$ are given by multiplication with χ_1 .

It is evident that the first inequality in (37) holds when $a_1^N \leq 1$, and that the second inequality holds when $a_0^N \leq 1$. When $a_1^N = n \geq 2$, the inclusions

$$\tau_1(H_{\mathfrak{p}}^0(\mathcal{N}^{\text{even}})^{2n-4}) \supseteq \chi_1((\Sigma^{-1} \mathcal{A}_{[n]})^{2n-3}) = (\Sigma^{-1} \mathcal{A}_{[n]})^{2n-1} = \mathcal{A}_{[n]}^{2n-2} \neq 0$$

give $H_{\mathfrak{p}}^0(\mathcal{N}^{\text{even}})^{2n-4} \neq 0$, hence $2a_1^N - 4 \leq 2a_0^N - 2$. When $a_0^N = n \geq 2$, from

$$\tau_1(H_{\mathfrak{p}}^0(\mathcal{N}^{\text{odd}})^{2n-3}) \supseteq \chi_1(\mathcal{A}_{[n]}^{2n-4}) = \mathcal{A}_{[n]}^{2n-2} \neq 0$$

we get $H_p^0(\mathcal{N}^{\text{odd}})^{2n-3} \neq 0$, hence $2a_0^N - 3 \leq 2a_1^N - 1$. Now (37) has been proved.

Let $(X^{[n]}, \mathfrak{p})$ now denote $(C^{[n]}, (\chi_2))$ or $(\lambda D^{[n]}, (\chi_1 + \lambda \chi_2))$. As above, we get

$$\begin{aligned} (\mathcal{L}^{\text{even}})_p &\cong \bigoplus_{n \geq 1} (\mathcal{R}_p / \mathfrak{p}^n \mathcal{R}_p)^{\mu_X^{\perp}(n,0)} \oplus (\mathcal{R}_p)^{\sum_{n \geq 1} (\mu_A^{\perp}(n,1) + \mu_B^{\perp}(n,0) + \mu_B^{\perp}(n,2))} \\ (\mathcal{L}^{\text{odd}})_p &\cong \bigoplus_{n \geq 1} (\mathcal{R}_p / \mathfrak{p}^n \mathcal{R}_p)^{\mu_X^{\perp}(n,1)} \oplus (\mathcal{R}_p)^{\sum_{n \geq 1} (\mu_A^{\perp}(0,n) + \mu_B^{\perp}(n,1))} \end{aligned}$$

as $\mathcal{R}_p$ -modules, and homomorphisms of $\mathcal{R}_p$ -modules

$$\alpha_p \cdot (\mathcal{L}^{\text{even}})_p \rightarrow (\mathcal{L}^{\text{odd}})_p \quad \text{and} \quad \beta_p \cdot (\mathcal{L}^{\text{odd}})_p \rightarrow (\mathcal{L}^{\text{even}})_p$$

such that $\alpha_p \beta_p$ and $\beta_p \alpha_p$ are given by multiplication with χ_1 . Since χ_1 is invertible in $\mathcal{R}_p$ when $\lambda \neq 0$, we get $(\mathcal{L}^{\text{even}})_p \cong (\mathcal{L}^{\text{odd}})_p$ as modules over the discrete valuation ring $\mathcal{R}_p$. The structure theorem for finitely generated modules over PIDs implies $\mu_X^{\perp}(n, 0) = \mu_X^{\perp}(n, 1)$ for $n \geq 1$. This proves (38) and (39). By feeding these equalities into (26), we obtain the expression for $\beta_{2u+p}^{\Lambda}(\mathbf{L})$ given in (41).

When $\text{rank}_k J / (\mathfrak{q}^3 \cap J) = 2$, by Theorem 3.1 we can also chose $\chi_2 = \tau_2^2$ for some $\tau_2 \in \mathcal{O}^1$, then apply the preceding argument to the pair $({}_0D^{[n]}, (\chi_1))$.

Remark 5.5 A reader skipping forward to the proof of the main theorem, in Sect. 8, will notice that formula (26) directly yields Case (b), while (41) implies that the assertion in Case (a) is equivalent to $\mu_B^{\perp}(0) = -2(\nu_B^{\perp} + \nu_{0D}^{\perp})$. The next two sections are devoted to proving the preceding equality, in the graded case.

6 Bigraded Exterior Algebras with Two Generators

In this section we assume that k is an algebraically closed field, and we refine the representation theory and homological algebra in Sect. 4, by accounting for the presence of an additional grading.

6.1 Direct Sum Decompositions of Bigraded Λ -Modules

Fix integers d_1 and d_2 that satisfy $1 \leq d_1 \leq d_2$ and set $d = d_2 - d_1$.

Let Λ be the exterior algebra $k\langle x_1, x_2 \rangle$, bigraded by $x_r \in \Lambda_{1,d_r}$ for $r = 1, 2$.

A *bigraded Λ -module* is a graded Λ -module equipped with a direct sum decomposition $\mathbf{L}_i = \bigoplus_{j \in \mathbb{Z}} \mathbf{L}_{i,j}$ satisfying $x_r \mathbf{N}_{i,j} \subseteq \mathbf{N}_{i+1,j+d_r}$.

For $(p, q) \in \mathbb{Z}^2$ the bigraded Λ -module $\Sigma^p \mathbf{N}(q)$ has $\Sigma^p \mathbf{N}(q)_{i,j} = \mathbf{N}_{i-p,j+q}$, and the product of x_r and $y \in \Sigma^p \mathbf{N}(q)_{i,j}$ is $(-1)^p x_r y \in \mathbf{N}_{i+1-p,j+q+d_r} = \Sigma^p \mathbf{N}(q)_{i+1,j+d_r}$.

Indecomposability of bigraded modules and the bigraded Krull-Remak-Schmidt property of algebras are defined by the obvious extensions of the notions from Sect. 4.1. The Gordon-Green results recalled there also extend; see [15, 9.6.6 and 9.6.7].

Lemma 6.1 *A complete set of indecomposable bigraded Λ -modules consists of*

$$\begin{aligned} \Sigma^p X^{[n]}(q) \quad & \text{with } (n, p, q) \in \mathbb{N} \times \mathbb{Z}^2 \quad \text{for } X \in \{A, B, C\}, \\ \Sigma^p_\lambda D^{[n]}(q) \quad & \text{with } (n, p, q) \in \mathbb{N} \times \mathbb{Z}^2 \quad \text{and } \begin{cases} \lambda \in k & \text{if } d_1 = d_2, \\ \lambda = 0 & \text{if } d_1 < d_2, \end{cases} \\ \Sigma^p \Lambda(q) \quad & \text{with } (p, q) \in \mathbb{Z}^2, \end{aligned} \quad (42)$$

where the modules listed above are obtained by bigrading those in Sect. 4.2 as follows:

$$\begin{aligned} u_i &\in A_{0,d(i-1)}^{[n]} \quad \text{for } 1 \leq i \leq n \quad \text{and} \quad v_i \in A_{1,d(i-1)+d_1}^{[n]} \quad \text{for } 1 \leq i \leq n+1. \\ u_i &\in B_{0,d(n-i)}^{[n]} \quad \text{for } 1 \leq i \leq n \quad \text{and} \quad v_i \in B_{1,d(n-i)+d_1}^{[n]} \quad \text{for } 1 \leq i \leq n-1. \\ u_i &\in C_{0,d(n-i)}^{[n]} \quad \text{for } 1 \leq i \leq n \quad \text{and} \quad v_i \in C_{1,d(n-i)+d_2}^{[n]} \quad \text{for } 1 \leq i \leq n. \\ u_i &\in {}_\lambda D_{0,d(i-1)}^{[n]} \quad \text{for } 1 \leq i \leq n \quad \text{and} \quad v_i \in {}_\lambda D_{1,d(i-1)+d_1}^{[n]} \quad \text{for } 1 \leq i \leq n. \end{aligned}$$

Proof It is easy to verify that the assigned bidegrees agree with the multiplications tables in Sect. 4.2. The only modules there that are not listed in the lemma occur when $d_1 < d_2$ and are isomorphic to $\Sigma^p_\lambda D^{[n]}(q)$ for some $\lambda \neq 0$. We finish the proof by showing that $D = \Sigma^p_\lambda D^{[n]}(q)$ admits no bigrading when $d_1 < d_2$.

Assume the contrary and set $j = \min\{j' \in \mathbb{Z} \mid D_{p,j'} \neq 0\}$. Multiplication by x_1 yields a bijective k -linear map $D_{p,*} \rightarrow D_{p+1,*}$, so $\min\{j' \in \mathbb{Z} \mid D_{p+1,j'} \neq 0\} = d_1 + j$. As $\lambda \neq 0$ multiplication by x_2 also is bijective, so $\min\{j' \in \mathbb{Z} \mid D_{p+1,j'} \neq 0\} = d_2 + j$. This is impossible because $d_1 \neq d_2$, so we have the desired contradiction.

Next we review the bigradings induced on cohomology.

6.2 Cohomology Operators for Bigraded Λ -Modules

Let N be a bigraded Λ -module such that $\text{rank}_k N$ is finite.

For each $i \in \mathbb{Z}$ the k -space $\mathcal{N}^i = \text{Ext}_\Lambda^i(N, k)$ decomposes as $\mathcal{N}^i = \bigoplus_{j \in \mathbb{Z}} \mathcal{N}^{ij}$; the index j is induced by the second gradings of the modules in the resolution.

Let $\mathcal{R}$ be the polynomial ring $k[\chi_1, \chi_2]$ with $\chi_r \in \mathcal{R}^{2,d_r}$ for $r = 1, 2$.

Composition products preserve bigradings: $\mathcal{R}^{ij} \mathcal{N}^{i'j'} \subseteq \mathcal{N}^{i+i', j+j'}$. They turn $\mathcal{R}$ into a bigraded k -algebra and $\mathcal{N}$ into a bigraded $\mathcal{R}$ -module.

For every $(p, q) \in \mathbb{Z}^2$ the bigraded $\mathcal{R}$ -module $\Sigma^p \mathcal{N}(q)$ has $\Sigma^p \mathcal{N}(q)^{i,j} = \mathcal{N}^{i+p, j+q}$, and the product of $\chi \in \mathcal{R}^{i,j}$ and $v \in (\Sigma^p \mathcal{N}(q))^{i',j'}$ equals χv . For each pair (p, q) there is a natural isomorphism of bigraded $\mathcal{R}$ -modules

$$\mathrm{Ext}_{\Lambda}^*(\Sigma^p \mathbf{N}(q), k) \cong \Sigma^{-p} \mathrm{Ext}_{\Lambda}^*(\mathbf{N}, k)(-q). \quad (43)$$

6.3 Indecomposable Bigraded $\mathcal{R}$ -Modules

For $n \in \mathbb{N}$ and admissible λ , see (42), define bigraded $\mathcal{R}$ -modules

$$\begin{aligned} \mathcal{A}_{[n]} &= \Sigma^{2-2n} \mathrm{Hom}_k \left(\frac{\mathcal{R}}{(\chi_1, \chi_2)^n}, k \right) (nd_2 - d_2), \quad \mathcal{B}_{[n]} = \Sigma^{2n-2} (\chi_1, \chi_2)^{n-1} (nd_1 - d_1), \\ \mathcal{C}_{[n]} &= \Sigma^{2n} \frac{(\chi_1, \chi_2)^n}{(\chi_2)^n} (nd_1), \quad {}^\lambda \mathcal{D}_{[n]} = \Sigma^{2n} \frac{(\chi_1, \chi_2)^n}{(\chi_1 + \lambda \chi_2)^n} (nd_1 + d). \end{aligned}$$

by endowing them with the bigradings induced from the bigrading of $\mathcal{R}$.

The bigraded $\mathcal{R}$ -modules $\mathcal{X}_{[n]}$ listed above are indecomposable and satisfy

$$\mathcal{X}_{[n]}^{-0,0} \cong k^n \quad \text{and} \quad \mathcal{X}_{[n]}^{-0,j} = 0 \quad \text{for } j < 0, \quad (44)$$

they are minimally generated by n elements in $\mathcal{X}_{[n]}^{-0,*}$. In particular, one has

$$\mathcal{A}_{[1]} = k \quad \text{and} \quad \mathcal{B}_{[1]} = \mathcal{R}.$$

6.4 Cohomology of Bigraded Λ -Modules

For each $n \geq 1$ and every admissible λ , see (42), there are isomorphisms

$$\begin{aligned} \mathrm{Ext}_{\Lambda}^*(\mathbf{A}^{[n]}, k) &\cong \mathcal{A}_{[n]} \oplus \Sigma^{-(2n+1)} \mathcal{R}(nd_1 + d), & \mathrm{Ext}_{\Lambda}^*(\mathbf{B}^{[n]}, k) &\cong \mathcal{B}_{[n]}, \\ \mathrm{Ext}_{\Lambda}^*(\mathbf{C}^{[n]}, k) &\cong \mathcal{C}_{[n]}, & \mathrm{Ext}_{\Lambda}^*(\lambda \mathbf{D}^{[n]}, k) &\cong {}^\lambda \mathcal{D}_{[n]}, \\ \mathrm{Ext}_{\Lambda}^*(\Lambda, k) &\cong k. \end{aligned}$$

of bigraded $\mathcal{R}$ -modules. Indeed, Sect. 4.4 shows that both sides of each formula are isomorphic up to some shift of internal degrees; that shift is identified from (44), since $\mathrm{Ext}_{\Lambda}^{0,j}(\mathbf{X}^{[n]}, k)$ is non-zero for $j = 0$ and is zero for $j < 0$.

7 Homogeneous Regular Sequences of Length Two

Throughout this section k denotes an algebraically closed field, Q a standard graded k -algebra, $\mathfrak{q}$ the ideal generated by Q_1 , and g_1, g_2 a regular sequence of forms. We set $d_h = \deg g_h$ for $h = 1, 2$ and assume $d_2 \geq d_1 \geq 2$.

Let R be the graded ring $Q/(g_1, g_2)$ and N a finitely generated graded R -module. When l is a non-negative integer, $\Omega_R^l(N)$ denotes the cokernel of the differential ∂_{l+1} in some minimal *graded* free resolution of N over R .

7.1 Direct Sum Decompositions

Set $\Lambda_{i,j} = \mathrm{Tor}_i^Q(R, k)_j$ and $\mathbf{N}_{i,j} = \mathrm{Tor}_i^Q(N, k)_j$, where the modules on the right come from resolutions of R and N consisting of graded free Q -modules and homomorphisms of degree zero. They yield natural direct sum decompositions

$$\mathrm{Tor}_i^Q(R, k) = \bigoplus_{j \in \mathbb{Z}} \Lambda_{i,j} \quad \text{and} \quad \mathrm{Tor}_i^Q(N, k) = \bigoplus_{j \in \mathbb{Z}} \mathbf{N}_{i,j}$$

that are compatible with homology products. The homology products of Sect. 5.1 are compatible with the bigradings and turn $\mathbf{N} = \mathrm{Tor}_*^Q(N, k)$ into a bigraded module over Λ , where $\Lambda = \mathrm{Tor}_*^Q(R, k)$. The graded analog of Sect. 4.2 identifies Λ and the bigraded exterior algebra $k\langle x_1, x_2 \rangle$ from Sect. 6.1, which has $x_r \in \Lambda_{1,d_r}$.

For $\mathbf{X} \in \{\mathbf{A}, \mathbf{B}, \mathbf{C}, {}_\lambda \mathbf{D}\}$ let $\mu_{\mathbf{X}}^{\mathbf{N}}(n, p; q)$ denote the multiplicities in $\mathbf{N}$ of the indecomposable bigraded Λ -module $\Sigma^p \mathbf{X}(q)$; let $\mu_{\Lambda}^{\mathbf{N}}(p; q)$ denote the multiplicity of $\Sigma^p \Lambda(q)$ in $\mathbf{N}$; see Lemma 6.1. In view of (42), λ may take the following values:

$$\lambda \in k \text{ if } d_1 = d_2 \quad \text{and} \quad \lambda = 0 \text{ if } d_1 < d_2. \quad (45)$$

Results transfer smoothly from the local context to the graded one:

7.2 Completions

Let $\widehat{Q}$ and $\widehat{R}$ denote the completions of Q and R at the ideal $\mathfrak{q}$, and set $\widehat{N} = \widehat{R} \otimes_R N$. Standard flatness arguments give $\mathrm{proj\,dim}_{\widehat{Q}} \widehat{N} = \mathrm{proj\,dim}_Q N$ and $\beta_i^{\widehat{R}}(\widehat{N}) = \beta_i^R(N)$, and yield canonical isomorphisms $\widehat{R} \otimes_R \Omega_R^l(N) \cong \Omega_{\widehat{R}}^l(\widehat{N})$ and

$$\bigoplus_{j \in \mathbb{Z}} \Lambda_{i,j} \cong \mathrm{Tor}_i^{\widehat{Q}}(\widehat{R}, k) \quad \text{and} \quad \bigoplus_{j \in \mathbb{Z}} \mathbf{N}_{i,j} \cong \mathrm{Tor}_i^{\widehat{Q}}(\widehat{N}, k).$$

The latter are compatible with homology products, and we use them to identify $\widehat{\text{Tor}}_*^Q(\widehat{R}, k)$ and $\widehat{\text{Tor}}_*^Q(\widehat{N}, k)$ and the graded objects underlying Λ and $\mathbf{N}$, respectively.

For $X \in \{A, B, C, {}_\lambda D\}$, the multiplicities in Sects. 5 and 7 are linked as follows:

$$\sum_q \mu_X^N(n, p; q) = \mu_X^N(n, p). \quad (46)$$

The theorems in this section uncover non-trivial relations among the invariants μ_X^N and ν_X^N , introduced in (18), and the numbers ρ_X^N defined by

$$\rho_X^N = \sum_{n,p,q} (-1)^p \mu_X^N(n, p; q) q. \quad (47)$$

This definition depends on the bigraded structure of $\text{Tor}_*^Q(N, k)$, so no local version exists for the relation in the next result.

Theorem 7.1 *If N is a finitely generated graded R -module such that $\text{proj dim}_Q N$ is finite and $N_j = 0$ for $j < 0$, then the following equality holds:*

$$\rho_B^N = \rho_A^N + d_1 \left(\nu_B^N + \sum_{\lambda} \nu_{\lambda D}^N \right) + d_2 (\nu_A^N + \nu_C^N). \quad (48)$$

Proof For a graded k -space V with each $\text{rank}_k V_j < \infty$ and $V_j = 0$ for $j < 0$, set

$$H_V(y) = \sum_{j \in \mathbb{N}_0} (\text{rank}_k V_j) y^j \in \mathbb{Z}[[y]].$$

If, in addition, V is bigraded and $V_{i,j} = 0$ for $i < 0$, then we also form

$$H_V(y, z) = \sum_{(i,j) \in \mathbb{N}_0^2} (\text{rank}_k V_{i,j}) y^i z^j \in \mathbb{Z}[[y, z]].$$

With $\dim$ denoting Krull dimension, the Hilbert-Serre Theorem gives an equality

$$H_N(y) = \frac{h_N(y)}{(1-y)^{\dim N}} \quad \text{for some } h_N(y) \in \mathbb{Z}[y] \quad \text{with } h_N(1) \neq 0$$

and a similar one for Q . A minimal Q -free resolution of N yields an exact sequence

$$0 \rightarrow Q \otimes_k \mathbf{N}_{s,*} \rightarrow \cdots \rightarrow Q \otimes_k \mathbf{N}_{1,*} \rightarrow Q \otimes_k \mathbf{N}_{0,*} \rightarrow N \rightarrow 0$$

of graded Q -modules. It provides the first equality in the string

$$\begin{aligned} H_N(y) &= \sum_{i=0}^s (-1)^i H_Q(y) \left(\sum_{j \in \mathbb{Z}} \text{rank}_k \mathbf{N}_{i,j} y^j \right) \\ &= \sum_{i=0}^s H_Q(y) \left(\sum_{j \in \mathbb{Z}} \text{rank}_k H_N(y, -1) \right) \\ &= \frac{h_Q(y)}{(1-y)^{\dim Q}} H_N(y, -1). \end{aligned}$$

Setting $m = \dim Q - \dim N$, we obtain in $\mathbb{Z}[y]$ an equality

$$H_N(y, -1) h_Q(y) = (1-y)^m h_N(y).$$

As $h_Q(1) \neq 0$, we see that $H_N(y, -1)$ is divisible by $(1-y)^m$, and hence by $(1-y)^2$ because $(g_1, g_2)N = 0$ implies $m \geq 2$. Thus, writing f' for $\partial f / \partial y$ we obtain

$$H_N(1, -1) = 0 \quad \text{and} \quad H'_N(1, -1) = 0. \quad (49)$$

By expressing N as a direct sum of indecomposable modules, we get an equality

$$H_N(y, z) = A(y, z) + B(y, z) + C(y, z) + \sum_{\lambda} \lambda D(y, z) + E(y, z) \quad (50)$$

in $\mathbb{Z}[y, z]$, where the summands have the following values:

$$X(y, z) = \begin{cases} \sum_{n,p,q} \mu_X^N(n, p; q) y^q H_{X^{[n]}}(y, z) z^p & \text{for } X = A, B, C, \lambda D, \\ \sum_{p,q} \mu_{\Lambda}^N(p; q) y^q H_{\Lambda}(y, z) z^p & \text{for } X = E. \end{cases} \quad (51)$$

In view of Lemma 6.1, the Hilbert series in (51) are given by the formulas

$$\begin{aligned} H_{A^{[n]}}(y, z) &= g_n(y) + y^{d_1} g_{n+1}(y) z, & H_{B^{[n]}}(y, z) &= g_n(y) + y^{d_2} g_{n-1}(y) z, \\ H_{C^{[n]}}(y, z) &= g_n(y) (1 + y^{d_2} z), & H_{\lambda D^{[n]}}(y, z) &= g_n(y) (1 + y^{d_1} z), \\ H_{\Lambda}(y, z) &= (1 + y^{d_1} z) (1 + y^{d_2} z), \end{aligned} \quad (52)$$

where we have set $g_n(y) = 1 + y^{d_2-d_1} + \dots + y^{(d_2-d_1)(n-1)}$. Note the equalities

$$g_n(1) = n \quad \text{and} \quad g'_n(1) = (d_2 - d_1) n(n-1)/2.$$

Differentiating the equalities (52) and evaluating the results at $(1, -1)$ yields

$$\begin{aligned}
 H'_{\Sigma^p A^{[n]}(q)}(1, -1) &= (-1)^{p+1}(nd_2 + d_1 + q) . \\
 H'_{\Sigma^p B^{[n]}(q)}(1, -1) &= (-1)^{p+1}(nd_1 - d_1 - q) . \\
 H'_{\Sigma^p C^{[n]}(q)}(1, -1) &= (-1)^{p+1}nd_2 . \\
 H'_{\Sigma^p_{\lambda} D^{[n]}(q)}(1, -1) &= (-1)^{p+1}nd_1 . \\
 H'_{\Sigma^p \Lambda(q)}(1, -1) &= 0 .
 \end{aligned} \tag{53}$$

By using first (50), (51) and (53), then (18) and (47), we obtain

$$\begin{aligned}
 -H'_N(1, -1) &= \sum_{n,p,q} (-1)^p \mu_A^N(n, p; q)(nd_2 + d_1 + q) \\
 &\quad + \sum_{n,p,q} (-1)^p \mu_B^N(n, p; q)(nd_1 - d_1 - q) \\
 &\quad + \sum_{n,p,q} (-1)^p \mu_C^N(n, p; q)(nd_2) \\
 &\quad + \sum_{\lambda} \sum_{n,p,q} (-1)^p \mu_{\lambda}^N(n, p; q)(nd_1) \\
 &= d_1 \left(\mu_A^N - \mu_B^N + v_B^N + \sum_{\lambda} v_{\lambda}^N \right) + d_2 (v_A^N + v_C^N) + \rho_A^N - \rho_B^N .
 \end{aligned}$$

We have $\mu_A^N = \mu_B^N$ by (19) and $H'_N(1, -1) = 0$ by (49), so (48) follows.

7.3 Composition Products

Let $\mathcal{E}$ be the graded k -algebra $\text{Ext}_R^*(k, k)$ and $\mathcal{N}$ its graded module $\text{Ext}_R^*(N, k)$; see Sect. 3.1. As in Sect. 7.1, resolutions of k and N by graded free R -modules and homomorphisms of degree zero define k -vector spaces $\mathcal{E}^{i,j} = \text{Ext}_R^i(k, k)^j$ and $\mathcal{N}^{i,j} = \text{Ext}_R^i(N, k)^j$, respectively. They yield natural decompositions

$$\mathcal{E}^i = \bigoplus_{j \in \mathbb{Z}} \mathcal{E}^{i,j} \quad \text{and} \quad \mathcal{N}^i = \bigoplus_{j \in \mathbb{Z}} \mathcal{N}^{i,j} \tag{54}$$

that are compatible with composition products. We review their properties.

It is clear that $\mathcal{E}^{0,*} = k$ and that $\mathcal{E}^{1,*}$ has a basis $\tau_1, \dots, \tau_e$ of elements in $\mathcal{E}^{1,1}$. The graded analog of Sect. 3.1 holds and yields a central bigraded polynomial subalgebra $\mathcal{R} = k[\chi_1, \chi_2]$ with $\chi_h \in \mathcal{E}^{2,d_h}$ for $h = 1, 2$. Theorem 3.1 also has a

graded analog, so we may assume $\chi_1 = \tau_1^2$ if $d_1 = 2$, and $\chi_h = \tau_h^2$ for $h = 1, 2$ if $d_2 = 2$.

The following graded analog of (21) holds: For each $(l, q) \in \mathbb{N}_0 \times \mathbb{Z}$ the bigraded Λ -module $\mathbf{N} = \text{Tor}_*^Q(N, k)$ yields an isomorphism of bigraded $\mathcal{B}$ -modules

$$\text{Ext}_R^*(\Omega^l(N)(-q), k) \cong \Sigma^l \text{Ext}_R^{\geq l}(N(-q), k) \cong \Sigma^l(\text{Ext}_\Lambda^{\geq l}(\mathbf{N}, k))(q). \quad (55)$$

Now we can use in the graded context results proved over local rings in Sect. 5.

Theorem 7.2 *Assume $\text{proj dim}_Q N = 2$ and let l be an integer satisfying*

$$l \geq \max\{2\beta_0^Q(N), 2(\beta_1^Q(N) - \beta_0^Q(N)) - 1\}.$$

If $d_1 = 2$, $L = \Omega^l(N)$, and $\mathbf{L} = \text{Tor}_^Q(L, k)$, then there is an isomorphism*

$$\bigoplus_{n,p,q} \mathbf{B}^{[n]}(q)^{\mu_{\mathbf{B}}^{\mathbf{L}}(n,p;q)} \cong \bigoplus_{i=1}^{\mu_{\mathbf{B}}^{\mathbf{L}}(0)} \mathbf{B}^{[n_i]}(q_i) \oplus \bigoplus_{i=1}^{\mu_{\mathbf{B}}^{\mathbf{L}}(0)} \Sigma \mathbf{B}^{[n'_i]}(q_i + 1) \quad (56)$$

of bigraded Λ -modules and there is an equality

$$\mu_{\mathbf{B}}^{\mathbf{L}}(0) = -2(v_{\mathbf{B}}^{\mathbf{L}} + v_{0\mathbf{D}}^{\mathbf{L}}). \quad (57)$$

Proof Set $s = \mu_{\mathbf{B}}^{\mathbf{L}}(0)$. We then have $\mu_{\mathbf{B}}^{\mathbf{L}}(1) = s$, by (25), so in view of the bigraded Krull-Remak-Schmidt property there is an isomorphism of Λ -modules

$$\bigoplus_{n,p,q} \mathbf{B}^{[n]}(q)^{\mu_{\mathbf{B}}^{\mathbf{L}}(n,p;q)} \cong \bigoplus_{j=1}^s \mathbf{B}^{[n_j]}(q_j) \oplus \bigoplus_{i=1}^s \Sigma \mathbf{B}^{[n'_i]}(q'_i) \quad (58)$$

with $n_j \neq 0 \neq n'_i$ for $1 \leq i, j \leq s$. We assume, as we may, the following inequalities:

$$q_1 \leq \cdots \leq q_s \quad \text{and} \quad q'_1 \leq \cdots \leq q'_s.$$

We will prove (56) by showing that $q'_i = q_i + 1$ holds for $i = 1, \dots, s$.

Choose χ_1 and χ_2 with $\chi_1 = \tau_1^2$ for some $\tau_1 \in \mathcal{E}^{1,1}$ (see Sect. 7.3) and set

$$\mathcal{B} = \mathcal{L} / (\text{H}_{(\chi_2)}^0(\mathcal{L}) + \sum_{\lambda} \text{H}_{(\chi_1 + \lambda \chi_2)}^0(\mathcal{L})).$$

The local cohomology modules in the preceding formulas are bigraded submodules of $\mathcal{L}$. In view of the isomorphisms in Sects. 6.2–6.4, formula (58) gives

$$\mathcal{B} \cong \bigoplus_{j=1}^s \mathcal{B}_{[n_j]}(-q_j) \oplus \bigoplus_{i=1}^s \Sigma^{-1} \mathcal{B}_{[n'_i]}(-q'_i).$$

Thus, we may identify the following bigraded $\mathcal{R}$ -modules:

$$\mathcal{B}^{\text{even},*} = \bigoplus_{j=1}^s \mathcal{B}_{[n_j]}(-q_j) \quad \text{and} \quad \mathcal{B}^{\text{odd},*} = \bigoplus_{i=1}^s \Sigma^{-1} \mathcal{B}_{[n'_i]}(-q'_i).$$

Multiplication by τ_1 induces $\mathcal{R}$ -linear maps

$$\mathcal{B}^{\text{even},*} \xrightarrow{\alpha} \mathcal{B}^{\text{odd},*} \xrightarrow{\beta} \mathcal{B}^{\text{even},*} \xrightarrow{\alpha} \mathcal{B}^{\text{odd},*}$$

of bidegree $(1, 1)$. The composed maps $\beta\alpha$ and $\alpha\beta$ are induced by multiplication by χ_1 , so they are injective because $\mathcal{B}_{[n]}$ is torsion-free; thus, α and β are injective.

Next we prove by induction that for each integer r with $0 \leq r < s$ we have

$$q'_1 = q_1 + 1, \dots, q'_r = q_r + 1.$$

The hypothesis is vacuous for $r = 0$, so we may suppose that the assertion has been proved for some $r \geq 0$. Assume, by way of contradiction, that $q_{r+1} + 1 < q'_{r+1}$ holds. For $1 \leq j \leq r+1$ and $r+1 \leq i \leq s$ we then have inequalities

$$q_j + 1 \leq q_{r+1} + 1 < q'_{r+1} \leq q'_i.$$

As $(\mathcal{B}_{[n_j]}(-q_j))^{0,q_j} \neq 0 = (\Sigma^{-1} \mathcal{B}_{[n'_i]}(-q'_i))^{1,q_j+1}$, see (44), the composed map

$$\alpha_{i,j}: \mathcal{B}_{[n_j]}(-q_j) \hookrightarrow \mathcal{B}^{\text{even}} \xrightarrow{\alpha} \mathcal{B}^{\text{odd}} \twoheadrightarrow \Sigma^{-1} \mathcal{B}_{[n'_i]}(-q'_i)$$

of bigraded $\mathcal{R}$ -modules satisfies $\text{Ker}(\alpha_{i,j})^{0,q_j} \neq 0$ for $1 \leq j \leq r+1 \leq i \leq s$.

Let F denote the field of fractions of $\mathcal{R}$. From Sect. 6.3 we have inclusions

$$\Sigma^{2-2n} \mathcal{B}_{[n]}(2-2n) = (\chi_1, \chi_2)^{n-1} \subseteq \mathcal{R}.$$

They induce canonical isomorphisms $F \otimes_{\mathcal{R}} \mathcal{B}_{[n]} \cong F$. Treating them as identifications leads to equalities $F \otimes_{\mathcal{R}} \mathcal{B}^{\text{even}} = F^s = F \otimes_{\mathcal{R}} \mathcal{B}^{\text{odd}}$. Accordingly, $F \otimes_{\mathcal{R}} \alpha$ is given by an $s \times s$ matrix (a_{ij}) with entries in F , and then $F \otimes_{\mathcal{R}} \alpha_{i,j}: F \rightarrow F$ is multiplication with $a_{i,j} \in F$. Thus, $\text{Ker}(F \otimes_{\mathcal{R}} \alpha_{i,j}) \neq 0$ implies $a_{i,j} = 0$, hence

$$F \otimes_{\mathcal{R}} \alpha = \begin{bmatrix} a_{1,1} & \cdots & a_{1,r+1} & a_{1,r+2} & \cdots & a_{1,s} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{r,1} & \cdots & a_{r,r+1} & a_{r,r+2} & \cdots & a_{r,s} \\ 0 & \cdots & 0 & a_{r+1,r+2} & \cdots & a_{r+1,s} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{s,r+2} & \cdots & a_{s,s} \end{bmatrix}.$$

The first $r + 1$ columns of $F \otimes_{\mathcal{R}} \alpha$ are linearly dependent over F , so we have $\text{rank}_F(F \otimes_{\mathcal{R}} \alpha) < s$. This is impossible, because α is injective, and hence so is $F \otimes_{\mathcal{R}} \alpha$. We obtained a contradiction, which shows that $q'_{r+1} \leq q_{r+1} + 1$ holds.

Set $L' = \Omega_R^{l+1}(N)$, and form the bigraded $\mathcal{R}$ -modules $\mathcal{L}' = \text{Ext}_R^*(L', k)$ and

$$\mathcal{B}' = \mathcal{L}' / (\text{H}_{(\chi_2)}^0 \mathcal{L}' + \sum_{\lambda} \text{H}_{(\chi_1 + \lambda \chi_2)}^0 \mathcal{L}').$$

In view of (55), we have isomorphisms of bigraded $\mathcal{R}$ -modules

$$\mathcal{B}'^{\text{even},*} \cong \bigoplus_{i=1}^s \mathcal{B}'_{[n'_i]}(-q'_i) \quad \text{and} \quad \mathcal{B}'^{\text{odd},*} \cong \bigoplus_{j=1}^s \Sigma^{-1} \mathcal{B}'_{[n_j+1]}(-q_j - 2).$$

The hypotheses of the theorem apply to L' . The induction assumption yields

$$q_1 + 2 = q'_1 + 1, \dots, q_r + 2 = q'_r + 1,$$

so the preceding stretch of the argument gives $q_{r+1} + 2 \leq q'_{r+1} + 1$. It follows that $q_{r+1} + 1 \leq q'_{r+1}$ holds, so we have proved the equality $q'_{r+1} = q_{r+1} + 1$. The induction step is now complete, so $q'_i = q_i + 1$ holds for $i = 1, \dots, s$.

We have proved the isomorphism (56). It gives the last equality in the string

$$\mu_{\mathcal{B}}^{\text{L}}(0) = s = \sum_{i=1}^s (q_i + 1) - \sum_{i=1}^s q_i = - \sum_{n,p,q} (-1)^p \mu_{\mathcal{B}}^{\text{L}}(n, p; q) q.$$

The last sum above equals $\rho_{\mathcal{B}}^{\text{L}}$; see (47). Formula (48) gives another expression for that number. It involves $\rho_{\mathcal{A}}^{\text{L}}$, $\nu_{\mathcal{A}}^{\text{L}}$, $\nu_{\mathcal{C}}^{\text{L}}$, and $\nu_{\lambda \mathcal{D}}^{\text{L}}$ with $\lambda \neq 0$; the first two terms are zero by (23), the other two are zero by (38) and (39), so (48) yields

$$- \sum_{n,p,q} (-1)^p \mu_{\mathcal{B}}^{\text{L}}(n, p; q) q = -\rho_{\mathcal{B}}^{\text{L}} = -2(\nu_{\mathcal{B}}^{\text{L}} + \nu_{0\mathcal{D}}^{\text{L}}).$$

We obtain (57) by concatenating the equalities in the last two displays.

8 Proof of the Main Theorem

In this section we prove Theorem 1.1. For starters, we recall its hypotheses.

Let Q be a standard graded algebra with $\text{rank}_k Q_2 = \binom{\text{rank}_k Q_1}{2}$.

Let J be a homogeneous ideal minimally generated by r elements of degree at least two, and set $R = Q/J$. We assume $\text{proj dim}_Q R = 2$.

Let N be a non-zero finitely generated graded R -module, and set

$$\begin{aligned}\beta_i &= \beta_i^R(N) \quad \text{for } i \geq 0. \\ d &= \text{depth } R - \text{depth}_R N. \\ l &= d + \max\{2\beta_d, 2\beta_{d+1} - 2\beta_d - 1\}.\end{aligned}$$

We assume that $\text{proj dim}_Q N$ is finite and set $M = \Omega_R^d(N)$. We then have

$$\text{proj dim}_Q M = \text{depth } Q - \text{depth}_Q M = \text{depth } R + 2 - \text{depth}_R M = d + 2.$$

by the Auslander-Buchsbaum Equality. By (20), we have $\text{proj dim}_Q M = 2$.

Set $L = \Omega_R^l(N)$, and note that we have $\text{proj dim}_Q L = 2$.

If $\text{proj dim}_R N$ is finite, then it is equal to d by the Auslander-Buchsbaum Equality. Since $l \geq d + 2$, we have $\beta_i = 0$ for $i \geq l$, so Theorem 1.1 holds trivially.

Let $\bar{k}$ denote an algebraic closure of k . For each $i \in \mathbb{Z}$ there is an isomorphism $\text{Tor}_i^{R \otimes_k \bar{k}}(N \otimes_k \bar{k}, \bar{k}) \cong \text{Tor}_i^R(N, k) \otimes_k \bar{k}$, which gives $\beta_i^R(N) = \beta_i^{R \otimes_k \bar{k}}(N \otimes_k \bar{k})$. Thus, we may also assume that k is algebraically closed.

After these adjustments, we treat one by one the three cases of the theorem.

Lemma 8.1 (Case (a)) *If $r = 2 = \text{grade } J$ and $J_2 \neq 0$, then*

$$\beta_i = (\beta_{l+1} - \beta_l)(i - l) + \beta_l \quad \text{for } i \geq l.$$

Proof Set $L = \text{Tor}_*^R(L, k)$. For every $u \geq 0$, from formula (41) we get

$$\beta_{2u+1}^\Lambda(L) - \beta_{2u}^\Lambda(L) = -(\nu_B^L + \nu_{0D}^L).$$

The same references as above, along with (57), also give

$$\beta_{2u+2}^\Lambda(L) - \beta_{2u+1}^\Lambda(L) = \mu_B^L(0) + \nu_B^L + \nu_{0D}^L = -(\nu_B^L + \nu_{0D}^L).$$

Due to (55), we have $\beta_i^\Lambda(L) = \beta_i^R(L) = \beta_{l+i}^R(N)$ for $i \geq 0$, whence

$$\beta_{l+i+1}^R(N) - \beta_{l+i}^R(N) = -(\nu_B^L + \nu_{0D}^L) \quad \text{for } i \geq 0.$$

This is equivalent to the assertion in (a).

Lemma 8.2 (Case (b)) *If $r = 2 = \text{grade } J$ and $J_2 = 0$, then*

$$\begin{aligned}\beta_i &= \frac{\beta_{l+2} - \beta_l}{2}(i - l) + \beta_l && \text{for even } i - l \geq 0. \\ \beta_i &= \frac{\beta_{l+2} - \beta_l}{2}(i - l - 1) + \beta_{l+1} && \text{for odd } i - l \geq 1.\end{aligned}$$

Proof For $p \in \{0, 1\}$ and $u \geq 0$, formula (26) yields

$$\beta_{2(u+1)+p}^\Lambda(\mathbf{L}) - \beta_{2u+p}^\Lambda(\mathbf{L}) = \mu_{\mathbf{B}}^{\mathbf{L}}(0).$$

Since $\beta_{l+i}^R(N) = \beta_i^\Lambda(\mathbf{L})$ holds for $i \geq 0$, by (55), we get

$$\beta_{l+i+2}^R(N) - \beta_{l+i}^R(N) = \mu_{\mathbf{B}}^{\mathbf{L}}(0) \quad \text{for } i \geq 0.$$

This finishes the proof of (b).

Lemma 8.3 (Case (c)) *If $r = 2 = \text{grade } J + 1$ or $r \geq 3$, then*

$$\beta_i = \beta_{d+3}\varphi_{i-d-3} + (r-1)\beta_{d+2}\varphi_{i-d-4} \quad \text{for } i \geq d+4,$$

where $(\varphi_i)_{i \geq 0}$ is defined recursively by

$$\varphi_{i+2} = \varphi_{i+1} + (r-1)\varphi_i \quad \text{with } \varphi_0 = \varphi_1 = 1.$$

Proof Set $L = \Omega_R^{d+2}(N)$. Iyengar [14, 3.4 and 1.8] gives the first equality below:

$$\sum_{i \geq 0} \beta_i^R(L)t^i = \frac{\beta_0^Q(L) + \beta_1^Q(L)t + \beta_2^Q(L)t^2}{1 - rt^2 - (r-1)t^3}.$$

This equality yields $\beta_i^R(L) = \beta_i^Q(L)$ for $i = 0, 1$, so we have

$$\begin{aligned} \beta_0^Q(L) + \beta_1^Q(L)t + \beta_2^Q(L)t^2 &= \beta_0^Q(L) + \beta_1^Q(L)t + (\beta_1^Q(L) - \beta_0^Q(L))t^2 \\ &= (1+t)(\beta_0^R(L) + (\beta_1^R(L) - \beta_0^R(L))t). \end{aligned}$$

As $\beta_i = \beta_{i-d-2}^R(L)$ for $i \geq d+2$, from the last two displays we obtain

$$\begin{aligned} \sum_{i \geq d+2} \beta_i t^i &= \frac{(1+t)(\beta_0^R(L) + (\beta_1^R(L) - \beta_0^R(L))t)}{(1+t)(1-t-(r-1)t^2)} \cdot t^{d+2} \\ &= \frac{\beta_{d+2} + (\beta_{d+3} - \beta_{d+2})t}{1-t-(r-1)t^2} \cdot t^{d+2}. \end{aligned}$$

Partial fractions decomposition yields the desired expression for β_i .

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Betti Diagrams with Special Shape

Mina Bigdeli and Jürgen Herzog

Abstract We consider classes of monomial ideals whose Betti diagrams have a special shape. Monomial ideals with such a Betti diagram satisfy the subadditivity condition for the maximal shifts in the resolution by obvious reasons, and they appear quite frequently in combinatorial contexts. Examples of ideals with special shape are the edge ideal as well as the vertex cover ideal of chordal graphs, whisker graphs and triangulated d -uniform hypergraphs.

1 Introduction

In this note we consider classes of monomial ideals whose Betti diagrams have a special shape. Let K be field, $S = K[x_1, \dots, x_n]$ the polynomial ring in n variables over K . Let $I \subset S$ be a graded ideal with $\text{proj dim}(S/I) = p$, and let $\mathbb{F}$ be the graded minimal free S -resolution of S/I . We denote by $t_i(I)$ the highest degree of a generator of F_i . The sequence $t_0(I), t_1(I), t_2(I), \dots, t_p(I)$ is called the t -sequence of the ideal I . With the notation introduced, the regularity of S/I is defined to be

$$\text{reg}(S/I) = \max\{t_i(I) - i : i = 1, \dots, p\}.$$

The ideal I is said to satisfy *subadditivity* if $t_{i+j}(I) \leq t_i(I) + t_j(I)$ for all positive integers i and j with $i + j \leq p$.

The special shape of the Betti diagram we have in mind is such that the sequence of numbers $r_0(I), r_1(I), \dots, r_p(I)$ with $r_i(I) = t_i(I) - i$ for $i = 0, \dots, p$ form a concave sequence until they reach the regularity and from this point on are non-increasing, see Definition 1. We call the sequence of the $r_i(I)$ the r -sequence of the ideal.

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There exist examples which show that not all graded ideals satisfy subadditivity, see [2], but it is an open problem whether all monomial ideals satisfy subadditivity. For edge ideals of graphs the inequality $t_{i+1}(I) \leq t_i(I) + t_1(I)$ was shown by Fernández-Ramos and Gimenez [10]. The same inequality has been shown later for any monomial ideal [14] by Srinivasan and the second author of this paper. Yazdan Pour independently proved the same result and presented it in a lecture at IPM [25]. In the meantime some more notable results regarding subadditivity have been obtained by Khoury and Srinivasan [18, Theorem 2.3], Abedelfatah and Nevo [1, Theorem 1.3] and Faridi [9, Theorem 3.7].

In this note we will show (Theorem 1) that the Betti diagram of the edge ideal of any chordal graph or any whisker graph has a special shape. The condition that the r -sequence is non-increasing after it reached the regularity is guaranteed if all extremal strands of the diagram are connected and overlap properly, which for example is the case if the ideal is componentwise linear, see Proposition 1. Examples of such ideals are the vertex cover ideals of chordal graphs, as shown in the proof of [11, Theorem 3.2]. Other examples are the vertex cover ideals of simplicial trees, see [8].

Edge ideals of chordal graphs and whisker graphs are special cases of edge ideals of certain hypergraphs. The triangulated d -uniform hypergraphs $\mathcal{C}$ of Hà and Van Tuyl, see [12], are natural generalizations of chordal graphs, and indeed it can be shown that the Betti diagram of $S/I(\mathcal{C})$ has special shape, see Theorem 2. These results are obtained from the fact that their non-empty strands are connected, and that the diagonal Betti numbers $\beta_{i,di}(S/I(\mathcal{C}))$ are non-zero for $i = 0, \dots, r$, where $rd = \text{reg}(S/I(\mathcal{C}))$. For a graph G , the maximal number i for which $\beta_{i,2i}(S/I(G)) \neq 0$ is just the induced matching number of the graph, see [17, Lemma 2.2]. A similar statement holds for d -uniform hypergraphs with a suitable definition of induced matching. It turns out, as shown in Lemma 5, that the induced matching number of d -uniform hypergraph is always less than or equal to the cardinality of a maximal set of pairwise $(d + 1)$ -disjoint edges of the hypergraph, and coincides with this cardinality if the hypergraph is properly-connected.

The edge ideal of a whisker graph is the polarization of a monomial ideal $I \subset S$ containing all the squares of the variables. More generally, one may consider any monomial ideal $I \subset S$ with $\dim(S/I) = 0$. Then this ideal contains a pure power of each variable. By using a result of Mermin et al. [21] it is shown that such ideals, under additional conditions on the pure powers, satisfy the subadditivity, see Corollary 4. These assumptions are for example satisfied, if I is generated in degree 2.

2 Betti Diagrams with Special Shape

Let $S = K[x_1, \dots, x_n]$ be the polynomial ring in the indeterminates $x_1, \dots, x_n$ over the field K , and let $I \subset S$ be a graded ideal with $p = \text{proj dim}(S/I)$. For each $i = 0, \dots, p$ let $r_i(I) = t_i(I) - i$. Note that the $r_i(I)$ -th row in the Betti diagram of S/I

Table 1 Betti diagram of the quotient ring associated to the 5-cycle

	0	1	2	3
0:	1	-	-	-
1:	-	5	5	-
2:	-	-	-	1

is the last row with non-zero entry in the i -th column. The integer $r_i(I)$ is sometimes called the i -regularity of S/I , and of course $\text{reg}(S/I) = \max\{r_i(I) : i = 1, \dots, p\}$.

It is clear that I satisfies subadditivity if and only if $r_{i+j}(I) \leq r_i(I) + r_j(I)$ for all positive integers i, j with $i + j \leq p$.

Definition 1 Let I be a graded ideal. The Betti diagram of S/I with regularity c is said to have a *special shape*, if

- (i) $r_0(I) \leq r_1(I) \leq \dots \leq r_g(I)$ and $r_{i+1}(I) - r_i(I) \leq r_i(I) - r_{i-1}(I)$ for $1 \leq i \leq g$, where g is the smallest integer such that $r_g(I) = c$.
- (ii) $r_{i+1}(I) \leq r_i(I)$ for $g \leq i \leq \text{proj dim}(S/I)$.

The first condition says that the r -sequence of the ideal is concave in the range from 0 to g , and the second condition guarantees that the r -sequence is non-increasing in the range from g to the projective dimension of S/I .

Not all Betti diagrams of monomial ideals have a special shape. This is not even the case for the edge ideal of a graph. A simplest such example is the edge ideal of the 5-cycle whose Betti diagram violates condition (i) of Definition 1 (see Table 1).

On the other hand, we do not know of any monomial ideal whose Betti diagram violates condition (ii) of Definition 1. However, we expect that such examples exist.

Lemma 1 Any graded ideal whose Betti diagram has a special shape satisfies subadditivity.

Proof If $i + j \leq g$, then (i) implies

$$r_{i+j}(I) - r_j(I) = \sum_{k=j+1}^{i+j} (r_k(I) - r_{k-1}(I)) \leq \sum_{k=1}^i (r_k(I) - r_{k-1}(I)) = r_i(I).$$

If $i + j \geq g$, and say $j \geq g$, then $r_i(I) + r_j(I) \geq r_i(I) + r_{i+j}(I) \geq r_{i+j}(I)$, by (ii). Finally, if $i + j > g$ and $i, j < g$, then we get

$$\begin{aligned} r_i(I) + r_j(I) &= (r_i(I) + r_{g-i}(I)) + (r_j(I) - r_{g-i}(I)) \geq r_g(I) + (r_j(I) - r_{g-i}(I)) \\ &\geq r_g(I) \geq r_{i+j}(I). \end{aligned}$$

Here we used the fact that $r_i(I) + r_{g-i}(I) \geq r_g(I)$, as we have seen before, further that $r_g(I) \geq r_{i+j}(I)$, by (ii) and that $r_j(I) - r_{g-i}(I) = \sum_{k=g-i+1}^j (r_k(I) - r_{k-1}(I)) \geq 0$, since each summand of this sum is ≥ 0 , by (i). $\square$

It is expected that all monomial ideals satisfy subadditivity, even if their Betti diagram does not have special shape.

In the following sections we will consider classes of ideals whose Betti diagram has a special shape.

3 Strand Connectedness

A graded Betti number $\beta_{p,p+q} \neq 0$ is said to be *extremal* if $\beta_{i,i+j} = 0$ for all $(p, q) \neq (i, j)$ such that $i \geq p$ and $j \geq q$. For each graded ideal I , the ring S/I has at least one extremal Betti number which is $\beta_{p_c(I), p_c(I)+c}(S/I)$, where $c = \text{reg}(S/I)$ and $p_c(I)$ is the biggest integer among all i with $\beta_{i,i+c}(S/I) \neq 0$. In particular,

$$\text{reg}(S/I) = \max\{j : \beta_{i,i+j}(S/I) \text{ is an extremal Betti number}\}.$$

Let I be a graded ideal. For each j , the j -strand of I is defined to be the set

$$j\text{-strand}(I) = \{i : \beta_{i,i+j}(S/I) \neq 0\}.$$

We denote by $p_j(I)$ the greatest integer belonging to the j -strand of I . A non-empty j -strand of I is said to be *connected*, if there exists $q_j(I) \leq p_j(I)$ with the property that $\beta_{i,i+j}(S/I) \neq 0$ if and only if $q_j(I) \leq i \leq p_j(I)$. In this case we show the j -strand of I with the interval $[q_j(I), p_j(I)]$. The ideal I is called *strand connected*, if each non-empty strand of I is connected. A j -strand of I is called an *extremal strand* if $\beta_{p_j(I), p_j(I)+j}(S/I)$ is an extremal Betti number.

The edge ideal I of the 5-cycle has one extremal Betti number which is $\beta_{3,5}(S/I) = 1$. Hence, the only extremal strand of I is 2-strand(I). Moreover, all the j -strands of I are connected, see Table 1.

The following lemma makes sure that condition (ii) of Definition 1 holds for the Betti diagram of a graded ideal, if certain conditions on the extremal strands of I are satisfied.

Lemma 2 *Let I be a graded ideal, and $\{j_1, \dots, j_s\}$ be the set of integers j for which the j -strand of I is extremal. Suppose that each of these strands is connected, say we have $j_k\text{-strand}(I) = [q_{j_k}(I), p_{j_k}(I)]$. Suppose further that for each $j_k \neq \text{reg}(S/I)$ there exists j_1 with $j_k < j_1$ and such that $q_{j_k}(I) \leq p_{j_1}(I)$. Then the Betti diagram of I satisfies condition (ii) of Definition 1.*

Proof We may assume that $j_1 < j_2 < \dots < j_s$. Then $j_s = \text{reg}(S/I)$, and we need to show that $r_i(I)$ is a non-increasing function on i for all $i \geq q_{j_s}(I)$. This is clear for $q_{j_s}(I) \leq i \leq p_{j_s}(I)$, because $r_i(I) = j_s$ for $i \in [q_{j_s}(I), p_{j_s}(I)]$ since the j_s -strand of I is connected.

Since the j_k -strands of I are extremal, it follows that $p_{j_1}(I) > p_{j_2}(I) > \dots > p_{j_s}(I)$, and for any other non-empty j -strand with $j_k < j \leq j_{k+1}$ it follows that $p_j(I) \leq p_{j_{k+1}}(I)$, because j -strand(I) is not an extremal strand.

We claim that $r_i(I) = j_k$ for i with $p_{j_{k+1}}(I) < i \leq p_{j_k}(I)$. The claim implies that $r_i(I)$ is a non-increasing function for all i with $p_{j_s}(I) < i \leq p_{j_1}(I)$. Together with the fact $r_i(I) = j_s$ for i with $q_{j_s}(I) \leq i \leq p_{j_s}(I)$ and that $j_s > j_{s-1}$, the desired conclusion will follow.

For the proof of the claim we let i be an integer with $p_{j_{k+1}}(I) < i \leq p_{j_k}(I)$. We first notice that our assumptions imply that $q_{j_k}(I) \leq p_{j_{k+1}}(I)$. Therefore, since j_k -strand(I) is connected, it follows that $\beta_{i, i+j_k}(I) \neq 0$, and this implies $r_i(I) \geq j_k$. Suppose that $r_i(I) > j_k$. Then the j -strand with $j = r_i(I)$ is non-empty. Let l be the integer with $j_l < j \leq j_{l+1}$. Then $k \leq l$ and $i \leq p_j(I) \leq p_{j_{l+1}}(I) \leq p_{j_{k+1}}(I)$, a contradiction. $\square$

For a Cohen–Macaulay ideal I the conditions described in Lemma 2 are satisfied. Indeed, in this case, I admits only one extremal strand, and this strand is connected, as follows from

Lemma 3 *Let M be a graded Cohen–Macaulay S -module of regularity c . Then the Betti number $\beta_{p_c(M), p_c(M)+c}(M)$ is the only extremal Betti number of M , and the c -strand of M is connected.*

Proof Note that $\text{Ext}_S^i(M, S) \neq 0$ if and only if $i = \text{projdim}(M)$, see [5, Proposition 3.3.3]. We first show that the c -strand of M is connected using this fact. Suppose on the contrary that there exists $i < p_c(M)$ such that $r_i(M) = c$ and $r_{i+1}(M) < c$, i.e. $\beta_{i, i+c}(M) \neq 0$, $\beta_{i+1, (i+1)+c}(M) = 0$. Let $\mathbb{F}$ be the graded minimal free resolution of M . Since $\beta_{i, i+c}(M) \neq 0$ and since c is the regularity of M , the highest degree of the basis homogeneous elements in F_i is $i + c$. Let $e_1, \dots, e_b$ be a homogeneous basis of F_i . We may assume that $\deg e_1 = i + c$. Let $\partial_{i+1} : F_{i+1} \rightarrow F_i$ be the $(i+1)$ -differential in $\mathbb{F}$. Since $\beta_{i+1, (i+1)+c}(M) = 0$ and since $\text{reg}(M) = c$, we have $\deg f \leq \deg e_1$ for all homogeneous basis elements f in F_{i+1} . Thus, since ∂_{i+1} is a graded map and since $\text{Im}(\partial_{i+1}) \subset \mathfrak{m}F_i$, it follows that for all basis elements f of F_{i+1} we have

$$\partial_{i+1}(f) = \sum_{e_l \neq e_1} a_l e_l \quad \text{with } a_l \in S. \quad (1)$$

Dualizing the resolution of M with respect to S we get the acyclic complex $\mathbb{F}^*$, since $\text{Ext}_S^i(M, S) = 0$ for $i < \text{projdim}(M)$. On the other hand (1) implies that $\partial_{i+1}^*(e_1^*) = 0$, while $e_1^* \notin \text{Im}(\partial_i^*)$ because $\text{Im}(\partial_i^*) \subset \mathfrak{m}F_i^*$. This contradicts the acyclicity of $\mathbb{F}^*$. Therefore the c -strand is connected.

We know that $\beta_{p_c(M), p_c(M)+c}(M)$ is an extremal Betti number. So it suffices to show that there is no other extremal Betti number. To see this, note that with the same argument as above, $\text{Ext}_S^{p_j(M)}(M, S) \neq 0$ if $\beta_{p_j(M), p_j(M)+j}(M)$ is an extremal Betti number. Since $\text{Ext}_S^i(M, S) \neq 0$ if and only if $i = \text{projdim}(M)$, there can be only one extremal Betti number. $\square$

A large class of strand connected ideals are the componentwise linear ideals. We first show

Lemma 4 *Let $J \subset I$ be two graded ideals equigenerated in degree d . Then*

- (i) $\beta_{i,i+d}(J) \leq \beta_{i,i+d}(I)$ for all i ;
- (ii) *suppose in addition that J, I have d -linear resolutions. If $\beta_i(J) = \beta_i(I)$ for some i , then $\beta_k(J) = \beta_k(I)$ for all $k \geq i$.*

Proof

- (i) See [4, Lemma 2.3(a)].
- (ii) We denote by $\text{Gin}(I)$ the generic initial ideal of I with respect to the reverse lexicographical order induced by $x_1 > x_2 > \cdots > x_n$. It follows from a theorem of Bayer and Stillman [3] (see also [13, Corollary 4.3.18(d)]) that $\text{Gin}(I)$ and $\text{Gin}(J)$ have again a linear resolution. Moreover, one has $\beta_k(J) = \beta_k(\text{Gin}(J))$ and $\beta_k(I) = \beta_k(\text{Gin}(I))$ for all k , see [13, Problem 4.5]. Since I, J have d -linear resolutions and hence are componentwise linear, it follows from [6, Lemma 1.4] that their generic initial ideal is stable. Consequently, we may assume that $J \subset I$ are stable ideals.

Let $G(I)$ denote the set of minimal monomial generators of I . Then $G(I) = G(J) \cup \{u_1, \dots, u_r\}$ for some $u_1, \dots, u_r \in G(I) \setminus G(J)$. It follows from the Eliahou–Kervaire resolution of a stable ideal (see [7] or [13, Corollary 7.2.3]) that for all k

$$\beta_k(I) = \sum_{u \in G(I)} \binom{m(u)-1}{k} = \beta_k(J) + \sum_{t=1}^r \binom{m(u_t)-1}{k},$$

where $m(u)$ denotes the largest number j such that x_j divides u . Therefore, $\beta_k(J) = \beta_k(I)$ if and only if $\sum_{t=1}^r \binom{m(u_t)-1}{k} = 0$. This is the case if and only if $k \geq \max\{m(u_t) : 1 \leq t \leq r\}$. Hence, by assumption, $i \geq \max\{m(u_t) : 1 \leq t \leq r\}$. This implies that $\sum_{t=1}^r \binom{m(u_t)-1}{k} = 0$ for all $k \geq i$ which yields the desired result. $\square$

Let I be a graded ideal and let $I_{(j)}$ be the ideal generated by all homogeneous polynomials of degree j belonging to I . The ideal I is called *componentwise linear* if $I_{(j)}$ has a linear resolution for all j . It follows from [13, Lemma 8.2.10] that, in this case, the ideal $\mathfrak{m}I_{(j)}$ has a linear resolution too, where $\mathfrak{m}$ denotes the graded maximal ideal of the ring S . Let I be a componentwise linear ideal. Then using [13, Proposition 8.2.13] one has $\beta_{i,i+j}(I) = \beta_i(I_{(j)}) - \beta_i(\mathfrak{m}I_{(j-1)})$ for all i and j . We use this fact in the proof of the following proposition.

Proposition 1 *Let I be a componentwise linear ideal. Then I is strand connected with each non-empty j -strand beginning in the homological degree 1 for $j > 0$. In particular, $r_1(I) \geq r_2(I) \geq \cdots$, and hence the Betti diagram of S/I has a special shape and subadditivity holds for I .*

Proof By [13, Proposition 8.2.13] the graded Betti numbers of I are given by the formula

$$\beta_{i,i+j}(I) = \beta_i(I_{(j)}) - \beta_i(\mathbf{m}I_{(j-1)}).$$

Thus for $j > 0$,

$$\beta_{i,i+j}(S/I) = \beta_{i-1}(I_{(j+1)}) - \beta_{i-1}(\mathbf{m}I_{(j)}).$$

It follows from this formula that for $j > 0$, a j -strand of I is non-empty if and only if I has a generator in degree $j + 1$, and this strand begins in homological degree 1. Furthermore, Lemma 4 implies that the non-empty strands are connected.

The remaining statements of the proposition follow from Lemma 2. $\square$

Remark 1 Proposition 1, in the case that the field K is of characteristic 0, can be also obtained from the following known results. Indeed, if $\text{char}(K) = 0$, then a graded ideal $I \subset S$ is componentwise linear if and only if $\beta_{i,i+j}(I) = \beta_{i,i+j}(\text{Gin}(I))$ for all i and j , (see [13, Theorem 8.2.22]). It follows from [13, Theorem 4.2.1] that $\text{Gin}(I)$ is Borel-fixed and [13, Proposition 4.2.4] implies that it is strongly stable. It follows from the Eliahou–Kervaire formula that for $j > 0$,

$$\beta_{i,i+j}(S/I) = \beta_{i-1,(i-1)+(j+1)}(\text{Gin}(I)) = \sum_{u \in G(I)_{j+1}} \binom{m(u)-1}{i-1},$$

where $G(I)_j$ denotes the set of monomials of degree j which belong to the unique minimal set of monomial generators of I . This implies that for $j > 0$, the j -strand(I) $\neq \emptyset$ if and only if $G(I)_{j+1} \neq \emptyset$. In this case we have j -strand(I) = $\{i: 1 \leq i \leq p_j(I)\}$, where $p_j(I) = \max\{m(u): u \in G(I)_{j+1}\}$.

As mentioned in the introduction, Faridi recently showed in [9] that the facet ideal of a simplicial tree Δ satisfies the subadditivity condition. As an application of Proposition 1 we see that the vertex cover ideal of a simplicial tree also satisfies the subadditivity condition.

Let Δ be a simplicial complex on $[n]$ with $\mathcal{F}(\Delta)$ as its facet set. The set $C \subset [n]$ is called a *vertex cover* of Δ if $C \cap F \neq \emptyset$ for all $F \in \mathcal{F}(\Delta)$. The *vertex cover ideal* of Δ is the Alexander dual of the facet ideal of Δ .

Corollary 1 *Let I be the vertex cover ideal of a simplicial forest. Then the Betti diagram of S/I has a special shape and hence I satisfies subadditivity condition.*

Proof Since the facet ideal of a simplicial forest is sequentially Cohen–Macaulay [8, Corollary 5.6], the ideal I is componentwise linear using [23, Theorem 3.8]. Thus Proposition 1 implies the desired conclusion. $\square$

4 Edge Ideals of Graphs

A set $\mathcal{S}$ of edges of a graph G is called a *matching* of G if no two edges of $\mathcal{S}$ have a common vertex and it is called an *induced matching* if it is a matching and no two edges of $\mathcal{S}$ are connected by an edge in G . The maximal size of an induced matching of G is called the *induced matching number* of G , denoted $\text{im}(G)$. It is known that $\text{im}(G) \leq \text{reg}(S/I(G))$, see [17, Lemma 2.2].

The induced matching number of a graph can be seen from the Betti diagram of its edge ideal. The following proposition is an immediate consequence of [17, Lemma 2.2] by Katzman. A hypergraph version of it will be shown in the next section.

Proposition 2 *Let G be a graph. Then $\beta_{i,2i}(S/I(G)) \neq 0$ if and only if $i \in \{0, 1, \dots, \text{im}(G)\}$.*

In the following we describe a situation which we will encounter throughout this section.

Corollary 2 *Let G be a graph satisfying the following properties:*

- (a) $\text{im}(G) = \text{reg}(S/I(G))$;
- (b) *the extremal strands of $I(G)$ are connected.*

Then the Betti diagram of $S/I(G)$ has a special shape. In particular, subadditivity holds for $I(G)$.

Proof Assumption (a) and Proposition 2 imply that $\beta_{i,2i}(S/I(G)) \neq 0$ for all $i = 0, \dots, \text{reg}(S/I(G))$. By [10, Corollary 1.9] we have $r_{i+1}(I) \leq r_i(I) + 1$. This implies that $r_i(I) = i$ for $i = 0, \dots, \text{reg}(S/I(G))$, and hence condition (i) of Definition 1 is satisfied. Moreover, each non-empty j -strand begins with j . It follows from Lemma 2 that condition (ii) is also satisfied. $\square$

Example 1 In view of Corollary 2 we discuss some families of graphs.

- (a) Let G be a bipartite graph. Kummini showed in [19, Theorem 1.1] that $\text{im}(G) = \text{reg}(S/I(G))$ if G is unmixed, and Van Tuyl showed in [22, Theorem 3.3] that $\text{im}(G) = \text{reg}(S/I(G))$ if G is sequentially Cohen–Macaulay. The unmixed bipartite graphs are classified by Villarreal in [24, Theorem 1.1]. In both cases, the unmixed and the sequentially Cohen–Macaulay case, condition (a) of Corollary 2 is satisfied. In particular, this happens if G is Cohen–Macaulay. In this case, by Lemma 3, condition (b) of Corollary 2 is also satisfied, and hence for Cohen–Macaulay bipartite graphs subadditivity holds.

The examples of sequentially Cohen–Macaulay bipartite graphs we considered all have the property that their Betti diagram has only one extremal strand, and this strand is connected. If this would be the case in general for bipartite graphs, then this would show that the Betti diagram of a sequentially Cohen–Macaulay bipartite graph has a special shape.

- (b) Let C_n be the cycle graph of length n . Jacques in his PhD thesis [16, Chapter 7] computed the graded Betti numbers of C_n . He showed that for all i, j with $i + j < n$,

$$\beta_{i,i+j}(S/I(C_n)) = \binom{j}{i-j} \beta_{j,2j}(S/I(C_n)),$$

where

$$\beta_{i,2i}(S/I(C_n)) = \begin{cases} \frac{n}{n-2i} \binom{n-2i}{i} & \text{for } 0 \leq i \leq \lfloor \frac{n}{3} \rfloor, \\ 0 & \text{for } \lfloor \frac{n}{3} \rfloor < i \leq \lfloor \frac{n}{2} \rfloor. \end{cases}$$

Moreover,

$$\begin{aligned} \beta_{2m+1,n}(S/I(C_n)) &= 1 && \text{if } n = 3m + 1 \text{ or } n = 3m + 2, \\ \beta_{2m,n}(S/I(C_n)) &= 2 && \text{if } n = 3m. \end{aligned}$$

The formulas show that all non-empty strands are connected, and that the smallest element in each non-empty j -strand is j . The fact that $\beta_{i,2i}(S/I(C_n)) \neq 0$ for $i = 0, \dots, \text{reg}(S/I(C_n))$ shows that $\text{im}(C_n) = \text{reg}(S/I(C_n))$. Hence, subadditivity holds for $I(C_n)$.

The following theorem describes two other prominent classes of graphs which satisfy the conditions of Corollary 2.

Theorem 1 *Let G be a chordal or a whisker graph. Then G satisfies the conditions of Corollary 2. In particular, the Betti diagram of $S/I(G)$ has a special shape, and hence subadditivity holds for $I(G)$.*

The theorem is a special case of a more general result which will be presented in the next section.

5 Generalizations

The results saying that chordal and whisker graphs satisfy subadditivity condition can be generalized to monomial ideals which are not only generated by monomials of degree 2.

First we will prove the analogue of Proposition 2 for hypergraphs. Let $V = \{v_1, \dots, v_n\}$ be a finite set, and let $\mathcal{C} = \{F_1, \dots, F_t\}$ be a family of distinct non-empty subsets of V . This family is called a *hypergraph* on the vertex set V . The elements of $\mathcal{C}$ are called the *edges* of $\mathcal{C}$. A hypergraph is called *d-uniform* if all of its edges have the same cardinality d . Let $\mathcal{C}$ be a d -uniform hypergraph on the vertex set $[n]$. The set of edges of $\mathcal{C}$ will be denoted by $E(\mathcal{C})$. With each edge $F = \{i_1, \dots, i_d\} \in E(\mathcal{C})$ we associate the monomial $\mathbf{x}_F = \prod_{j=1, \dots, d} x_{i_j}$. The ideal

$I(\mathcal{C})$, generated by the monomials $\mathbf{x}_F$ with F an edge of $\mathcal{C}$, is called the *edge ideal* of $\mathcal{C}$.

Let $V \subset [n]$. The hypergraph $\mathcal{C}_V$ induced by V is the hypergraph with $E(\mathcal{C}_V) = \{F \in \mathcal{C} : F \subset V\}$. Let $\{F_1, \dots, F_i\} \subset E(\mathcal{C})$ and set $V = \bigcup_{j=1}^i F_j$. The set $\{F_1, \dots, F_i\}$ is called an *induced matching* of $\mathcal{C}$, if $E(\mathcal{C}_V) = \{F_1, \dots, F_i\}$ and the edges of $\mathcal{C}_V$ are pairwise disjoint. The maximal number of edges of $\mathcal{C}$ which form an induced matching of $\mathcal{C}$ is called the *induced matching number* of $\mathcal{C}$, denoted $\text{im}(\mathcal{C})$. Note that in the case of graphs this is the classical definition of the induced matching number of a graph.

In view of Lemma 5, the following proposition may be considered as a generalization of [12, Theorem 6.5].

Proposition 3 *Let $\mathcal{C}$ be a d -uniform hypergraph. Then $\beta_{i,di}(S/I(\mathcal{C})) \neq 0$ if and only if $i \in \{0, 1, \dots, \text{im}(\mathcal{C})\}$.*

Proof Let $E(\mathcal{C}) = \{F_1, \dots, F_r\}$ and let $\{F_{i_1}, \dots, F_{i_c}\}$ be an induced matching of $\mathcal{C}$. Since the F_{i_j} are pairwise disjoint, it follows that the edge ideal of $\mathcal{C}_V$ for $V = \bigcup_{j=1}^c F_{i_j}$ is generated by the regular sequence $\mathbf{x}_{F_{i_1}}, \dots, \mathbf{x}_{F_{i_c}}$. Thus the Taylor complex on this sequence provides a minimal free resolution of $S/I(\mathcal{C}_V)$. It follows that $\beta_{i,di}(S/I(\mathcal{C}_V)) \neq 0$ for $i = 0, \dots, c$. By the restriction lemma [15, Lemma 4.4] we have $\beta_{i,j}(S/I(\mathcal{C}_V)) \leq \beta_{i,j}(S/I(\mathcal{C}))$ for all i and j . This implies that $\beta_{i,di}(S/I(\mathcal{C})) \neq 0$ if $i \in \{0, 1, \dots, c\}$.

Now let $i > c$. We must show that $\beta_{i,di}(S/I(\mathcal{C})) = 0$. To prove this, we apply the strategy of Katzman [17] and use the Taylor complex to compute $\beta_{i,di}(S/I(\mathcal{C}))$. The Taylor complex $\mathbb{T}$ provides a graded free S -resolution of $S/I(\mathcal{C})$ which is rarely minimal. Nevertheless it can be used to compute the (multi)graded Betti numbers of $S/I(\mathcal{C})$. The module T_1 is a free S -module with basis $e_1, \dots, e_r$, and $T_i = \bigwedge^i T_1$ for $i = 0, \dots, r$. The differential $\partial_i : T_i \rightarrow T_{i-1}$ is given by

$$\partial_i(e_{k_1, \dots, k_i}) = \sum_{j=1}^i (-1)^{j+1} \mu_j e_{k_1, \dots, \widehat{k_j}, \dots, k_i},$$

where for $k_1 < k_2 < \dots < k_i$ the element $e_{k_1, \dots, k_i}$ stands for $e_{k_1} \wedge \dots \wedge e_{k_i}$ and $\mu_j = \mathbf{x}_{G_j}$, where $G_j = \bigcup_{l=1}^i F_{k_l} \setminus \bigcup_{l \neq j} F_{k_l}$.

Suppose now that $\beta_{i,di}(S/I(\mathcal{C})) \neq 0$. It follows from the Taylor complex that the multigraded shifts (monomial notation) of the resolution of $S/I(\mathcal{C})$ are squarefree and are least common multiples of the monomial generators of $I(\mathcal{C})$. Therefore there exist edges $F_{k_1}, \dots, F_{k_i}$ of $\mathcal{C}$ which are pairwise disjoint and there is a non-zero Betti number $\beta_{i, \mathbf{a}_F}(S/I(\mathcal{C}))$, where $\mathbf{a}_F(l) = 1$ if $l \in F := \bigcup_{j=1}^i F_{k_j}$ and 0 otherwise. Let $e = e_{k_1, \dots, k_i}$. Then $\partial_i(e) = \sum_{j=1}^i (-1)^{j+1} \mathbf{x}_{F_{k_j}} e_{k_1, \dots, \widehat{k_j}, \dots, k_i}$, and hence $\partial_i(e) = 0$ in $\mathbb{T}/\mathfrak{m}\mathbb{T}$, where $\mathfrak{m}$ is the graded maximal ideal of S . We denote by $\bar{e}$ the residue class of e in $\mathbb{T}/\mathfrak{m}\mathbb{T}$. Then $\bar{e}$ is a cycle in $\mathbb{T}/\mathfrak{m}\mathbb{T}$. Any other cycle in $\mathbb{T}/\mathfrak{m}\mathbb{T}$ of multidegree $\mathbf{a}_F$ arises like $\bar{e}$ from a decomposition of F as a union of pairwise disjoint edges. Thus we may as well assume that the homology class of $\bar{e}$ gives the non-zero contribution

to $\text{Tor}_i^S(S/I(\mathcal{C}), K)_{id} \cong H_i(\mathbb{T}/m\mathbb{T})_{id}$. We will obtain a contradiction by showing next that $\bar{e}$ is a boundary in $\mathbb{T}/m\mathbb{T}$. Indeed, since $i > \text{im}(\mathcal{C})$, $\{F_{k_1}, \dots, F_{k_i}\}$ is not an induced matching. In other words there exists an edge F_{k_l} with $F_{k_l} \neq F_{k_i}$ for $l = 1, \dots, i$ and $F_{k_l} \subset \bigcup_{l=1}^i F_{k_l}$. Let $f = e_{k_l} \wedge e$. It follows from the definition of F_{k_l} that $\partial_{i+1}(f) = e - e_{k_l} \wedge \partial_i(e)$. Hence $\partial_{i+1}(\bar{f}) = \bar{e}$. Thus $\bar{e}$ is a boundary. $\square$

Let, as before, $\mathcal{C}$ be a d -uniform hypergraph on $[n]$. A subset $D \subset [n]$ is called an *independent set* of $\mathcal{C}$, if D does not contain any edge of $\mathcal{C}$. We call the maximal cardinality of an independent set of $\mathcal{C}$ the *independence number* of $\mathcal{C}$ and denote it by $\text{ind}(\mathcal{C})$.

Proposition 4 *Let $\mathcal{C}$ be a d -uniform hypergraph. Then*

$$\text{im}(\mathcal{C})(d-1) \leq \text{reg}(S/I(\mathcal{C})) \leq \text{ind}(\mathcal{C}).$$

Proof The lower bound for $\text{reg}(S/I(\mathcal{C}))$ follows from Proposition 3. To see the upper bound, let $c = \text{reg}(S/I(\mathcal{C}))$. Then there exists an integer i and a non-zero homology class $[z] \in H_i(\mathbf{x}; S/I(\mathcal{C}))_{i+c}$, where $H_i(\mathbf{x}; S/I(\mathcal{C}))$ is the i -th Koszul homology module of $S/I(\mathcal{C})$ with respect to $x_1, \dots, x_n$. Since for a squarefree monomial ideal all shifts in the resolution are squarefree, the cycle z is of the form $\sum_I \lambda_I u_I e_I$, where each I is a subset of $[n]$ of cardinality i , $\lambda_I \in K$, u_I is the residue class of a squarefree monomial v_I modulo $I(\mathcal{C})$ and $e_I = e_{j_1} \wedge \dots \wedge e_{j_i}$ for $I = \{j_1 < \dots < j_i\}$. It follows that $c = \deg v_I$ for all I with $u_I \neq 0$. Since $z \neq 0$, there exists at least one $u_I \neq 0$. This set I must be an independent set, otherwise $u_I = 0$. $\square$

In Proposition 5 we give an example of a class of hypergraphs for which the Betti diagram has a special shape. The proof is based on a recursive formula, Formula (2) below, for the graded Betti numbers of hypergraphs, given by Há and Van Tuyl [12, Theorem 4.16] under certain conditions on the hypergraphs. First we recall this formula. See [12] for more details.

Let $\mathcal{C}$ be a d -uniform hypergraph and F, F' two edges of $\mathcal{C}$. The *distance* between F and F' , denoted by $\text{dist}_{\mathcal{C}}(F, F')$, is the smallest integer t such that there exists a chain $L_0, L_1, \dots, L_t \in \mathcal{C}$ with $F = L_0$, $F' = L_t$ and $|L_i \cap L_{i+1}| = d-1$ for all $0 \leq i \leq t-1$. If no such chain exists, we set $\text{dist}_{\mathcal{C}}(F, F') = \infty$.

A d -uniform hypergraph $\mathcal{C}$ is said to be *properly-connected* if for any two edges F and F' of $\mathcal{C}$ with the property that $F \cap F' \neq \emptyset$, we have $\text{dist}_{\mathcal{C}}(F, F') = d - |F \cap F'|$. The class of simple graphs is an obvious class of properly-connected hypergraphs. The 3-uniform hypergraph $\mathcal{C}$ shown in Fig. 1 is properly-connected while $\mathcal{D}$ is not, because for $F = \{2, 4, 5\}$ and $F' = \{3, 4, 6\}$, one has $\text{dist}_{\mathcal{D}}(F, F') = 3$.

By a *splitting edge* F of a hypergraph $\mathcal{C}$ we mean an edge such that $I(\mathcal{C}) = (\mathbf{x}_F) + I(\mathcal{C} \setminus F)$ is a splitting of $I(\mathcal{C})$ in the sense of [7].

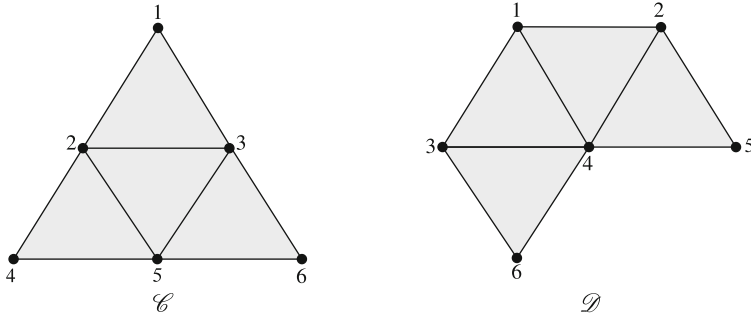


Fig. 1 $\mathcal{C}$ is properly connected, while $\mathcal{D}$ is not

Now let $\mathcal{C}$ be a d -uniform properly-connected hypergraph with a splitting edge F . Then the recursive formula for the graded Betti numbers of $S/I(\mathcal{C})$, given by Há and Van Tuyl, is as follows:

$$\beta_{i,i+j}(S/I(\mathcal{C})) = \beta_{i,i+j}(S/I(\mathcal{C} \setminus F)) + \alpha_{i,j}(\mathcal{C}) + \sum_{l=0}^{i-2} \binom{t}{l} \beta_{i-1-l, (i-1-l) + (j-d+1)}(S/I(\mathcal{C}')), \quad (2)$$

where

$$\alpha_{i,j}(\mathcal{C}) = \begin{cases} \binom{t}{i-1} & \text{if } j = d-1, \\ 0 & \text{if } j \neq d-1. \end{cases}$$

Here, $\mathcal{C} \setminus F$ is a hypergraph obtained from $\mathcal{C}$ by removing the edge F , t is the cardinality of the set

$$\bigcup_{\substack{F' \in \mathcal{C} \\ F' \cap F \neq \emptyset}} F' \setminus F,$$

and $\mathcal{C}'$ is the hypergraph

$$\mathcal{C}' = \{F' \in \mathcal{C} : \text{dist}_{\mathcal{C}}(F, F') \geq d+1\}.$$

A d -uniform properly-connected hypergraph $\mathcal{C}$ with the vertex set $[n]$ is said to be *triangulated* if for every non-empty subset $V \subset [n]$, the induced hypergraph $\mathcal{C}_V$ contains a vertex $v \in V$ such that the induced hypergraph of $\mathcal{C}_V$ on the set $N_{\mathcal{C}_V}[v]$ is a d -complete hypergraph of order $|N_{\mathcal{C}_V}[v]|$. Here, $N_{\mathcal{C}_V}[v] = \{v\} \cup \{u \in [n] : \{u, v\} \subset F \text{ for some } F \in \mathcal{C}_V\}$. Há and Van Tuyl showed in [12, Theorem 5.6] that properly-connected f -forests are examples of triangulated hypergraphs. The hypergraph $\mathcal{C}$ in Fig. 1 is an f -forest, so triangulated. Figure 2 illustrates a 3-uniform triangulated hypergraph which is not an f -forest.

Fig. 2 A triangulated 3-uniform hypergraph $\mathcal{C}$

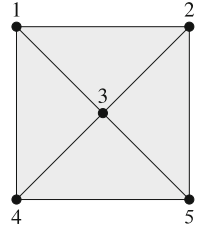
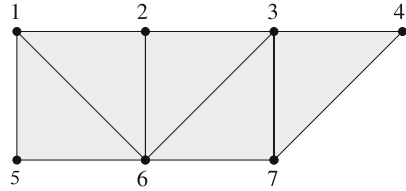


Fig. 3 A 3-uniform hypergraph which is not properly-connected



Há and Van Tuyl also showed in [12, Theorem 5.8] that for triangulated hypergraphs one can compute the graded Betti numbers recursively by using Formula (2), that is, if $\mathcal{C}$ is a triangulated hypergraph, then there exists a splitting edge F in $\mathcal{C}$ such that $\mathcal{C} \setminus F$ and $\mathcal{C}'$ are both triangulated, [12, Lemma 5.7]. We make use of this fact to prove Proposition 5. Before we state this proposition, we first compute the integers $r_i(I)$, where I is the edge ideal of a triangulated d -uniform hypergraph.

Let $\mathcal{C}$ be a d -uniform properly-connected hypergraph. Two edges F, F' of $\mathcal{C}$ are called t -disjoint if $\text{dist}_{\mathcal{C}}(F, F') \geq t$. A set of edges E of $\mathcal{C}$ is pairwise t -disjoint if every pair of edges of E is t -disjoint. We denote by $\mathfrak{p}(\mathcal{C})$ the cardinality of the biggest set of pairwise $(d+1)$ -disjoint edges of $\mathcal{C}$. In Lemma 5 we show that for a d -uniform hypergraph $\mathcal{C}$ we have $\text{im}(\mathcal{C}) \leq \mathfrak{p}(\mathcal{C})$. The equality does not hold in general. However, in the same lemma, we prove the equality for properly-connected d -uniform hypergraphs. Figure 3 illustrates a 3-uniform hypergraph which is not properly-connected. The set $\{\{1, 5, 6\}, \{3, 4, 7\}\}$ is pairwise 4-disjoint, while all the induced matchings of this clutter are of size 1.

Lemma 5 *Let $\mathcal{C}$ be a d -uniform hypergraph. Then $\text{im}(\mathcal{C}) \leq \mathfrak{p}(\mathcal{C})$. Suppose in addition that $\mathcal{C}$ is properly-connected. Then $\text{im}(\mathcal{C}) = \mathfrak{p}(\mathcal{C})$.*

Proof Suppose that $\mathcal{D} \subset \mathcal{C}$ is an induced matching of $\mathcal{C}$. We show that $\mathcal{D}$ is a $(d+1)$ -pairwise disjoint set. It is enough to show that for any pair $F, F' \in \mathcal{D}$, $\text{dist}_{\mathcal{C}}(F, F') \geq d+1$. Let $\text{dist}_{\mathcal{C}}(F, F') = t$. Thus there exists a chain $L_0, \dots, L_t$ in $\mathcal{C}$ such that $F = L_0$, $F' = L_t$, and $|L_i \cap L_{i+1}| = d-1$ for all $0 \leq i \leq t-1$. Since $\mathcal{D}$ is an induced matching, we have $F \cap F' = \emptyset$. It follows that $|F \cap L_{t-1}| \leq 1$ because $|L_{t-1} \cap F'| = d-1$. If $|F \cap L_{t-1}| = 1$, then $L_{t-1} \subset F \cup F'$. This implies that $L_{t-1} \in \mathcal{C}_{F \cup F'}$, a contradiction. Thus $|F \cap L_{t-1}| = 0$. On the other hand by using induction we have $|F \cap L_i| \geq d-i$ for all i . In particular, $|F \cap L_{t-1}| \geq d-(t-1)$. Therefore $t \geq d+1$.

Now suppose that $\mathcal{C}$ is properly-connected. It follows from [12, Theorem 6.5] that $\beta_{i,di}(S/I(\mathcal{C})) \neq \emptyset$ if and only if $i \leq p(\mathcal{C})$. Thus $\text{im}(\mathcal{C}) = p(\mathcal{C})$ using Proposition 3. $\square$

Lemma 6 *Let $\mathcal{C}$ be a triangulated d -uniform hypergraph and let $I := I(\mathcal{C})$ be its edge ideal. Then $\text{reg}(S/I) = \text{im}(\mathcal{C})(d-1)$ and $r_i(I) = i(d-1)$ for all $i \leq \text{im}(\mathcal{C})$.*

Proof Since $\mathcal{C}$ is properly-connected, by Lemma 5, $\text{im}(\mathcal{C}) = p(\mathcal{C})$. Thus $\text{reg}(S/I) = \text{im}(\mathcal{C})(d-1)$ using [12, Theorem 6.8]. Proposition 3 implies that $\beta_{i,di}(S/I) \neq 0$ for all $i \leq \text{im}(\mathcal{C})$. Thus $r_i(I) \geq i(d-1)$ for all $i \leq \text{im}(\mathcal{C})$. By [14, Corollary 4] we have $r_i(I) \leq r_{i-1}(I) + d-1$ for all $i \geq 2$. Since $r_1(I) = d-1$ we conclude that $r_i(I) = i(d-1)$ for all $i \leq \text{im}(\mathcal{C})$. $\square$

Proposition 5 *Let $\mathcal{C}$ be a triangulated d -uniform hypergraph and let $I := I(\mathcal{C})$. Let j be an integer such that $0 \leq j \leq \text{reg}(S/I)$. Then*

- (a) $j\text{-strand}(I) \neq \emptyset$ if and only if $j = k(d-1)$ for some positive integer k ;
- (b) if $j = k(d-1)$ for some positive integer k , then k is the smallest integer belonging to the j -strand of I , and
- (c) if the j -strand of I is non-empty, then it is connected.

Proof Since $\mathcal{C}$ is triangulated [12, Lemma 5.7] implies that it has a splitting edge, say F , and both $\mathcal{C} \setminus F$ and $\mathcal{C}'$ are triangulated. Using induction on $|\mathcal{C}|$ we may assume that the statements (a), (b) and (c) hold for $I(\mathcal{C} \setminus F)$ and $I(\mathcal{C}')$.

We first prove the statements for $j = d-1$. It is clear that (a) and (b) hold if $j = d-1$. By the definition of $p_{d-1}(I)$ we have $(d-1)\text{-strand}(I) \subset [1, p_{d-1}(I)]$. Let i be an integer with $i \in [1, p_{d-1}(I)]$. If $i \in [1, p_{d-1}(I(\mathcal{C} \setminus F))]$, then since (c) holds for $I(\mathcal{C} \setminus F)$, $\beta_{i,i+(d-1)}(S/I(\mathcal{C} \setminus F)) \neq 0$. Thus (2) implies that $i \in (d-1)\text{-strand}(I)$. Suppose $i \notin [1, p_{d-1}(I(\mathcal{C} \setminus F))]$. Hence $p_{d-1}(I(\mathcal{C} \setminus F)) < p_{d-1}(I)$. Since $\beta_{p_{d-1}(I), p_{d-1}(I)+(d-1)}(S/I) \neq 0$, using (2), we have $\alpha_{p_{d-1}(I), d-1}(\mathcal{C}) \neq 0$ or $\sum_{l=0}^{p_{d-1}(I)-2} \binom{l}{i} \beta_{p_{d-1}(I)-1-l, p_{d-1}(I)-1-l}(S/I(\mathcal{C}')) \neq 0$. The latter can not happen because otherwise there exists an integer l with $0 \leq l \leq p_{d-1}(I) - 2$ such that $\binom{l}{i} \beta_{p_{d-1}(I)-1-l, p_{d-1}(I)-1-l}(S/I(\mathcal{C}')) \neq 0$. It follows that $p_{d-1}(I) = l+1 \leq p_{d-1}(I)-1$, a contradiction. So $\alpha_{p_{d-1}(I), d-1}(\mathcal{C}) \neq 0$. Then $0 \neq \binom{l}{p_{d-1}(I)-1} \leq \binom{l}{i-1} = \alpha_{i, d-1}(\mathcal{C})$ which implies that $\beta_{i, i+(d-1)}(S/I) \neq 0$ and so $i \in (d-1)\text{-strand}(I)$. Therefore, $(d-1)\text{-strand}(I) = [1, p_{d-1}(I)]$. In particular, the $(d-1)$ -strand of I is connected. So (c) holds for $j = d-1$.

Now let $j > d-1$. Then $\alpha_{i,j}(\mathcal{C}) = 0$ for all i in Formula (2). First let $j = k(d-1)$ for some $k > 1$. Since by Lemma 6 we have $r_k(I) = j$, it follows that $\beta_{k, k+j}(S/I) \neq 0$. Thus the j -strand is non-empty in this case. Let $(k-1)(d-1) < j < k(d-1)$ for some $k > 1$. Since by induction hypothesis (a) holds for $I(\mathcal{C} \setminus F)$ and $I(\mathcal{C}')$, the j -strand of $I(\mathcal{C} \setminus F)$ and $(j-d+1)$ -strand of $I(\mathcal{C}')$ are empty. Hence the Há-Vantuyt formula (2) implies that the j -strand of I is also empty. This proves (a).

Note that if $j = k(d - 1)$ and $\beta_{i,i+j}(S/I) \neq 0$ for some $i < k$, then $r_i(I) = i(d - 1) \leq (k - 1)(d - 1) < j$ which is impossible. Therefore k is the smallest integer such that $\beta_{k,k+j}(S/I) \neq 0$. So (b) holds for I .

Now we prove (c). Suppose $j\text{-strand}(I) \neq \emptyset$ for some $j > d - 1$. Then $j = k(d - 1)$ and $j\text{-strand}(I) \subset [k, p_j(I)]$ for some $k > 1$. Suppose on contrary that $j\text{-strand}(I) \neq [k, p_j(I)]$. So there exists $i \in [k, p_j(I)]$ such that $\beta_{i,i+j}(S/I) = 0$ while $\beta_{i+1,(i+1)+j}(S/I) \neq 0$. Using Formula (2) we have $\beta_{i,i+j}(S/I(\mathcal{C} \setminus F)) = 0$. So induction hypothesis implies that $i > p_j(I(\mathcal{C} \setminus F))$. Therefore,

$$\beta_{i+1,(i+1)+j}(S/I) = \sum_{l=0}^{i-1} \binom{t}{l} \beta_{i-l,(i-l)+(j-d+1)}(S/I(\mathcal{C}')) \neq 0. \quad (3)$$

Thus there exists $0 \leq l' \leq i - 1$ such that

$$\binom{t}{l'} \beta_{i-l',(i-l')+(j-d+1)}(S/I(\mathcal{C}')) \neq 0. \quad (4)$$

Hence $l' \leq t$ and $i - l' \in (j - d + 1)\text{-strand}(I(\mathcal{C}'))$. Using induction hypothesis we have $i - l' \in [k - 1, p_{j-d+1}(I(\mathcal{C}'))]$.

On the other hand $\beta_{i,i+j}(S/I) = 0$ implies that for all $0 \leq l \leq i - 2$,

$$\binom{t}{l} \beta_{i-l-l,(i-l-l)+(j-d+1)}(S/I(\mathcal{C}')) = 0. \quad (5)$$

It follows that for all $0 \leq l \leq i - 2$ either $l > t$ or $i - 1 - l \notin [k - 1, p_{j-d+1}(I(\mathcal{C}'))]$. If $l' = i - 1$, then $i - l' \in [k - 1, p_{j-d+1}(I(\mathcal{C}'))]$ implies that $k = 2$. Thus $i - l' = k - 1$. Now suppose $l' < i - 1$. Since $l' \leq t$ we have $i - 1 - l' \notin [k - 1, p_{j-d+1}(I(\mathcal{C}'))]$. Consequently, $i - l' = k - 1$ also in this case. Since $i \geq k$ we have $l' > 0$. Now putting $l = l' - 1$ in (5) we have either $l' - 1 > t$ or $i - l' \notin [k - 1, p_{j-d+1}(I(\mathcal{C}'))]$. Both are impossible because $l' \leq t$ and $i - l' = k - 1$. Therefore there does not exist such l' , which by (3) implies that $\beta_{i+1,(i+1)+j}(S/I) = 0$, a contradiction. So the j -strand of I is connected. $\square$

Theorem 2 *Let $\mathcal{C}$ be a triangulated d -uniform hypergraph and let I be its edge ideal. Then the Betti diagram of S/I has a special shape, and hence I satisfies subadditivity.*

Proof It follows from Lemma 6 that condition (i) of Definition 1 is satisfied. We show that (ii) is also satisfied. Let g be the smallest integer with $\beta_{g,g+\text{reg}(S/I)}(S/I) \neq 0$. By the same Lemma we have $\text{reg}(S/I) = r_g(I) = g(d - 1)$. Since $\text{reg}(S/I) = \text{im}(\mathcal{C})(d - 1)$, we conclude that $g = \text{im}(\mathcal{C})$. Let $i \geq g$ with $r_i(I) < r_{i+1}(I)$. It follows that $i > g$, because $r_j(I) \leq \text{reg}(S/I)$ for all j . Since $r_i\text{-strand}(I) \neq \emptyset$ by Proposition 5(a) we have $r_i(I) = k(d - 1)$ for some k . Moreover, since $r_i(I) \leq \text{reg}(S/I) = g(d - 1)$, we have $k \leq g$. Therefore $i > k$.

Fig. 4 A triangulated 3-uniform hypergraph $\mathcal{C}$

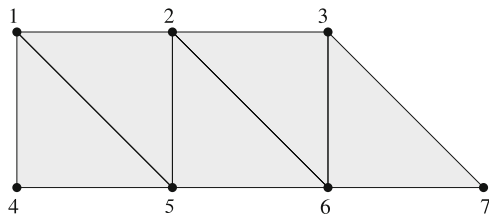


Table 2 Betti diagram of $S/I(\mathcal{C})$ with $\mathcal{C}$ as in Fig. 4

	0	1	2	3
0 :	1	-	-	-
1 :	-	-	-	-
2 :	-	5	4	-
3 :	-	-	-	-
4 :	-	-	1	1

Proposition 5 implies that there exists an integer k' such that $r_{i+1}(I) = k'(d-1)$ and $r_{i+1}\text{-strand}(I) = [k', p_{r_{i+1}(I)}(I)]$. Since, by [14, Corollary 3], $r_{i+1}(I) \leq r_i(I) + d - 1$, we have $k' \leq k + 1$. It follows that $k' = k + 1$, because $r_i(I) < r_{i+1}(I)$. Therefore $i \geq k'$. Since $i + 1 \in r_{i+1}\text{-strand}(I)$ we have $i < p_{r_{i+1}(I)}(I)$. Therefore $i \in r_{i+1}\text{-strand}(I)$, i.e. $\beta_{i, i+r_{i+1}(I)}(I) \neq 0$, a contradiction to the definition of $r_i(I)$. Hence, $r_i(I) \geq r_{i+1}(I)$ for all $i \geq g$. $\square$

Figure 4 shows a triangulated 3-uniform hypergraph $\mathcal{C}$. The Betti diagram of the quotient ring $S/I(\mathcal{C})$ is given in Table 2.

As seen in Theorem 2, this Betti diagram has a special shape.

A *simplicial vertex* in a graph G is a vertex v such that $G_{N_G[v]}$ is a complete graph. It is a well known fact that any chordal graph admits a simplicial vertex. As mentioned in Theorem 1, chordal graphs have a special shape:

Corollary 3 *Let G be a chordal graph. Then $I(G)$ has a special shape, and hence it satisfies subadditivity.*

Proof Each simple graph is properly-connected. Since any induced subgraph of a chordal graph is again chordal and since any chordal graph has a simplicial vertex, we conclude that any chordal graph is triangulated. Thus the assertion follows from Theorem 2. $\square$

Next we consider generalizations of whisker graphs. Let G be a graph on $[n]$ and G^* its whisker graph. The graph G^* is obtained from G by attaching to each vertex i of G a new edge $\{i, i'\}$. Then $I(G^*) = I(G) + (x_1y_1, \dots, x_ny_n)$, and hence $I(G^*)$ is the polarization of the 0-dimensional ideal $(I(G), x_1^2, \dots, x_n^2)$.

We now consider more generally a monomial ideal $I \subset S$ with $\dim(S/I) = 0$. Then I contains pure powers of all the variables. Thus we can write $I = (J, x_1^{a_1}, \dots, x_n^{a_n})$, where J is the monomial ideal generated by monomials $u = x_1^{b_1} \cdots x_n^{b_n}$ with $b_i < a_i$ for $i = 1, \dots, n$. The graded Betti numbers of I are determined by those of J and certain colon ideals, see Formula (6).

For a standard graded K -algebra $R = S/I$, we denote by

$$P_R = \sum_{i,j} \dim_K \operatorname{Tor}_i^S(R, K)_j t^i s^j$$

the graded Poincaré series of R . Under our assumptions on I and J we will use a result of Mermin, Peeva and Stillman [21, Theorem 2.1] in the more general version as quoted by Mermin and Murai [20, Equation (1.3)], according to which

$$P_{S/I} = \sum_{T \subset [n]} t^{|T|} s^{a_T} P_{S/J_T}, \quad (6)$$

where $a_T = \sum_{i \in T} a_i$ and where $J_T = J : \prod_{i \in T} x_i^{a_i}$.

Let Δ be a simplicial complex on the vertex set $[n]$. To each vertex i of Δ we attach a number a_i and for $F \in \Delta$ we set $a_F = \sum_{i \in F} a_i$. Finally, we let

$$t_j(\mathbf{a}, \Delta) = \max\{a_F : F \in \Delta, |F| = j\}.$$

Here $\mathbf{a}$ stands for the sequence $a_1, \dots, a_n$.

Proposition 6 *Let $J \subset S$ be a monomial ideal whose minimal monomial generating set does not contain any pure power of the variables. and let $I = (J, x_1^{a_1}, \dots, x_n^{a_n})$. Assume that the highest degree of a minimal monomial generator of J is less than or equal to $\min\{a_1, \dots, a_n\}$. Furthermore, let Δ be the simplicial complex with $I_\Delta = \sqrt{J}$, and let $\dim \Delta = d - 1$. Then*

- (a) $t_j(I) \geq t_j(\mathbf{a}, \Delta)$ for $j = 1, \dots, d$. If Δ is unmixed or if $a_1 = a_2 = \dots = a_n$, then equality holds.
- (b) $\operatorname{reg}(S/I) = t_d(\mathbf{a}, \Delta)$, if J is squarefree.

Proof

- (a) It follows from formula (6) that

$$t_j(I) = \max_T \{t_{j-|T|}(J_T) + a_T\}, \quad (7)$$

where the maximum is taken over all $T \subset [n]$ with $|T| \leq j$ and such that $J_T \neq S$. We claim that $J_T \neq S$ if and only if $T \in \Delta$. Indeed, let $P_1, \dots, P_m$ be the minimal prime ideals of J . Then these prime ideals are also the minimal prime ideals of I_Δ . Thus, if $F_1, \dots, F_m$ are the facets of Δ , then $P_i = P_{[n] \setminus F_i}$ for $i = 1, \dots, m$. It follows that $J_T \neq S$ if and only if $T \subset F_i$ for some i , in other words, if $T \in \Delta$.

It follows that

$$t_j(I) = \max\{t_{j-|T|}(J_T) + a_T : T \in \Delta, |T| \leq j\} \quad (8)$$

Since we assume that $j \leq d$, there exists $T \in \Delta$ with $|T| = j$. Thus

$$t_j(I) \geq \max\{t_0(J_T) + a_T : T \in \Delta, |T| = j\} = \max\{a_T : T \in \Delta, |T| = j\} = t_j(\mathbf{a}, \Delta).$$

Assume now that Δ is unmixed, and let $T \in \Delta$ with $|T| \leq j$. Since Δ is unmixed there exists $F \in \Delta$ with $|F| = j$ and $T \subset F$. Let g be the maximal degree of a generator of J . Then [14, Corollary 4] implies that

$$t_{j-|T|}(J_T) + a_T \leq (j - |T|)g + a_T \leq a_{F \setminus T} + a_T = a_F \leq t_j(\mathbf{a}, \Delta).$$

Finally assume that $a = a_i$ for all i . Then

$$t_{j-|T|}(J_T) + a_T \leq (j - |T|)g + a|T| \leq ja = t_j(\mathbf{a}, \Delta).$$

Thus in both cases $t_j(I) \leq t_j(\mathbf{a}, \Delta)$.

(b) A K -basis of S/I consists of the monomials $\prod_{i \in F} x_i^{b_i}$ with $F \in \Delta$ and $b_i < a_i$ for $i = 1, \dots, n$. Since $\dim(S/I) = 0$, the regularity of S/I is given by the highest degree of one of these basis elements. Basis elements of largest degree are of the form $\prod_{i \in F} x_i^{a_i-1}$, where F is a face with $|F| = d$. This yields the desired result. $\square$

Theorem 1 for whisker graphs is contained as special case of the following result. We maintain the notation of Proposition 6. After relabeling we may assume in addition that $a_1 \leq a_2 \leq \dots \leq a_n$.

Corollary 4

- (a) Let Δ be a simplicial complex and let $J = I_\Delta$. If Δ is unmixed or if $a_1 = a_2 = \dots = a_n$, then I satisfies subadditivity.
- (b) Suppose in addition that there exists a facet F of Δ with $a_F = \sum_{j=1}^d a_{n-j+1}$. Then the Betti diagram of S/I has a special shape.

Proof

(a) It is clear from the definition of the numbers $t_i(\mathbf{a}, \Delta)$ that $t_{i+j}(\mathbf{a}, \Delta) \leq t_i(\mathbf{a}, \Delta) + t_j(\mathbf{a}, \Delta)$ for all i and j with $i + j \leq \dim \Delta + 1$. Thus as long as $i + j \leq d$ subadditivity holds, and for i with $d \leq i \leq \text{proj dim}(S/I)$, we have that $t_i(I) = t_d(\mathbf{a}, \Delta)$ by Proposition 6(b). Thus subadditivity holds for S/I .

(b) The assumption implies that $t_i(\mathbf{a}, \Delta) = \sum_{j=1}^i a_{n-j+1}$. This yields the desired conclusion. $\square$

The following example demonstrates Corollary 4. Here $J = (x_1, x_2) \cap (x_2, x_3) \cap (x_3, x_4) \cap (x_4, x_5)$ and $a_1 = 2, a_2 = 3, a_3 = 4, a_4 = 5$. The dimension of S/J is 3.

In this example, as seen in Table 3, $r_1(I) - r_0(I) = 4 < r_2(I) - r_1(I) = 3 < r_3(I) - r_2(I) = 2$.

Table 3 Betti diagram of S/J

	0	1	2	3	4	5
0:	1	-	-	-	-	-
1:	-	2	-	-	-	-
2:	-	7	14	7	1	-
3:	-	1	9	15	8	1
4:	-	1	2	4	3	1
5:	-	-	6	7	2	-
6:	-	-	1	4	4	1
7:	-	-	1	1	-	-
8:	-	-	-	2	3	1
9:	-	-	-	1	2	1

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Koszul Algebras Defined by Three Relations

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Abstract This work concerns commutative algebras of the form $R = Q/I$, where Q is a standard graded polynomial ring and I is a homogenous ideal in Q . It has been proposed that when R is Koszul the i th Betti number of R over Q is at most $\binom{g}{i}$, where g is the number of generators of I ; in particular, the projective dimension of R over Q is at most g . The main result of this work settles this question, in the affirmative, when $g \leq 3$.

1 Introduction

This work is about the homological properties of homogeneous affine algebras, that is to say, algebras R of the form Q/I where $Q = k[x_1, \dots, x_e]$, with each x_i of degree one, and I is a homogenous ideal in Q . The emphasis is on *Koszul* algebras: algebras R with the property that $\mathrm{Tor}_i^R(k, k)_j = 0$ whenever $i \neq j$; equivalently, the minimal graded free resolution of k over R is linear. We are interested in the connection between the Koszul property of R and invariants of R as a Q -module; in particular, the graded Betti numbers, namely, the numbers $\beta_{i,j}^Q(R) := \mathrm{rank}_k \mathrm{Tor}_i^Q(R, k)_j$, and the total Betti numbers $\beta_i^Q(R) := \sum_j \beta_{i,j}^Q(R)$.

It has long been known that there is such connection: Backelin [5] and Kempf [18] proved that when R is Koszul $\beta_{i,j}^Q(R) = 0$ when $j > 2i$. Said otherwise, in the Betti table of R viewed as an Q -module, the nonzero entries all lie on or above the diagonal line. In [3, 4] these results have been refined to obtain more stringent constraints on the shape of this Betti table. Our focus is on the Betti numbers themselves and in particular the following intriguing question formulated in [4].

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Question 1.1 Set $g := \beta_1^Q(R)$, the number of relations defining R . If R is a Koszul algebra, are there inequalities

$$\beta_i^Q(R) \leq \binom{g}{i} \quad \text{for each } i?$$

In particular, is the projective dimension of R over Q at most g ?

This conjecture holds when the ideal I of relations can be generated by monomials, for then the Taylor resolution of R over Q furnishes the desired bounds. It follows by standard arguments that the same is true when I has a Gröbner basis of quadrics; or even if it has one after a suitable deformation; see Remark 4.1. As far as we are aware, no other families of Koszul algebras are known that satisfy this conjecture. On the other hand, it would be remiss of us not to mention that the only known example of a Koszul algebra that does not have a Gröbner basis of quadrics is [13, Example 1.20].

The main result of this work is that Question 1.1 has an affirmative answer when $g \leq 3$; in particular, Koszul algebras defined by 3 equations have projective dimension at most 3; see Corollary 4.6. Combining this with earlier work of Backelin and Fröberg [6] and Conca [11], we arrive at a similar conclusion when the embedding dimension of R is at most 3; see Remark 4.7.

This may seem scant evidence for an affirmative answer to Question 1.1 but there exist ideals I generated by three quadrics with $\text{pdim}_Q(Q/I) = 4$, and that is the largest it can be, by a result of Eisenbud and Huneke; see [20]. So the Koszul property is reflected already in this special case. Another reason the preceding result is not without interest is that, by a result of Bruns [8], essentially every free resolution over Q is the free resolution of an ideal that can be generated by three elements; these will not, in general, be quadrics.

Our proof of Corollary 4.6 is based on some general results concerning the *multiplicative structure* of $\text{Tor}_*^Q(R, k)$, which is nothing but the Koszul homology algebra of R . We prove that when R is Koszul, the diagonal k -subalgebra $\oplus_i \text{Tor}_i^Q(R, k)_{2i}$ is the quotient of the exterior algebra on $\text{Tor}_1^Q(R, k)_2$, modulo quadratic relations that depend only on the first syzygies of I ; see Theorem 3.3 and Remark 3.5. One consequence of this is that, for Koszul algebra, one has $\beta_{i,2i}^Q(R) \leq \binom{g}{i}$ for all i . Moreover, if equality holds for *some* $2 \leq i \leq g$ then R is a complete intersection; the proof of this latter also uses a characterization of complete intersections in terms of the product in the Koszul homology algebra, due to Bruns [9]. These results are proved in Sect. 3. The arguments exploit the machinery of minimal models of algebras, developed by Avramov [2]. The relevant details are recalled in Sect. 2, where they are also used to establish results on almost complete intersection rings, which also play a key role in the proof of our main result.

Throughout the paper k will be a field and $R := \{R_i\}_{i \geq 0}$ a standard graded finitely generated k -algebra; in other words, R is generated as a k -algebra by R_1 and $\text{rank}_k R_1$ is finite. Let Q be the symmetric k -algebra on R_1 and $Q \rightarrow R$ the canonical projection. In particular, Q is a standard graded finitely generated polynomial ring and the ideal $I := \text{Ker}(Q \rightarrow R)$ is homogenous and contained in $Q_{\geq 2}$.

2 Betti Numbers and Deviations

In this section we recollect the construction of the minimal model of R , and certain numerical invariants based on them; namely the deviations of R and the Betti numbers of R over Q . There is little new, except Theorem 2.7.

Since $Q \rightarrow R$ is a morphism of graded rings, resolutions of R over Q , and invariants based on them, have an *internal* degree induced by the grading on Q , in addition to the usual homological degree. In what follows, given an element a in such a bigraded object we write $\deg(a)$ for the internal degree and $|a|$ for the homological degree; the tacit assumption is that only homogeneous elements of graded objects are considered.

2.1 Minimal Models

Let $Q[X]$ be a *minimal model* for R over Q . Thus, X is a bigraded set such that for each n , the set $X_n := \{x \in X \mid |x| = n\}$ is finite and the graded algebra underlying $Q[X]$ is $Q \otimes_k \bigotimes_{n=1}^{\infty} k[X_n]$ where $k[X_n]$ is the symmetric algebra on the graded K -vector space kX_n when n is even, and the exterior algebra on that space when n is odd. In particular, $Q[X]$ is strictly graded-commutative with respect to homological degree: For a, b in $Q[X]$ one has

$$ab = (-1)^{|a||b|}ba \quad \text{and} \quad a^2 = 0 \quad \text{if } |a| \text{ is odd.}$$

The differential on $Q[X]$ satisfies the Leibniz rule and is *decomposable*, in that

$$d(x) \subseteq (Q_{\geq 1} + (X))^2 Q[X] \quad \text{for all } x \in X. \quad (1)$$

Thus, $Q[X]$ is a DG(=Differential Graded) Q -algebra. There is a morphism of Q -algebras $Q[X] \rightarrow R$ that is a quasiisomorphism, so that $Q[X]$ is a DG algebra resolution of R . For details of the construction see [2, §7.2]. The first steps can be described explicitly.

Remark 2.1 Let $f := f_1, \dots, f_g$ be a minimal generating set for the ideal I and let $Q[X_1]$ be the Koszul complex on f . Thus, $X_1 := \{x_1, \dots, x_g\}$ is a set of indeterminates with $|x_i| = 1$ and $\deg(x_i) = \deg(f_i)$ for each i , and $Q[X_1]$ is the

exterior algebra QX_1 . The differential on $Q[X_1]$ is determined by $d(x_i) = f_i$. The canonical augmentation $Q[X_1] \rightarrow R$ induces an isomorphism $H_0(Q[X_1]) \cong R$.

Let $z := z_1, \dots, z_l$ be cycles in $Q[X_1]$ that form a minimal generating set for the R -module for $H_1(Q[X_1])$. The next step in the construction of the minimal model is to kill these cycles. In detail: Let $X_2 := \{y_1, \dots, y_l\}$ be a graded set with $|y_i| = 2$ and $\deg(y_i) = \deg(z_i)$ for each i . With $k[X_2]$ the symmetric algebra on kX_1 , one has $Q[X_{\leq 2}] = Q[X_1] \otimes_k k[X_2]$. The differential on $Q[X_{\leq 2}]$ extends the one on $Q[X_1]$ and is determined by $d(y_i) = z_i$. Thus $Q[X_1]$ is a DG subalgebra of $Q[X_{\leq 2}]$, the augmentation $Q[X_1] \rightarrow R$ extends to $Q[X_{\leq 2}] \rightarrow R$, and satisfies

$$H_0(Q[X_{\leq 2}]) \cong R \quad \text{and} \quad H_1(Q[X_{\leq 2}]) = 0.$$

The next step of the construction is to kill the cycles in $H_2(Q[X_{\leq 2}])$, and so on. One thus gets a tower of DG Q -algebras $Q[X_1] \subseteq Q[X_{\leq 2}] \subseteq \dots$ whose union is $Q[X]$. By construction, for each $n \geq 1$ one has

$$H_i(Q[X_{\leq n}]) = \begin{cases} R & \text{for } i = 0 \\ 0 & \text{for } 1 \leq i \leq n - 1. \end{cases}$$

Moreover, $d(X_{n+1})$ is a minimal generating set for the R -module $H_n(Q[X_{\leq n}])$.

Henceforth, we fix a minimal model $Q[X]$ for R and set $k[X] := k \otimes_Q Q[X]$.

Remark 2.2 It follows from (1) that the differential on $k[X]$ satisfies $d(X) \subseteq (X)^2$. In particular, $d(X_1) = 0 = d(X_2)$, so that $H_1(k[X]) \cong kX_1$ as bigraded k -vector spaces, and the k -vector subspace of the cycles in $k[X]_2$ is $kX_2 \oplus \wedge^2 kX_1$. Moreover, one has

$$H_2(k[X]) \cong kX_2 \oplus (\wedge^2 kX_1 / d(kX_3)) \quad \text{and} \quad \text{Ker}(kX_3 \xrightarrow{d} \wedge^2 kX_1) \subseteq H_3(k[X]) \quad (2)$$

as bigraded k -vector spaces.

Next we recall some numerical invariants that can be read off the minimal model.

2.2 Betti Numbers and Deviations

Since $Q[X]$ is a DG algebra resolution of R over Q , there is an isomorphism

$$H(k[X]) \cong \text{Tor}^Q(k, R)$$

of bigraded k -algebras. Thus the graded Betti numbers of R over Q are given by

$$\beta_{ij}^Q(R) := \text{rank}_k H_i(k[X])_j \quad \text{and} \quad \beta_i^Q(R) := \sum_j \beta_{ij}^Q(R).$$

The notation notwithstanding, these Betti numbers are invariants of R alone, for they correspond to ranks of Koszul homology modules of R ; see Remark 3.1. The same is the case with the *deviations*, $\{\varepsilon_{ij}(R)\}$, of R which are the integers

$$\varepsilon_{ij}(R) := \text{card}(X_{i-1,j}) \quad \text{and} \quad \varepsilon_i(R) := \sum_j \varepsilon_{ij}(R) \quad \text{for } i \geq 1.$$

Typically, these invariants are derived from the generating series for $\text{Tor}_i^R(k, k)_j$; the definition above is justified by [2, Theorem 7.2.6].

Remark 2.3 The deviations can be estimated in terms of the homology of the DG subalgebras $Q[X_{\leq i}]$ of $Q[X]$ generated by the graded set $X_{\leq i}$, for various i . Indeed, the exact sequence of complexes

$$0 \longrightarrow Q[X_{\leq i}] \longrightarrow Q[X_{\leq i+1}] \longrightarrow Q[X_{\leq i+1}]/Q[X_{\leq i}] \longrightarrow 0$$

yields, in homology, the exact sequence of graded R -modules

$$H_{i+1}(Q[X_{\leq i+1}]) \longrightarrow H_{i+1}(Q[X_{\leq i+1}]/Q[X_{\leq i}]) \longrightarrow H_i(Q[X_{\leq i}]) \longrightarrow 0.$$

By construction, the term in the middle is RX_{i+1} and the map $RX_{i+1} \rightarrow H_i(Q[X_{\leq i}])$ is a minimal presentation. Thus, the exact sequence above yields a presentation:

$$RX_{i+2} \xrightarrow{d} RX_{i+1} \longrightarrow H_i(Q[X_{\leq i}]) \longrightarrow 0. \quad (3)$$

where $d(X_{i+2}) \subseteq R_{\geq 1}X_{i+1}$. This discussion justifies the following result.

Lemma 2.4 *For all $i \geq 1$ and $j \in \mathbb{Z}$, there are (in)equalities*

$$\varepsilon_{i+2,j}(R) = \beta_{0,j}^R(H_i(Q[X_{\leq i}])) \quad \text{and} \quad \varepsilon_{i+3,j}(R) \geq \beta_{1,j}^R(H_i(Q[X_{\leq i}])) \quad \square$$

The next result explains why deviations have a bearing on Question 1.1.

Proposition 2.5 *There are inequalities*

$$-\varepsilon_3(R) \leq \binom{\beta_1^Q(R)}{2} - \beta_2^Q(R) \leq \varepsilon_4(R) - \varepsilon_3(R);$$

When $\varepsilon_{ij}(R) = 0$ for $i \leq 4$ and $j > i$, equality holds on the right iff $\beta_{34}^Q(R) = 0$.

Proof From (2) one gets an equality

$$\binom{\varepsilon_2(R)}{2} - \text{rank}_k H_2(k[X]) = \text{rank}_k d(kX_3) - \varepsilon_3(R).$$

The inequalities follow, since $\varepsilon_2(R) = \beta_1^Q(R)$ and $\text{rank}_k H_i(k[X]) = \beta_2^Q(R)$.

The stated hypothesis on $\varepsilon_{ij}(R)$ implies $\text{Ker}(kX_3 \rightarrow \wedge^2 kX_1) = H_3(k[X])_4$. This justifies the last assertion.

The inequalities in Proposition 2.5 can be strict.

Remark 2.6 Recall that $I = \text{Ker}(Q \rightarrow R)$. Assume that for some prime $\mathfrak{p} \supseteq I$ the ideal $I_{\mathfrak{p}}$ is generated by a regular sequence of length $\beta_1^Q(R)$. For example, this is so if R is Cohen-Macaulay and a residual intersection; see [16, 5.8]. There are inequalities

$$\binom{\beta_1^Q(R)}{2} - \beta_2^Q(R) \leq 0 \quad \text{and} \quad 0 \leq \varepsilon_4(R) - \varepsilon_3(R).$$

Indeed, the hypotheses on $I_{\mathfrak{p}}$ yields, for each i , the equalities below.

$$\beta_i^Q(R) \geq \beta_i^{Q_{\mathfrak{p}}}(R_{\mathfrak{p}}) = \binom{\beta_1^{Q_{\mathfrak{p}}}(R_{\mathfrak{p}})}{i} = \binom{\beta_1^Q(R)}{i}.$$

The inequality is standard. As $H_1(Q[X_1])_{\mathfrak{p}} = 0$, the assertion about the deviations is obtained by localizing the exact sequence (3) for $i = 1$, at the prime ideal $\mathfrak{p}$.

Both inequalities can be strict: For $Q = k[x_1, \dots, x_4]$ and $I = (x_1^2 - x_2^2, x_1x_2, x_3x_4)$, the Betti table of R over Q is

	0	1	2	3
0	1	—	—	—
1	—	3	—	—
2	—	—	4	2

On the other hand, a direct computation shows that $\varepsilon_3(R) = 1$ and $\varepsilon_4(R) = 2$.

2.3 Almost Complete Intersections

The ring R is *almost complete intersection* if it satisfies $\beta_1^Q(R) = \dim Q - \dim R + 1$; in words, if the minimal number of generators for $\text{Ker}(Q \rightarrow R)$ is precisely one more than its codimension. In what follows ω_R is the canonical module of R , namely, the R -module $\text{Ext}_Q^c(R, Q)$, where $c = \dim Q - \dim R$.

Theorem 2.7 *If R is an almost complete intersection, then $\varepsilon_3(R) \leq \varepsilon_4(R)$.*

Proof Kunz [19, Proposition 1.1] proved that the canonical module of an almost complete intersection is not free. In the minimal model $Q[X]$, the DG algebra $Q[X_1]$ is the Koszul complex on I . As R is almost complete intersection $H_1(Q[X_1]) \cong \omega_R$; see [19, Proposition 2.1]. The desired result thus follows from Lemmas 2.4 and 2.8 below.

Lemma 2.8 *If $\beta_0^R(\omega_R) > \beta_1^R(\omega_R)$, then ω_R is free, of rank one.*

Proof The grading plays no role in what follows, so is ignored. One has $\omega_R \cong \omega_{Q/J}$, where J is the intersection of the primary component of I with height equal to the height of I . It is then easy to check that there are (in)equalities

$$\beta_1^R(\omega_R) \geq \beta_1^{Q/J}(\omega_R) \quad \text{and} \quad \beta_0^R(\omega_R) = \beta_0^{Q/J}(\omega_R).$$

Thus, it suffices to verify the result for Q/J , and so, replacing R by Q/J , we may assume that I has no embedded associated primes. Consider a minimal presentation of ω_R .

$$0 \longrightarrow Z \longrightarrow R^{\beta_1} \longrightarrow R^{\beta_0} \longrightarrow \omega_R \longrightarrow 0.$$

For prime $\mathfrak{p} \in \text{Ass } R$, the $R_{\mathfrak{p}}$ -module $(\omega_R)_{\mathfrak{p}} \cong \omega_{R_{\mathfrak{p}}}$ is the canonical module of $R_{\mathfrak{p}}$, which is a local ring of dimension zero. With $\ell(-)$ denoting length, from the exact sequence above one then gets

$$\ell(Z_{\mathfrak{p}}) = (\beta_1 - \beta_0 + 1)\ell(R_{\mathfrak{p}}) < 0.$$

The inequality is by the hypothesis. Thus, $Z_{\mathfrak{p}} = 0$ for each $\mathfrak{p} \in \text{Ass } R$, which yields $Z = 0$. Thus, ω_R has finite projective dimension. Aoyama's result [1, Theorem 3] then implies that ω_R is free of rank one as desired.

Remark 2.9 Jorgensen and Leuschke [17, Question 2.6] ask: Is a Cohen-Macaulay ring R with $\beta_0^R(\omega_R) \geq \beta_1^R(\omega_R)$ Gorenstein? Lemma 2.8 settles it when the inequality is strict. The Cohen-Macaulay assumption is needed; consider $R = k[[x, y]]/(x^2, xy)$.

3 The Koszul Homology Algebra

This section concerns the Koszul homology algebra of R . The main result, Theorem 3.3, describes its diagonal subalgebra, in the sense explained further below, when the resolution of k over R is linear for the first few steps.

Let K^R be the Koszul complex of R ; see [10, §1.6]. By construction, K^R is a DG R -algebra whose underlying graded algebra is $R \otimes_k \bigwedge_k V$, where $V = \Sigma R_1$, the k -vector space R_1 in (homological) degree one. The differential on K^R is R -linear and defined by $d(\Sigma v) = v$ for $v \in R_1$ and the Leibniz rule. Observe that K^R is bigraded, with internal degree inherited from R , and strictly graded-commutative with respect to the homological degree. Its homology algebra, $H(K^R)$, inherits these properties. The next remark is well known.

Remark 3.1 Let $Q[X]$ be the minimal model for R over Q , introduced in Sect. 2, and set $k[X] := k \otimes_Q Q[X]$. There are isomorphisms of bigraded k -algebras

$$H(K^R) \cong H(k[X]) \cong \mathrm{Tor}^Q(k, R).$$

Indeed, the Koszul complex K^Q of Q is a free resolution of k over Q , and $Q[X]$ is free resolution of R . Hence there are quasiisomorphisms of DG algebras

$$k \otimes_Q Q[X] \xleftarrow{\simeq} K^Q \otimes_Q Q[X] \xrightarrow{\simeq} K^Q \otimes_Q R \cong K^R.$$

In homology, this yields the stated isomorphism between $H(K^R)$ and $H(k[X])$. The second isomorphism has been commented on earlier.

It follows from the preceding isomorphisms that $\mathrm{rank}_k H_i(K^R)_j = \beta_{ij}^Q(R)$. The focus of this section is the k -subalgebra

$$\Delta(R) = \bigoplus_{i \geq 0} H_i(K^R)_{2i}$$

that we call the *diagonal subalgebra* of $H(K^R)$. It is supported on the main diagonal of the Betti table of R over Q . Being a subalgebra of $H(K^R)$, the k -algebra $\Delta(R)$ is also strictly graded-commutative, with $\Delta(R)_1 = \Delta(R)_{1,2} = H_1(R)_2$. Hence, by the universal property of exterior algebras, there is a morphism of graded k -algebras

$$\rho: \bigwedge_k H_1(R)_2 \longrightarrow \Delta(R). \quad (4)$$

This map is surjective when R is Koszul, and then one has a concrete description of its kernel; see Theorem 3.3 below.

3.1 Koszul Algebras and Regularity

Recall that the k -algebra R is *Koszul* if $\mathrm{Tor}_i^R(k, k)_j = 0$ unless $i = j$; equivalently, if the minimal resolution of k over R is linear. We need a weakening of this condition, and to this end recall an invariant introduced in [4, §4]: The *n th partial regularity* of the R -module k is the number

$$\mathrm{reg}_n^R(k) := \sup\{j - i \mid \text{where } i \leq n \text{ and } \mathrm{Tor}_i^R(k, k)_j \neq 0\}.$$

Thus, R is Koszul precisely when $\mathrm{reg}_n^R(k) \leq 0$ for each n . We are particularly interested in the condition $\mathrm{reg}_n^R(k) = 0$ that translates to the condition that the minimal resolution of k over R is linear up to degree n ; equivalently, $\varepsilon_{ij}(R) = 0$ for $i \leq n$ and $j \neq i$. Given the description of the deviations in terms of the minimal

model $Q[X]$ of R , described in Sect. 2, one gets:

$$\operatorname{reg}_{n+1}^R(k) = 0 \text{ if and only if } X_{i,j} = \emptyset \text{ for } i \leq n \text{ and } j \neq i + 1. \quad (5)$$

This equivalence plays an important role in the sequel. Roos [23] has constructed, for each integer $n \geq 2$, a k -algebra R that is *not* Koszul and has $\operatorname{reg}_{n+1}^R(k) = 0$.

We also repeatedly use the following facts established in [4, Theorem 4.1].

Remark 3.2 When $\operatorname{reg}_{n+1}^R(k) = 0$ for an integer n , then for each $0 \leq i \leq n$ one has:

1. $\beta_{i,j}^Q(R) = 0$ for $j > 2i$;
2. $\Delta(R)_i = \Delta_1(R)^i$.

The statement, and proof, of the next result is an elaboration of the remark above.

Theorem 3.3 Assume $\operatorname{reg}_{n+1}^R(k) = 0$ for $n = \operatorname{edim} R - \operatorname{depth} R$. With $Q[X]$ denoting the minimal model of R , the differential on $k[X] := k \otimes_Q Q[X]$ satisfies $d(X_3) \subseteq k[X_1]_{2,4}$ and there is an isomorphism of k -algebras

$$\Delta(R) \cong k[X_1]/(d(X_3)).$$

Proof Since $d(X_1) = 0$, as noted in Remark 2.2, the differential on $k[X_1]$ is zero as well. Consider the inclusion $k[X_1] \subseteq k[X]$ of DG algebras. The map (4) is realized as the induced map on homology:

$$\rho: k[X_1] \longrightarrow \bigoplus_{i \geq 0} H_i(k[X])_{2i}.$$

For any monomial $x_1^{d_1} \cdots x_s^{d_s}$ in $k[X]$ with $x_n \in X$, the hypothesis on regularity, in the form (5), yields

$$\deg(x_1^{d_1} \cdots x_s^{d_s}) = \sum_{n=1}^s (|x_n| + 1) d_n.$$

Given this, an elementary computation yields that $\deg(x_1^{d_1} \cdots x_s^{d_s}) = 2|x_1^{d_1} \cdots x_s^{d_s}|$ precisely when $|x_n| = 1$ for $n = 1, \dots, s$. Thus, the diagonal subalgebra $\oplus_i k[X]_{i,2i}$ of $k[X]$ is $k[X_1]$. Since $H(k[X])$ is subquotient of $k[X]$ and it follows that ρ is surjective.

To verify the claim about its kernel, it suffices to verify that there is an equality

$$d(k[X]) \cap k[X_1] = d(X_3)k[X_1].$$

Again from Remark 2.2 one gets $d(X_2) = 0$ and $d(X_3) \subseteq k[X_1]$. Thus it suffices to consider the differential of monomials $x_1^{d_1} \cdots x_s^{d_s}$ where $|x_n| \geq 3$ for some n in

$1, \dots, s$. For such a monomial, it is easy to verify that

$$\deg(x_1^{d_1} \cdots x_s^{d_s}) = 2(|x_1^{d_1} \cdots x_s^{d_s}| - 1)$$

if and only if $|x_n| = 3$ for exactly one n in $1, \dots, s$, and then $d_n = 1$, and $x_n = 1$ for the rest. This is the desired result.

The following corollary will be useful in the sequel.

Corollary 3.4 *Assume $\operatorname{reg}_{n+1}(R) = 0$ for $n = \operatorname{edim} R - \operatorname{depth} R$. If $\beta_{i,2i}^Q(R) \geq (\beta_i^Q(R))$ for some $2 \leq i \leq \beta_1^Q(R)$, then R is a complete intersection.*

Proof By Theorem 3.3 the stated inequality holds if and only if $\operatorname{map} d(X_3) = 0$; equivalently, when the map ρ is injective. A theorem of Bruns [9, Theorem 2], see also [10, Theorem 2.3.14], then yields that R is complete intersection.

Next we give an elementary description of $d(kX_3)$ appearing in Theorem 3.3. This is based on the construction of the model $Q[X]$, described in Remark 2.1.

Remark 3.5 The set up and hypotheses are as in Theorem 3.3.

Suppose $X_1 = \{x_1, \dots, x_g\}$ and set $f_i = d(x_i)$. By construction, $\mathbf{f} := f_1, \dots, f_g$ is a minimal generating set for the ideal $\operatorname{Ker}(Q \rightarrow R)$, and $Q[X_1]$ is the Koszul complex on $\mathbf{f}$. Let $l_1, \dots, l_r$ be a generating set for the linear syzygies of $\mathbf{f}$; that is to say, for the k -vector space of cycles in $Q[X_1]_{1,3}$. Set $L = \sum_i Q_1 l_i$, the k -vector subspace of cycles in $Q[X_1]_{1,4}$ that are generated by the linear syzygies. In the same vein, let M be the k -vector subspace of $Q[X_1]_{1,4}$ spanned by the syzygies $\{f_i x_j - f_j x_i\}$, where $1 \leq i < j \leq g$. Thus $V = L \cap M$ is the k -vector space generated by the nonminimal Koszul syzygies. Choose a basis $b_1, \dots, b_p$ of the k -vector space V and write

$$b_h = \sum_{1 \leq i < j \leq g} c_{hij} (f_i x_j - f_j x_i).$$

It follows from the construction of the minimal model that the image of d is the ideal of $k[X_1]$ generated by the quadratic forms

$$\sum_{ij} c_{hij} x_i x_j \quad h = 1, \dots, p.$$

Remark 3.6 By Remark 3.2, the subalgebra $\Delta(R)$ of $H(K^R)$ is generated by its linear part. The same is true of the larger subalgebra $\oplus_{j \geq 2i-1} H_{i,j}$; this is the content of [7, Theorem 3.1]. Its proof follows the lines of the argument for that of Theorem 3.3.

4 Algebras Defined by Three Relations

In this section we investigate the Koszul homology of Koszul algebras defined by three relations, answering Question 1.1 along the way for this class of rings. We begin by collecting some observations which are surely well known. In what follows R and Q are as before; recall that $I = \text{Ker}(Q \rightarrow R)$.

Remark 4.1 Let J be the initial ideal of I with respect to a term ordering. The semicontinuity of flat degenerations yields inequalities

$$\beta_{i,j}(R) \leq \beta_{i,j}(Q/J) \quad \text{for all } i, j. \quad (6)$$

The ideal J is monomial and its Taylor resolution [22, Construction 26.5] yields

$$\beta_i^Q(Q/J) \leq \binom{\beta_1^Q(Q/J)}{i}.$$

Combining the preceding inequalities yields an upper bound for the Betti numbers of R over Q in the spirit of Question 1.1, and settles it in a special case, namely, when k -algebra R is G -quadratic; that is to say, when in some system of coordinates and for some choice of term order the initial ideal J is generated by quadratics. In that case $\beta_1^Q(Q/J) = \beta_1^Q(R)$, so the estimates above settle Question 1.1. Since the Betti numbers of R over Q are invariant under change of coordinates, and quotient by a regular sequence of linear forms, Question 1.1 has an affirmative answer when R is LG -quadratic; see the paragraph preceding [3, Question 6.5] for details.

Proposition 4.2 *Assume I is generated by quadrics and set $n := \beta_{1,2}^Q(R)$. The following statements hold.*

- (1) $\beta_{i,i+1}^Q(R) \leq \binom{n}{i}$ for each i , and if equality holds for $i = 2$, then I has codimension one and a linear resolution of length n ;
- (2) $\beta_{i,i+1}^Q(R) \leq 2$ for some $i \geq 2$ implies $\beta_{i+1,i+2}^Q(R) = 0$;
- (3) $\beta_2^Q(R) \leq 2\binom{n}{2}$ when $\text{reg}_3^R(k) = 0$.

When J is the initial ideal of I with respect to a term ordering and $m := \beta_{1,3}^Q(J)$, the following inequalities hold

- (4) $\beta_{2,3}^Q(R) + \beta_{2,4}^Q(R) \leq \binom{n+m}{2}$;
- (5) $\beta_{2,3}^Q(R) + m \leq \binom{n}{2}$.

Proof For ease of notation we write $\beta(-)$ for $\beta^Q(-)$.

(2): Set $b := \beta_{i-1,i}(R)$ and $c := \beta_{i,i+1}(R)$. Choosing bases, the linear strand of the resolution of R over Q in homological degrees i and $i+1$ reads

$$\cdots \longrightarrow R(-i-1)^c \xrightarrow{\phi} R(-i)^b \xrightarrow{\psi} \cdots,$$

where ϕ is a $b \times c$ matrix of linear forms. The hypothesis is that $c \leq 2$ and the desired conclusion is that ϕ has no linear syzygies. This is clear when $c = 1$.

Assume $c = 2$. A linear syzygy of ϕ is a linear dependence among its two columns. This means that, with $[l_1, \dots, l_b]$ and $[m_1, \dots, m_b]$ denoting the columns, there exist distinct linear forms r, s such that

$$r[l_1, \dots, l_b] = s[m_1, \dots, m_b]$$

Now, whenever l_i is nonzero, so must m_i (and vice-versa). Then since $r \neq s$ we have that $r|m_i$ and $s|l_i$ for each i . But this implies that

$$\begin{aligned} [l_1, \dots, l_b] &= s[l'_1, \dots, l'_b] \\ [m_1, \dots, m_b] &= r[m'_1, \dots, m'_b] \end{aligned}$$

with l'_i and m'_i in k . Since $i \geq 2$, the image of ψ is a submodule of a free module, so it follows that $[l'_i]$ and $[m'_i]$ are themselves syzygies of ψ , which is absurd.

The initial ideal J of I plays a role in the remainder of the proof. We repeatedly use, without comment, (6).

(1): From Remark 4.1 one gets the inequalities below:

$$\beta_{i,i+1}(R) \leq \beta_{i,i+1}(Q/J) = \beta_{i,i+1}(Q/J_2) \leq \beta_i(Q/J_2) \leq \binom{\beta_1(Q/J_2)}{i} = \binom{n}{i}.$$

The last equality holds as $\beta_{1,2}(Q/J_2) = \beta_{1,2}(R)$. This justifies the first claim in (1).

Suppose that $\beta_{2,3}(R) = \binom{n}{2}$; we may assume $n \geq 2$. One then has equalities

$$\beta_{2,3}(R) = \beta_{2,3}(Q/J) = \binom{n}{2}.$$

By [22, Theorem 22.12], as I is generated by quadrics, there is an equality

$$\beta_{2,3}(R) = \beta_{2,3}(Q/J) - \beta_{1,3}(Q/J),$$

and hence $\beta_{1,3}(Q/J) = 0$. Given Buchberger's algorithm, it follows that J is generated in degree two, that is to say, $J = J_2$. In particular, $\beta_1(Q/J) = n$. Given this and the equality $\beta_{2,3}(Q/J) = \binom{n}{2}$ one deduces that the differential in the Taylor resolution of Q/J is minimal (meaning, it has coefficients in $Q_{\geq 1}$) in degrees 2 and 3. Given the description of the differential in the Taylor resolution, it follows that any pair of monomial generators of J has a common factor, and that the l.c.m. of any three of the generators is not equal to the l.c.m. of a proper subset. A simple computation then yields that the g.c.d. of the generators has degree one and thus J , and hence also I , has codimension one. Since I is generated by quadrics, it must

then be of the form $f(g_1, \dots, g_n)$, where f is a linear form and $g_1, \dots, g_n$ is a regular sequence of linear forms. This implies that I has a linear resolution of length n .

(3): Assume $\text{reg}_3^R(k) = 0$. Then $\beta_{2,j}(R) = 0$ for $j \neq 3, 4$, by Remark 3.2(1), so one gets the equality below

$$\beta_2(R) = \beta_{2,3}(R) + \beta_{2,4}(R) \leq \binom{n}{2} + \binom{n}{2}.$$

The inequalities hold by (1) above and Remark 3.2(2), which yields $\beta_{2,4}(R) \leq \binom{n}{2}$.

(4): This is by Remark 4.1 as $\beta_{2,3}(Q/J)$ and $\beta_{2,4}(Q/J)$ depend only on $J_{\leq 3}$.

(5): Since Q/J is a flat degeneration of R , the Betti table of R is obtained from that of Q/J via consecutive cancellations; see Peeva [21] and [22, Theorem 22.12]. Thus one has

$\beta(R)$	0	1	2	3	...
0	1	—	—	—	...
1	—	n	c	?	...
2	—	—	d	?	...
3					

$\beta(Q/J)$	0	1	2	3	...
0	1	—	—	—	...
1	—	n	$c + m$	?	...
2	—	m	$d + v$	?	...
3		...			

This justifies the inequality in (5), given (1).

Lemma 4.3 Assume that R is defined by 3 quadratic relations. If $\sum_{i \geq 4} \beta_{2,i}^Q(R) \leq 2$ then the projective dimension of R is at most 3.

Proof As before, we write $\beta_{i,j}$ for $\beta_{i,j}^Q(R)$. Proposition 4.2(1) yields $\beta_{2,3} \leq 3$ and also that when equality holds $\text{pdim}_Q R = 3$. When $\beta_{2,3} \leq 2$ the hypothesis yields $\beta_2 \leq 4$, so $\text{pdim}_Q R \leq 3$, by the Syzygy Theorem [10, Theorem 9.5.6].

In what follows we prove the following are the only possible Betti tables for Koszul algebras defined by three relations. The first one is the Betti table of a complete intersection of three quadrics; the second is that of the ring $k[x, y, z]/(x^2, y^2, xz)$; the third is defined by the ideal of minors of a 3×2 matrix of linear forms, and of rank two; for example, $k[x, y]/(x, y)^2$. The last is the Betti table of an ideal with linear resolution; for example, $k[x, y, z]/(x^2, xy, xz)$.

Remark 4.4 Assume $\text{edim } R = 3$. It follows from D’Ali’s classification [14, Theorem 3.1] of quadratic algebras that the ones that are *not* Koszul have Betti table

	0	1	2	3
0	1	—	—	—
1	—	3	—	—
2	—	—	4	2

This remark is used in proving (3) $\Rightarrow$ (1) and (4) $\Rightarrow$ (1) in the result below.

$\begin{array}{c cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 1 & - & - & - \\ 1 & - & 3 & - & - \\ 2 & - & - & 3 & - \\ 3 & - & - & - & 1 \end{array}$	$\begin{array}{c cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 1 & - & - & - \\ 1 & - & 3 & 1 & - \\ 2 & - & - & 2 & 1 \end{array}$	$\begin{array}{c ccc} & 0 & 1 & 2 \\ \hline 0 & 1 & - & - \\ 1 & - & 3 & 2 \end{array}$	$\begin{array}{c cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 1 & - & - & - \\ 1 & - & 3 & 3 & 1 \end{array}$
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Fig. 1 The Betti tables for Koszul algebras defined by three relations

Theorem 4.5 *When R is generated by 3 quadrics, the following conditions are equivalent.*

- (1) R is Koszul;
- (2) $\text{reg}_{n+1}^R(k) = 0$ for $n = \text{edim} R - \dim R$;
- (3) The Betti table for R over Q is one of those listed in Fig. 1.
- (4) $H_*(K^R)$ is generated, as a k -algebra, by its linear strand.

Proof We write $\beta_{i,j}$ for $\beta_{i,j}^Q(R)$. We remark that if R is a complete intersection there is little to prove, so we assume that it is not.

(1) $\Rightarrow$ (2): This is a tautology.

(2) $\Rightarrow$ (3): When (2) holds, Remark 3.2 and Corollary 3.4 show that $\beta_{2,4} \leq 2$ and $\beta_{2,j} = 0$ for $j > 4$. Now Lemma 4.3 guarantees that the projective dimension of R is at most 3. The proof of Proposition 4.2(1) shows that if $\beta_{2,3} = 3$ then the Betti table is the last one in the list above. If $\beta_{2,3} \leq 2$ then Proposition 4.2(2) yields $\beta_{3,4} = 0$. Finally $\beta_{3,6} = 0$ by Corollary 3.4, as R is not a complete intersection. Hence $\beta_3 = \beta_{3,5}$. The inequalities

$$\beta_{2,3} \leq 2 \quad \text{and} \quad \beta_{2,4} \leq 2$$

allow few possibilities for Betti tables. A computation using Boij-Söderberg theory [15] confirms that the only options are the middle two Betti tables in Fig. 1.

(3) $\Rightarrow$ (1): Evidently, if (3) holds R is quadratic and $\text{pdim}_Q R \leq 3$. Thus, one can pass to a quotient of R by a regular sequence of linear forms and assume that $\text{edim } R \leq 3$. It then follows from Remark 4.4 that R must be Koszul.

(1) $\Rightarrow$ (4): We have already verified that (1) $\Rightarrow$ (3); the desired implication thus follows from an inspection of the Betti tables in Fig. 1, and Remark 3.6.

(4) $\Rightarrow$ (1): Extending the field k , we can assume it is algebraically closed. We first prove that the projective dimension of R must be at most 3.

Recall from the proof of Proposition 4.2(1) that if $\beta_{2,3} = 3$ then R has a linear resolution of length three. Since $H_{1,2}(R)$ generates the diagonal by assumption, we know that $\beta_{2,4} \leq 3$. Assume $\text{pdim}_Q R \geq 4$. Then the Syzygy Theorem [10, Theorem 9.5.6] implies $\beta_2 = 5$ so that $\beta_{2,3} = 2$ and $\beta_{2,4} = 3$. Let J be the initial ideal of I with respect to some term order. By Proposition 4.2(5), the ideal J has one

cubic generator so the Betti table of Q/J is

	0	1	2	3
0	1	–	–	–
1	–	3	3	1
2	–	1	γ	?
3	–	?	?	?

Since $\beta_{3,4}(R) = 0$, by Proposition 4.2(2), we get that $\gamma \geq 4$, so the 1 in this table can cancel. This contradicts 4.2(4). This completes the proof that $\text{pdim}_Q R \leq 3$.

Given this, we can reduce to the case $\text{edim} R = 3$, as in the proof of (3) $\Rightarrow$ (1). Now again we apply Remark 4.4: if R is not Koszul, then it is clear from the Betti table in *op. cit.* that $H_*(K^R)$ is not generated by its linear strand.

The implication (1) $\Rightarrow$ (3) in the theorem above settles Question 1.1 when $g \leq 3$. It is worth noting that the proof of that implication does not use Remark 4.4.

Corollary 4.6 *If R is Koszul and $\beta_1^Q(R) \leq 3$, then*

$$\beta_i^Q(R) \leq \binom{\beta_1^Q(R)}{i} \quad \text{for each } i \geq 0. \quad \square$$

To wrap up, we note that the results in this work, combined with those available in the literature, settle Question 1.1 also for algebras of embedding dimension at most three; that is to say, for algebras with at most three generators.

Remark 4.7 Assume $\text{rank}_k(R_1) \leq 3$ so that $\text{rank}_k(R_2) \leq 6$. A theorem of Backelin and Fröberg [6, Theorem 4.8] shows that if $\text{rank}_k(R_2) \leq 2$, the ring R is Koszul, and then Conca [11] proves that R is LG-quadratic, with essentially one exception, and observes in [12] that that too is LG-quadratic. The inequality in Question 1.1 follows; see Remark 4.1. When $\text{rank}_k(R_2) \geq 3$ and R is Koszul, Corollary 4.6 leads to the same conclusion.

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Some Algebras with the Weak Lefschetz Property

David Cook II and Uwe Nagel

Abstract Using a connection to lozenge tilings of triangular regions, we establish an easily checkable criterion that guarantees the weak Lefschetz property of a quotient by a monomial ideal in three variables. It is also shown that each such ideal also has a semistable syzygy bundle.

1 Introduction

Recently, there have been many investigations of the presence of the weak Lefschetz property (see, e.g., [1, 2, 4, 9, 13–21]). A standard graded Artinian algebra A over a field K is said to have the *weak Lefschetz property* if there is a linear form $\ell \in A$ such that the multiplication map $\times \ell : [A]_i \rightarrow [A]_{i+1}$ has maximal rank for all i (i.e., it is injective or surjective). The name is a reminder that the Hard Lefschetz Theorem guarantees this property for the cohomology ring of a compact Kähler manifold. It is a desirable property as its presence constrains, e.g., the Hilbert function (see [11]). Many algebras are expected to have the weak Lefschetz property. However, establishing this fact is often very challenging. The recent lecture notes of Harima, Maeno, Morita, Numata, Wachi, and Watanabe [10] and the survey [19] provide an excellent review of the state of knowledge on the Lefschetz properties.

The authors have developed a combinatorial approach towards deciding the presence of the weak Lefschetz property for monomial algebras in three variables in [5, 7]. It relies on a study of lozenge tilings of so-called triangular regions. The needed results are recalled in Sect. 2.1. This approach has been used, for example,

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to investigate the weak Lefschetz property of monomial algebras of type two [7] and quotients by ideals with four generators [8]. Furthermore, we showed in [8] that there is a connection between lozenge tilings and semistability of syzygy bundles (see Theorem 4). Here we use these methods to relate the weak Lefschetz property and semistability of syzygy bundles. We establish sufficient conditions for the presence of the weak Lefschetz property as well as for the equivalence of the weak Lefschetz property and the semistability of a syzygy bundle (see Theorem 6). These conditions are easily expressed and checked using triangular regions (see Remark 3). As a consequence, we describe infinite families of ideals which satisfy these conditions. The number of generators of these ideals can be arbitrarily large. As a simple example, consider the following ideal

$$I = (x^{12}, x^6y^2z^3, x^3y^2z^7, xy^7z^3, xy^5z^5, xyz^9, y^{12}, z^{12}).$$

Our criterion immediately gives that its quotient has the weak Lefschetz property.

2 Algebraic Triangular Regions

We recall facts needed to establish our main results in the following section.

Let $R = K[x, y, z]$ be the standard graded polynomial ring over the field K . Unless specified otherwise, K is always a field of characteristic zero. All R -modules are assumed to be finitely generated and graded.

Let I be a monomial ideal of R . Then $A = R/I = \bigoplus_{j \geq 0} [A]_j$ is the sum of finite vector spaces, called the *homogeneous components (of A) of degree j* . The *Hilbert function* of A is the function $h_A : \mathbb{Z} \rightarrow \mathbb{Z}$ given by $h_A(j) = \dim_K [A]_j$. The *socle* of A , denoted $\text{soc } A$, is the annihilator of $\mathfrak{m} = (x, y, z)$, the homogeneous maximal ideal of R , that is, $\text{soc } A = \{a \in A \mid a \cdot \mathfrak{m} = 0\}$.

2.1 Triangular Regions Labeled by Monomials

Now we briefly review a connection between monomial ideals and triangular regions; for a more thorough discussion see [5].

For an integer $d \geq 1$, the *triangular region (of R) in degree d* , denoted $\mathcal{T}_d$, is an equilateral triangle of side length d composed of $\binom{d}{2}$ downward-pointing (∇) and $\binom{d+1}{2}$ upward-pointing ($\triangle$) equilateral unit triangles. These triangles are labeled by the monomials in $[R]_{d-2}$ and $[R]_{d-1}$, respectively, as follows: place x^{d-1} at the top y^{d-1} at the bottom-left, and z^{d-1} at the bottom-right; the remaining labels are found via interpolation. See Fig. 1a for an illustration.

The *triangular region (of R/I) in degree d* , denoted by $T_d(I)$, is the part of $\mathcal{T}_d$ that is obtained after removing the triangles labeled by monomials in I . Note that the labels of the downward- and upward-pointing triangles in $T_d(I)$ form K -bases of $[R/I]_{d-2}$ and $[R/I]_{d-1}$, respectively. See Fig. 1b for an example.

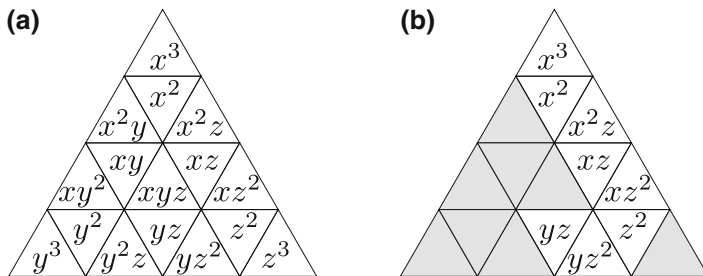


Fig. 1 A triangular region with respect to R and with respect to R/I . (a) $\mathcal{T}_4$. (b) $T_4(xy, y^2, z^3)$

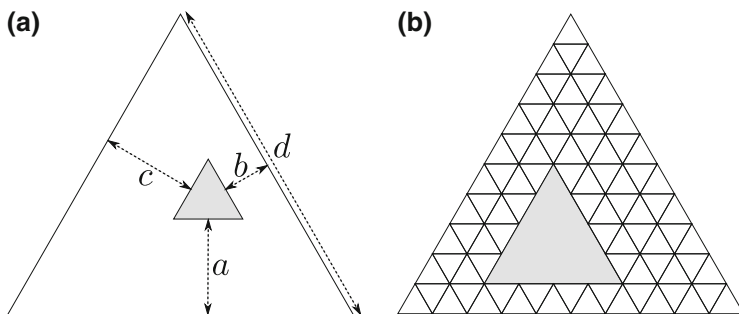


Fig. 2 $T_d(I)$ as constructed by removing punctures. (a) $T_d(x^a y^b z^c)$. (b) $T_{10}(xy^3 z^2)$

Notice that the regions missing from $\mathcal{T}_d$ in $T_d(I)$ can be viewed as a union of (possibly overlapping) upward-pointing triangles of various side lengths that include the upward- and downward-pointing triangles inside them. Each of these upward-pointing triangles corresponds to a minimal generator $x^a y^b z^c$ of I that has, necessarily, degree at most $d - 1$. The value $d - (a + b + c)$ is the *side length of the puncture associated to $x^a y^b z^c$* , regardless of possible overlaps with other punctures. See Fig. 2 for an example.

We say that two punctures *overlap* if they share at least an edge. Two punctures are said to be *touching* if they share precisely a vertex.

2.2 Tilings and the Weak Lefschetz Property

A *lozenge* is a union of two unit equilateral triangles glued together along a shared edge, i.e., a rhombus with unit side lengths and angles of 60° and 120° . Lozenges are also called calissons and diamonds in the literature. See Fig. 3.

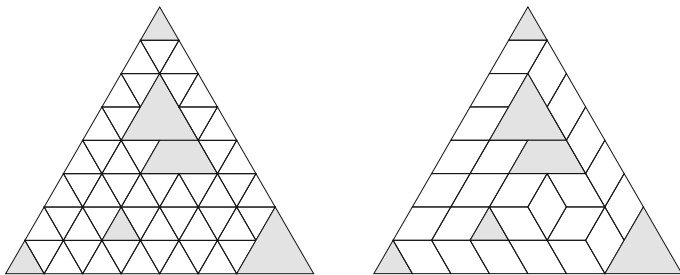


Fig. 3 A triangular region $T \subset \mathcal{T}_8$ together with one of its 13 tilings

Fix a positive integer d and consider the triangular region $\mathcal{T}_d$ as a union of unit triangles. Thus a *subregion* $T \subset \mathcal{T}_d$ is a subset of such triangles. We retain their labels. We say that a subregion T is ∇ -heavy, $\triangle$ -heavy, or *balanced* if there are more downward pointing than upward pointing triangles or less, or if their numbers are the same, respectively. A subregion is *tileable* if either it is empty or there exists a tiling of the region by lozenges such that every triangle is part of exactly one lozenge. Since a lozenge in $\mathcal{T}_d$ is the union of a downward-pointing and an upward-pointing triangle, and every triangle is part of exactly one lozenge, a tileable subregion is necessarily balanced.

Let $T \subset \mathcal{T}_d$ be any subregion. Given a monomial $x^a y^b z^c$ with degree less than d , the *monomial subregion* of T associated to $x^a y^b z^c$ is the part of T contained in the triangle a units from the bottom edge, b units from the upper-right edge, and c units from the upper-left edge (see Fig. 2). In other words, this monomial subregion consists of the triangles that are in T and the puncture associated to the monomial $x^a y^b z^c$.

We previously established a characterization of tileable triangular regions associated to monomial ideals.

Theorem 1 ([5, Theorem 2.2]) *Let $T = T_d(I)$ be a balanced triangular region, where $I \subset R$ is any monomial ideal. Then T is tileable if and only if T has no ∇ -heavy monomial subregions.*

A subregion $T \subset \mathcal{T}_d$ can be associated to a bipartite planar graph G that is an induced subgraph of the honeycomb graph. Lozenge tilings of T can be then related to perfect matchings on G (see Fig. 4). The connection was used by Kuperberg in [12], the earliest citation known to the authors, to study symmetries on plane partitions.

Using this connection, the *bi-adjacency matrix* of T is the bi-adjacency matrix $Z(T) := Z(G)$ of the graph G associated to T . If $T = T_d(I)$ for some Artinian ideal I , it is shown in [7, Proposition 4.5] that $Z(T)$ is the transpose of the matrix defined by $\times(x + y + z) : [R/I]_{d-2} \rightarrow [R/I]_{d-1}$ using monomial bases in the reverse-lexicographic order. Based on results by Migliore, Miró-Roig, and the second author [17], we established the following criterion for the presence of the weak Lefschetz property.

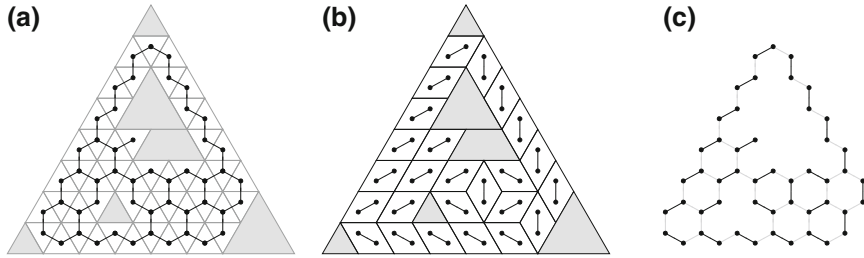


Fig. 4 A graph G , its region, and a perfect matching. (a) The region T to G . (b) A lozenge tiling of T . (c) A perfect matching of G

Theorem 2 ([7, Corollary 4.7]) *Let I be an Artinian monomial ideal in $R = K[x, y, z]$. Then R/I has the weak Lefschetz property if and only if, for each positive integer d , the matrix $Z(T_d(I))$ has maximal rank.*

Sometimes it is enough to consider only one matrix to check for the weak Lefschetz property. The criterion below adjusts a misprint in [7].

Proposition 1 ([7, Corollary 4.8(i)]) *Let I be an Artinian monomial ideal in $R = K[x, y, z]$, and suppose the degrees of the socle generators of R/I are at least $d - 2$. If $0 \neq h_{R/I}(d - 2) = h_{R/I}(d - 1)$, then R/I has the weak Lefschetz property if and only if $\det Z(T_d(I))$ is not zero in K .*

It is well-known that the permanent of $Z(G)$ is equal to the number of perfect matchings of G . A perfect matching of G can be signed via the permutation it generates; thus a lozenge tiling τ of T can be similarly signed (see [5]). Hence the signed tilings of the region are related to the determinant of the bi-adjacency matrix.

Theorem 3 ([5, Theorem 3.5]) *If $T \subset \mathcal{T}_d$ is a non-empty balanced subregion, then the signed lozenge tilings of T are enumerated by $\det Z(T)$, that is, the sum of the signs over all lozenge tilings of T is equal to $\det Z(T)$.*

We recursively define a puncture of $T \subset \mathcal{T}_d$ to be a *non-floating* puncture if it touches the boundary of $\mathcal{T}_d$ or if it overlaps or touches a non-floating puncture of T . Otherwise we call a puncture a *floating* puncture. The following result indicates that it is easier to enumerate the signed lozenge tilings of regions with only even floating punctures.

Corollary 1 ([5, Corollary 4.7]) *Let T be a tileable triangular region, and suppose all floating punctures of T have an even side length. Then all lozenge tilings of T have the same sign, and so $\text{perm } Z(T) = |\det Z(T)| > 0$.*

2.3 Tileability and Semistability of the Syzygy Bundle

Let I be a monomial ideal of R whose punctures in $\mathcal{T}_d$ (corresponding to the minimal generators of I having degree less than d) have side lengths that sum to m . Then we define the *over-puncturing coefficient* of I in degree d to be $\mathfrak{o}_d(I) = m - d$. If $\mathfrak{o}_d(I) < 0$, $\mathfrak{o}_d(I) = 0$, or $\mathfrak{o}_d(I) > 0$, then we call I *under-punctured*, *perfectly-punctured*, or *over-punctured* in degree d , respectively.

Let now $T = T_d(I)$ be a triangular region with punctures whose side lengths sum to m . Then we define similarly the *over-puncturing coefficient* of T to be $\mathfrak{o}_T = m - d$. If $\mathfrak{o}_T < 0$, $\mathfrak{o}_T = 0$, or $\mathfrak{o}_T > 0$, then we call T *under-punctured*, *perfectly-punctured*, or *over-punctured*, respectively. Note that $\mathfrak{o}_T \leq \mathfrak{o}_d(I)$. The inequality can be strict; for example, this is the case if there is an ideal J that is generated by monomials whose degrees are bounded above by $d - 1$, strictly contains I , and satisfies $T = T_d(J)$.

Perfectly-punctured regions admit a numerical tileability criterion.

Proposition 2 ([8, Corollary 2.4]) *Let $T = T_d(I)$ be a triangular region. Then any two of the following conditions imply the third:*

1. T is perfectly-punctured;
2. T has no over-punctured monomial subregions; and
3. T is tileable.

Let I be an Artinian ideal of $S = K[x_1, \dots, x_n]$ that is minimally generated by forms $f_1, \dots, f_m$. The *syzygy module* of I is the graded module $\text{syz } I$ that fits into the exact sequence

$$0 \rightarrow \text{syz } I \rightarrow \bigoplus_{i=1}^m S(-\deg f_i) \rightarrow I \rightarrow 0.$$

Its sheafification $\widetilde{\text{syz } I}$ is a vector bundle on $\mathbb{P}^{n-1}$, called the *syzygy bundle* of I . It has rank $m - 1$.

Let E be a vector bundle on projective space. The *slope* of E is defined as $\mu(E) := \frac{c_1(E)}{\text{rk}(E)}$. Furthermore, E is said to be *semistable* if the inequality $\mu(F) \leq \mu(E)$ holds for every coherent subsheaf $F \subset E$.

Using the characterization of semistability for monomial ideals given by Brenner [3], we previously established a connection to tileability.

Theorem 4 ([8, Theorem 3.3]) *Let I be an Artinian ideal in $R = K[x, y, z]$ generated by monomials whose degrees are bounded above by d , and let $T = T_d(I)$. If T is non-empty, then any two of the following conditions imply the third:*

1. I is perfectly-punctured;
2. T is tileable; and
3. $\widetilde{\text{syz } I}$ is semistable.

We note that Brenner and Kaid showed in [4, Theorem 3.3] that, for almost complete intersections in three variables, nonsemistability implies the weak Lefschetz property in characteristic zero.

3 The Criterion

In this section we will establish sufficient conditions that guarantee the presence of the weak Lefschetz property. We use this result to exhibit explicit infinite families of ideals such that their quotients have the weak Lefschetz property.

We begin with a necessary condition for the presence of the weak Lefschetz property.

Proposition 3 *Let I be a monomial ideal such that $T_d(I)$ is a balanced region that is not tileable. Put $J = I + (x^d, y^d, z^d)$. Then R/J never has the weak Lefschetz property, regardless of the characteristic of K .*

Proof Since $T_d(I) = T_d(j)$ is not tileable, Theorem 3 gives $\det Z(T_d(J)) = 0$. Thus, $Z(T_d(J))$ does not have maximal rank. Now we conclude by Theorem 2. $\square$

We illustrate the preceding proposition with an example.

Example 1 Consider the regions depicted in Fig. 5. These regions are both balanced, but non-tileable as they contain ∇ -heavy monomial subregions (see Theorem 1). In particular, the monomial subregion associated to xy^2z in T and the monomial subregion associated to xy^2z^2 in T' are both ∇ -heavy. Thus, the algebras $R/(x^6, y^7, z^8, xy^5z, xy^2z^3, x^3y^2z)$ and $R/(x^6, y^7, z^7, xy^4z^2, xy^2z^4, x^2y^2z^2)$ both do not have the weak Lefschetz property, regardless of the characteristic of the base field.

Now we use Proposition 3 in order to relate the weak Lefschetz property and semistability of syzygy bundles (see Sect. 2.3). In preparation, we record the following observation. Recall that the monomial ideal of a triangular region $T \subset \mathcal{T}_d$

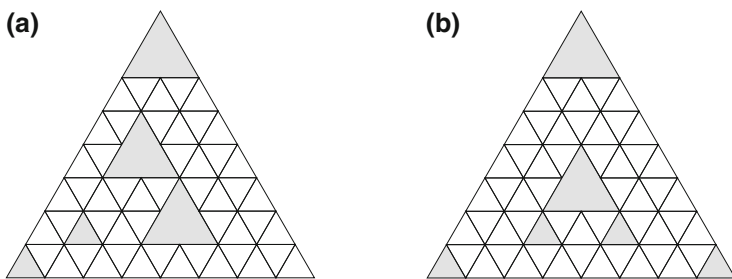


Fig. 5 Two balanced non-tileable triangular regions. (a) $T = T_8(x^6, y^7, z^8, xy^5z, xy^2z^3, x^3y^2z)$. (b) $T' = T_8(x^6, y^7, z^7, xy^4z^2, xy^2z^4, x^2y^2z^2)$

is the largest ideal J whose minimal generators have degrees less than d such that $T = T_d(J)$.

Lemma 1 *Let $J \subset R$ be the monomial ideal of a triangular region $T \subset \mathcal{T}_d$. Then:*

1. *The region T has no overlapping punctures if and only if each degree of a least common multiple of two distinct minimal generators of J is at least d .*
2. *The punctures of T are not overlapping nor touching if and only if each degree of a least common multiple of two distinct minimal generators of J is at least $d + 1$.*

Proof Let m_1 and m_2 be two distinct minimal generators of J . Then their associated punctures overlap if and only if there is a monomial of degree $d - 1$ that is a multiple of m_1 and m_2 . The existence of such a monomial means precisely that the degree of the least common multiple of m_1 and m_2 is at most $d - 1$. Now claim (i) follows.

Assertion (ii) is shown similarly by observing that the punctures to m_1 and m_2 touch if and only if there is a monomial of degree d that is a multiple of m_1 and m_2 . $\square$

The following consequence is useful later on.

Corollary 2 *Assume $T \subset \mathcal{T}_d$ is a triangular region whose punctures are not overlapping nor touching, and let J be the monomial ideal of T . Then R/J does not have non-zero socle elements of degree less than $d - 1$.*

Proof Since J is a monomial ideal, every minimal first syzygy of J corresponds to a relation $m_i n_i - m_j n_j = 0$ for suitable monomials n_i and n_j , where m_i and m_j are distinct monomial minimal generators of J . Applying Lemma 1 to the equality $m_i n_i = m_j n_j$, we conclude that the degree of each first syzygy of J is at least $d + 1$. It follows that the degree of every second syzygy of J is at least $d + 2$. Each minimal second syzygy of J corresponds to a socle generator of R/J . Hence, the degrees of the socle generators of R/J are at least $d - 1$. $\square$

Remark 1 The converse of Corollary 2 is not true in general. For example, the socle generators of $R/(x^6, y^7, z^8, xy^5z, xy^2z^3, x^3y^2z)$ have degrees greater than 7, but two punctures of $T_8(x^6, y^7, z^8, xy^5z, xy^2z^3, x^3y^2z)$ touch each other (see Fig. 5).

Recall that perfectly-punctured regions were defined above Proposition 2. This concept is used in the proof of the following result.

Theorem 5 *Let $I \subset R$ be an Artinian ideal whose minimal monomial generators have degrees $d_1, \dots, d_t$. Set*

$$d := \frac{d_1 + \dots + d_t}{t - 1}.$$

Assume the following conditions are satisfied:

1. *The number d is an integer.*
2. *For all $i = 1, \dots, t$, one has $d > d_i$.*
3. *Each degree of a least common multiple of two distinct minimal generators of I is at least d .*

Then the syzygy bundle of I is semistable if R/I has the weak Lefschetz property.

Proof Consider the triangular region $T = T_d(I)$. By assumption (iii) and Lemma 1, we obtain that the punctures of T do not overlap. Recall that the side length of the puncture to a minimal generator of degree d_i is $d - d_i$. The definition of d is equivalent to

$$d = \sum_{i=1}^t (d - d_i).$$

We conclude that the region T is balanced and perfectly-punctured. Since the ideal I is Artinian, assumptions (i) and (ii) imply that x^d, y^d , and z^d are in I . Hence Proposition 3 shows that T is tileable. Now Theorem 4 gives the semistability of the syzygy bundle of I . $\square$

Remark 2 Note that Theorem 5 does not contradict the fact that, for almost complete intersections in three variables, nonsemistability implies the weak Lefschetz property in characteristic zero [4, Theorem 3.3]. In fact, if $t = 4$, conditions (i)–(iii) of Theorem 5 always imply that the syzygy bundle is semistable in characteristic zero. This follows also directly from [3, Proposition 2.2 and Corollary 6.4].

The converse of the above result is not true, in general.

Example 2 Consider the ideal

$$J = (x^7, x^4y^2z^2, xy^3z^3, y^7, z^7).$$

It is perfectly-punctured, and its triangular region $T_9(J)$ is tileable. Thus, the syzygy bundle of J is semistable by Theorem 4. However, R/J does not have the weak Lefschetz property. Notice though that R/I has the weak Lefschetz property, where I is the very similar ideal

$$I = (x^7, x^5yz, xy^3z^3, y^7, z^8).$$

Both regions, $T_9(I)$ and $T_9(J)$ are tileable and symmetric. In fact, they are examples of mirror symmetric regions that are studied in [6].

Under stronger assumptions the converse to Theorem 5 is indeed true.

Theorem 6 *Let $I \subset R$ be an Artinian ideal with minimal monomial generators $m_1, \dots, m_t$. Set*

$$d := \frac{d_1 + \dots + d_t}{t - 1},$$

where $d_i = \deg m_i$. Assume the following conditions are satisfied:

1. The number d is an integer.
2. For all $i = 1, \dots, t$, one has $d > d_i$.

3. If $i \neq j$, then the degree of the least common multiple of m_i and m_j is at least $d + 1$.
4. If m_i is not a power of x, y , or z , then $d - d_i$ is even.

Then the syzygy bundle of I is semistable if and only if R/I has the weak Lefschetz property.

Proof By Theorem 5, it is enough to show that R/I has the weak Lefschetz property if the syzygy bundle of I is semistable.

Consider the region $T = T_d(I)$. In the proof of Theorem 5 we showed that T is balanced and perfectly-punctured. Hence T is tileable by Theorem 4. Since all floating punctures of T have an even side length by assumption (iv), Theorem 3 and Corollary 1 give that $Z(T)$ has maximal rank.

Assumption (iii) means that the punctures of T are not overlapping nor touching (see Lemma 1). Hence, Corollary 2 yields that the degrees of the socle generators of R/I are at least $d - 1$. Therefore, Proposition 1 proves that R/I has the weak Lefschetz property. $\square$

Remark 3 The assumptions of Theorem 6 have the following interpretation using the triangular region $T_d(I)$. Condition (ii) means that each minimal generator of I gives a puncture of positive side length of $T_d(I)$. Condition (i) expresses the fact that the side lengths of all punctures add up to d , that is, $T_d(I)$ is perfectly punctured. Assumption (iv) says that all non-corner punctures have an even side length, and Condition (iii) is the requirement that no punctures of $T_d(I)$ touch or overlap.

We now show that, for all positive integers $d_1, \dots, d_t$ with $t \geq 3$ that satisfy the numerical assumptions (i), (ii), and (iv) of Theorem 6, there is a monomial ideal I whose minimal generators have degrees $d_1, \dots, d_t$ to which Theorem 6 applies and guarantees the weak Lefschetz property of R/I .

Example 3 Let $d_1, \dots, d_t$ be $t \geq 3$ positive integers satisfying the following numerical conditions:

1. The number $d := \frac{d_1 + \dots + d_t}{t-1}$ is an integer.
2. For all $i = 1, \dots, t$, one has $d > d_i$.
3. At most three of the integers $d - d_i$ are not even.

Re-indexing if needed, we may assume that $d_3 \leq \min\{d_1, d_2\}$ and that $d - d_i$ is even whenever $4 \leq i \leq t$. Consider the following ideal

$$I = (x^{d_1}, y^{d_2}, z^{d_3}, m_4, \dots, m_t),$$

where $m_4 = x^{d-d_3}y z^{-d-1+d_3+d_4}$ if $t \geq 4$, $m_5 = x^{2d-d_3-d_4}y^2 z^{-2d-2+d_3+d_4+d_5}$ if $t \geq 5$, and

$$m_i = \begin{cases} x^{d-d_3}y^{1+\sum_{k=4}^{i-1}(d-d_k)}z^{-d(i-3)-1+\sum_{k=3}^i d_k} & \text{if } 6 \leq i \leq t \text{ and } i \text{ is even} \\ x^{-1+\sum_{k=3}^{i-1}(d-d_k)}y^2 z^{-d(i-3)-1+\sum_{k=3}^i d_k} & \text{if } 7 \leq i \leq t \text{ and } i \text{ is odd.} \end{cases}$$

Note that $\deg m_i = d_i$ for all i . One easily checks that the degree of the least common multiple of any two distinct minimal generators of I is at least $d + 1$, that is, the punctures of $T_d(I)$ do not overlap nor touch each other.

Corollary 3 *If I is an ideal as defined in Example 3, then R/I has the weak Lefschetz property and the syzygy bundle of I is semistable.*

Proof By construction, the considered ideals satisfy assumptions (i)–(iv) of Theorem 6. By inspection, one sees that the region $T_d(I)$ has no over-punctured monomial subregions. Hence, it is tileable by Proposition 2. (Alternatively, one can exhibit a family of non-intersecting lattice paths to check tileability.) By Theorem 4, it follows that the syzygy bundle of I is semistable, and hence R/I has the weak Lefschetz property by Theorem 6. $\square$

Remark 4 Given an integer $t \geq 3$, there are many choices for the integers $d_1, \dots, d_t$, and thus for the ideals exhibited in Example 3. A convenient choice, for which the description of the ideal becomes simpler, is $d_1 = 2t - 4$, $d_2 = d_3 = d - 1$, and $d_4 = \dots = d_t = d - 2$, where d is any integer satisfying $d \geq 2t - 3$. Then the corresponding ideal is

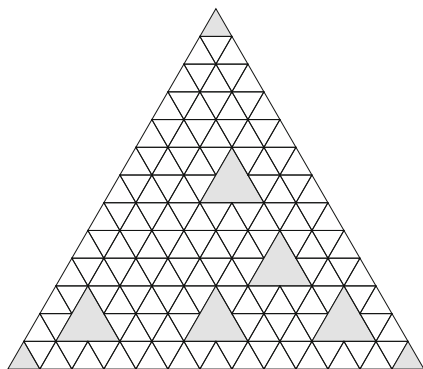
$$I = (x^{2t-4}, y^{d-1}, z^{d-1}, xyz^{d-4}, x^3y^2z^{d-7}, m_6, \dots, m_t),$$

where

$$m_i = \begin{cases} xy^{2i-7}z^{d+4-2i} & \text{if } 6 \leq i \leq t \text{ and } i \text{ is even} \\ x^{2i-8}y^2z^{d+4-2i} & \text{if } 7 \leq i \leq t \text{ and } i \text{ is odd.} \end{cases}$$

If $d = 13$ and $t = 8$, then this gives the ideal I mentioned in the introduction. Its triangular region $T_{13}(I)$ is depicted in Fig. 6.

Fig. 6 The region corresponding to $d_1 = d_2 = d_3 = 12$ and $d_4 = \dots = d_8 = 11$ in Example 3



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Multigraded Generic Initial Ideals of Determinantal Ideals

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Abstract Let I be either the ideal of maximal minors or the ideal of 2-minors of a row graded or column graded matrix of linear forms L . In previous work we showed that I is a Cartwright-Sturmfels ideal, that is, the multigraded generic initial ideal $\operatorname{gin}(I)$ of I is radical (and essentially independent of the term order chosen). In this paper we describe generators and prime decomposition of $\operatorname{gin}(I)$ in terms of data related to the linear dependences among the row or columns of the submatrices of L . In the case of 2-minors we also give a closed formula for its multigraded Hilbert series.

1 Introduction

Ideals of minors of matrices of linear forms are widely studied within commutative algebra and algebraic geometry. For example, they arise in classical constructions in invariant theory and define certain Veronese and Segre varieties. One often concentrates on matrices whose entries satisfy extra conditions, such as generic matrices (whose entries are distinct variables), generic symmetric matrices, catalecticant matrices, and 1-generic matrices. In this paper, we study ideals of minors of matrices which are homogeneous with respect to a multigrading. More precisely, given a standard $\mathbb{Z}^m$ -graded polynomial ring we study the ideals of maximal minors or 2-minors of $m \times n$ matrices with the property that all the entries in the i -th row are homogeneous of degree $e_i \in \mathbb{Z}^m$. We call such matrices row graded. Similarly one can define column graded matrices.

Gröbner bases of ideals of minors have been extensively studied, and a wealth of results is available on ideals of minors of generic matrices, generic symmetric

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matrices, and catalecticant matrices for specific term orders. In particular, in [2] and [10] Bernstein, Sturmfels, and Zelevinsky showed that the maximal minors of a matrix of variables are a universal Gröbner basis of the ideal that they generate. Sturmfels [9] and Villarreal [11] produced a universal Gröbner basis of the ideal of 2-minors of a matrix of variables. These results were generalized in [1, 3–5, 8]. In [6] and [7] we studied universal Gröbner bases of the ideals of maximal minors and 2-minors of row or column graded matrices. In [7] we introduced two new families of ideals, that we call Cartwright-Sturmfels and Cartwright-Sturmfels*. The families are defined in terms of properties of their multigraded generic initial ideals, and they are instrumental to our study of universal Gröbner bases. Some of the main results of [6] and [7] consist of showing that the ideals of minors that we study are Cartwright-Sturmfels, and that some of them are also Cartwright-Sturmfels*. In particular, we showed that their multigraded generic initial ideals are radical and do not depend on the term order.

In [6, Theorems 3.2 and 4.1] we described the monomial generators of the multigraded generic initial ideal of the ideal of maximal minors of a column graded matrix and of a row graded matrix respectively, the latter under the assumption that the ideal has maximal height. We recall these results in Sects. 3 and 4. In Theorem 2 we describe a system of generators and the prime decomposition of the multigraded generic initial ideal of the ideal of maximal minors of any row graded matrix. In Theorem 4 we do the same for the ideal of 2-minors of a row graded matrix. In addition we give a closed formula for the multigraded Hilbert series of the ideals of 2-minors.

2 Preliminaries

Let S be a polynomial ring over a field K , endowed with a standard $\mathbb{Z}^v$ -graded structure, i.e., the degree of every indeterminate is an element of the canonical basis $\{e_1, \dots, e_v\}$ of $\mathbb{Z}^v$. For $i = 1, \dots, v$ let u_i be the number of indeterminates of S of degree e_i . We denote them by $x_{i1}, \dots, x_{iu_i}$. We assume that $u_i > 0$ for all i .

The group $G = \mathrm{GL}_{u_1}(K) \times \dots \times \mathrm{GL}_{u_v}(K)$ acts on S as the group of $\mathbb{Z}^v$ -graded K -algebras automorphisms. Let $B = B_{u_1}(K) \times \dots \times B_{u_v}(K)$ be the *Borel subgroup* of G , consisting of the upper triangular invertible matrices. An ideal $I \subset S$ is *Borel fixed* if $g(I) = I$ for every $g \in B$. In analogy with the standard $\mathbb{Z}$ -graded situation, the property of being Borel fixed can be characterized combinatorially. Indeed one has that an ideal I of S is Borel fixed with (respect to the given $\mathbb{Z}^v$ -graded structure) if and only if it satisfies the following conditions:

1. I is generated by monomials,
2. For every monomial generator m of I one has that $(x_{ik}/x_{ij})^d m \in I$ for every $i = 1, \dots, v$, for every $1 \leq k < j \leq u_i$ and every $0 \leq d \leq c$ such that $\binom{c}{d} \neq 0$ in K where c is the exponent of x_{ij} in m .

Given a term order τ and a $\mathbb{Z}^v$ -graded homogeneous ideal I of S , one can consider its *multigraded generic initial ideal* $\text{gin}_\tau(I)$ defined as $\text{in}_\tau(g(I))$, where g is a general element in G . Notice that we always assume that

$$x_{ij} > x_{ik} \text{ if } 1 \leq j < k \leq u_i.$$

As in the standard graded setting, multigraded generic initial ideals are Borel fixed and can be obtained as $\text{in}_\tau(b(I))$ for a general $b \in B$.

In [7], we introduced the following two families of multigraded ideals. Let $T = K[x_{11}, x_{21}, \dots, x_{v1}] \subset S$ be endowed with the $\mathbb{Z}^v$ -graded structure induced by that of S . Notice that a $\mathbb{Z}^v$ -graded homogeneous ideal of T is a monomial ideal of T . Hence a $\mathbb{Z}^v$ -graded ideal of S which is extended from T is an ideal of S which is generated by monomials in the variables $x_{11}, x_{21}, \dots, x_{v1}$. We denote by $\text{HS}(M, y)$ the multigraded Hilbert series of a finitely generated $\mathbb{Z}^v$ -graded S -module M .

Definition 1 Let I be a $\mathbb{Z}^v$ -graded ideal of S . We say that I is a *Cartwright-Sturmfels ideal* if there exists a radical Borel fixed ideal J of S such that $\text{HS}(I, y) = \text{HS}(J, y)$. We say that I is a *Cartwright-Sturmfels* ideal* if there exists a $\mathbb{Z}^v$ -graded ideal J of S extended from T such that $\text{HS}(I, y) = \text{HS}(J, y)$.

In [6, Theorem 3.5] we have showed that if I, J are Borel fixed ideals and $\text{HS}(I, y) = \text{HS}(J, y)$, then $I = J$ as soon as I (or J) is radical. In particular this implies that the multigraded generic initial ideal of a Cartwright-Sturmfels ideal does not depend on the term order (but only on the total order given on the indeterminates). In [7, Proposition 1.9] we gave a characterization of Cartwright-Sturmfels* ideals that implies that the multigraded generic initial ideal of a Cartwright-Sturmfels* ideal is independent of the term order as well. Therefore in the sequel the multigraded generic initial ideal of a Cartwright-Sturmfels ideal I will be simply denoted by $\text{gin}(I)$ and similarly for Cartwright-Sturmfels* ideals. Moreover we will sometimes call it the generic initial ideal, as we will always deal with the multigraded version.

3 Maximal Minors: The Row Graded Case

Given integers $m \leq n$, let $S = K[x_{ij} : 1 \leq i \leq m, 1 \leq j \leq n]$ with the $\mathbb{Z}^m$ -graded structure induced by the assignment $\deg x_{ij} = e_i \in \mathbb{Z}^m$. For $a \in \mathbb{N}^m$ let P_a be the associated Borel fixed prime ideal, i.e.,

$$P_a = (x_{ij} : 1 \leq i \leq m \text{ and } 1 \leq j \leq a_i).$$

Let $L = (\ell_{ij})$ be a row graded $m \times n$ matrix of linear forms, i.e. the entries of L are homogeneous and $\deg \ell_{ij} = e_i \in \mathbb{Z}^m$. Equivalently,

$$\ell_{ij} = \sum_{k=1}^m \lambda_{ijk} x_{ik}$$

where $\lambda_{ijk} \in K$. Let $I_m(L)$ be the ideal of maximal minors of L . Under the assumption that $I_m(L)$ has the largest possible codimension in [6, Sect.4] we proved that:

Theorem 1 *Assume that $I_m(L)$ has codimension $n - m + 1$. Then:*

1. $I_m(L)$ is Cartwright-Sturmfels.
2. Its generic initial ideal is

$$\text{gin}(I_m(L)) = (x_{1a_1} \cdots x_{ma_m} : \sum_{i=1}^m a_i \leq n).$$

3. One has the following irredundant prime decomposition:

$$\text{gin}(I_m(L)) = \bigcap_{a \in C} P_a$$

where $C = \{a \in \mathbb{N}^m : \sum_{i=1}^m a_i = n - m + 1\}$.

In [7] we generalized the first assertion by proving that $I_m(L)$ is Cartwright-Sturmfels independently of its codimension. Our goal here is identifying its generic initial ideal and the corresponding prime decomposition. To this end we introduce the following invariants. For every subset A of $[m]$ let $b_L(A)$ be the dimension of the K -vector subspace of $\bigoplus_{i \in A} S_{e_i}$ generated by the columns of the matrix $L_A = (\ell_{ij})$ with $i \in A$ and $j \in [n]$. Then we have:

Theorem 2 *With the notation above one has:*

1. $I_m(L)$ is Cartwright-Sturmfels.
2. The generic initial ideal of $I_m(L)$ is

$$\text{gin}(I_m(L)) = (x_{1a_1} \cdots x_{ma_m} : \sum_{i \in A} a_i \leq b_L(A) \text{ for every } A \subseteq [m]).$$

3. Furthermore

$$\text{gin}(I_m(L)) = \bigcap_{a \in C} P_a$$

where C is the set of the elements $a = (a_1, \dots, a_m) \in \mathbb{N}^m$ such that for some $A \subseteq [m]$ one has $a_i = 0$ for $i \in [m] \setminus A$ and

$$\sum_{i \in A} a_i = b_L(A) - |A| + 1.$$

In the proof of the theorem we need the following lemma. Let $I \subset S$ be a $\mathbb{Z}^m$ -graded ideal, let $h \in S_{e_k}$ such that the coefficient of x_{kn_k} in h is non-zero. We may identify $S/(h)$ with the polynomial subring S' of S generated by all the variables of S with the exception of x_{kn_k} . Then $I + (h)/(h)$ is identified with an ideal I' of S' .

Lemma 1 *Let I be a Cartwright-Sturmfels ideal. With the notation and identification above one has that*

$$\text{gin}(I')S \subseteq \text{gin}(I)$$

for all h .

Proof In [7, Theorem 1.16] we have already proved that I' is a Cartwright-Sturmfels ideal of S' and that $\text{gin}(I')S = \text{gin}(I'S)$. By construction I' is the image of I under the K -algebra map $\phi_h : S \rightarrow S'$ which sends x_{ij} to itself if $(i, j) \neq (k, n_k)$ and x_{kn_k} to $-\lambda^{-1} \sum_{j=1}^{n_k-1} \mu_j x_{kj}$ where $h = \lambda x_{kn_k} + \sum_{j=1}^{n_k-1} \mu_j x_{kj}$. Denote by $\phi^h : S \rightarrow S$ the K -algebra automorphism of S which sends x_{ij} to itself if $(i, j) \neq (k, n_k)$ and x_{kn_k} to $-\lambda^{-1}h$. Since by construction $\phi_h = \phi_{x_{kn_k}} \phi^h$ and since I and $\phi^h(I)$ have the same gin, we may assume $h = x_{kn_k}$. Using a revlex order with x_{kn_k} as smallest variable one has $\text{in}(I') = \text{in}(I) \cap S'$, hence $\text{in}(I')S \subseteq \text{in}(I)$. Computing the generic initial ideal on both sides one obtains $\text{gin}(\text{in}(I')S) \subseteq \text{gin}(\text{in}(I))$. Finally, since I and I' are Cartwright-Sturmfels ideals, then $\text{gin}(\text{in}(I)) = \text{gin}(I)$ and $\text{gin}(\text{in}(I')) = \text{gin}(I')$. Hence we conclude that $\text{gin}(I')S \subseteq \text{gin}(I)$, as desired. $\square$

Remark 1 Without the assumption that I is a Cartwright-Sturmfels ideal the statement in Lemma 1 does not hold. For example if $I = (x_1x_2, x_1x_3, x_1^2 + x_4^2)$ in $K[x_1, x_2, x_3, x_4]$ with the standard $\mathbb{Z}$ -grading and $h = x_4$, then $I' = (x_1x_2, x_1x_3, x_1^2)$ and $\text{gin}(I') = I'$ since I' is Borel fixed, while $\text{gin}(I) = (x_1^2, x_1x_2, x_2^2, x_1x_3^2)$. Here the gins are computed with respect to the revlex order. The argument given above breaks down when we state that $\text{gin}(\text{in}(I)) = \text{gin}(I)$: This is true for Cartwright-Sturmfels ideals and false in general.

Proof (Theorem 2) As said above, (1) has been proved already in [7].

As for (2) let

$$U = (x_{1a_1} \cdots x_{ma_m} : \sum_{i \in A} a_i \leq b_L(A) \text{ for every } A \subseteq [m]).$$

We start by proving the inclusion $\text{gin}(I_m(L)) \subseteq U$. For any $A = \{i_1, \dots, i_v\} \subseteq [m]$ one has $I_m(L) \subseteq I_v(L_A)$. Up to column operations, L_A is equivalent to a row graded matrix L'_A of size $v \times b_L(A)$. Hence $\text{gin}(I_m(L)) \subseteq \text{gin}(I_v(L'_A))$. Since L'_A can be seen as a multigraded linear section of a matrix Y_A of variables of the same size, applying Lemma 1 and Theorem 1 we have

$$\text{gin}(I_m(L)) \subseteq \text{gin}(I_v(L'_A)) \subseteq \text{gin}(I_v(Y_A))$$

and

$$\text{gin}(I_v(Y_A)) = (x_{i_1a_1} \cdots x_{i_v a_v} : a_1 + \cdots + a_v \leq b_L(A)).$$

Therefore,

$$\text{gin}(I_m(L)) \subseteq \bigcap_{A \subseteq [m]} \text{gin}(I_{|A|}(L_A)) \subseteq \bigcap_{A \subseteq [m]} \text{gin}(I_{|A|}(Y_A))$$

and

$$\bigcap_{A \subseteq [m]} \text{gin}(I_{|A|}(Y_A)) = \bigcap_{A \subseteq [m], A = \{i_1, \dots, i_v\}} (x_{i_1 a_1} \cdots x_{i_v a_v} : a_1 + \cdots + a_v \leq b_L(A))$$

and the latter can be proved to be equal to U by a straightforward verification.

In order to prove the reverse inclusion $\text{gin}(I_m(L)) \supseteq U$, let $M = x_{1a_1} \cdots x_{ma_m}$ be a monomial such that $\sum_{i \in A} a_i \leq b_L(A)$ for every $A \subseteq [m]$. We will show that $M \in \text{gin}(I_m(L))$. Denote by b_i the number $b_L(\{i\})$. By assumption $a_i \leq b_i$ and, up to a multigraded linear transformation (which does not affect the gin) we may assume that only the first b_i variables of multidegree e_i are actually used in L . We argue by induction on $N = \sum_{i=1}^m (b_i - a_i)$. Note that $N \geq 0$ by assumption.

If $N = 0$, i.e., $a_i = b_i$ for all i , then, after suitable column operations, the matrix L can be brought in the form

$$\begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1a_1} & 0 & \cdots & \cdots & \cdots & \cdots & \cdots & 0 \\ 0 & \cdots & \cdots & 0 & x_{21} & x_{22} & \cdots & x_{2a_2} & 0 & \cdots & \cdots & 0 \\ \vdots & & & & & & & & & & \vdots \\ 0 & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & 0 & x_{m1} & x_{m2} & \cdots & x_{ma_m} \end{pmatrix}.$$

Therefore $I_m(L) = \prod_{i=1}^m (x_{i1}, x_{i2}, \dots, x_{ia_i})$ and $\text{gin}(I_m(L)) = I_m(L)$ because $I_m(L)$ is Borel fixed. Hence $M \in I_m(L) = \text{gin}(I_m(L))$ as required.

Assume now that $N > 0$, that is $a_j < b_j$ for some j in $[m]$, say $a_1 < b_1$. Let h be a generic linear combination of ℓ_{1j} with $j = 1, \dots, n$. By Lemma 1 it suffices to show that $M \in \text{gin}(I_m(L'))$, where L' is the image of L in $S/(h)$. Set $b'_i = b_{L'}(\{i\})$ and notice that, by construction, $b'_i = b_i$ for $i > 1$ and $b'_1 = b_1 - 1$. Then we may conclude by induction provided that we show that

$$\sum_{i \in A} a_i \leq b_{L'}(A), \quad (1)$$

for all $A \subseteq [m]$ and for h generic. As we will see, inequality (1) follows essentially from Grassmann's formula. To this end let V_i be the kernel of the K -linear map $f_i : K^n \rightarrow S_{e_i}$ defined by $f((\lambda_1, \dots, \lambda_n)) = \sum_{j=1}^n \lambda_j L_{ij}$ and let V'_i be the corresponding objects associated to L' . By construction, $V'_i = V_i$ for $i > 1$, $V'_1 = V_1 + \langle h' \rangle$ with h' generic. Furthermore $b_L(A) = n - \dim_K \cap_{i \in A} V_i$ and $b'_{L'}(A) = n - \dim_K \cap_{i \in A} V'_i$. If $1 \notin A$ then $b_L(A) = b'_{L'}(A)$ and (1) holds by assumption. If $1 \in A$ let $W = \cap_{i \in A, i \neq 1} V_i$. Then, by Grassmann's formula we have:

$$b_L(A) = n - \dim W - \dim V_1 + \dim(V_1 + W)$$

and

$$b'_L(A) = n - \dim W - \dim(V_1 + \langle h' \rangle) + \dim(V_1 + \langle h' \rangle + W).$$

Now, if $V_1 + W$ is strictly contained in K^n , then $\dim(V_1 + \langle h' \rangle) = \dim V_1 + 1$ and $\dim(V_1 + \langle h' \rangle + W) = \dim(V_1 + W) + 1$ for a generic h' . Hence it follows that $b_L(A) = b'_L(A)$ and (1) holds by assumption.

Finally if $V_1 + W = K^n$ then we have

$$b'_L(A) = n - \dim W - \dim(V_1 + \langle h' \rangle) + n = b_L(A \setminus \{1\}) + b_1 - 1$$

and then

$$\sum_{i \in A} a_i = a_1 + \sum_{i \in A, i \neq 1} a_i < b_1 + b_L(A \setminus \{1\}) = b_{L'}(A) + 1$$

that is,

$$\sum_{i \in A} a_i \leq b_{L'}(A)$$

as desired.

We now prove assertion (3) of the theorem. In the proof of (2) we have shown that

$$\text{gin}(I_m(L)) = \bigcap_{A \subseteq [m]} \text{gin}(I_{|A|}(Y_A)).$$

Moreover, by Theorem 1

$$\text{gin}(I_{|A|}(Y_A)) = \bigcap_{a \in C_A} P_a$$

where C_A is the set of the $a \in \mathbb{Z}^m$ with $a_i = 0$ for $i \in [m] \setminus A$ and $\sum_{i \in A} a_i = b_L(A) - |A| + 1$. Combining the two equalities we get the desired prime decomposition. $\square$

4 Maximal Minors: The Column Graded Case

For completeness, in this section we recall the results proved in [6] concerning ideals of maximal minors in the column graded case. Given integers $m \leq n$, let $S = K[x_{ij} : 1 \leq i \leq m, 1 \leq j \leq n]$ graded by $\deg x_{ij} = e_j \in \mathbb{Z}^n$. For $a \in \mathbb{N}^n$ with $a_i \leq m$ let P_a be the associated Borel fixed prime ideal, i.e.,

$$P_a = (x_{ij} : 1 \leq i \leq a_i \text{ and } 1 \leq j \leq n).$$

Let $L = (\ell_{ij})$ be a column graded $m \times n$ matrix of linear forms, i.e. the ℓ_{ij} 's are homogeneous of degree e_j . Equivalently,

$$\ell_{ij} = \sum_{k=1}^m \lambda_{ijk} x_{kj}$$

where $\lambda_{ijk} \in K$.

Let $I_m(L)$ be the ideal of maximal minors of L . In [6, Section 3] we have proved the following statement. We rephrase the result using the terminology introduced in [7].

Theorem 3 *With the notation above one has:*

1. $I_m(L)$ is a Cartwright-Sturmfels ideal as well as a Cartwright-Sturmfels* ideal.
2. The generic initial ideal of $I_m(L)$ is

$$\text{gin}(I_m(L)) = (x_{1a_1} \cdots x_{1a_m} : [a_1, \dots, a_m]_L \neq 0)$$

where $[a_1, \dots, a_m]_L$ denotes the m -minor of L corresponding to columns indices $a_1, \dots, a_m$.

3. The prime decomposition of $\text{gin}(I_m(L))$ is

$$\text{gin}(I_m(L)) = \bigcap_{c \in C} P_c$$

where C is the set of the elements $c = (c_1, \dots, c_n) \in \{0, 1\}^n$ whose support $\text{Supp}(c) = \{i \in [n] : c_i = 1\}$ is minimal with respect to the property that $\text{Supp}(c) \cap \{a_1, \dots, a_m\} \neq \emptyset$ for every $(a_1, \dots, a_m)$ such that $[a_1, \dots, a_m]_L \neq 0$. In other words, the minimal associated primes of $I_m(L)$ are exactly the ideals $(x_{1b_1}, \dots, x_{1b_k})$ where $\{b_1, \dots, b_k\}$ is a minimal vertex cover of the simplicial complex whose facets are $\{\{a_1, \dots, a_m\} \mid [a_1, \dots, a_m]_L \neq 0\}$.

Remark 2 If all the minors of L are nonzero, then

$$\text{gin}(I_m(L)) = (x_{1a_1} \cdots x_{1a_m} : 1 \leq a_1 < \dots < a_m \leq n) = \bigcap_{c \in C} P_c$$

where C is the set of all subsets of $[n]$ of cardinality $n - m + 1$. Notice moreover that

$$\text{gin}(I_m(L)) = \text{gin}(I_m(X_{m \times n})) = \text{gin}(I_{n-m+1}(X_{(n-m+1) \times n}))^*$$

where $X_{u \times v}$ denotes a matrix of indeterminates of size $u \times v$ and $*$ denotes the Alexander dual.

5 A Family of Hilbert Series of Multigraded Algebras

In this section we discuss a combinatorially defined family of formal power series. We will show that every such power series is the Hilbert series of a multigraded K -algebra defined by a radical Borel fixed ideal. In Sect. 6 we will apply these results to ideals of 2-minors of multigraded matrices.

Let $n, m \in \mathbb{N}_{>0}$ and let

$$\Phi : 2^{[m]} \rightarrow \{0, \dots, n\}$$

be a map such that:

1. $\Phi(\emptyset) = n$,
2. $\Phi(A) \geq \Phi(B)$ whenever $A \subseteq B$.

Consider the formal power series

$$H_\Phi(y_1, \dots, y_m) = \sum_{A \subseteq [m]} \sum_{\substack{a \in \mathbb{N}^m \\ \text{Supp}(a) = A}} \binom{\Phi(A) - 1 + |a|}{\Phi(A) - 1} y^a \in \mathbb{Q}[[y_1, \dots, y_m]]$$

where for $a = (a_1, \dots, a_m) \in \mathbb{N}^m$ one sets

$$\text{Supp}(a) = \{i \in [m] : a_i > 0\} \quad \text{and} \quad |a| = a_1 + \dots + a_m.$$

We now prove that the series $H_\Phi(y_1, \dots, y_m)$ is the multigraded Hilbert series of S/I , where S is a multigraded polynomial ring and I is a radical Borel fixed ideal. Let $d_i = \Phi(\{i\})$ and let $S = K[x_{ij} : i = 1, \dots, m \text{ and } 1 \leq j \leq d_i]$ endowed with the $\mathbb{Z}^m$ -multigraded structure induced by $\deg x_{ij} = e_i \in \mathbb{Z}^m$. For every $\emptyset \neq A = \{a_1, \dots, a_i\} \subseteq [m]$ let

$$I_A = \left(\prod_{i \in A} x_{ib_i} : 1 \leq b_i \leq d_i \text{ for every } i \in A \text{ and } \sum_{i \in A} (d_i - b_i) \geq \Phi(A) \right).$$

Lemma 2 *The ideal $I = \sum_{A \subseteq [m]} I_A$ is radical Borel fixed and S/I has multigraded Hilbert series equal to $H_\Phi(y_1, \dots, y_m)$.*

Proof It is clear that each I_A is radical and Borel fixed. It then follows that also I is radical and Borel fixed. It remains to prove that S/I has multigraded Hilbert series equal to $H_\Phi(y_1, \dots, y_m)$. In other words, we have to prove that the number of monomials of S not in I and of multidegree $a \in \mathbb{N}^m$ is exactly $\binom{\Phi(A) - 1 + |a|}{\Phi(A) - 1}$ where $A = \text{Supp}(a)$. Whenever $A \subseteq B$ the homogeneous components of I_A of multidegrees bigger than or equal to $\sum_{i \in B} e_i$ are already contained in I_B because, by assumption, $\Phi(A) \geq \Phi(B)$. Therefore it suffices to prove that for every $a \in \mathbb{N}^m$ and $A = \text{Supp}(a)$ the number of monomials not in I_A and of multidegree a is exactly $\binom{\Phi(A) - 1 + |a|}{\Phi(A) - 1}$.

Without loss of generality we may also assume that $A = \{1, \dots, m\}$ and then set $J = I_A$ and $u = \Phi(A)$. Then

$$J = (x_{1b_1} \cdots x_{mb_m} : 1 \leq b_i \leq d_i \text{ and } b_1 + \cdots + b_m \leq d_1 + \cdots + d_m - u)$$

and we have to prove that the number of monomials not in J and of multidegree $a \in \mathbb{N}^m$ and full support $\text{Supp}(a) = \{1, \dots, m\}$ is exactly $\binom{u-1+|a|}{u-1}$. Let $c = (c_1, \dots, c_m) \in \mathbb{N}^m$ with $1 \leq c_i \leq d_i$ such that

$$\sum_{i=1}^m c_i > \sum_{i=1}^m d_i - u. \quad (2)$$

For such a vector c and given $a \in \mathbb{N}_{>0}^m$ we consider the set X_c of monomials M of multidegree a such that for every i we have $\min\{j : x_{ij}|M\} = c_i$. By construction, the set of the monomials of multidegree a that are not in J is the disjoint union of the X_c 's where c runs in the set of vectors specified above. Since

$$\#X_c = \prod_{i=1}^m \binom{d_i - c_i + a_i - 1}{a_i - 1}$$

it is enough to prove the following identity:

$$\sum_c \prod_{i=1}^m \binom{d_i - c_i + a_i - 1}{a_i - 1} = \binom{u - 1 + |a|}{u - 1} \quad (3)$$

where the sum runs over all the vectors c described above. Setting $w_i = d_i - c_i$ we have to prove that

$$\sum_w \prod_{i=1}^m \binom{w_i + a_i - 1}{a_i - 1} = \binom{u - 1 + |a|}{u - 1} \quad (4)$$

where the sum runs over all the vectors $w \in \mathbb{N}^m$ such that $0 \leq w_i \leq d_i - 1$ and

$$\sum_{i=1}^m w_i \leq u - 1. \quad (5)$$

Since, by construction, $u \leq d_i$ for every i , the restriction $w_i \leq d_i - 1$ is subsumed by (2). Now the identity (4) is easy: Both the left and the right side count the number of monomials of total degree $\leq u - 1$ in a set of variables which is a disjoint union of subsets of cardinality $a_1, a_2, \dots, a_m$. $\square$

One can also identify the prime decomposition of the ideal described in Lemma 2. For $v = (v_1, \dots, v_m) \in \mathbb{N}^m$ with $v_i \leq d_i$ let P_v be the prime Borel fixed ideal associated to v , that is,

$$P_v = (x_{ij} : 1 \leq i \leq m \text{ and } 1 \leq j \leq v_i).$$

Lemma 3 *Let I be the ideal of Lemma 2. For every $v \in \mathbb{N}^m$ with $v_i \leq d_i$ set $A(v) = \{i : v_i < d_i\}$. The following are equivalent:*

1. $P_v \supseteq I$,
2. $\sum_{i=1}^m v_i \geq \sum_{i=1}^m d_i - \Phi(A(v)) - |A(v)| + 1$.

Proof For the implication 1. $\Rightarrow$ 2. we assume, by contradiction, that there exists a v such that $P_v \supseteq I$ and v does not satisfy condition 2. of the lemma. Setting $A = A(v)$, one has that

$$\sum_{i \in A} v_i \leq \sum_{i \in A} d_i - \Phi(A) - |A|.$$

Now let $b_i = v_i + 1$ for every $i \in A$. By construction, $b_i \leq d_i$ and

$$\sum_{i \in A} b_i = \sum_{i \in A} v_i + |A| \leq \sum_{i \in A} d_i - \Phi(A).$$

Hence $\prod_{i \in A} x_{ib_i} \in I$ and, by construction, $\prod_{i \in A} x_{ib_i} \notin P_v$, contradicting the assumption.

To prove the converse, assume that v satisfies 2. and let $M = \prod_{i \in A} x_{ib_i}$ be a generator of I with $A \subset [m]$. Let $B = A(v)$. If there exists $i \in A \setminus B$, then $v_i = d_i$ and hence $x_{ib_i} \in P_v$, that is, $M \in P_v$. If instead $A \subseteq B$ and $b_i > v_i$ for every $i \in A$ we have:

$$\begin{aligned} \sum_{i \in A} d_i - \Phi(A) &\geq \sum_{i \in A} b_i \geq \sum_{i \in A} v_i + |A| = \sum_{i \in B} v_i - \sum_{i \in B \setminus A} v_i + |A| \\ &\geq \sum_{i \in B} d_i - \Phi(B) - |B| + 1 + |A| - \sum_{i \in B \setminus A} v_i. \end{aligned}$$

Hence

$$\begin{aligned} -\Phi(A) &\geq \sum_{i \in B \setminus A} (d_i - v_i) - \Phi(B) - |B| + 1 + |A| \geq \\ &\geq |B \setminus A| - \Phi(B) - |B| + 1 + |A| = -\Phi(B) + 1 \end{aligned}$$

that is,

$$\Phi(B) \geq \Phi(A) + 1$$

a contradiction. Therefore there exists an index $i \in A$ such that $v_i \geq b_i$. This implies that the monomial M is in P_v . $\square$

Lemma 4 *The ideal I of Lemma 2 has the following irredundant prime decomposition:*

$$I = \bigcap_v P_v$$

where $v = (v_1, \dots, v_m)$ runs over the vectors of $\mathbb{N}^m$ such that:

1. $0 \leq v_i \leq d_i$ for every $i = 1, \dots, m$,
2. $\sum_{i=1}^m v_i = \sum_{i=1}^m d_i - \Phi(A(v)) - |A(v)| + 1$ where $A(v) = \{i : v_i < d_i\}$,
3. $\Phi(B) < \Phi(A(v))$ for every $B \supsetneq A(v)$.

Proof Since I is radical and Borel fixed, it is the intersection of the P_v 's that contain it. They are described in Lemma 3 and we have to prove that, among them, those that are minimal over I are exactly those described by conditions 2. and 3. of the lemma. First observe that if v satisfies

$$\sum_{i=1}^m v_i > \sum_{i=1}^m d_i - \Phi(A(v)) - |A(v)| + 1 \quad (6)$$

then P_v is not minimal over I . To this end we set $w = v - e_i$ where $i \in A(v)$ and $v_i > 0$. Such an i exists, since otherwise from (6) one would deduce that

$$0 > \sum_{i \in A(v)} d_i - \Phi(A(v)) - |A(v)| + 1$$

which is a contradiction because $d_i > 0$ for every $i \in A(v)$ and $\Phi(A(v)) \leq \min\{d_i \mid i \in A(v)\}$. For such a w we have $A(w) = A(v)$ and hence

$$\sum_{i=1}^m w_i \geq \sum_{i=1}^m d_i - \Phi(A(w)) - |A(w)| + 1$$

that implies $I \subseteq P_w \subsetneq P_v$.

Secondly we show that if v satisfies 2. but not 3. then P_v is not minimal over I . By assumption there exists $i \in [m]$, $i \notin A(v)$, such that if we set $B = A(v) \cup \{i\}$ we have $\Phi(B) = \Phi(A(v))$. If $d_i > 0$ then we may set $w = v - e_i$ so that $B = A(w)$. Then one checks that w satisfies (1) and hence $I \subseteq P_w \subsetneq P_v$. If instead $d_i = 0$ then $\Phi(B) = 0$ because $i \in B$ and hence $\Phi(A(v)) = 0$. Then by 2.

$$|A(v)| = \sum_{i \in A(v)} (d_i - v_i) + 1$$

which is a contradiction since $d_i > v_i$ for every $i \in A(v)$.

Finally we have to check that every v satisfying 2. and 3. corresponds to a prime which is minimal over I . By contradiction, let v satisfy 2. and 3. and assume that there exists w satisfying 2. such that $w \leq v$ componentwise and $w_j < v_j$ for some $j \in [m]$. Then $A(v) \subseteq A(w)$ and $j \in A(w)$. If $A(v) = A(w)$ then

$$\sum_{i=1}^m d_i - \Phi(A(w)) - |A(w)| - 1 = \sum_{i=1}^m w_i < \sum_{i=1}^m v_i = \sum_{i=1}^m d_i - \Phi(A(v)) - |A(v)| - 1,$$

a contradiction because the first and last expression are equal. If instead $A(v) \subsetneq A(w)$ then

$$\sum_{i=1}^m (v_i - w_i) = |A(w)| - |A(v)| + \Phi(A(w)) - \Phi(A(v))$$

because both v and w satisfy condition 2. of the lemma. On the other hand,

$$\sum_{i=1}^m (v_i - w_i) \geq |A(w)| - |A(v)|$$

because for every $j \in A(w) \setminus A(v)$ one has $d_j = v_j > w_j$. Summing up,

$$\Phi(A(w)) - \Phi(A(v)) \geq 0$$

that is,

$$\Phi(A(w)) = \Phi(A(v))$$

contradicting the assumption that v satisfies 3. of the lemma. $\square$

6 Ideals Generated by 2-Minors

We now apply Lemmas 2 and 4 to obtain a description of the generic initial ideal for ideals of 2-minors of multigraded matrices. Up to a transposition of the matrix, it is not restrictive to assume that the matrix is row graded.

For integers m, n let $S = K[x_{ij} : i = 1, \dots, m \text{ and } j = 1, \dots, n]$ be endowed with the multigrading induced by $\deg x_{ij} = e_i \in \mathbb{Z}^m$. Let $L = (\ell_{ij})$ be a $m \times n$ matrix of linear forms with $\deg \ell_{ij} = e_i$. We know by [7] that the ideal $I_2(L)$ of 2-minors of L is a Cartwright-Sturmfels ideal and we want to describe its generic initial ideal.

Consider the surjective map of K -algebras

$$\psi : K[x_{ij}] \rightarrow K[\ell_{ij}] \text{ defined by } \psi(x_{ij}) = \ell_{ij}. \quad (7)$$

Its kernel is of the form $\sum_{i=1}^m W_i$, where W_i is the K -subspace of S_{e_i} corresponding to the K -linear dependence relations on the i -th row of L . Going modulo the 2-minors on each side of the map (7) we get an isomorphism

$$K[x_{ij}]/(I_2(X) + \ker(\psi)) \simeq K[\ell_{ij}]/I_2(L) \quad (8)$$

where $X = (x_{ij})$. Identifying $K[x_{ij}]/I_2(X)$ with the Segre product

$$R = K[x_i y_j : i = 1, \dots, m \text{ and } j = 1, \dots, n]$$

via the map sending x_{ij} to $x_i y_j$, we obtain an isomorphism

$$R/J \simeq K[\ell_{ij}]/I_2(L) \quad (9)$$

where $J = \sum_{i=1}^m (W'_i x_i)$ where each W'_i is a space of linear forms in $y_1, \dots, y_n$. Now for a subset A of $[m]$ we set

$$\phi(A) = n - \dim_K \sum_{a \in A} W'_a.$$

Since the ideal J is monomial in the variables x_i , it is easy to compute the multigraded Hilbert series of the quotient it defines. It turns out that the multigraded Hilbert series of R/J , and hence of $K[\ell_{ij}]/I_2(L)$, is given by

$$\sum_{A \subseteq [m]} \sum_{\substack{a \in \mathbb{Z}^m \\ A = \text{Supp}(a)}} \binom{\phi(A) - 1 + |a|}{\phi(A) - 1} y^a.$$

Hence, by Lemmas 2 and 4 we conclude

Theorem 4 *With the notation above one has:*

1. $I_2(L)$ is a Cartwright-Sturmfels ideal.
2. The generic initial ideal of $I_2(L)$ is generated by the monomials of the form $\prod_{i \in A} x_{ib_i}$ such that

$$A \subseteq [m], \quad 1 \leq b_i \leq n - \dim_K V_i \text{ for every } i \in A, \text{ and}$$

$$\sum_{i \in A} b_i \leq n(|A| - 1) + \dim_K V_A - \sum_{i \in A} \dim_K V_i$$

where $V_i = \{\lambda \in K^n \mid \sum_{j=1}^n \lambda_j L_{ij} = 0\}$ and $V_A = \sum_{i \in A} V_i$.

3. The irredundant prime decomposition of $\text{gin}(I_2(L))$ is given by

$$\text{gin}(I_2(L)) = \cap_v P_v$$

where v varies among the vectors $v \in \mathbb{N}^m$ satisfying the following conditions:

$$\begin{aligned} 0 &\leq v_i \leq n - \dim_K V_i \text{ for every } i = 1, \dots, m, \\ \sum_{i=1}^m v_i &= n(m-1) + 1 - |A| + \dim_K V_A - \sum_{i \in A} \dim_K V_i, \\ \dim_K V_B &> \dim_K V_A \text{ for every } B \supsetneq A, \end{aligned}$$

where $A = \{i : v_i < n - \dim_K V_i\}$.

Remark 3 Notice that the argument above gives another proof of the fact that the ideal of 2-minors of a multigraded matrix is Cartwright-Sturmfels.

Remark 4 In the case of non-maximal minors of size at least 3, the corresponding ideals are not Cartwright-Sturmfels.

For example, let X be a matrix of variables of size 4×4 and let τ be the revlex order. Then $\text{gin}_\tau(I_3(X))$ has $x_{1,1}^2, x_{2,2}, x_{3,2}, x_{4,2}$ among its minimal generators.

In addition, let X be a matrix of variables of size 4×5 and let τ be the lex order associated to the following ordering of the variables:

$$\begin{aligned} x_{11}, x_{22}, x_{33}, x_{44}, x_{12}, x_{23}, x_{34}, x_{45}, x_{21}, x_{32}, x_{43}, \\ x_{13}, x_{24}, x_{35}, x_{31}, x_{42}, x_{14}, x_{25}, x_{41}, x_{15}. \end{aligned}$$

Then $\text{in}(I_3(X))$ has $x_{12}x_{23}x_{31}x_{45}^2$ and $x_{12}x_{23}x_{31}x_{44}^2$ among its minimal generators and it does not define a Cohen-Macaulay ring.

Nevertheless it would be interesting to compute the multigraded generic initial ideal in these cases as well.

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A Stronger Local Monomialization Theorem

Steven Dale Cutkosky

Abstract In this article we prove stronger versions of local monomialization.

1 Introduction

In this note we derive extensions of the monomialization theorems for morphisms of varieties in [8] and [10]. I thank Jan Denef for conversations on this topic, suggesting that I make improvements of this type, and explaining applications of these theorems. A global “weak” monomialization theorem is established in [2] by Abramovich, Denef and Karu, generalizing an earlier theorem by Abramovich and Karu in [1]. A monomialization is “weak” if the modifications used have no further requirements; in a monomialization all modifications must be products of blow ups of nonsingular sub varieties. In this note we show that a local monomialization can be found which satisfies the extra local statements obtained in [2]. In [13] and [14] some comments are made about how the results of this paper can be used.

The techniques in this paper come from the theory of resolution of singularities. Some basic references in this subject are [3–7, 15–18]. In this paper we assume that the ground field has characteristic zero. Counterexamples to local monomialization in positive characteristic are given in [11]. A proof of local monomialization, within the context of analytic geometry, is given in [12] for germs of real and complex analytic maps.

2 A Stronger Local Monomialization Theorem for Algebraic Morphisms

In this section we state and prove an extension, Theorem 2.1, of the local monomialization theorem Theorem 1.4 [10].

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Suppose that K is an algebraic function field over a field k . A local ring R is an algebraic local ring of K if R is a subring of K , the quotient field of R is K and R is essentially of finite type over k .

Theorem 2.1 *Suppose that k is a field of characteristic zero, $K \rightarrow K^*$ is a (possibly transcendental) extension of algebraic function fields over k , and that v^* is a valuation of K^* which is trivial on k . Further suppose that R is an algebraic local ring of K and S is an algebraic local ring of K^* such that S dominates R and v^* dominates S . Suppose that I is a nonzero ideal of S . Then there exist sequences of monoidal transforms $R \rightarrow R'$ and $S \rightarrow S'$ along v^* such that R' and S' are regular local rings, S' dominates R' , there exist regular parameters $(y_1, \dots, y_n)$ in S' , $(x_1, \dots, x_m)$ in R' , units $\delta_1, \dots, \delta_m \in S'$ and an $m \times n$ matrix (c_{ij}) of nonnegative integers such that (c_{ij}) has rank m , and*

$$x_i = \prod_{j=1}^n y_j^{c_{ij}} \delta_i$$

for $1 \leq i \leq m$. Further, we have that

- 1) S' is a local ring of the blowup of an ideal J of S such that $JS' = (y_1^{\alpha_1} \cdots y_n^{\alpha_n})$ for some $\alpha_1, \dots, \alpha_n \in \mathbf{N}$.
- 2) $IS' = (y_1^{\beta_1} \cdots y_n^{\beta_n})$ for some $\beta_1, \dots, \beta_n \in \mathbf{N}$.

The proof of Theorem 1.4 [10] is given in [8] and [10]. Also see [9] for some errata. The new part of Theorem 1.1 is the addition of the conclusions 1) and 2). We now explain the changes in this proof which must be made to obtain the stronger result Theorem 2.1. Theorem 2.1 is a consequence of Theorem 2.5 which is proven at the end of this section.

We first indicate changes required in the proofs of [8] to obtain Theorem 2.1 in the case of a finite extension of function fields. In the construction of Theorem 5.1 [8], we have that $v(y'_1), \dots, v(y'_s)$ are rationally independent. We first settle the case when the quotient field of S is finite over the quotient field of R and the valuation v has rank 1.

Theorem 2.2 *Suppose that $R \rightarrow S$ and $R' \rightarrow S'$ satisfy the assumptions and conclusions of Theorem 5.1 [8] and $h \in S'$ is nonzero. Then there exist sequences of monoidal transforms $R' \rightarrow R''$ and $S' \rightarrow S''$ along v such that S'' dominates R'' , R'' has regular parameters $x''_1, \dots, x''_n$, S'' has regular parameters $y''_1, \dots, y''_n$ having the monomial form of the conclusions of Theorem 5.1 [8] and*

$$h = (y''_1)^{e_1} \cdots (y''_s)^{e_s} \bar{u}$$

where $e_1, \dots, e_s \in \mathbf{N}$ and $\bar{u} \in S''$ is a unit.

Proof This is an immediate consequence of Theorems 4.8 and 4.10 of [8]. □

Corollary 2.3 *Suppose that $R \rightarrow S$ and $R' \rightarrow S'$ satisfy the assumptions and conclusions of Theorem 5.1 [8] and $I \subset S$ is a nonzero ideal. Then there exist sequences of monoidal transforms $R' \rightarrow R''$ and $S' \rightarrow S''$ along v such that S'' dominates R'' , R'' has regular parameters $x''_1, \dots, x''_n$, S'' has regular parameters $y''_1, \dots, y''_n$ having the monomial form of the conclusions of Theorem 5.1 [8] and the following holds.*

- 1) S'' is a local ring of the blowup of an ideal J of S such that $JS'' = ((y''_1)^{a_1} \cdots (y''_s)^{a_s})$ for some $a_1, \dots, a_s \in \mathbf{N}$.
- 2) $IS'' = ((y''_1)^{b_1} \cdots (y''_s)^{b_s})$ for some $b_1, \dots, b_s \in \mathbf{N}$.

Proof Let K be an ideal in S such that S' is a local ring of the blow up of K . Let $KS' = (f_0)$ and $IS' = (f_1, \dots, f_i)$. Let $h = \prod_{j=0}^i f_j$. By Theorem 2.2, there exist sequences of monoidal transforms $R' \rightarrow R''$ and $S' \rightarrow S''$ along v such that $R'' \rightarrow S''$ and h satisfy the conclusions of Theorem 2.2. Now by Lemma 4.2 and Remark 4.1 [8], there exists a sequence of monoidal transforms $S'' \rightarrow S(1)$ such that $R'' \rightarrow S(1)$ has a monomial form as in the conclusion of Theorem 5.1 [8], $KS(1) = (y_1(1)^{e_1} \cdots y_s(1)^{e_s})$ and $IS(1) = (y_1(1)^{c_1} \cdots y_s(1)^{c_s})$ for some e_i and c_j in $\mathbf{N}$.

Now $S' \rightarrow S(1)$ is a product of Perron transforms (as defined in Section 4.1 [8]). Thus $S(1)$ is a local ring of the blow up of an ideal L in S' such that

$$LS(1) = (y_1(1)^{g_1(i+1)} \cdots y_s(1)^{g_s(i+1)})$$

for some $g_j(i+1) \in \mathbf{N}$. There exists a positive integer β such that $S(1)$ is a local ring of the blow up of an ideal J of S such that $JS(1) = K^\beta LS(1)$ (this can be verified using the universal property of blowing up) and the corollary follows.

We now prove the case when the quotient field of S is finite over the quotient field of R and the valuation ring has arbitrary rank. We use the notation of Theorem 5.3 [8].

Theorem 2.4 *Suppose that $R \rightarrow S$ satisfies the assumptions of Theorem 5.3 [8] and I is a nonzero ideal in S . Then there exist sequences of monoidal transforms $R \rightarrow R'$ and $S \rightarrow S'$ such that $R' \rightarrow S'$ satisfies the conclusions of Theorem 5.3 [8] and*

- 1) S' is a local ring of the blowup of an ideal J of S such that

$$JS' = \left(\prod_{j=0}^{r-1} \prod_{l=1}^{s_{j+1}} (w_{t_1+\dots+t_j+l})^{\varepsilon_{jl}} \right)$$

for some $\varepsilon_{jl} \in \mathbf{N}$ (with the convention that $t_1 + \dots + t_0 = 0$).

- 2) $IS' = \left(\prod_{j=0}^{r-1} \prod_{l=1}^{s_{j+1}} (w_{t_1+\dots+t_j+l})^{\gamma_{jl}} \right)$ for some $\gamma_{jl} \in \mathbf{N}$.

To obtain Theorem 2.4, we must modify the proof of Theorem 5.3 [8] as follows.

On line 1 of page 117, replace the reference to Theorem 5.1 [8] with Corollary 2.3. Insert the following at the end of line 15 on page 119 (after “for $1 \leq i \leq$

λ''): “By our construction, S'' is a local ring of the blow up of an ideal J of S such that

$$JS''_{q''_{r-1}} = \left(\prod_{j=0}^{r-2} \prod_{l=1}^{s_{j+1}} (y''_{t_1+\dots+t_j+l})^{\alpha_{jl}} \right)$$

for some $\alpha_{jl} \in \mathbb{N}$ so

$$JS'' = H'' \left(\prod_{j=0}^{r-2} \prod_{l=1}^{s_{j+1}} (y''_{t_1+\dots+t_j+l})^{\alpha_{jl}} \right)$$

where H'' is an ideal in S'' such that $H'' S''_{q''_{r-1}} = S''_{q''_{r-1}}$. By our construction,

$$IS''_{q''_{r-1}} = \left(\prod_{j=0}^{r-2} \prod_{l=1}^{s_{j+1}} (y''_{t_1+\dots+t_j+l})^{\beta_{jl}} \right)$$

for some $\beta_{jl} \in \mathbb{N}$ so

$$IS'' = K'' \left(\prod_{j=0}^{r-2} \prod_{l=1}^{s_{j+1}} (y''_{t_1+\dots+t_j+l})^{\beta_{jl}} \right)$$

where K'' is an ideal in S'' such that $K'' S''_{q''_{r-1}} = S''_{q''_{r-1}}$.

After line 11 of page 120 (after “for $1 \leq i \leq \lambda''$ ”), insert: “We also may obtain, using Theorem 2.2 and the argument of the proof of Corollary 2.3 (above) that

$$H'' U' = (\bar{y}_{\lambda+1}^{\gamma_1} \cdots \bar{y}_{\lambda+s_r}^{\gamma_{s_r}})$$

and

$$K'' U' = (\bar{y}_{\lambda+1}^{\delta_1} \cdots \bar{y}_{\lambda+s_r}^{\delta_{s_r}}).$$

Insert the following at the end of line -2 of page 121 (at the end of the proof): We have

$$H'' \bar{S}(m' + 1) = (\bar{y}_{\lambda+1}(m' + 1)^{\gamma_1} \cdots \bar{y}_{\lambda+s_r}(m' + 1)^{\gamma_{s_r}} + \Sigma)$$

and

$$\begin{aligned} K'' \bar{S}(m' + 1) &= (\bar{y}_{\lambda+1}(m' + 1)^{\delta_1} \cdots \bar{y}_{\lambda+s_r}(m' + 1)^{\delta_{s_r}} + \Psi_1, \dots, \bar{y}_{\lambda+1}(m' + 1)^{\delta_1} \\ &\quad \cdots \bar{y}_{\lambda+s_r}(m' + 1)^{\delta_{s_r}} + \Psi_e) \end{aligned}$$

where $\Sigma, \Psi_1, \dots, \Psi_e \in (\bar{y}_1(m' + 1), \dots, \bar{y}_\lambda(m' + 1))$ and $\gamma_1, \dots, \delta_{s_r} \in \mathbb{N}$.

Let $\varepsilon_i = \max\{\gamma_i, \delta_i\}$ for $1 \leq i \leq s_r$. Define a MTS $\bar{S}(m' + 1) \rightarrow \bar{S}(m' + 2)$ by

$$\bar{y}_i(m' + 1) = \begin{cases} \prod_{j=1}^{s_r} \bar{y}_{\lambda+j}(m' + 2)^{\varepsilon_j} \bar{y}_i(m' + 2) & \text{for } 1 \leq i \leq \lambda \\ \bar{y}_i(m' + 2) & \text{for } \lambda < i \leq n. \end{cases}$$

Further, by our construction, $\bar{S}(m' + 2)$ is a local ring of the blow up of an ideal B of S'' such that

$$B\bar{S}(m' + 2) = \left(\prod_{j=0}^{r-1} \prod_{l=1}^{s_j+1} (\bar{y}_{t_1+\dots+t_j+l}(m' + 2))^{\varepsilon_{jl}} \right)$$

for some $\varepsilon_{jl} \in \mathbb{N}$. Thus there exists an ideal J^* of S and $\beta > 0$ such that $\bar{S}(m' + 2)$ is a local ring of the blow up of J^* and $J^*\bar{S}(m' + 2) = J^\beta B\bar{S}(m' + 2)$. We have thus achieved the conclusions of Theorem 2.4 in $R(m') \rightarrow \bar{S}(m' + 2)$.

We now indicate the changes which need to be made in the statements of [10] to obtain the proof of Theorem 2.1.

We first extend Theorem 2.2 to arbitrary extensions of characteristic zero algebraic function fields. We will call this “Extended Theorem 2.2”.

In the statement and proof of Theorem 2.2, we replace references to Theorem 5.1 [8] with Theorem 10.1 [10], Theorems 4.8 and 4.10 [8] with Theorems 9.1 and 9.3 [10]. Also replace n with m when referring to regular parameters in birational extensions of R and s with $\bar{s}$. We must add the following sentence to the first line of the proof: “First suppose that $\text{rank}(\nu) > 0$, so that $\text{rank}(\nu) = 1$ (here ν is the restriction of the given rank 1 valuation ν^* of the quotient field of S to the quotient field K of R)”. At the end of the proof, add: “If $\text{rank}(\nu) = 0$, then ν is trivial so $R = K$ and the proof is a substantial simplification.”

We now extend Corollary 2.3. We will call this “Extended Corollary 2.3”. In the statement and proof of Corollary 2.3, replace references to Theorem 5.1 [8] with Theorem 10.1 [10] and Section 4.1 [8] with Section 5 [10]. Replace references to Theorem 2.2 with “Extended Theorem 2.2”. Replace n with m when referring to regular parameters in birational extensions of R . Replace s with $\bar{s}$. We must add the following sentence to the first line of the proof: “First suppose that $\text{rank}(\nu) > 0$, so that $\text{rank}(\nu) = 1$ (here ν is the restriction of the given rank 1 valuation ν^* of the quotient field of S to the quotient field K of R)”. At the end of the proof, add: “If $\text{rank}(\nu) = 0$, then ν is trivial so $R = K$ and the proof is a substantial simplification.”

We now adopt the notation on valuation rings introduced on page 1579–1581 of [10], which we will use below.

We extend Theorem 2.4 to arbitrary extensions of algebraic function fields in the following Theorem 2.5.

Theorem 2.5 *Suppose that $R \rightarrow S$ satisfies the assumptions of Theorem 10.5 [10] and I is a nonzero ideal in S . Then there exist sequences of monoidal transforms*

$R \rightarrow R'$ and $S \rightarrow S'$ such that $R' \rightarrow S'$ satisfies the conclusions of Theorem 10.5 [10] and

1) S' is a local ring of the blowup of an ideal J of S such that

$$JS' = \left(\prod (w_{\bar{\tau}_0 + \dots + \bar{\tau}_{i-1} + \bar{\tau}_{i,1} + \dots + \bar{\tau}_{i,j-1} + l})^{\varepsilon_{ijl}} \right)$$

where the product is over

$$0 \leq i \leq \beta, 1 \leq j \leq \sigma(i), 1 \leq l \leq \bar{s}_{ij}$$

(with $\bar{s}_{i1} = \bar{s}_i$) and $\varepsilon_{ijl} \in \mathbf{N}$ for all i, j, l .

2) We have that

$$IS' = \left(\prod (w_{\bar{\tau}_0 + \dots + \bar{\tau}_{i-1} + \bar{\tau}_{i,1} + \dots + \bar{\tau}_{i,j-1} + l})^{\gamma_{ijl}} \right)$$

where the product is over

$$0 \leq i \leq \beta, 1 \leq j \leq \sigma(i), 1 \leq l \leq \bar{s}_{ij}$$

and $\gamma_{ijl} \in \mathbf{N}$ for all i, j, l .

Proof In the proof of Theorem 2.4, replace references to Theorem 5.3 [8] with Theorem 10.5 [10]. Replace references to Theorem 2.4 with Theorem 2.5, references to Corollary 2.3 with Extended Corollary 2.3. and references to Theorem 2.2 with Extended Theorem 2.2. The indexing of the variables must be changed (as in the statement of Theorem 2.5). The prime ideal q_{r-1} in the valuation ring of the quotient field of S is replaced by the prime ideal $q_{\beta-1, \sigma(\beta-1)}$ of the valuation ring of the quotient field of S , so $q''_{r-1} = q_{r-1} \cap S''$ becomes $q''_{\beta-1} = q_{\beta-1, \sigma(\beta-1)} \cap S''$.

We must add the following sentences as the first lines of the proof: “We prove the theorem by induction on rank V^* . If rank $V^* = 1$ then the theorem is immediate from Extended Corollary 2.3.

By induction on $\gamma = \text{rank } V^*$, we may assume that the theorem is true whenever rank $V^* = \gamma - 1$. We are reduced to proving the theorem in the following two cases:

Case 1. $\sigma(\beta) = 1$

Case 2. $\sigma(\beta) > 1$.

Suppose that we are in Case 1, $\sigma(\beta) = 1$. Then $V^*/q_{\beta-1, \sigma(\beta-1)}$ is a rank 1 valuation ring which dominates the rank 1 valuation ring $V/p_{\beta-1}$. $V^*/q_{\beta-1, \sigma(\beta-1)}$ has rational rank $\bar{s}_\beta$ and $V/p_{\beta-1}$ has rational rank $\bar{r}_\beta$.”

We must add the following sentences as the last lines of the proof: “Now suppose that we are in Case 2, $\sigma(\beta) > 1$. Then $V^*/q_{\beta, \sigma(\beta)-1}$ is a rank 1, rational rank $\bar{s}_{\beta, \sigma(\beta)}$ valuation ring which dominates the rank 0 valuation ring V/p_β , which is a field. The proof in Case 2 is thus a substantial simplification of the proof in Case 1.” $\square$

3 A Geometric Local Monomialization Theorem

The main result in this section is Theorem 3.2. Uniformizing parameters on an affine k -variety V are a set of elements $u_1, \dots, u_s \in A = \Gamma(V, \mathcal{O}_V)$ such that $du_1, \dots, du_s$ is a free basis of $\Gamma(V, \Omega_{V/k})$ as an A -module.

Theorem 3.1 *Suppose that k is a field of characteristic zero, $\varphi : Y \rightarrow X$ is a dominant morphism of k -varieties and v is a zero dimensional valuation of the function field $k(Y)$ (the residue field of the valuation ring of v is algebraic over k) which has a center on Y (the valuation ring of v dominates a local ring of Y). Further suppose that $\mathcal{I} \subset \mathcal{O}_Y$ is a nonzero ideal sheaf. Then there exists a commutative diagram of morphisms of k -varieties*

$$\begin{array}{ccc} Y_v & \xrightarrow{\varphi_v} & X_v \\ \beta_v \downarrow & & \downarrow \alpha_v \\ Y & \xrightarrow{\varphi} & X \end{array} \quad (1)$$

where the vertical arrows are proper morphisms which are products of blow ups of nonsingular sub varieties, and if q' is the center of v on Y_v and p' is the center of v on X_v , then there exists an affine open neighborhood V_v of q' in Y_v and an affine open neighborhood U_v of p' in $X_{v,p'}$, regular parameters $y_1, \dots, y_n$ in $\mathcal{O}_{Y_v, q'}$ which are uniformizing parameters on V_v and regular parameters $x_1, \dots, x_m$ in $\mathcal{O}_{X_v, p'}$ which are uniformizing parameters on U_v (where $m = \dim X$, $n = \dim Y$) and units $\delta_1, \dots, \delta_m \in \Gamma(V_v, \mathcal{O}_{Y_v})$ such that

- 1) $x_i = \prod_{j=1}^n y_j^{c_{ij}} \delta_i$ with $c_{ij} \in \mathbf{N}$ for $1 \leq i \leq m$ and $\text{rank}(c_{ij}) = m$.
- 2) $V_v \setminus Z(y_1 \cdots y_n) \rightarrow Y$ is an open immersion;
- 3) $\mathcal{I} \mathcal{O}_{V_v} = y_1^{a_1} \cdots y_n^{a_n} \mathcal{O}_{V_v}$ for some $a_1, \dots, a_n \in \mathbf{N}$
- 4) $\varphi_v : V_v \rightarrow U_v$ is toroidal (Section 2.2 [2]) with respect to the locus of the product of the y_j and the locus of the product of the x_i .

Proof Let p be the center of v on X and q be the center of v on Y . Let $v^* = v$, $R = \mathcal{O}_{X,p}$, $S = \mathcal{O}_{Y,q}$, $I = \mathcal{I}_q$, $K^* = k(Y)$ and $K = k(X)$. Let

$$\begin{array}{ccc} R' & \rightarrow & S' \\ \uparrow & & \uparrow \\ R & \rightarrow & S \end{array}$$

be the diagram of the conclusions of Theorem 2.1. There exists a commutative diagram

$$\begin{array}{ccc} Y_v & \xrightarrow{\varphi_v} & X_v \\ \beta_v \downarrow & & \downarrow \alpha_v \\ Y & \xrightarrow{\varphi} & X \end{array}$$

where the vertical arrows are products of blow ups of nonsingular sub varieties and if p' is the center of ν on X_ν and q' is the center of ν on Y_ν then $\mathcal{O}_{Y_\nu, q'} = S'$ and $\mathcal{O}_{X_\nu, p'} = R'$. Since k has characteristic zero, there exists an affine open neighborhood U_ν of p' such that $x_1, \dots, x_m$ are uniformizing parameters on U_ν . Let V_ν be an affine neighborhood of q' in Y_ν such that $y_1, \dots, y_n$ are uniformizing parameters on V_ν and $\delta_1, \dots, \delta_m \in \Gamma(V_\nu, \mathcal{O}_{Y_\nu})$ are units. Let $\overline{C}$ be an $m \times m$ sub matrix of $C = (c_{ij})$ of rank m . Without loss of generality,

$$\overline{C} = \begin{pmatrix} c_{11} & \cdots & c_{1m} \\ \vdots & & \vdots \\ c_{m1} & \cdots & c_{mm} \end{pmatrix}.$$

Let $d = \text{Det}(\overline{C})$. Let

$$A = \Gamma(V_\nu, \mathcal{O}_{Y_\nu})$$

and

$$B = A[z_1, \dots, z_m] / (z_1^d - \delta_1, \dots, z_m^d - \delta_m) = A[\bar{z}_1, \dots, \bar{z}_m]$$

where $\bar{z}_i$ is the class of z_i .

Let $V'_\nu = \text{Spec}(B)$ with natural finite étale morphism $\pi : V'_\nu \rightarrow V_\nu$. Define $\bar{y}_j$ for $1 \leq j \leq n$ by

$$y_j = \bar{y}_j \prod_{l=1}^m \bar{z}_l^{db_{jl}}$$

where

$$\overline{B} = (b_{jl}) = \begin{pmatrix} -\overline{C}^{-1} \\ 0 \end{pmatrix}$$

is an $n \times m$ matrix with coefficients in $\frac{1}{d}\mathbf{Z}$. We have that for $1 \leq i \leq m$,

$$x_i = \prod_{j=1}^n y_j^{c_{ij}} \delta_i = \left(\prod_{j=1}^n \bar{y}_j^{c_{ij}} \right) \left(\prod_{l=1}^m \bar{z}_l^{\sum_{j=1}^n c_{ij} db_{jl}} \right) \delta_i = \prod_{j=1}^n \bar{y}_j^{c_{ij}}. \quad (2)$$

The morphism $\pi : V'_\nu \rightarrow V_\nu$ is étale so $y_1, \dots, y_n$ are uniformizing parameters on V'_ν or equivalently, $dy_1, \dots, dy_n$ are a free basis of $\Omega_{B/k}^1$ as a B -module. Let

$$\varepsilon_j = \left(\prod_{l=1}^m \bar{z}_l^{db_{jl}} \right)^{-1} \in B$$

and $\bar{y}_j = \varepsilon_j y_j$ for $1 \leq j \leq n$. Since $y_1, \dots, y_n$ are regular parameters in $\mathcal{O}_{V'_v, q''}$ for all $q'' \in \pi^{-1}(q')$ and $d\bar{y}_j = \varepsilon_j dy_j + y_j d\varepsilon_j$, we have that

$$\left(\Omega_{V'_v/k}^1 / d\bar{y}_1 \mathcal{O}_{V'_v} + \dots + d\bar{y}_n \mathcal{O}_{V'_v} \right)_{q''} = 0$$

for all $q'' \in \pi^{-1}(q')$. Let

$$Z = \text{Supp} \left(\Omega_{V'_v/k}^1 / d\bar{y}_1 \mathcal{O}_{V'_v} + \dots + d\bar{y}_n \mathcal{O}_{V'_v} \right).$$

Z is a closed subset of V'_v which is disjoint from $\pi^{-1}(q')$. The morphism $V'_v \rightarrow V_v$ is finite, so $\pi(Z)$ is a closed subset of V_v which does not contain q' . After replacing V_v with an affine neighborhood of q' in $V_v \setminus \pi(Z)$, we have that $\bar{y}_1, \dots, \bar{y}_n$ are uniformizing parameters on V'_v giving the monomial expression (2) and so $V_v \rightarrow U_v$ is toroidal with respect to the locus of the products of the y_j and the products of the x_i . $\square$

Theorem 3.2 *Suppose that k is a field of characteristic zero, $\varphi : Y \rightarrow X$ is a dominant morphism of k -varieties and $\mathcal{J} \subset \mathcal{O}_Y$ is a nonzero ideal sheaf. Then there exists a finite number of commutative diagrams*

$$\begin{array}{ccc} Y_i & \xrightarrow{\varphi_i} & X_i \\ \beta_i \downarrow & & \downarrow \alpha_i \\ Y & \xrightarrow{\varphi} & X \end{array}$$

for $1 \leq i \leq t$ such that the vertical arrows are products of blow ups of nonsingular sub varieties and there are affine open subsets $V_i \subset Y_i$ and $U_i \subset X_i$ such that $\varphi_i(V_i) \subset U_i$, $\cup_{i=1}^t \beta_i(V_i) = Y$ and the restriction $\varphi_i : V_i \rightarrow U_i$ is toroidal with respect to strict normal crossings divisor E_i on V_i and D_i on X_i such that the restriction of φ_i to $V_i \setminus E_i$ is an open immersion and $\mathcal{J} \mathcal{O}_{V_i}$ is a divisor on V_i whose support is contained in E_i .

Proof Let Ω be the Zariski Riemann manifold of Y (Section 17 of Chapter VI [19]). The points of Ω are equivalence classes of valuations of $k(Y)$ which dominate a local ring of Y . There are natural continuous surjections $\rho_Z : \Omega \rightarrow Z$ for any birational proper morphism $Z \rightarrow Y$. Let Σ be the subset of Ω of zero dimensional valuations which dominate a local ring of Y . For each $v \in \Sigma$, we construct a diagram (1) with corresponding open subset V_v of Y_v .

Suppose that ω is a valuation of $k(Y)$ which dominates a local ring of Y . If ω is not zero dimensional, then there exists $v \in \Sigma$ such that v is composite with ω (Theorem 7, page 16 [19]), so that $\omega \in \rho_{Y_v}^{-1}(V_v)$. Thus $\{\rho_{Y_v}^{-1}(V_v) \mid v \in \Sigma\}$ is an open cover of Ω , and thus there is a finite sub cover since Ω is quasi compact (Theorem 40 page 113 [19]). $\square$

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The Cayley Trick for Tropical Hypersurfaces with a View Toward Ricardian Economics

Michael Joswig

Abstract The purpose of this survey is to summarize known results about tropical hypersurfaces and the Cayley Trick from polyhedral geometry. This allows for a systematic study of arrangements of tropical hypersurfaces and, in particular, arrangements of tropical hyperplanes. A recent application to the Ricardian theory of trade from mathematical economics is explored.

1 Introduction

The main motivation for this text are the applications of tropical geometry to economics, that came up recently. In particular, Shiozawa gave an explanation of the Ricardian theory of international trade in terms of tropical combinatorics [17]. The purpose of that theory is to study the relationship between wages and prices on the world market. Our goal here is to put some of Shiozawa's results into the wider context of polyhedral and tropical geometry.

The Cayley Trick explains a special class of subdivisions of the Minkowski sum of finite point configurations in terms of a lifting to higher dimensions. Those subdivisions are called *mixed*. Mixed subdivisions of Minkowski sums and mixed volumes play a key role in Bernstein's method for solving systems of polynomial equations. Triangulations and more general polytopal subdivisions are the topic of the monograph [3] by De Loera et al. In Section 1.3 of that book the relationship between systems of polynomials and mixed subdivisions is discussed. Tropical geometry studies the images of algebraic varieties over fields with a discrete non-archimedean valuation under the valuation map; see Maclagan and Sturmfels [15]. Section 4.6 of that reference deals with a tropical version of Bernstein's Theorem, and this employs the Cayley Trick, too; see also Jensen's recent work on tropical homotopy continuation [11]. A first version of the Cayley Trick was obtained by Sturmfels [18]. In its full generality it was proved by Huber et al. [10].

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As its key contribution to tropical geometry the Cayley Trick explains how unions of tropical hypersurfaces work out. It says that the union of two tropical hypersurfaces is dual to the mixed subdivision of the regular subdivisions which are dual to the two components. This has been exploited by Develin and Sturmfels for the study of arrangements of tropical hyperplanes in the context of tropical convexity [4]. More recently, those results have been extended by Fink and Rincón [7] and by Loho and the author [13]. It is this perspective which proves useful for applications to Ricardian economics.

Another recent application of tropical geometry to economics is Baldwin and Klemperer's study of "product-mix auctions" [1]. There are n indivisible goods, which are auctioneered in a one-round auction. Each bidder gives bids (real numbers) for finitely many bundles of such goods (integer vectors of length n). Aggregating all bundles of all bidders together with their bids leads to a mixed subdivision which is known as the "demand complex". In contrast to the situation for the Ricardian economy, which is about tropical hyperplanes, i.e., tropical hypersurfaces of degree one, the tropical hypersurfaces that occur in product-mix auctions may have arbitrarily high degree. While some of the results presented here do apply, product mix auctions themselves are beyond the scope of this survey. In addition to the original [1] the interested reader should consult Tran and Yu [19]. In a similar vein Crowell and Tran studied applications of tropical geometry to mechanism design [2].

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2 Regular and Mixed Subdivisions

We will start out with an explanation of the Cayley Trick. Let A be a finite set of points in $\mathbb{R}^d$. A (*polyhedral*) *subdivision* of A is a finite polytopal complex whose vertices lie in the set A and that covers the convex hull $\text{conv } A$. For basic facts on the subject we refer to [3]. If λ is any function that assigns a real number to each point in A , then the set

$$U(A, \lambda) := \text{conv} \{ (a, \lambda(a)) \in \mathbb{R}^d \times \mathbb{R} \mid a \in A \} + \mathbb{R}_{\geq 0}(\mathbf{0}, 1) \quad (1)$$

is an unbounded polyhedron in $\mathbb{R}^{d+1}$; here "+" is the Minkowski addition, and $(\mathbf{0}, 1)$ is the unit vector that indicates the "upward" direction. Those faces of $U(A, \lambda)$ that are bounded admit an outward normal vector which points down, i.e., its scalar product with $(\mathbf{0}, 1)$ is strictly negative. Note that the outward normal vector of a facet is unique up to scaling. On the other hand each lower-dimensional face has an entire cone of outward normal vectors, which is positively spanned by the outward normal vectors of the facets containing that face. Projecting the bounded faces to

$\mathbb{R}^d$ by omitting the last coordinate defines a subdivision of A , which we denote as $\Sigma_\downarrow(A, \lambda)$. A subdivision that arises in this way is called *regular*. The example where $\lambda(a) = \|a\|^2$ is the Euclidean norm squared is the *Delaunay* subdivision of A .

Now consider two finite subsets, A and B , in $\mathbb{R}^d$. In the sequel we will be interested in special subdivisions of the Minkowski sum $A + B$. Yet it will be important to address points in $A + B$ by their *labels*. That is, for distinct $a, a' \in A$ and distinct $b, b' \in B$ it may happen that $a + b = a' + b'$. Nonetheless the label (a, b) differs from the label (a', b') . This means that the various labels of each point in $A + B$ keep track of the possibly many ways in which that point originates from A and B . For $A' \subset A$ and $B' \subset B$ the *mixed cell* of $A + B$ with label set $A' \times B'$ is the polytope

$$M(A', B') := \text{conv} \{a + b \mid a \in A', b \in B'\} . \quad (2)$$

Notice that the label set may also record points that are not vertices of $M(A', B')$. A polyhedral subdivision of $A + B$ is *mixed* if it is formed from mixed cells.

The *Cayley embedding* of the point configurations A and B in $\mathbb{R}^d$ is the point configuration

$$\mathcal{C}(A, B) := \{(a, -1) \mid a \in A\} \cup \{(b, 1) \mid b \in B\} \quad (3)$$

in $\mathbb{R}^{d+1}$. Any polytope of the form $\text{conv}(\mathcal{C}(A', B'))$ for subsets $A' \subset A$ and $B' \subset B$ is a *Cayley cell*. Intersecting the Cayley cell $\text{conv}(\mathcal{C}(A', B'))$ with the hyperplane $x_{d+1} = 0$ yields the *Minkowski cell* $M(A', B')$ with labeling $A' \times B'$. Note that formally this does not quite agree with (2), not only because we identify $\mathbb{R}^d$ with a linear hyperplane in $\mathbb{R}^{d+1}$, but also because the intersection of the Cayley cell with that hyperplane needs to be scaled by a factor of two to arrive at (2). However, to avoid cumbersome notation, we ignore these details. Let Σ be any polyhedral subdivision of $\mathcal{C}(A, B)$. Then the set

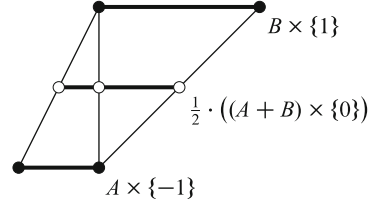
$$M(\Sigma) := \{M(A', B') \mid \text{conv}(\mathcal{C}(A', B')) \in \Sigma\}$$

is a subdivision of the scaled Minkowski sum $\frac{1}{2}(A + B)$, and it is called the *mixed subdivision* induced by Σ . Again we will ignore the scaling factor, i.e., we will view $M(\Sigma)$ as a subdivision of $A + B$.

Example 1 Let $A = \{0, 1\}$ and $B = \{1, 3\}$ be two pairs of points on the real line. The Cayley embedding $\mathcal{C}(A, B)$ are the four vertices of the trapezoid shown in Fig. 1. The two triangles

$$\begin{aligned} \text{conv} \mathcal{C}(A, \{1\}) &= \text{conv}\{(0, -1), (1, -1), (1, 1)\} \quad \text{and} \\ \text{conv} \mathcal{C}(\{1\}, B) &= \text{conv}\{(1, -1), (1, 1), (3, 1)\} \end{aligned}$$

Fig. 1 Cayley embedding of the two intervals $A = [0, 1]$ and $B = [1, 3]$



are Cayley cells, and they generate a subdivision of $\mathcal{C}(A, B)$ which induces a mixed subdivision of the Minkowski sum $A + B = \{0 + 1, 1 + 1, 0 + 3, 1 + 3\}$.

Notice that two Minkowski cells $M(A', B')$ and $M(A'', B'')$ intersect in the Minkowski cell $M(A' \cap A'', B' \cap B'')$ with labeling $(A' \cap A'') \times (B' \cap B'')$. This consistency among the labels of the cells in a mixed subdivision allows to uniquely lift back any mixed subdivision to a subdivision of the Cayley embedding.

Theorem 2 (Cayley Trick [10, 18]) *The map M from the set of subdivisions of $\mathcal{C}(A, B)$ to the set of mixed subdivisions of $A + B$ is a bijection that preserves refinement. Moreover, M maps regular subdivisions to regular subdivisions.*

Minkowski sums, mixed subdivisions and the Cayley Trick generalize to any finite number of point sets. To this end assume that we have n sets $A_1, A_2, \dots, A_n$ in $\mathbb{R}^d$. Then we pick an affine basis $u_1, u_2, \dots, u_n$ of $\mathbb{R}^{n-1}$, i.e., the vertices of a full-dimensional simplex. We define the *Cayley embedding*

$$\mathcal{C}(A_1, A_2, \dots, A_n) := \{(a_1, u_1) \mid a_1 \in A_1\} \cup \dots \cup \{(a_n, u_n) \mid a_n \in A_n\}$$

in $\mathbb{R}^d \times \mathbb{R}^{n-1}$. A particularly interesting case arises if we take n copies of the same point set $A \in \mathbb{R}^d$. Then the Cayley embedding satisfies

$$\mathcal{C}(\underbrace{A, A, \dots, A}_{n \text{ times}}) \cong A \times \Delta_{n-1} \quad , \quad (4)$$

where $\Delta_{n-1} = \text{conv}(e_1, e_2, \dots, e_n)$ is the $(n-1)$ -dimensional standard simplex in $\mathbb{R}^n$. Notice that we write “ $\cong$ ” instead of “ $=$ ” since (4) is only an affinely isomorphic image of what we defined in (3). An in-depth explanation of the Cayley Trick can be found in [3, §9.2].

3 Tropical Hypersurfaces

Now we want to take a look into a few basic concepts from tropical geometry. The Cayley Trick will prove useful to understanding unions of tropical hypersurfaces.

The *tropical semiring* is the set $\mathbb{T} = \mathbb{R} \cup \{\infty\}$ equipped with $\min$ as the addition and $+$ as the multiplication. The neutral element of the addition is ∞ , and the

multiplicative neutral element is 0. The tropical semiring behaves like a ring—with the lack of additive inverses as the crucial exception. If we want to stress the systematic role of these two arithmetic operations we write $\oplus$ instead of $\min$ and $\odot$ instead of $+$. For further details on tropical geometry we refer to the monograph [15] and the forthcoming book [12].

A *tropical polynomial* is a formal linear combination of finitely many monomials (with integer exponents that may also be negative) in, say, d variables with coefficients in $\mathbb{T}$. In this way a tropical polynomial F gives rise to a function

$$F(x) = \bigoplus_{m \in I} c_m \odot x^{\odot m} = \min_{m \in I} (c_m + m \cdot x) , \quad (5)$$

where I is a finite subset of $\mathbb{Z}^d$ and the coefficients c_m are elements of $\mathbb{T}$. By construction (5) is a piecewise linear and concave function from $\mathbb{R}^d$ to $\mathbb{R}$. The set $\text{supp}(F) = \{m \in I : c_m \neq \infty\}$ is the *support* of F . Occasionally we will distinguish between formal tropical polynomials and tropical polynomial functions. The set of formal tropical polynomials has a semiring structure where the addition and the multiplication is induced by $\oplus$ and $\odot$.

We may read the support of a tropical polynomial as a point configuration that is equipped with a height function given by the coefficients, and this is what gives us a connection to the previous section. The *extended Newton polyhedron* of a tropical polynomial is a special case of (1). More precisely, if F is defined as in (5), then we have

$$\widetilde{\mathcal{N}}(F) := U(\text{supp}(F), c) .$$

Projecting the faces of $\widetilde{\mathcal{N}}(F)$ down yields the regular subdivision $\Sigma_{\downarrow}(F) := \Sigma_{\downarrow}(\text{supp}(F), c)$ of the support, and the convex hull is the *Newton polytope* $\mathcal{N}(F) := \text{conv}(\text{supp}(F))$. It is worth noting that *any* lifting function on *any* finite set of lattice points can be read as a tropical polynomial.

One purpose of tropical geometry is to study classical algebraic varieties via their tropicalizations, which can be described in polyhedral terms. Here we will restrict our attention to tropical hypersurfaces, which are the tropical analogs of the vanishing locus of a single classical polynomial. The tropical polynomial F *vanishes* at $x \in \mathbb{R}^d$ if the minimum in (5) is attained at least twice, and the set

$$\mathcal{T}(F) := \{x \in \mathbb{R}^d \mid F \text{ vanishes at } x\}$$

is the *tropical hypersurface* defined by F . It is immediate that $\mathcal{T}(F)$ is a polyhedral complex in $\mathbb{R}^d$. What may be less obvious is that this is a meaningful definition. Yet the Fundamental Theorem of Tropical Geometry says that the tropical hypersurfaces are the images of classical varieties over a field with a non-Archimedean valuation (into the reals) under the valuation map; see Theorem 5 below. However, we wish to postpone this discussion for a short moment, as we first want to introduce another

polyhedron that we can associate with F ; this is the *dome*

$$\begin{aligned} \mathcal{D}(F) &:= \{(p, s) \in \mathbb{R}^{d+1} \mid p \in \mathbb{R}^d, s \in \mathbb{R}, s \leq F(p)\} \\ &= \bigcap_{m \in \text{supp}(F)} \{(p, s) \in \mathbb{R}^{d+1} \mid s \leq c_m + m \cdot p\} , \end{aligned} \quad (6)$$

which is unbounded in the negative e_{d+1} -direction and of full dimension $d + 1$. Let $\mathcal{D}^\uparrow(F)$ be the polyhedral complex that arises from $\partial\mathcal{D}(F)$ by omitting the last coordinate, and we call this the *normal complex* of the extended Newton polyhedron $\widetilde{\mathcal{N}}(F)$, or the *normal complex* of F , for short. This is a polyhedral subdivision of $\mathbb{R}^d$ which is piecewise-linearly isomorphic to the boundary $\partial\mathcal{D}(F)$ of the dome. Now the tropical hypersurface $\mathcal{T}(F)$ is the codimension-1-skeleton of the normal complex $\mathcal{D}^\uparrow(F)$, i.e., it corresponds to the codimension-2-skeleton of the polyhedron $\mathcal{D}(F)$. The latter is the set of faces whose dimension does not exceed $d - 1$. Summing up we have the following observation.

Lemma 3 *The facet defining inequalities of $\mathcal{D}(F)$ correspond to certain points in the support of F . Furthermore, the facets of $\mathcal{D}(F)$ are in bijection with the maximal cells of $\mathcal{D}^\uparrow(F)$ as well as with the connected components of the complement of $\mathcal{T}(F)$ in $\mathbb{R}^d$.*

More precisely, using the notation of (5) and (6), the point $m \in \text{supp } F$ yields a facet of $\mathcal{D}(F)$ if there exists an $x \in \mathbb{R}^d$ such that $F(x) = c_m \odot x^{\odot m}$ and $F(x) < c_{m'} \odot x^{\odot m'}$ for all $m' \neq m$. In that case the inequality $s \leq c_m + m \cdot x$ is facet defining. Now we want to relate the dome with the extended Newton polyhedron and the induced regular subdivision.

Proposition 4 *There is an inclusion reversing bijection between the face poset of $\mathcal{D}(F)$ and the poset of bounded faces of $\widetilde{\mathcal{N}}(F)$. This entails that the tropical variety $\mathcal{T}(F)$ is dual to the regular subdivision $\Sigma_\downarrow(F)$ of $\text{supp } F$.*

Essentially this is a consequence of cone polarity. Notice that the face poset of $\mathcal{D}(F)$ is isomorphic with the poset of cells of the normal complex $\mathcal{D}^\uparrow(F)$.

To explore the relationship of tropical with algebraic geometry here it suffices to consider one fixed field with a non-Archimedean valuation. Its elements look as follows. A *formal Puiseux series* with complex coefficients is a power series of the form

$$\gamma(t) = \sum_{k=m}^{\infty} a_k \cdot t^{k/N} ,$$

where $m, N \in \mathbb{Z}$, $N > 0$ and $a_k \in \mathbb{C}$. These formal power series with rational exponents can be added and multiplied in the usual way to yield an algebraically closed field of characteristic zero, which we denote as $\mathbb{C}\{\{t\}\}$. As a key feature there

is a map $\text{val} : \mathbb{C}\{\{t\}\} \rightarrow \mathbb{Q}$ that sends a Puiseux series to its lowest exponent. This *valuation map* satisfies

$$\begin{aligned} \text{val}(\gamma(t) + \delta(t)) &\geq \min(\text{val}(\gamma(t)), \text{val}(\delta(t))) = \text{val}(\gamma(t)) \oplus \text{val}(\delta(t)) \quad \text{and} \\ \text{val}(\gamma(t) \cdot \delta(t)) &= \text{val}(\gamma(t)) + \text{val}(\delta(t)) = \text{val}(\gamma(t)) \odot \text{val}(\delta(t)) . \end{aligned}$$

We abbreviate $\mathbb{C}\{\{t\}\}$ by $\mathbb{K}$. For any Laurent polynomial $f = \sum_{i \in I} \gamma_i(t) x^i$ in the ring $\mathbb{K}[x_1^\pm, x_2^\pm, \dots, x_d^\pm]$ its *tropicalization* is the tropical polynomial

$$\text{trop}(f) := \bigoplus_{i \in I} \text{val}(\gamma_i) \odot x^{\odot i} .$$

The vanishing locus of f is the hypersurface $V(f) := \{x \in (\mathbb{K} \setminus \{0\})^d \mid f(x) = 0\}$ in the algebraic torus $(\mathbb{K} \setminus \{0\})^d$. In the sequel we consider a Laurent polynomial $f \in \mathbb{K}[x_1^\pm, x_2^\pm, \dots, x_d^\pm]$ and its tropicalization $F = \text{trop}(f)$. The following key result has been obtained by Kapranov; see [6].

Theorem 5 (Fundamental Theorem of Tropical Geometry) *For every Laurent polynomial $f \in \mathbb{K}[x_1^\pm, x_2^\pm, \dots, x_d^\pm]$ we have*

$$\overline{\text{val}(V(f))} = \mathcal{T}(\text{trop}(f)) . \quad (7)$$

Here the valuation map val is applied element-wise and coordinate-wise to the points in the hypersurface $V(f)$, and here $\overline{}$ denotes the topological closure in $\mathbb{R}^d$. It should be noted that the Fundamental Theorem admits a generalization to arbitrary ideals in $\mathbb{K}[x_1^\pm, x_2^\pm, \dots, x_d^\pm]$; see [15, §3.2]. The hypersurface case corresponds to the principal ideals. Now let us consider two Laurent polynomials $f, g \in \mathbb{K}[x_1^\pm, x_2^\pm, \dots, x_d^\pm]$ with tropicalizations $F = \text{trop}(f)$ and $G = \text{trop}(g)$.

Lemma 6 *We have*

$$\mathcal{T}(F \odot G) = \mathcal{T}(\text{trop}(f \cdot g)) = \mathcal{T}(F) \cup \mathcal{T}(G) .$$

Proof A direct computation shows that $\text{trop}(f \cdot g)$ equals $F \odot G$. As $V(f \cdot g) = V(f) \cup V(g)$ holds classically, the claim follows from Theorem 5.

The next result says that tropicalization commutes with forming unions of (tropical) hypersurfaces.

Proposition 7 *The diagram*

$$\begin{array}{ccccc}
 V(f) & \longrightarrow & V(f \cdot g) & \longleftarrow & V(g) \\
 \downarrow \text{val} & & \downarrow \text{val} & & \downarrow \text{val} \\
 \mathcal{T}(F) & \longrightarrow & \mathcal{T}(F \odot G) & \longleftarrow & \mathcal{T}(G) \\
 \downarrow \text{id} \times F & & \downarrow \text{id} \times (F \odot G) & & \downarrow \text{id} \times G \\
 \partial \mathcal{D}(F) & \xrightarrow{\odot G} & \partial \mathcal{D}(F \odot G) & \xleftarrow{\odot F} & \partial \mathcal{D}(G)
 \end{array} \tag{8}$$

commutes. The map $\odot G$ sends a point $(w, s) \in \mathbb{R}^{d+1}$ to $(w, s + G(w))$, and $\odot F$ is similarly defined. The unmarked horizontal arrows are embeddings of subsets.

Proof The upper two squares in the diagram commute due to the Fundamental Theorem. We focus on the lower left square; the lower right one is similar. Let $w \in \mathcal{T}(F)$. The latter is contained in $\mathcal{T}(F \odot G) = \mathcal{T}(F) \cup \mathcal{T}(G)$ by Lemma 6. Evaluating F at w yields the point $(w, F(w))$ in the codimension-2-skeleton of the dome $\mathcal{D}(F) \subset \mathbb{R}^{d+1}$, which is part of the boundary. Any point in the boundary of $\mathcal{D}(F)$ has the form $(v, F(v))$ for some $v \in \mathbb{R}^d$. We can check that

$$\odot G(v, F(v)) = (v, F(v) + G(v)) = (v, F \odot G(v)) , \tag{9}$$

and thus $\odot G$, indeed, maps arbitrary points in the boundary of $\mathcal{D}(F)$ to boundary points of $\mathcal{D}(F \odot G)$. Setting $v = w$ in (9) now finishes the proof.

Remark 8 From (9) it also follows that the normal complex $\mathcal{D}^\uparrow(F \odot G)$ is the common refinement of the normal complexes $\mathcal{D}^\uparrow(F)$ and $\mathcal{D}^\uparrow(G)$.

The vertices of the regular subdivision $\Sigma_\downarrow(F \odot G)$ are sums of one point in $\text{supp}(F)$ with one point in $\text{supp}(G)$, i.e., they correspond to products of a monomial in f with a monomial in g . Altogether the Cayley embedding of the monomials of the factors, seen as configurations of lattice points, project to the monomials in the product. Now, via the Cayley Trick, any regular subdivision of the Cayley embedding induces a coherent subdivision of the Minkowski sum.

Corollary 9 *The regular subdivision $\Sigma_\downarrow(F \odot G)$ is a coherent mixed subdivision of $\text{supp}(F) + \text{supp}(G)$.*

Classically, varieties defined by homogeneous polynomials are studied in the projective space. Here the situation is similar. A tropical polynomial F is *homogeneous* of degree δ if its support is contained in the affine hyperplane $\sum x_i = \delta$. For such F the tropical hypersurface $\mathcal{T}(F)$ can be seen as a subset of the *tropical projective torus* $\mathbb{R}^d / \mathbb{R}\mathbf{1}$. That set is homeomorphic with $\mathbb{R}^{d-1}$, and it has a natural

compactification, the *tropical projective space*

$$\mathbb{TP}^{d-1} := \left((\mathbb{R} \cup \{\infty\})^d \setminus \{\infty \mathbf{1}\} \right) / \mathbb{R} \mathbf{1} .$$

Below we will also look into the set $\mathbb{T}_{\max} = \mathbb{R} \cup \{-\infty\}$ with $\max$ as the addition instead of $\min$. These two versions of the tropical semiring are isomorphic via $-\min(x, y) = \max(-x, -y)$. Hence the results of this section, suitably adjusted, also hold for $\max$ -tropical polynomials. To avoid confusion we will use “ $\min$ ” or “ $\max$ ” as subscripts wherever necessary. As far as the regular subdivisions are concerned, for $\min$ we look at the lifted points from “below”, while for $\max$ we look from “above”. To mark this difference we write $\Sigma^\uparrow(H)$ and $\mathcal{D}_\downarrow(H)$, if H is a $\max$ -tropical polynomial, and we have

$$\Sigma^\uparrow(H) = \Sigma_\downarrow(-H) \quad \text{and} \quad \mathcal{D}_\downarrow(H) = -\mathcal{D}^\uparrow(-H) , \quad (10)$$

where $-H$ is the $\min$ -tropical polynomial that arises from H by replacing each coefficient by its negative, and the minus in front of the polyhedral complex on the right refers to reflection at the origin of $\mathbb{R}^d$.

4 Arrangements of Tropical Hyperplanes

The simplest kind of algebraic hypersurfaces are the hyperplanes, i.e., the linear ones. Arrangements of hyperplanes is a classical topic with a rich connection to algebraic geometry, group theory, topology and combinatorics. A standard reference is the monograph [16] by Orlik and Terao. The tropicalization of hyperplane arrangements was pioneered by Develin and Sturmfels [4]. The Cayley Trick will sneak into the discussion through Corollary 9.

Let $V = (v_{ik}) \in \mathbb{T}_{\min}^{d \times n}$ be a matrix whose coefficients are real numbers or ∞ . Throughout we will assume that each column contains at least one finite entry. Writing $v^{(k)}$ for the k -column this means that $v^{(k)} + \mathbb{R} \mathbf{1}$ is a point in the tropical projective space $\mathbb{TP}_{\min}^{d-1}$.

The *negative* column vector $-v^{(k)}$ is an element of $(\mathbb{R} \cup \{-\infty\})^d$, and it defines the homogeneous *max*-tropical linear form

$$\begin{aligned} (-v_{1k}) \odot x_1 \oplus_{\max} (-v_{2k}) \odot x_2 \oplus_{\max} \cdots \oplus_{\max} (-v_{dk}) \odot x_d \\ = \max\{x_1 - v_{1k}, x_2 - v_{2k}, \dots, x_d - v_{dk}\} , \end{aligned}$$

which we will identify with $-v^{(k)}$. Since we assumed that $v^{(k)}$ has at least one finite coefficient that tropical linear form is not trivial. The tropical variety $\mathcal{T}_{\max}(-v^{(k)})$

is a max-tropical hyperplane and, by Lemma 6, the tropical variety associated with the product of linear forms

$$\begin{aligned}\mathcal{T}_{\max}(-V) &:= \mathcal{T}_{\max}((-v^{(1)}) \odot (-v^{(2)}) \odot \cdots \odot (-v^{(n)})) \\ &= \mathcal{T}_{\max}(-v^{(1)}) \cup \mathcal{T}_{\max}(-v^{(2)}) \cup \cdots \cup \mathcal{T}_{\max}(-v^{(n)})\end{aligned}$$

is a union of tropical hyperplanes.

The support $\text{supp}(-v^{(k)})$ is a subset of the vertices of the ordinary standard simplex $\Delta_{d-1} = \text{conv}(e_1, e_2, \dots, e_d)$ in $\mathbb{R}^d$. So the Newton polytope of the tropical linear form $-v^{(k)}$ is a face of Δ_{d-1} . If that Newton polytope is the entire simplex of dimension $d - 1$ then the normal complex $\mathcal{D}^\uparrow(v^{(k)})$ has a unique vertex that is contained in d maximal cells which are dual to the vertices of Δ_{d-1} . Notice that, following standard practice in polyhedral geometry [3], even in the max-tropical setting, we usually prefer to look at regular subdivisions from “below”, and (10) takes care of the translation. Thus we study $\mathcal{D}^\uparrow(v^{(k)})$ rather than its image $\mathcal{D}_\downarrow(-v^{(k)})$ under reflection. The following result is a consequence of the Cayley Trick in the guise of Corollary 9.

Proposition 10 *The regular subdivision $\Sigma_\downarrow(V)$, which is dual to the max-tropical hypersurface $\mathcal{T}_{\max}(-V)$, coincides with the mixed subdivision*

$$M(\Sigma_\downarrow(v^{(1)}), \Sigma_\downarrow(v^{(2)}), \dots, \Sigma_\downarrow(v^{(n)})) .$$

Example 11 For

$$V = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 4 & 3 & 0 \\ 0 & 1 & 3 & 2 \end{pmatrix}$$

the tropical hypersurface $\mathcal{T}_{\max}(-V)$ is a union of four tropical lines in the tropical projective 2-torus. The arrangement and the normal complex $\mathcal{D}_\downarrow(-V)$ is displayed in Fig. 2. In that drawing each point $(x_1, x_2, x_3) + \mathbb{R}\mathbf{1} \in \mathbb{R}^3/\mathbb{R}\mathbf{1}$ occurs as $(x_2 - x_1, x_3 - x_1)$.

The Newton polytope of each max-tropical linear form $-v^{(k)}$ is the standard triangle Δ_2 . The fourfold Minkowski sum is the dilated triangle $4 \cdot \Delta_2$. The mixed subdivision $\Sigma^\uparrow(-V)$ is shown in Fig. 3. That picture also shows the tropical line arrangement from Fig. 2 embedded into the dual graph of the subdivision.

According to Lemma 3 each connected component of the complement of $\mathcal{T}_{\max}(-V)$ is marked with the corresponding element from the support set of the

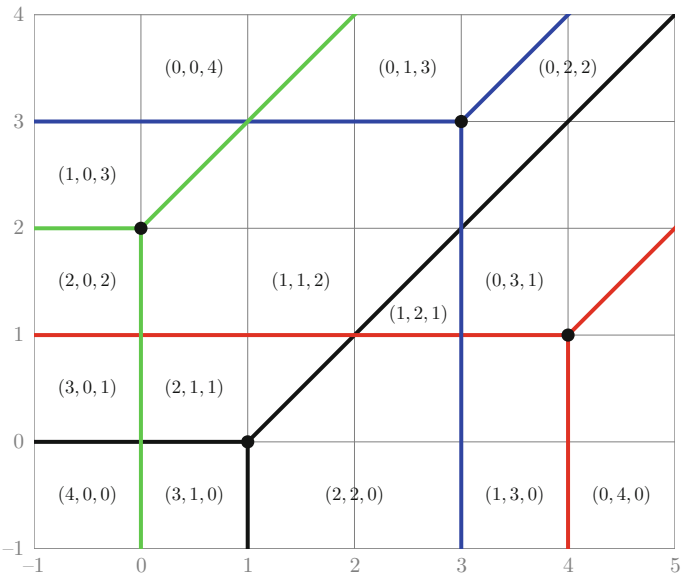


Fig. 2 Four max-tropical hyperplanes in $\mathbb{R}^3/\mathbb{R}\mathbf{1}$

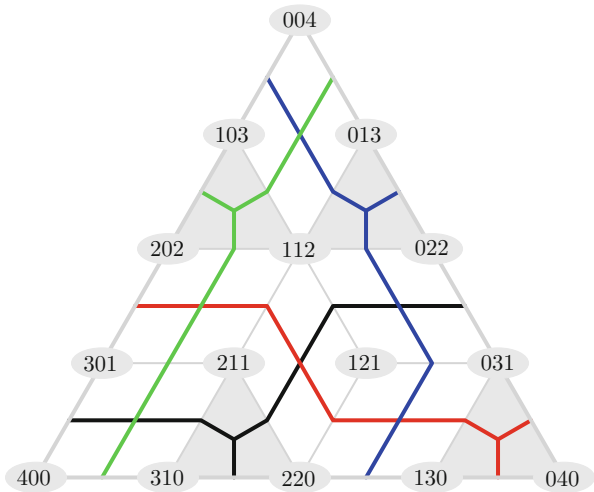


Fig. 3 Mixed subdivision of $4\cdot\Delta_2$

max-tropical polynomial

$$\begin{aligned}
 & (-v^{(1)}) \odot (-v^{(2)}) \odot (-v^{(3)}) \odot (-v^{(4)}) \\
 &= \max(4x_1, 3x_1 + x_2, 3x_1 + x_3, 2x_1 + 2x_2 - 1, 2x_1 + x_2 + x_3, 2x_1 + 2x_3 - 1, \\
 &\quad x_1 + 3x_2 - 4, x_1 + 2x_2 + x_3 - 2, x_1 + x_2 + 2x_3 - 1, x_1 + 3x_3 - 3, \\
 &\quad 4x_2 - 8, 3x_2 + x_3 - 5, 2x_2 + 2x_3 - 4, x_2 + 3x_3 - 4, 4x_3 - 6) .
 \end{aligned} \tag{11}$$

For instance evaluating the max-tropical polynomial (11) at the point $(0, 2, 0)$ yields the maximum 3, which is attained at the tropical monomial $2x_1 + 2x_3 - 1$ with exponent vector $(2, 2, 0)$. In the language of [5] that vector is the “coarse type” of the corresponding (maximal) cell of $\Sigma^\uparrow(-V)$; see Remark 16 below. In the dual mixed subdivision we actually see $(2, 2, 0)$ as the coordinates of the lattice point dual to the maximal cell of $\mathcal{D}_\downarrow(-V)$ that contains the point $(0, 2, 0)$ in its interior.

We now turn to investigating a tropical version of convexity. The set

$$\text{tpos}_{\min}(V) := \left\{ \bigoplus_{k=1}^n \lambda_k \odot v^{(k)} \mid \lambda_k \in \mathbb{T} \right\} = \{V \odot_{\min} \lambda \mid \lambda \in \mathbb{T}^n\}$$

is the *min-tropical cone* spanned by (the columns of) V . It satisfies $\text{tpos}_{\min}(V) = \text{tpos}_{\min}(V) + \mathbb{R}\mathbf{1}$, which is why it can be studied as a subset of the tropical projective space $\mathbb{TP}_{\min}^{d-1}$. The image $\text{tconv}_{\min}(V)$ of $\text{tpos}_{\min}(V)$ under the canonical projection $\mathbb{T}_{\min}^d \rightarrow \mathbb{TP}_{\min}^{d-1}$ is called a *tropical polytope*. In the sequel we will concentrate on the intersection

$$\text{tconv}_{\min}^\circ(V) := \text{tconv}_{\min}(V) \cap (\mathbb{R}^d / \mathbb{R}\mathbf{1}) ,$$

which comprises the points with finite coordinates in the tropical polytope $\text{tconv}_{\min}(V)$.

Theorem 12 *The set $\text{tconv}_{\min}^\circ(V)$ is a union of cells of the polyhedral complex*

$$\mathcal{D}_\downarrow((-v^{(1)}) \odot (-v^{(2)}) \odot \cdots \odot (-v^{(n)}))$$

in $\mathbb{R}^d / \mathbb{R}\mathbf{1}$. If all coefficients of V are finite then $\text{tconv}_{\min}^\circ(V) = \text{tconv}_{\min}(V)$ is the union of those cells that are bounded.

Theorem 12 was proved by Develin and Sturmfels [4] for finite coefficients. Extensions to the general case have been obtained by Fink and Rincón [7] and by Loho and the author [13]. As a consequence of Remark 8 the normal complex $\mathcal{D}_\downarrow((-v^{(1)}) \odot (-v^{(2)}) \odot \cdots \odot (-v^{(n)}))$ in $\mathbb{R}^d / \mathbb{R}\mathbf{1}$ is the common refinement of the n normal complexes $\mathcal{D}^\downarrow(-v^{(1)})$, $\mathcal{D}^\downarrow(-v^{(2)})$, $\dots$, $\mathcal{D}^\downarrow(-v^{(n)})$; see Fig. 2.

Example 13 For the matrix V from Example 11 the tropical polytope $\text{tconv}_{\min}(V)$ agrees with $\text{tconv}_{\min}^\circ(V)$, and it is shown in Fig. 4. There are four bounded cells in the

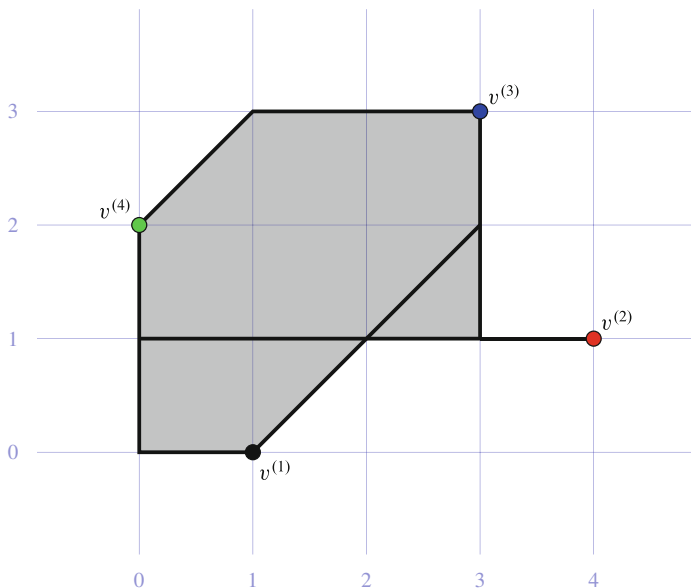


Fig. 4 Tropical polytope generated by four points in $\mathbb{R}^3/\mathbb{R}\mathbf{1}$

subdivision $\Sigma^\uparrow(-V)$, shown in Fig. 2, which are maximal with respect to inclusion, three of dimension two and one of dimension one. These form the tropical convex hull.

We want to come back to the Cayley Trick. If all entries of the matrix $V \in \mathbb{T}^{d \times n}$ are finite then the Newton polytope of the linear forms corresponding to each of the n columns is the simplex Δ_{d-1} . In this case the Cayley embedding $\mathcal{C}(\Delta_{d-1}, \Delta_{d-1}, \dots, \Delta_{d-1})$ is isomorphic with the product $\Delta_{d-1} \times \Delta_{n-1}$ of simplices.

Corollary 14 *If all entries of V are finite then the regular mixed subdivision of $n \cdot \Delta_{d-1}$ from Proposition 10 is piecewise linearly isomorphic with a slice of the regular subdivision of $\Delta_{d-1} \times \Delta_{n-1}$ where the vertex (e_i, e_k) is lifted to v_{ik} .*

Clearly, when we talk about subdivisions of products of simplices it makes sense to think about exchanging the factors. A direct computation shows that this corresponds to changing from $V \in \mathbb{R}^{d \times n}$ to the transpose $V^\top \in \mathbb{R}^{n \times d}$. Figure 5 shows the max-tropical hyperplane arrangement in $\mathbb{R}^4/\mathbb{R}\mathbf{1}$ and the min-tropical convex hull arising from the 4×3 -matrix $V^\top$ for V as in Example 11. The mixed subdivision $\Sigma^\uparrow(-V^\top)$ is displayed in Fig. 6.

If the matrix V contains at least one coefficient ∞ then Corollary 14 holds for the proper subpolytope

$$\text{conv} \{ (e_i, e_k) \in \mathbb{R}^d \times \mathbb{R}^n \mid v_{ik} \in \mathbb{R} \}$$

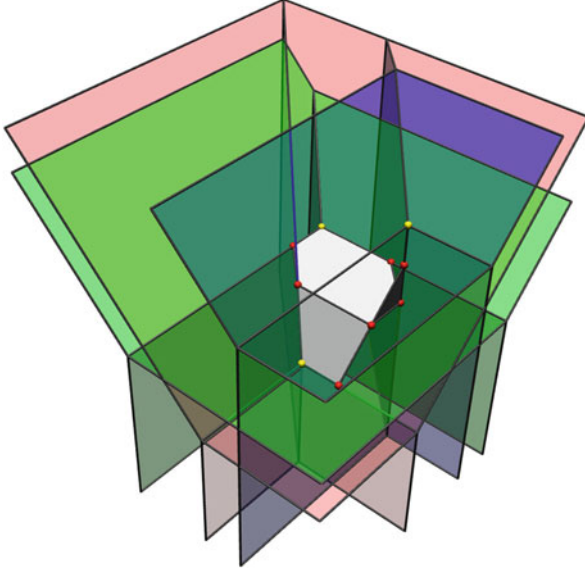


Fig. 5 Three max-tropical hyperplanes in $\mathbb{R}^4/\mathbb{R}\mathbf{1}$. Compare with Fig. 4

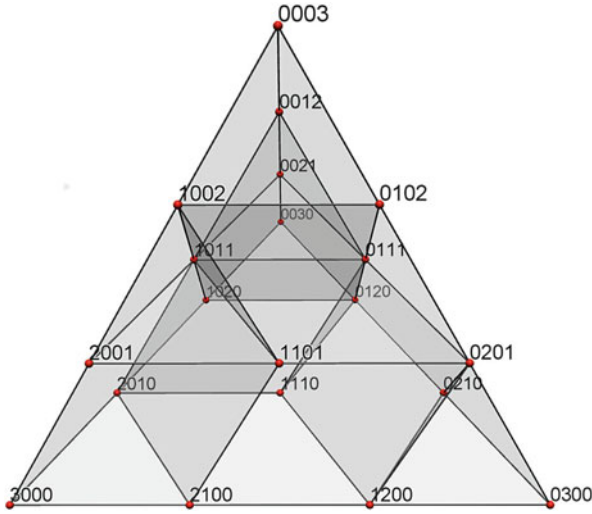


Fig. 6 Mixed subdivision of $3 \cdot \Delta_3$ dual to max-tropical hyperplane arrangement from Fig. 5

of $\Delta_{d-1} \times \Delta_{n-1}$. These subpolytopes and their subdivisions are studied in [7] and [13]. The *tropical covector* of a point $z \in \mathbb{R}^d$ with respect to the matrix V is defined as

$$\text{tc}(z) := \{(i, k) \in [d] \times [n] \mid v_{ik} - z_i = \min(v_{1k} - z_1, v_{2k} - z_2, \dots, v_{dk} - z_d)\} \quad .$$

That is to say, the pair (i, k) lies in $\text{tc}(z)$ if and only if the minimum of the coordinates of the difference $v^{(k)} - z$ is attained at i . This encodes the relative position of z with respect to the columns of V . It is immediate that $\text{tc}(z)$ is constant on the set $z + \mathbb{R}\mathbf{1}$. Thus it is well-defined for points in the tropical projective torus. In the language of [4] the tropical covectors occur as “types”, and these are called “fine types” in [5]. The term “covector” was first used in [7] and subsequently in [13].

Example 15 Again we consider $V \in \mathbb{R}^{3 \times 4}$ as in Example 11. For instance, for $z = (0, 1, 3)^\top$ we have

$$\text{tc}(z) = \{(3, 1), (3, 2), (1, 3), (3, 3), (2, 4), (3, 4)\} . \quad (12)$$

It is instrumental to locate the point z in Fig. 2: It is the unique point in the intersection of the green hyperplane (column 4) with the blue hyperplane (column 3). In the mixed subdivision picture in Fig. 3 the point z corresponds to the maximal cell with vertices $(0, 0, 4)$, $(0, 1, 3)$, $(1, 0, 3)$ and $(1, 1, 2)$.

Remark 16 Consider the four max-tropical linear forms $(-v^{(k)})$ corresponding to the columns of our running example matrix V for *separate variables*. That is, we choose a new set of variables for each column. More precisely, we consider

$$\begin{aligned} (-v^{(1)}) &= \max\{y_{11}, y_{21} - 1, y_{31}\} & (-v^{(2)}) &= \max\{y_{12}, y_{22} - 4, y_{32} - 1\} \\ (-v^{(3)}) &= \max\{y_{13}, y_{23} - 3, y_{33} - 3\} & (-v^{(4)}) &= \max\{y_{14}, y_{24}, y_{34} - 2\} \end{aligned} .$$

Now we can look at the max-tropical polynomial W in the 12 variables $y_{11}, y_{12}, \dots, y_{34}$ which arises as the tropical product of these four tropical linear forms; this is shown in Table 1. The tropical covector $\text{tc}(z)$ of a point $z \in \mathbb{R}^3$ agrees with the least common multiple of those monomials of W at which the maximum $W(z, z, z, z)$ is attained; here we substitute $y_{11} = y_{12} = y_{13} = y_{14} = z_1$, $y_{21} = y_{22} = y_{23} = y_{24} = z_2$ etc. by real numbers. For instance, letting $z = (0, 1, 3)^\top$ the maximum $W(z, z, z, z) = 6$ is attained at the four terms underlined in Table 1. Observe that the four marked terms in Table 1 are precisely those which correspond to subsets of the tropical covector shown in (12). If we substitute $y_{11} = y_{12} = y_{13} = y_{14} = x_1$, $y_{21} = y_{22} = y_{23} = y_{24} = x_2$ etc. by indeterminates $x = (x_1, x_2, x_3)$ the resulting expression $W(x, x, x, x)$ is precisely the tropical polynomial in (11). This latter substitution explains the relationship between the “fine types” and the “coarse types” discussed in [5] or, equivalently, the relationship between the tropical covectors and the coordinates in the mixed subdivision picture.

Table 1 Max-tropical polynomial which arises as the product of four tropical linear forms with separate variables y_{ij} for $(i, j) \in [3] \times [4]$

$W(y_{11}, y_{12}, \dots, y_{34}) = \max($
$y_{11} + y_{12} + y_{13} + y_{14}, y_{11} + y_{12} + y_{13} + y_{24}, -2 + y_{11} + y_{12} + y_{13} + y_{34},$
$-3 + y_{11} + y_{12} + y_{14} + y_{23}, -3 + y_{11} + y_{12} + y_{14} + y_{33}, -3 + y_{11} + y_{12} + y_{23} + y_{24},$
$-5 + y_{11} + y_{12} + y_{23} + y_{34}, -3 + y_{11} + y_{12} + y_{24} + y_{33}, -5 + y_{11} + y_{12} + y_{33} + y_{34},$
$-4 + y_{11} + y_{13} + y_{14} + y_{22}, -1 + y_{11} + y_{13} + y_{14} + y_{32}, -4 + y_{11} + y_{13} + y_{22} + y_{24},$
$-6 + y_{11} + y_{13} + y_{22} + y_{34}, -1 + y_{11} + y_{13} + y_{24} + y_{32}, -3 + y_{11} + y_{13} + y_{32} + y_{34},$
$-7 + y_{11} + y_{14} + y_{22} + y_{23}, -7 + y_{11} + y_{14} + y_{22} + y_{33}, -4 + y_{11} + y_{14} + y_{23} + y_{32},$
$-4 + y_{11} + y_{14} + y_{32} + y_{33}, -7 + y_{11} + y_{22} + y_{23} + y_{24}, -9 + y_{11} + y_{22} + y_{23} + y_{34},$
$-7 + y_{11} + y_{22} + y_{24} + y_{33}, -9 + y_{11} + y_{22} + y_{33} + y_{34}, -4 + y_{11} + y_{23} + y_{24} + y_{32},$
$-6 + y_{11} + y_{23} + y_{32} + y_{34}, -4 + y_{11} + y_{24} + y_{32} + y_{33}, -6 + y_{11} + y_{32} + y_{33} + y_{34},$
$-1 + y_{12} + y_{13} + y_{14} + y_{21}, y_{12} + y_{13} + y_{14} + y_{31}, -1 + y_{12} + y_{13} + y_{21} + y_{24},$
$-3 + y_{12} + y_{13} + y_{21} + y_{34}, y_{12} + y_{13} + y_{24} + y_{31}, -2 + y_{12} + y_{13} + y_{31} + y_{34},$
$-4 + y_{12} + y_{14} + y_{21} + y_{23}, -4 + y_{12} + y_{14} + y_{21} + y_{33}, -3 + y_{12} + y_{14} + y_{23} + y_{31},$
$-3 + y_{12} + y_{14} + y_{31} + y_{33}, -4 + y_{12} + y_{21} + y_{23} + y_{24}, -6 + y_{12} + y_{21} + y_{23} + y_{34},$
$-4 + y_{12} + y_{21} + y_{24} + y_{33}, -6 + y_{12} + y_{21} + y_{33} + y_{34}, -3 + y_{12} + y_{23} + y_{24} + y_{31},$
$-5 + y_{12} + y_{23} + y_{31} + y_{34}, -3 + y_{12} + y_{24} + y_{31} + y_{33}, -5 + y_{12} + y_{31} + y_{33} + y_{34},$
$-5 + y_{13} + y_{14} + y_{21} + y_{22}, -2 + y_{13} + y_{14} + y_{21} + y_{32}, -4 + y_{13} + y_{14} + y_{22} + y_{31},$
$-1 + y_{13} + y_{14} + y_{31} + y_{32}, -5 + y_{13} + y_{21} + y_{22} + y_{24}, -7 + y_{13} + y_{21} + y_{22} + y_{34},$
$-2 + y_{13} + y_{21} + y_{24} + y_{32}, -4 + y_{13} + y_{21} + y_{32} + y_{34}, -4 + y_{13} + y_{22} + y_{24} + y_{31},$
$-6 + y_{13} + y_{22} + y_{31} + y_{34}, \underline{-1 + y_{13} + y_{24} + y_{31} + y_{32}}, \underline{-3 + y_{13} + y_{31} + y_{32} + y_{34}},$
$-8 + y_{14} + y_{21} + y_{22} + y_{23}, -8 + y_{14} + y_{21} + y_{22} + y_{33}, -5 + y_{14} + y_{21} + y_{23} + y_{32},$
$-5 + y_{14} + y_{21} + y_{32} + y_{33}, -7 + y_{14} + y_{22} + y_{23} + y_{31}, -7 + y_{14} + y_{22} + y_{31} + y_{33},$
$-4 + y_{14} + y_{23} + y_{31} + y_{32}, -4 + y_{14} + y_{31} + y_{32} + y_{33}, -8 + y_{21} + y_{22} + y_{23} + y_{24},$
$-10 + y_{21} + y_{22} + y_{23} + y_{34}, -8 + y_{21} + y_{22} + y_{24} + y_{33}, -10 + y_{21} + y_{22} + y_{33} + y_{34},$
$-5 + y_{21} + y_{23} + y_{24} + y_{32}, -7 + y_{21} + y_{23} + y_{32} + y_{34}, -5 + y_{21} + y_{24} + y_{32} + y_{33},$
$-7 + y_{21} + y_{32} + y_{33} + y_{34}, -7 + y_{22} + y_{23} + y_{24} + y_{31}, -9 + y_{22} + y_{23} + y_{31} + y_{34},$
$-7 + y_{22} + y_{24} + y_{31} + y_{33}, -9 + y_{22} + y_{31} + y_{33} + y_{34}, -4 + y_{23} + y_{24} + y_{31} + y_{32},$
$-6 + y_{23} + y_{31} + y_{32} + y_{34}, \underline{-4 + y_{24} + y_{31} + y_{32} + y_{33}}, \underline{-6 + y_{31} + y_{32} + y_{33} + y_{34}}) .$

Evaluating at $y_{1j} = 0, y_{2j} = 1, y_{3j} = 3$, for $j = 1, 2, 3, 4$, yields the value 6, which is the maximum taken over $81 = 3^4$ terms. The four terms for which that maximum is attained are marked

5 Ricardian Theory of International Trade

There is a recent interest to apply techniques from tropical geometry to questions studied in economics. Here we focus on Shiozawa's work on international trade theory, and we summarize one part of the paper [17]. Shiozawa suggests to describe the Ricardian theory of international trade in terms of tropical hyperplane arrangements and tropical convexity. Since the underpinnings of that theory rely on the Cayley Trick in an essential way, it is obvious that it can be exploited.

A *Ricardian economy* is described by a pair (R, q) where $R = (r_{ik})$ is a $d \times n$ -matrix of positive real numbers and q is a vector of d positive reals. The d rows of R represent commodities, and its n columns are countries. The *production coefficient* r_{ik} measures how much labor is required to produce one unit of commodity i in country k , and the number q_k is the total available work force of the country k . These parameters are fixed. The purpose of this highly abstract economic model is to study the interaction of prices for the commodities and wages for the labor. In fact, here we will focus on just a single aspect of Ricardian trade theory, which is why subsequently we will even ignore the work force vector q .

A *wage-price system* in this economy is a pair (w, p) , where w is a column vector of length n and p is a column vector of length d . Again all entries are positive. The number w_k is the wage in country k , and p_i is the international price for commodity i . Now a wage-price system (w, p) is *admissible* if

$$r_{ik}w_k \geq p_i \quad \text{for all } i \in [d] \text{ and } k \in [n] . \quad (13)$$

These inequalities reflect the fundamental assumption that the countries compete freely among one another on the world market. This is supposed to say that the prices are low enough to avoid excess profit. Notice that the Ricardian economic model neglects any transport costs.

In Eq. (13), for every fixed commodity i , we can form the minimum over all countries to obtain a total of m consolidated inequalities, one for each commodity. If we now assume that the prices are as large as possible without violating the admissibility constraints, we arrive at the equations

$$\min \{r_{i1}w_1, r_{i2}w_2, \dots, r_{in}w_n\} = p_i \quad \text{for all } i \in [d] . \quad (14)$$

Going from the inequalities (13) to Eq. (14) imposes an extra condition. The wages are said to be *sharing* for the given prices if that condition is satisfied.

It is of interest for which countries the minimum on the left of (14) is attained. The pair $(i, k) \in [d] \times [n]$ is called *competitive* for the admissible wage-price system (w, p) if

$$r_{ik}w_k = p_i \leq r_{i\ell}w_\ell \quad \text{for all } \ell \in [n] .$$

This means that k belongs to those countries that are efficient enough to produce commodity i at the international price p_i . The condition that the prices are sharing means that for each commodity k there is at least one country i such that (i, k) is competitive for (w, p) .

If we now rewrite $\bar{a}_{ij} = \log r_{ij}$ as well as $\bar{w}_k = \log w_k$ and $\bar{p}_i = \log p_i$ then (14) becomes a system of tropical linear equations which read

$$\begin{aligned}\bar{p}_i &= \min \{ \bar{r}_{i1} + \bar{w}_1, \bar{r}_{i2} + \bar{w}_2, \dots, \bar{r}_{in} + \bar{w}_n \} \\ &= (\bar{r}_{i1} \odot \bar{w}_1) \oplus (\bar{r}_{i2} \odot \bar{w}_2) \oplus \dots \oplus (\bar{r}_{in} \odot \bar{w}_n) \quad \text{for all } i \in [d] .\end{aligned}\tag{15}$$

That is, $\bar{p}$ is sharing if and only if $\bar{p}$ is contained in the tropical cone $\text{tpos}_{\min}(\bar{R})$. Notice that in this translation we make use of the fact that the logarithm function is monotone. Letting $\bar{R} = (\bar{r}_{ij})_{i,j} \in \mathbb{R}^{d \times n}$ and similarly $\bar{w} = (\bar{w}_1, \dots, \bar{w}_n)^\top$ as well as $\bar{p} = (\bar{p}_1, \dots, \bar{p}_d)^\top$ we obtain (15) in matrix form

$$\bar{R} \odot \bar{w} = \bar{p} .\tag{16}$$

Example 17 Let us consider as $\bar{R}$ the matrix V from Example 11, i.e., $d = 3$ and $n = 4$. This way, e.g., the coefficient $v_{24} = \bar{r}_{24} = 0$ is interpreted as the logarithmic cost to produce one unit of commodity 2 in country 4. For instance, the logarithmic wage–price system

$$\bar{w} = (5, 5, 1, 2)^\top \quad \text{and} \quad \bar{p} = (1, 2, 4)^\top\tag{17}$$

satisfies Eq. (16). Notice that $(1, 2, 4)^\top$ and, e.g., $(0, 1, 3)^\top$ are the same modulo $\mathbb{R}\mathbf{1}$. That is, multiplying the prices and the wages by a global constant does not change Eq. (16). For this particular wage–price system the pairs

$$(1, 3), \quad (2, 4), \quad (3, 3) \quad \text{and} \quad (3, 4)$$

are competitive. That is, the commodity 1 can only be produced sufficiently efficient in country 3, while commodity 2 is best produced in country 4. The third commodity can be produced efficiently in countries 3 and 4. For these wages and prices countries 1 and 2 cannot compete at all. The logarithmic wage vector $\bar{w} = (5, 5, 1, 2)^\top$ is sharing for the logarithmic price vector $\bar{p} = (1, 2, 4)^\top$: For each commodity there is at least one country that can produce sufficiently efficient to meet the prices on the world market. Notice that the competitive pairs form a subset of the tropical covector of the point $(0, 1, 3)^\top$ given in (12). In fact, they correspond to the covector of the point $\bar{w} \in \mathbb{R}^4/\mathbb{R}\mathbf{1}$ with respect to $V^\top$. Conceptually, this information allows to locate $\bar{w}$ in Fig. 5. Practically, however, it is a bit tedious to accomplish in a flat picture.

In the Ricardian economy there is a built-in symmetry between prices and wages, and this is what we want to elaborate now. We can rewrite the admissibility condition (13) as

$$w_k \geq r_{ik}^{-1} p_i \quad \text{for all } i \in [d] \text{ and } k \in [n] .$$

This works as we assumed that the production coefficients r_{ik} are strictly positive. We can define the matrix $R^\# = (r_{ik}^{-1})_{ki} \in \mathbb{R}^{n \times d}$ and its logarithm

$$\bar{R}^\# = (\log r_{ik}^{-1})_{ki} = (-\log r_{ik})_{ki} = -\bar{R}^\top .$$

Notice that, in contrast to the definition of $\bar{R}$, for the construction of $\bar{R}^\#$ we are taking *negative* logarithms of the coefficients, and this corresponds to changing to max as the tropical addition. Also observe that $\bar{R}^{\#\#} = \bar{R}$. In this way the admissibility condition, in its logarithmic form, is equivalent to

$$\bar{w} \geq \bar{R}^\# \odot_{\max} \bar{p} .$$

As before we impose an extra condition, namely equality in the above:

$$\bar{w} = \bar{R}^\# \odot_{\max} \bar{p} . \tag{18}$$

In that case the prices are called *covering* for the given wages. This is dual to the sharing condition for the wages. That is, in this case, each country can produce at least one commodity efficiently enough to be able to afford maximum wages. In this way a wage–price system (w, p) that is both sharing and covering yields the pair of equalities

$$\bar{w} = \bar{R}^\# \odot_{\max} (\bar{R} \odot_{\min} \bar{w}) \tag{19}$$

$$\bar{p} = \bar{R} \odot_{\min} (\bar{R}^\# \odot_{\max} \bar{p}) . \tag{20}$$

Let us define the *Shapley operator* of the Ricardian economy as the map $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ that sends a logarithmic price vector $\bar{p}$ to $\bar{R} \odot_{\min} (\bar{R}^\# \odot_{\max} \bar{p})$. Then (20) says that $\bar{p}$ is a fixed point of the Shapley operator T . The name “Shapley operator” is borrowed from the theory of stochastic games; see [20, §2.2]. Below we will characterize the fixed points of the Shapley operator.

For two vectors $x, y \in \mathbb{R}^d$ we can define

$$\delta(x, y) := \max_{i,j} |x_i + y_j - x_j - y_i| ,$$

and this yields a metric on the tropical projective torus $\mathbb{R}^d/\mathbb{R}\mathbf{1}$. This is sometimes called *Hilbert’s projective metric*. For an arbitrary tropical polytope $P \subset \mathbb{R}^d/\mathbb{R}\mathbf{1}$ and an arbitrary vector $v \in \mathbb{R}^d$ among all vectors $w \in \mathbb{R}^d$ with $w \geq v$ and $w + \mathbf{1} \in P$

there is a unique coordinatewise minimal vector w' ; see [14, Proposition 7]. The point $w' + \mathbb{R}\mathbf{1}$ is the *nearest point* to v in P . The value $\delta(v, w')$ minimizes the distance between v and all points in P . However, in general, that minimum may be attained for other points, too; see [14, Example 9].

Theorem 18 *The Shapley operator $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ maps a vector $\bar{p}$ to its nearest point in the tropical polytope $\text{tconv}_{\min}(\bar{R})$. In particular, the points in $\text{tconv}_{\min}(\bar{R})$ are precisely the fixed points of T .*

Proof The i th coefficient of the vector $T(\bar{p})$ is the real number

$$\begin{aligned} (\bar{r}_{i1}, \dots, \bar{r}_{in}) \odot_{\min} \left(\max_{j \in [d]} \{-\bar{r}_{j1} + \bar{p}_j\}, \dots, \max_{j \in [d]} \{-\bar{r}_{jn} + \bar{p}_j\} \right)^{\top} \\ = \min_{k \in [n]} \max_{j \in [d]} \{\bar{r}_{ik} - \bar{r}_{jk} + \bar{p}_j\} . \end{aligned}$$

This agrees with the formula in [14, Lemma 8], from which we infer that the Shapley operator sends $\bar{p}$ to the nearest point in the min-tropical convex hull $P := \text{tconv}(\bar{R})$ of the columns of the matrix $\bar{R}$. For each point $x \in P$ its nearest point in P is x itself. As an immediate consequence the Shapley operator T is idempotent, i.e., for all logarithmic price vectors $\bar{p}$ we have

$$T(\bar{p}) = T(T(\bar{p})) .$$

In the above we first analyzed the prices and then deduced the wages. However, this reasoning can be reversed. The *dual Shapley operator* is the map $T^{\#} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that maps a logarithmic wage vector $\bar{w}$ to $\bar{R}^{\#} \odot_{\max} (\bar{R} \odot_{\min} \bar{w})$. The wages in (19) can be analyzed directly by studying $T^{\#}$ instead of T , as in Theorem 18.

Example 19 We continue the Example 17. Again we look at the logarithmic wage-price system $(\bar{w}, \bar{p})$ from (17) for the (logarithmic) production coefficients given by the 3×4 -matrix V from Example 11. We saw that the wages are sharing, but the prices are not covering since the countries 1 and 2 cannot successfully compete on the world market. Applying the dual Shapley operator $T^{\#}$ to $\bar{w} = (5, 5, 1, 2)^{\top}$ gives the new logarithmic wage vector

$$\bar{w}' = (4, 3, 1, 2)^{\top} ,$$

which now yields

$$V \odot_{\min} \bar{w}' = (1, 2, 4)^{\top} = \bar{p}$$

from (16). That is, lowering the logarithmic wages from $\bar{w}$ to $\bar{w}'$ gives the logarithmic wage-price system $(\bar{w}', \bar{p})$ that is both sharing and covering. Notice that this only affects the wages in the countries 1 and 2 which could not compete

previously. The competitive pairs are given by the covector of the point $(1, 2, 4)^\top$, which agrees with $(0, 1, 3)^\top$ modulo $\mathbb{R}\mathbf{1}$, in (12).

Corollary 20 *The wage–price systems that are both sharing and covering bijectively correspond to pairs of points in $\text{tconv}_{\min}(\bar{R})$ and $\text{tconv}_{\min}(\bar{R}^\top) = -\text{tconv}_{\max}(\bar{R}^\#)$ that are linked via (16) and (18).*

In the language of [17] the tropical polytope $\text{tconv}_{\min}(\bar{R})$ is the “spanning core in the price-simplex”, whereas $\text{tconv}_{\max}(\bar{R}^\#)$ is the “spanning core in the wage-simplex”. In that paper the mixed subdivisions of dilated simplices occur as the “McKenzie–Minabe diagrams”; see [17, §9]. Points are interpreted as “production scale vectors”. These describe which percentage of the total work force of a country produces which commodities. This allows, e.g., to read off the total world production.

The Cayley Trick allows for four ways to describe and to visualize the same data. For our running example 3×4 -matrix we have the initial arrangement of four tropical hyperplanes in $\mathbb{R}^3/\mathbb{R}\mathbf{1}$ in Fig. 2 and its transpose of three tropical hyperplanes in $\mathbb{R}^4/\mathbb{R}\mathbf{1}$ in Fig. 5. The first arrangement is dual to the mixed subdivision of $4 \cdot \Delta_2$ in Fig. 3, while the second is dual to the mixed subdivision of $3 \cdot \Delta_3$ in Fig. 6.

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Ideals Associated to Poset Homomorphisms: A Survey

Martina Juhnke-Kubitzke and Sara Saeedi Madani

Abstract In this survey, we give an overview to the various known algebraic properties and invariants of ideals of poset homomorphisms. A particular attention lies on classical related notions that occur as special cases.

1 Introduction

In 1987, Hibi [17] introduced a special class of algebras—nowadays known as *Hibi rings*—that are associated to distributive lattices. An element α in a distributive lattice L that is not the minimum of L , is called *join-irreducible* if, whenever $\alpha = \beta \vee \gamma$ for elements $\beta, \gamma \in L$, it follows that $\alpha = \beta$ or $\alpha = \gamma$. By a classical theorem of Birkhoff [3], any finite distributive lattice L is isomorphic to the ideal lattice $\mathcal{I}(P)$ of its poset P of join-irreducible elements.

Given a field $\mathbb{K}$ and a distributive lattice L , we let S be the polynomial ring over $\mathbb{K}$ in the indeterminates x_α , where $\alpha \in L$. The *join-meet ideal* I_L of L is defined to be the toric ideal generated by the quadratic binomials $x_\alpha x_\beta - x_{\alpha \wedge \beta} x_{\alpha \vee \beta}$ with $\alpha, \beta \in L$ incomparable [6]. The Hibi ring $\mathbb{K}[L]$ of L is a $\mathbb{K}$ -algebra isomorphic to S/I_L . More precisely, a toric representation of $\mathbb{K}[L]$ is given by

$$\mathbb{K}\left[\prod_{p \in \alpha} x_p \prod_{p \notin \alpha} y_p : \alpha \in \mathcal{I}(P)\right] \subset R, \quad (1)$$

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where $R = \mathbb{K}[x_p, y_p : p \in P]$ and P is the poset of join-irreducible elements of L . Alternatively, the Hibi ring of L has a presentation of the form

$$\mathbb{K}\left[s \prod_{p \in \alpha} t_p : \alpha \in \mathcal{I}(P)\right] \subset T,$$

where $T = \mathbb{K}[s, t_p : p \in P]$.

Often, algebraic properties of the Hibi ring $\mathbb{K}[L]$ are studied either in terms of combinatorial properties of L or even of the underlying poset P (of join-irreducible elements of L). We now provide a (historical) overview of some of the most important results for Hibi rings. In [17], Hibi showed that $\mathbb{K}[L]$ is a normal Cohen-Macaulay domain with an ASL structure. Using a result by Stanley [22, p. 82], he constructed a $\mathbb{K}$ -basis of the canonical module of $\mathbb{K}[L]$ in terms of the strictly order reversing functions on $\hat{P}$, where $\hat{P}$ is the poset obtained from P by adding unique minimal and maximal elements. In addition, in the same paper, Hibi provided a characterization of the Gorenstein property of $\mathbb{K}[L]$. Namely, he showed that $\mathbb{K}[L]$ is a Gorenstein ring if and only if the poset P of join-irreducible elements of L is pure (i.e., all maximal chains of P are of the same length). Extending this result, in 2015, Ene et al. [8] classified all distributive lattices whose Hibi rings are Pseudo-Gorenstein. The condition that was provided there is also in terms of the chains of the poset P . In the same paper, the authors gave a necessary condition imposed on P if $\mathbb{K}[L]$ is level. For some special classes of lattices, the level property was characterized completely. Before, Miyazaki [20] found a sufficient condition for $\mathbb{K}[L]$ being level that is different from the condition of [8].

In [1], it was shown that the defining ideal I_L of $\mathbb{K}[L]$ has a quadratic Gröbner basis, which implies that $\mathbb{K}[L]$ is a Koszul algebra. In [9], the second author together with Ene and Herzog showed that the Castelnuovo-Mumford regularity of I_L is given by $|P| - \text{rank } P$. Here, if $I_L = (0)$ (which happens whenever P is a chain), the regularity is set to be one. Note that since $\mathbb{K}[L]$ is a Cohen-Macaulay algebra of Krull dimension $|P| + 1$, it follows from the Auslander-Buchsbaum formula that the projective dimension of I_L equals $|L| - |P| - 2$.

In 2005, Herzog and Hibi [11] associated to a poset P its so-called *Hibi ideal*, which is the monomial ideal I_P generated by the generators of the $\mathbb{K}$ -algebra, given in (1), namely

$$I_P = \left(\prod_{p \in \alpha} x_p \prod_{p \notin \alpha} y_p : \alpha \in \mathcal{I}(P) \right).$$

Using Hibi ideals, they characterized those bipartite graphs whose edge ideals are Cohen-Macaulay. Indeed, the set of those Cohen-Macaulay edge ideals coincides with the set of Alexander duals of Hibi ideals. In 2011, Ene, Herzog and Mohammadi [7] considered so-called *generalized Hibi ideals* and studied some of their algebraic properties. In 2014, Fløystad et al. [10] introduced a further generalization of such ideals corresponding to order-preserving maps between two

posets. Those ideals, which are the main subjects of this survey, are called *ideals of poset homomorphisms*. We provide their precise definition and an overview of their known algebraic properties in the next sections.

The article is organized as follows. In Sect. 2, we fix some notation and define ideals of poset homomorphisms. In Sect. 3, we focus on the description of the minimal prime ideals of this class of ideals and classify them with respect to the unmixed property. In Sect. 4, we summarize some of the important results on the minimal graded free resolutions of such ideals, including the ones on their Castelnuovo-Mumford regularity, their projective dimension, their (multigraded) Betti numbers and the linear strand of their minimal graded free resolution. Finally, in Sect. 5, we list the known characterizations of ideals of poset homomorphisms with respect to various algebraic properties such as being Buchsbaum, Cohen-Macaulay, level, Gorenstein, a complete intersection, as well as satisfying Serre's condition S_r and having a linear resolution.

2 Ideals of Poset Homomorphisms

First, we introduce the focus of interest of this survey article; namely, ideals of poset homomorphisms. Given two finite posets P and Q , a map $\phi : P \rightarrow Q$ is called *order-preserving* (or *isotone*) if for all $a, b \in P$ with $a \geq b$ it holds that $\phi(a) \geq \phi(b)$. Sometimes, those maps are also referred to as poset homomorphisms, and therefore, the set of all those maps from P to Q is denoted by $\text{Hom}(P, Q)$.

To each $\phi \in \text{Hom}(P, Q)$, one associates the monomial

$$u_\phi := \prod_{p \in P} x_{p, \phi(p)}$$

in the polynomial ring $S_{P, Q} := \mathbb{K}[x_{p, q}; p \in P, q \in Q]$, where $\mathbb{K}$ is an arbitrary field. The ideals of interest for this survey, which were introduced in [10], are the monomial ideals generated by u_ϕ for $\phi \in \text{Hom}(P, Q)$. They are denoted by $L(P, Q)$, i.e.,

$$L(P, Q) := (u_\phi; \phi \in \text{Hom}(P, Q)) \subseteq S_{P, Q}.$$

For simplicity, we set $\mathbb{K}[P, Q] := S_{P, Q}/L(P, Q)$. Moreover, we denote by $\Delta(P, Q)$ the simplicial complex on the vertex set $P \times Q$ with Stanley-Reisner ring $\mathbb{K}[P, Q]$.

If P or Q is an n -element chain, denoted by $[n]$, then $L(P, Q)$ is called *letterplace* or *co-letterplace* ideal, respectively. It is easily seen that a co-letterplace ideal $L(P, [n])$ coincides with the generalized Hibi ideal of P , as defined in [7]. In particular, a co-letterplace ideal $L(P, [n])$ with $n = 2$ is nothing else than the Hibi ideal I_P of P .

A further motivation to study these ideals comes from the fact that several well-known classes of ideals arise as quotients of letterplace ideals by a regular sequence.

This connection was discovered in [10], where it was shown that in particular the following classes of monomial ideals arise in this way:

- Initial ideals of determinantal ideals;
- Initial ideals of the ideals of 2-minors of a symmetric matrix;
- Ferrers ideals;
- Strongly stable ideals.

We want to remark, that it is natural to consider the $\mathbb{K}$ -algebra generated by the monomials u_ϕ as a generalization of the classical Hibi ring. The study of these so-called *isotonic algebras* has been initiated recently in [2]. In that article, Herzog et al. provided a formula for the Krull dimension of those algebras. Moreover, they conjectured that—analogueous to Hibi rings—isotonic algebras are normal Cohen-Macaulay domains, and, in certain cases, they could verify this conjecture. In [16], it was studied under which conditions the defining ideal of an isotonic algebra is generated by squarefree binomials. Since isotonic algebras are not in the focus of this survey, we refer to the mentioned articles [2, 16] for more details.

In the following sections, we focus on the ideals $L(P, Q)$ and their various studied algebraic properties and invariants.

3 Minimal Primes and Unmixedness of $L(P, Q)$

We start with the classification of all minimal primes of $L(P, Q)$. This yields a characterization of those posets P and Q for which $L(P, Q)$ is unmixed. In order to do so, we first need to introduce multivalued order-preserving maps.

For $\psi \subseteq Q \times P$ and $q \in Q$, we define

$$\psi(q) := \{p \in P : (q, p) \in \psi\} \text{ and } \mathfrak{p}_\psi := (x_{p,q} : (q, p) \in \psi).$$

We denote by $\text{MHom}(Q, P)$ the collection of sets $\psi \subseteq Q \times P$ satisfying the following conditions:

- (a) $\psi(q) \neq \emptyset$ for all $q \in Q$, and
- (b) for all $q, q' \in Q$ with $q < q'$, there exist $p \in \psi(q)$ and $p' \in \psi(q')$ such that $p < p'$.

The just introduced notation $\text{MHom}(Q, P)$ is motivated by the fact that elements of $\text{MHom}(Q, P)$ can be interpreted as multivalued order-preserving maps from Q to P . In particular, any order-preserving map from Q to P belongs to $\text{MHom}(Q, P)$. Multivalued order-preserving maps can be used to describe the set of minimal primes $\text{Min}(L(P, Q))$ of $L(P, Q)$.

Proposition 3.1 ([19, Proposition 3.5]) *Let P and Q be posets. Then the following hold:*

- (a) *Let $\mathfrak{p} \in \text{Min}(L(P, Q))$. Then there exists $\psi \in \text{MHom}(Q, P)$ with $\mathfrak{p} = \mathfrak{p}_\psi$.*
- (b) *Let $\psi \in \text{MHom}(Q, P)$. Then $L(P, Q) \subseteq \mathfrak{p}_\psi$ if and only if for each $\phi \in \text{Hom}(P, Q)$, there exists an element $p \in P$ with $p \in \psi(\phi(p))$.*

Under the additional assumption that the poset P has a unique minimal or maximal element, minimal primes of $L(P, Q)$ have been studied in [10] before and the above result is a generalization of Proposition 1.5 in that article. As a consequence of the previous result, we have the following.

Corollary 3.2 (See [19, Corollary 3.6]) *Let P and Q be posets, and let $\psi \in \text{Hom}(Q, P)$. Then $\mathfrak{p}_\psi \in \text{Min}(L(P, Q))$ if and only if for each $\phi \in \text{Hom}(P, Q)$, the map $\psi \circ \phi$ has a fixed point. Moreover,*

$$\text{height}(L(P, Q)) = |Q|,$$

and

$$\dim \mathbb{K}[P, Q] = |Q| \cdot (|P| - 1).$$

Proof Note that by Proposition 3.1 (b) it is enough to show that every $\mathfrak{p} \in \text{Min}(L(P, Q))$ has height at least $|Q|$ and that there exists a prime $\mathfrak{p}$ with $\text{height}(\mathfrak{p}) = |Q|$. Indeed, for any $\mathfrak{p} \in \text{Min}(L(P, Q))$, Proposition 3.1 (a) implies that there exists a multivalued map $\psi \in \text{MHom}(Q, P)$ such that $\mathfrak{p} = \mathfrak{p}_\psi$. As $\psi(q) \neq \emptyset$ for all $q \in Q$, the prime $\mathfrak{p}$ contains at least $|Q|$ variables. Hence, $\text{height}(\mathfrak{p}) \geq |Q|$.

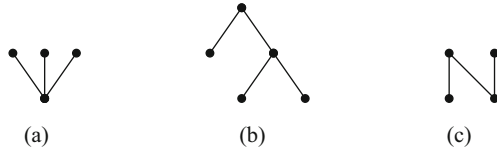
Next, we prove that there exists a prime $\mathfrak{p} \in \text{Min}(L(P, Q))$ with $\text{height}(\mathfrak{p}) = |Q|$. Since any constant map $\psi \in \text{Hom}(Q, P)$ satisfies the condition of Proposition 3.1 (b), we have that $L(P, Q) \subseteq \mathfrak{p}_\psi$. As $\mathfrak{p}_\psi$ is of height $|Q|$, it follows that it is indeed a minimal prime of $L(P, Q)$. $\square$

We want to remark that the formula for the height was also deduced in [10, Corollary 1.2 (a)] from [10, Proposition 1.5] in the special case that P has a unique minimal or maximal element.

We are now ready to give a criteria for the unmixedness of $L(P, Q)$ from [19]. Recall that a monomial ideal I in a polynomial ring S is *unmixed* if its associated primes are of the same height, (see, e.g., [12, A.5]). We first introduce some special classes of posets, which are crucial for the desired description.

Following [15] we call a poset P *rooted* (resp. *co-rooted*) if for any two incomparable elements $p_1, p_2 \in P$, there does not exist any element $p \in P$ such that $p > p_1, p_2$. (resp. $p < p_1, p_2$). Observe that a connected poset P is rooted (resp. co-rooted) if and only if its Hasse diagram (resp. the Hasse diagram of its dual) is a rooted tree as a directed graph (where edges are oriented from the smaller to the bigger element along a cover relation). In addition, a disconnected poset is rooted (resp. co-rooted) if and only if each of its connected components is rooted (resp. co-rooted). Examples of posets that are rooted, co-rooted and neither rooted nor co-rooted are shown in Fig. 1a,b,c, respectively.

Fig. 1 Hasse diagram of some posets. (a) A rooted poset; (b) a co-rooted poset; (c) a poset that is neither rooted nor co-rooted



Given posets P and Q we denote by $L(P, Q)^\tau$ the ideal obtained from $L(P, Q)$ by switching the indices of the variables. Moreover, we use $L(P, Q)^\vee$ to denote the Alexander dual of $L(P, Q)$. One interesting question is under which conditions one has the equality $L(P, Q)^\vee = L(Q, P)^\tau$. In [7], it was shown that this holds if one of the posets is a chain. Later, in [15], Herzog et al. provided a complete classification of pairs (P, Q) of posets satisfying $L(P, Q)^\vee = L(Q, P)^\tau$.

Proposition 3.3 ([15, Corollary 1.5]) *Let P and Q be posets. Then $L(P, Q)^\vee = L(Q, P)^\tau$ if and only if P or Q is connected and one of the following conditions holds:*

- (a) *Both, P and Q , are rooted.*
- (b) *Both, P and Q , are co-rooted.*
- (c) *P is connected and Q is a disjoint union of chains.*
- (d) *Q is connected and P is a disjoint union of chains.*
- (e) *P or Q is a chain.*

We now state the mentioned characterization from [19] of those pairs (P, Q) such that $L(P, Q)$ is unmixed. The proof of this characterization is strongly based on the previous proposition from [15] and on the proof of Theorem 1.4 in [15].

Proposition 3.4 ([19, Proposition 3.8]) *Let P and Q be posets. Then $L(P, Q)$ is unmixed if and only if one of the following conditions holds:*

- (a) *Both, P and Q , are rooted.*
- (b) *Both, P and Q , are co-rooted.*
- (c) *P or Q is a disjoint union of chains.*

Moreover, if P or Q is connected and $L(P, Q)$ is unmixed, then $L(P, Q)^\vee = L(Q, P)^\tau$.

4 The Minimal Graded Free Resolution of $L(P, Q)$

The focus of this section lies in the study of the minimal graded free resolution of $L(P, Q)$. In particular, we summarize results concerning the Castelnuovo-Mumford regularity, the projective dimension and the Betti numbers of those ideals from [4, 19] and [5].

We start with a discussion of known results for the regularity and projective dimension.

An *antichain* of a poset P is a subset $A \subseteq P$ such that all of its elements are pairwise incomparable. In the following, given a poset P , we denote by $\mathfrak{a}(P)$ the maximal cardinality of an antichain of P . In the special case that $L(P, Q)$ is unmixed,

the following result provides explicit formulas for the regularity and the projective dimension of $\mathbb{K}[P, Q]$.

Theorem 4.1 ([19, Theorem 4.3]) *Let P and Q be posets such that $L(P, Q)$ is unmixed. Let further s and r be the number of connected components of P and Q , respectively. Then*

- (a) $\text{reg } \mathbb{K}[P, Q] = \alpha(Q) \cdot (|P| - s) + s - 1$,
- (b) $\text{pdim } \mathbb{K}[P, Q] = \alpha(P) \cdot (|Q| - r) + s(r - 1) + 1$.

Sketch of the Proof The claim in (a) is shown using induction on $|P| + |Q|$. In the first step, one reduces the problem to the situation that the posets P and Q are connected (see [19, Corollary 3.1]). In the next step, several cases have to be distinguished. If one of the posets is a chain, then the claim follows from Corollaries 2.5 and 3.3 in [7]. By Proposition 3.4, the remaining cases to consider are those when P and Q are both rooted or both co-rooted. Since in both cases, the proofs use the same line of arguments, we only sketch the proof in the rooted case. Let v and w , respectively denote the roots of P and Q , respectively. First, one shows that $L(P, Q) = J_1 \cap J_2$, where

$$J_1 = \bigcap_{\substack{\psi \in \text{Hom}(Q, P) \\ \psi(w) = v}} \mathfrak{p}_\psi \quad \text{and} \quad J_2 = \bigcap_{\substack{\psi \in \text{Hom}(Q, P) \\ \psi(w) \neq v}} \mathfrak{p}_\psi.$$

One considers the following exact sequence of graded $S_{P, Q}$ -modules

$$0 \rightarrow \mathbb{K}[P, Q] \rightarrow S_{P, Q}/J_1 \oplus S_{P, Q}/J_2 \rightarrow S_{P, Q}/(J_1 + J_2) \rightarrow 0.$$

It then follows from [21, Corollary 18.7 (2)], that

$$\text{reg } \mathbb{K}[P, Q] \leq \max\{\text{reg}(S_{P, Q}/J_1), \text{reg}(S_{P, Q}/J_2), \text{reg}(S_{P, Q}/(J_1 + J_2)) + 1\}. \quad (2)$$

One can show that

- $J_1 = (x_{v, w}) + L(P, Q \setminus w)S_{P, Q}$,
- $J_2 = L(P \setminus v, Q)S_{P, Q}$, and
- $J_1 + J_2 = L(P \setminus v, Q)S_{P, Q} + (x_{v, w})$.

Finally, the claim follows by the induction hypothesis.

The formula in (b) follows from a theorem of Terai (see [12, Proposition 8.1.8]).

□

We would like to remark that for arbitrary posets the regularity of $\mathbb{K}[P, Q]$ is always bounded from below by the formula in Theorem 4.1 (a). An almost immediate upper bound for the regularity is given by $|P| \cdot (|Q| - 1)$ (see [19, Proposition 4.1] (a)). With respect to the projective dimension, it is easy to see that

$$|Q| \leq \text{pdim } \mathbb{K}[P, Q] \leq |P| \cdot (|Q| - 1) + 1,$$

see also [19, Proposition 4.1] (b). To complete the discussion of these two invariants, we would like to mention the following conjecture that was proposed in [19].

Conjecture 4.2 ([19, Conjecture 6.1]) *Let P and Q be posets with s and r connected components, respectively. Then*

- (a) $\text{reg}(\mathbb{K}[P, Q]) = \mathfrak{a}(Q) \cdot (|P| - s) + s - 1$.
- (b) $\text{pdim}(\mathbb{K}[P, Q]) \leq \mathfrak{a}(P) \cdot (|Q| - r) + s(t - 1) + 1$.

We want to emphasize that, as shown by Example 4.4 in [19], the inequality in part (b) of the above conjecture can be strict.

Now, we restrict our attention to letterplace ideals that is we are in the situation that the poset P is a chain of length n . We list known results for the (multigraded) Betti numbers of $L([n], Q)$ and in particular for its linear strand. The results are based on [5] and [19].

We first need to introduce some notation. Any subset $R \subseteq [n] \times Q$ can be written as a disjoint union $R = \bigcup_{i=1}^n \{i\} \times R_i$, where $R_i = \{(i, q) : (i, q) \in R\}$. Note that R_i inherits a poset structure from Q . Let $\min(R_i)$ and $\max(R_i)$ denote the set of minimal and maximal elements of R_i , respectively. Given this decomposition of R , for each $1 \leq i \leq n-1$, one associates a bipartite graph $G_i(R)$ to R on vertex set $(\{i\} \times R_i) \cup (\{i+1\} \times R_{i+1})$ in the following way: $\{(i, q_1), (i+1, q_2)\}$ is an edge if and only if $q_1 \leq q_2$. We denote by $\Delta_i(R)$ the simplicial complex whose Stanley-Reisner ideal is the edge ideal of the bipartite graph $G_i(R)$. Moreover, we set $X_i(R) = (\Delta_i(R))_{\max(R_i) \cup \min(R_{i+1})}$, i.e., $X_i(R)$ is the induced subcomplex of $\Delta_i(R)$ on vertex set $\max(R_i) \cup \min(R_{i+1})$.

For $i \in \mathbb{N}$ and $R \subseteq [n] \times Q$, we denote by $\beta_{i,R} = \beta_{i,R}(L([n], Q))$ the multigraded Betti number of $L([n], Q)$ in homological degree i and internal degree R . Here, the multidegree R naturally refers to the characteristic vector of R (inside $\mathbb{N}^{[n] \times Q}$). We let

$$\beta(R, t) = t^r \beta_{0,R} + t^{r-1} \beta_{1,R} + \cdots + t \beta_{r-1,R}$$

denote the *Betti polynomial* of $L([n], Q)$ (corresponding to multidegree R), and we set

$$\widetilde{H}(X_i(R), t) = \sum_{\ell \geq -1} \dim_{\mathbb{K}} \widetilde{H}^{\ell}(X_i(R); \mathbb{K}) t^{\ell},$$

where $\widetilde{H}^{\ell}(X_i(R); \mathbb{K})$ denotes the ℓ th reduced cohomology group of $X_i(R)$.

The following theorem gives a possibility to explicitly compute the multigraded Betti numbers of a letterplace ideal.

Theorem 4.3 ([5, Proposition 3.10]) *Let Q be a poset and let $R \subseteq [n] \times Q$. Then*

$$\beta(R, t) = t^n \prod_{i=1}^{n-1} \widetilde{H}(X_i(R), t).$$

Since letterplace and co-letterplace ideals are dual to each other (see Proposition 3.3), it is natural to ask, whether one can also compute the multigraded Betti numbers of the latter type of ideals. As the minimal graded free resolution of a co-letterplace ideal $L(P, [n])$ is known to be always linear ([7], see also Sect. 5), one could expect that the situation is easier than for letterplace ideals. Indeed, in [7] a description of the minimal free resolution is provided using decomposition functions. Another, more explicit description, was found very recently in [4]. Since the construction in [4] requires a lot of definitions and additional terminology, we omit it here and just refer to this article.

As the last topic of this section, we consider the length of the linear strand of the minimal graded free resolution of $\mathbb{K}[P, Q]$. In the following, let

$$\text{lin}(\mathbb{K}[P, Q]) := \max\{i : \beta_{i, i+|P|-1}(\mathbb{K}[P, Q]) \neq 0\}$$

denote the length of the linear strand of the minimal graded free resolution of $\mathbb{K}[P, Q]$. Further, let $\bar{\alpha}(P)$ and $\underline{\alpha}(P)$ be the number of maximal and minimal elements of P , respectively.

The following result provides lower bounds on $\text{lin}(\mathbb{K}[P, Q])$ without imposing any restrictions on the posets P and Q .

Proposition 4.4 ([19, Proposition 4.5]) *Let P and Q be posets, and let $Q' \subseteq Q$ be a subposet. Then:*

- (a) *If Q' has a unique minimal element, then $\text{lin}(\mathbb{K}[P, Q]) \geq \bar{\alpha}(P) \cdot (|Q'| - 1) + 1$.*
- (b) *If Q' has a unique maximal element, then $\text{lin}(\mathbb{K}[P, Q]) \geq \underline{\alpha}(P) \cdot (|Q'| - 1) + 1$.*
- (c) *If Q' is bounded (i.e., there exists a unique maximal and a unique minimal element), then $\text{lin}(\mathbb{K}[P, Q]) \geq \alpha(P) \cdot (|Q'| - 1) + 1$.*

For special classes of posets, one can show that the length of the linear strand is as long as it can possibly be:

Corollary 4.5 *Let P and Q be posets. Then:*

- (a) [19, Corollary 4.6 (a)] *If Q is connected, and P and Q are both rooted or both co-rooted, then*

$$\text{pdim } \mathbb{K}[P, Q] = \text{lin}(\mathbb{K}[P, Q]) = \alpha(P) \cdot (|Q| - 1) + 1.$$

- (b) [5, Proposition 3.17, Remark 3.19] *If P is a chain and Q has an element that is comparable to any other element of Q , then*

$$\text{pdim } \mathbb{K}[P, Q] = \text{lin}(\mathbb{K}[P, Q]) = |Q|.$$

We want to remark that a special case of part (b)—assuming that Q has a unique maximal or minimal element—was also shown in [19, Corollary 4.6 (b)].

5 Algebraic Properties of $L(P, Q)$

In Sect. 3 we already considered the characterization of the unmixed property for the ideals $L(P, Q)$. The aim of this section is to continue the study of algebraic properties of those ideals. This includes e.g., characterizations of Cohen-Macaulayness, the level property and Gorensteinness.

We start with a classification of those pairs of posets P and Q such that $L(P, Q)$ has a linear resolution. For this, recall that a monomial ideal I in a polynomial ring $S = \mathbb{K}[x_1, \dots, x_n]$, which is generated in a single degree, is called *weakly polymatroidal* if it satisfies the following condition: If $u = x_1^{a_1} \cdots x_n^{a_n}$ and $v = x_1^{b_1} \cdots x_n^{b_n}$ are monomial minimal generators of I such that there exists an integer t with $a_1 = b_1, \dots, a_{t-1} = b_{t-1}$ and $a_t > b_t$, then one has $x_t(v/x_t) \in I$ for some $l > t$.

Theorem 5.1 ([19, Theorem 5.1]) *Let P and Q be posets. Then the following conditions are equivalent:*

- (a) $L(P, Q)^k$ is weakly polymatroidal, for each positive integer k .
- (b) $L(P, Q)$ has linear quotients.
- (c) $L(P, Q)$ has a linear resolution.
- (d) $\beta_{1, |P|+i}(L(P, Q)) = 0$ for $i \neq 1$.
- (e) P is an antichain or Q is a chain.

We want to remark that it was shown in [7] before that co-letterplace ideals satisfy all properties stated in the previous theorem.

Using the previous theorem, in [19] a characterization of the Cohen-Macaulay property for $L(P, Q)$ was provided. In order to formulate this characterization, we recall the definitions of Serre's condition and the sequentially Cohen-Macaulay property. Given a Noetherian graded $\mathbb{K}$ -algebra R and a finitely generated graded R -module M , we say that M satisfies *Serre's condition* (S_r) if $\text{depth}(M_{\mathfrak{p}}) \geq \min\{r, \text{height}(\mathfrak{p})\}$ for all $\mathfrak{p} \in \text{Spec}(R)$. Moreover, a graded R -module M is called *sequentially Cohen-Macaulay* if there exists a finite filtration of graded R -modules $0 = M_0 \subset M_1 \subset \cdots \subset M_r = M$ such that M_i/M_{i-1} is Cohen-Macaulay for each $i = 1, \dots, r$, and the Krull dimensions of these quotients M_i/M_{i-1} are increasing.

Theorem 5.2 ([19, Theorem 5.2]) *Let P and Q be posets such that $|Q| \geq 2$. Then the following conditions are equivalent:*

- (a) $\Delta(P, Q)$ is (pure) shellable.
- (b) $\mathbb{K}[P, Q]$ is Cohen-Macaulay.
- (c) $\mathbb{K}[P, Q]$ satisfies Serre's condition (S_r) for some $r \geq 2$.
- (d) $\mathbb{K}[P, Q]$ is sequentially Cohen-Macaulay.
- (e) P is connected, and, in addition, P is a chain or Q is an antichain.

In the special case that P is a chain, Cohen-Macaulayness of $L(P, Q)$ was proven in [7] (see also [10]).

As an almost immediate consequence of the previous result, one obtains a characterization of Gorensteinness of $\mathbb{K}[P, Q]$ as follows.

Corollary 5.3 ([19, Corollary 5.3]) *Let P and Q be posets such that $|Q| \geq 2$. Then the following conditions are equivalent:*

- (a) $\mathbb{K}[P, Q]$ is a complete intersection.
- (b) $\mathbb{K}[P, Q]$ is Gorenstein.
- (c) P is connected and Q is an antichain.

The characterization of the Cohen-Macaulay property of $\mathbb{K}[P, Q]$ also allows to derive a classification of those pairs P and Q such that $\mathbb{K}[P, Q]$ is Buchsbaum (though this is not as immediate as the previous corollary). A Stanley-Reisner ring $\mathbb{K}[\Delta]$ of a simplicial complex Δ is called *Buchsbaum* if it is locally Cohen-Macaulay and unmixed. Here, *locally Cohen-Macaulay* means that every localization of $\mathbb{K}[\Delta]$ is Cohen-Macaulay, except possibly the localization at the irrelevant ideal.

Corollary 5.4 ([19, Corollary 5.5]) *Let P and Q be posets such that $|Q| \geq 2$. Then the following conditions are equivalent:*

- (a) $\mathbb{K}[P, Q]$ is Buchsbaum, but not Cohen-Macaulay.
- (b) $\mathbb{K}[P, Q]$ is locally Cohen-Macaulay, but not Cohen-Macaulay.
- (c) P is an antichain with 2 elements.

In [5, Proposition 3.17], a criterion for the non-vanishing of the multigraded Betti numbers of $L([n], Q)$ in the last step of its minimal graded free resolution is provided. In particular, it is shown that any non-vanishing multigraded Betti number in the highest homological degree equals 1. As a consequence of this result, a characterization of those letterplace ideals, which are level, is obtained. Recall that for a homogeneous ideal I in a polynomial ring S , S/I is *level* if and only if S/I is Cohen-Macaulay and the last step of its minimal graded free resolution is pure, namely $\beta_p(S/I) = \beta_{p,j}(S/I)$, for some j , where $p = \text{pdim } S/I$, cf. [23, Chapter III.3].

Proposition 5.5 ([5, Corollary 3.18]) *Let P and Q be posets. Then $\mathbb{K}[P, Q]$ is level if and only if all maximal antichains in Q have the same cardinality. If this cardinality is c , then the unique nonzero graded Betti number in maximal homological degree is $\beta_{|Q|-1, |Q|+(|P|-1)c}(L(P, Q))$.*

The previous proposition indeed solves a problem suggested in [19, Problem 6.2].

To complete the discussion of algebraic properties of $L(P, Q)$, we want to remark that it was shown in [19, Proposition 5.6] that $\mathbb{K}[P, Q]$ is *Golod* if and only if P is disconnected or Q is a chain. The only-if part of this result follows from [18, Lemma 7.4]. For the if-part, it suffices to notice that if Q is a chain, then $\mathbb{K}[P, Q]$ has a linear resolution and hence is Golod (cf., [14]). If P is disconnected, one uses that $L(P, Q)$ decomposes as the product of ideals on different sets of variables. In this situation, Golodness of $\mathbb{K}[P, Q]$ follows from a result in [13].

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How to Flatten a Soccer Ball

Kaie Kubjas, Pablo A. Parrilo, and Bernd Sturmfels

Abstract This is an experimental case study in real algebraic geometry, aimed at computing the image of a semialgebraic subset of 3-space under a polynomial map into the plane. For general instances, the boundary of the image is given by two highly singular curves. We determine these curves and show how they demarcate the “flattened soccer ball”. We explore cylindrical algebraic decompositions, by working through concrete examples. Maps onto convex polygons and connections to convex optimization are also discussed.

1 Introduction

Computational tools for real algebraic geometry have numerous applications. This article offers a case study, focused on the following very simple scenario. We consider a compact semialgebraic subset of real 3-space that is defined by one polynomial h in three variables:

$$\mathcal{B} = \{ (u, v, w) \in \mathbb{R}^3 : h(u, v, w) \geq 0 \}. \quad (1)$$

We think of $\mathcal{B}$ as our “soccer ball”. See Fig. 1 for an illustration. A *flattening* of $\mathcal{B}$ is its image under a polynomial map

$$\phi : \mathbb{R}^3 \rightarrow \mathbb{R}^2, (u, v, w) \mapsto (f(u, v, w), g(u, v, w)). \quad (2)$$

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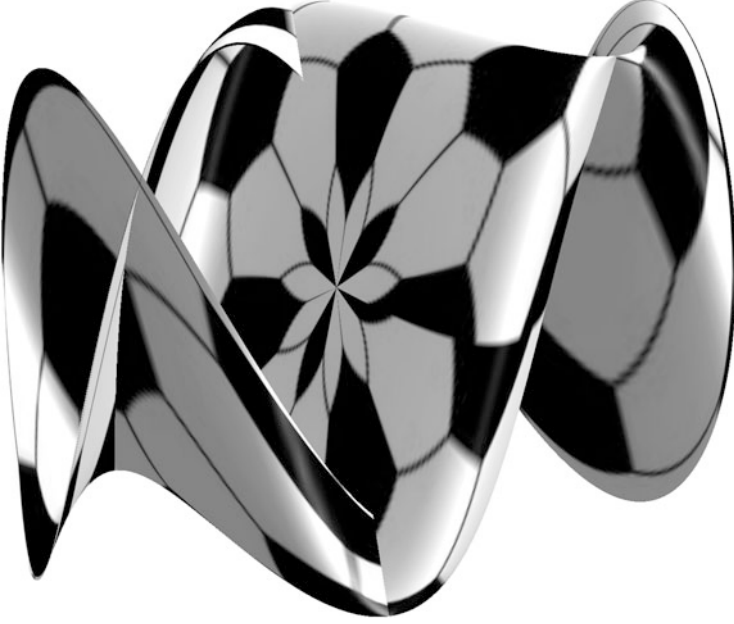


Fig. 1 A soccer ball is flattened and folded into a square

Using quantifiers, the “flattened soccer ball” can be expressed as

$$\phi(\mathcal{B}) = \{(x, y) \in \mathbb{R}^2 : \exists u, v, w : x = f(u, v, w) \text{ and } y = g(u, v, w) \\ \text{and } h(u, v, w) \geq 0\}.$$

By Tarski’s theorem on quantifier elimination, the image is a semialgebraic set in the plane $\mathbb{R}^2$, so it can be described as a Boolean combination of polynomial inequalities. Cylindrical algebraic decomposition [8] can be used to compute a quantifier-free representation. This is an active research area and several implementations are available [5, 7, 11, 14]. Our aim is to explore the main ingredients in such a representation of $\phi(\mathcal{B})$. A related problem is the computation of the convex hull $\text{conv}(\phi(\mathcal{B}))$, whose boundary points represent optimal points for the optimization problem of maximizing $\lambda f + \mu g$ over $\mathcal{B}$, where λ, μ are parameters.

This project started in November 2014 at the Simons Institute for the Theory of Computing in Berkeley, during the workshop *Symbolic and Numerical Methods for Tensors and Representation Theory*. The following example was part of its “Algebraic Fitness Session”.

Example 1 Consider the map given by the two elementary symmetric polynomials,

$$\phi : \mathbb{R}^3 \rightarrow \mathbb{R}^2, (u, v, w) \mapsto (uv + vw + uw, uvw).$$

We seek to compute the image under ϕ of the unit ball

$$\mathcal{B} = \{ (u, v, w) \in \mathbb{R}^3 : u^2 + v^2 + w^2 \leq 1 \}. \quad (3)$$

The flattened soccer ball $\phi(\mathcal{B})$ is the compact region in $\mathbb{R}^2$ that is depicted in Fig. 2. In particular, $\phi(\mathcal{B})$ is not convex. If the second coordinate uvw were replaced by a homogeneous quadric then $\phi(\mathcal{B})$ would be convex, by a theorem of Brickman [6].

We can quickly get an impression of the flattened ball $\phi(\mathcal{B})$ by sampling points from the ball $\mathcal{B}$ and plotting their images in $\mathbb{R}^2$. These are the red points in Fig. 2. We next sample points from the sphere $\partial\mathcal{B}$ and we plot these in blue. Figure 2 shows the existence of two small regions with many red points but no blue points at all. This means that the image of the sphere is strictly contained in the image of the ball. In symbols, $\phi(\partial\mathcal{B}) \subset \phi(\mathcal{B})$. The Zariski closure of the boundary of the image $\phi(\mathcal{B})$ is given by the polynomials

$$p = x^3 - 27y^2 \text{ and } q = (2x+1)(4x^6 - 4x^5 - 92x^3y^2 + x^4 + 6x^2y^2 + 729y^4 + 48xy^2 - 16y^2)$$

while q vanishes on the blue boundary. ◇

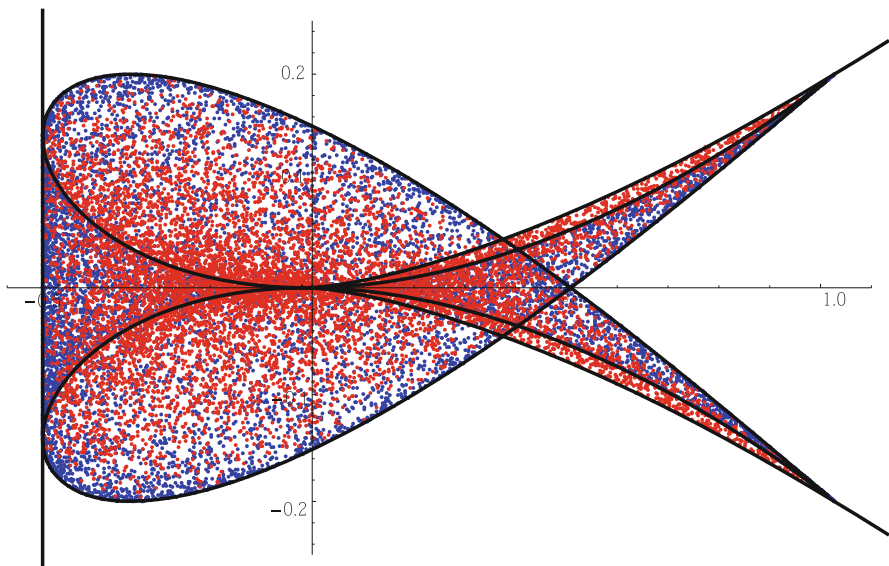


Fig. 2 Flattening of the unit ball $\mathcal{B}$ under the map $(u, v, w) \mapsto (uv + vw + uw, uvw)$. Red points are randomly sampled from the interior of $\mathcal{B}$, and blue points are sampled from the boundary of $\mathcal{B}$ (Color figure online)

For a triple (f, g, h) of polynomials in $\mathbb{R}[u, v, w]$, representing the pair $(\mathcal{B}, \phi)$, we define the *algebraic boundary* of $\phi(\mathcal{B})$ to be the Zariski closure in $\mathbb{C}^2$ of the topological boundary of $\phi(\mathcal{B})$. In addition to $\mathcal{B}$ being compact, we also assume that $\mathcal{B}$ is regular, i.e. the closure of the interior of $\mathcal{B}$ contains $\mathcal{B}$. This excludes examples where lower-dimensional pieces stick out, like the Whitney umbrella. With these hypotheses, we can apply results in real algebraic geometry, found in [13, Lemma 3.1] and [19, Lemma 4.2], to conclude that the algebraic boundary is pure of dimension 1 in $\mathbb{C}^2$. It is defined by the product of two squarefree polynomials p and q in $\mathbb{R}[x, y]$. The curve $V(p)$ is the branch locus of the map ϕ itself. It depends only on f and g but not on h . The curve $V(q)$ is the branch locus of the restriction of ϕ to the surface $V(h)$. It depends on h . Note that q is reducible in Example 1.

This paper is organized as follows. In Sect. 2 we study the algebraic geometry underlying our problem. If the data f, g, h are generic polynomials then the curves $V(p)$ and $V(q)$ are irreducible. We determine their Newton polygons and singularities. In Sect. 3 we explore the global topology of the flattened soccer ball $\phi(\mathcal{B})$. We present upper and lower bounds on the number of connected components in its complement. Section 4 introduces tools from symbolic computation for deriving an exact representation of $\phi(\mathcal{B})$. Section 5 offers connections to convexity and to sum-of-squares techniques in polynomial optimization.

2 Algebraic Curves

A standard approach in algebraic geometry is to focus on the generic instance in a family of problems. This then leads to an upper bound for the algebraic complexity of the output that is valid for all special instances. In what follows we pursue that standard approach.

Suppose that f, g and h are generic inhomogeneous polynomials of degrees d_1, d_2 and e in $\mathbb{R}[u, v, w]$. The soccer ball $\mathcal{B}$ and the map ϕ are defined as in (1) and (2). Let p denote the squarefree polynomial that defines the branch locus of ϕ , and let q be the squarefree polynomial that defines the branch locus of $\phi|_{\{h=0\}}$. These polynomials are unique up to scaling. They represent the algebraic boundary of $\phi(\mathcal{B})$. Both curves are in fact irreducible:

Theorem 1 *For generic polynomials f, g, h in $\mathbb{R}[u, v, w]$, the boundary polynomials p and q of the flattened soccer ball $\phi(\mathcal{B})$ are irreducible. Their Newton polygons are the triangles*

$$\begin{aligned} \text{Newt}(p) &= D_p \cdot \text{conv}\{(0, 0), (0, d_1), (d_2, 0)\} \text{ where } D_p = d_1^2 + d_1 d_2 + d_2^2 - 3d_1 - 3d_2 + 3; \\ \text{Newt}(q) &= D_q \cdot \text{conv}\{(0, 0), (0, d_1), (d_2, 0)\} \text{ where } D_q = e(d_1 + d_2 + e - 3). \end{aligned}$$

The irreducible complex curves $V(p)$ and $V(q)$ are highly singular, with genera

$$\begin{aligned}\text{genus}(p) &= \frac{1}{2}(2d_1^3 + 3d_1^2d_2 + 3d_1d_2^2 + 2d_2^3 - 13d_1^2 - 16d_1d_2 - 13d_2^2 + 27d_1 + 27d_2 - 20), \\ \text{genus}(q) &= \frac{1}{2}(d_1^2e + 2d_1d_2e + 3d_1e^2 + d_2^2e + 3d_2e^2 + 2e^3 - 10d_1e - 10d_2e - 13e^2 + 21e + 2).\end{aligned}$$

The numbers of singular points of these curves in the complex affine plane $\mathbb{C}^2$ are

$$\begin{aligned}\#\text{Sing}(V(p)) &= \frac{1}{2}((D_p \cdot d_1 - 1)(D_p \cdot d_2 - 1) - D_p \cdot \gcd(d_1, d_2) + 1) - \text{genus}(p), \\ \#\text{Sing}(V(q)) &= \frac{1}{2}((D_q \cdot d_1 - 1)(D_q \cdot d_2 - 1) - D_q \cdot \gcd(d_1, d_2) + 1) - \text{genus}(q).\end{aligned}$$

In this statement, $\text{genus}(p)$ denotes the genus of the Riemann surface that is obtained by resolving the singularities of the curve $V(p)$. Equivalently, this is the *geometric genus*. The proof of Theorem 1 realizes the plane curves $V(p)$ and $V(q)$ as generic projections of smooth curves in 3-space. This implies that all their singular points are nodes (cf. [12]), and these are counted by the difference between the arithmetic genus and the geometric genus.

Table 1 underscores how singular our curves are. For instance, the last row concerns a general map ϕ of degree 4. The branch locus $V(q)$ of that map restricted to the boundary surface $V(h)$ has degree 56. A general plane curve of that same degree has genus 1485. However, the genus of our curve $V(q)$ is only 36, so it has $1485 - 36 = 1449$ singular points.

From the polygon $\text{Newt}(p)$ in Theorem 1 we see that the curve $V(p)$ has degree $D_p \cdot \max(d_1, d_2)$, and similarly for $V(q)$. When the input polynomials f, g, h of degrees d_1, d_2, e are not generic but special, these numbers serve as an upper bound. We take the sum of these numbers to get

Corollary 1 *For any f, g, h , the algebraic boundary of $\phi(\mathcal{B})$ has degree at most*

$$(d_1^2 + d_1d_2 + d_2^2 - 3d_1 - 3d_2 + 3 + e(d_1 + d_2 + e - 3)) \cdot \max(d_1, d_2).$$

This bound is tight when the polynomials f, g, h are generic relative to their degrees.

Remark 1 If $d_1 \leq d_2 = 2$ and e is arbitrary then the branch curve $V(p)$ of the map ϕ has genus 0. This means the curve admits a parametrization by rational functions.

Table 1 The numerical values in Theorem 1 for input polynomials of low degree

(d_1, d_2, e)	degree(p)	genus(p)	# Sing($V(p)$)	degree(q)	genus(q)	# Sing($V(q)$)
(1, 2, 2)	2	0	0	8	1	8
(1, 3, 2)	12	1	14	18	4	36
(2, 2, 2)	6	0	10	12	4	51
(2, 3, 2)	21	5	122	24	9	160
(2, 4, 2)	52	21	604	40	16	345
(3, 3, 2)	36	17	578	30	16	390
(3, 4, 2)	76	43	2048	48	25	792
(4, 4, 2)	108	82	5589	56	36	1449

The two cases given in the third and fourth row of Table 1 will be of most interest to us. For each of them, we may assume that $\mathcal{B}$ is the unit ball (3), but ϕ is arbitrary.

Example 2 If we flatten the unit ball (3) via a quadratic map ($d_1 = d_2 = 2$) then the branch locus of ϕ is the rational sextic curve $V(p)$, with 10 singular points. The branch curve of the restriction of ϕ to $V(h)$ is the curve $V(q)$ of degree 12 and genus 4, so it has 51 singular points. These two curves make up the boundary of $\phi(\mathcal{B})$.

If both f and g are homogeneous quadrics then the image of $\mathcal{B}$ under ϕ is convex. This follows from [6, Theorem 2.1]. More precisely, $\phi(\mathcal{B})$ is a spectrahedral shadow, bounded by a curve of degree six. This scenario corresponds to the case $p = n = 3$ in Table 1 of [20]. The image $\phi(\mathcal{B})$ is generally not convex when one of the quadrics f, g is not homogeneous. For instance, the image of the unit ball under the map $(u, v) \mapsto (u^2 - v, v^2)$ is not convex. $\diamond$

Example 3 Let $d_1 = 2, d_2 = 3$ as in Example 1, but with f and g generic. The picture of $\phi(\mathcal{B})$ is now much more complicated than that in Fig. 2. The red boundary $V(p)$ is a curve of degree 21 with 122 complex singular points, and the blue boundary $V(q)$ is a curve of degree 24 with 160 complex singular points. This is worked out in Example 8. $\diamond$

Proof (of Theorem 1) We consider two curves in affine 3-space $\mathbb{C}^3$. The curve C_1 is defined by the 2×2 -minors of the Jacobian matrix of (f, g) with respect to (u, v, w) . This 2×3 -matrix has general entries of degree $d_1 - 1$ in the first row and general entries of degree $d_2 - 1$ in the second row. By the Thom-Porteous-Giambelli Formula, we have $\deg(C_1) = (d_1 - 1)^2 + (d_1 - 1)(d_2 - 1) + (d_2 - 1)^2$. This expression equals D_p . The curve C_2 is the complete intersection defined by the polynomial h , which has degree e , and the Jacobian determinant of (f, g, h) with respect to (u, v, w) , which has degree $d_1 + d_2 + e - 3$. By Bézout's Theorem, $\deg(C_2) = e(d_1 + d_2 + e - 3)$. The hypothesis that f, g and h are generic ensures that C_1 and C_2 are smooth and irreducible. Their degrees are the quantities D_p and D_q in the statement.

Both of the results from algebraic geometry that were used in the previous paragraph (Thom-Porteous-Giambelli and Bézout) require certain genericity hypotheses on the geometric data to which they apply. These hypotheses are satisfied in our case because the given polynomials f, g and h are assumed to have generic coefficients. See e.g. [17, Section 3.5.4].

The curves defined by p and q are the images of C_1 and C_2 under the map $\phi = (f, g)$ from $\mathbb{C}^3$ to $\mathbb{C}^2$. Our first claim states that, for $i = 1, 2$, the Newton polygon of the plane curve is the triangle $r \cdot \text{conv}\{(0, 0), (0, d_i), (d_i, 0)\}$, where $r = \deg(C_i)$.

We prove this using *tropical geometry* [15]. By genericity of f, g and h , the tropical curve $\text{trop}(C_i)$ in $\mathbb{R}^3$ is the 1-dimensional fan with rays spanned by $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ and $(-1, -1, -1)$, where each ray has multiplicity r . Our goal is to compute the tropical curve $\text{trop}(\phi(C_i))$ in $\mathbb{R}^2$. This contains the image of $\text{trop}(C_i)$ under the tropicalization of the map ϕ . This is the piecewise-linear map that takes (U, V, W) in $\mathbb{R}^3$ to $(\min\{d_1 U, d_1 V, d_1 W, 0\}, \min\{d_2 U, d_2 V, d_2 W, 0\})$. Its image is the weighted ray in $\mathbb{R}^2$ spanned by $(-d_1 r, -d_2 r)$. The other rays of the tropical curve

$\text{trop}(\phi(C_i))$ arise from the points of C_i at which f and g vanish. We derive these using the method of Geometric Tropicalization, specifically [15, Theorem 6.5.11]. The relevant very affine curve is $C_i \setminus \{uvwfg = 0\}$, and the normal crossing boundary in the SNC pair is the divisor defined by $uvwfg$ on C_i .

The surface $\{f = 0\}$ meets the curve C_i in d_1r points, and the divisorial valuations at these points map to the weighted ray $(d_1r, 0)$ in $\mathbb{R}^2$. Likewise, the surface $\{g = 0\}$ meets C_i in d_2r points, and their divisorial valuations create the weighted ray $(0, d_2r)$ in $\mathbb{R}^2$. Hence the tropical plane curve $\text{trop}(\phi(C_i))$ consists of the three weighted rays specified by $(-d_1r, -d_2r)$, $(d_1r, 0)$ and $(0, d_2r)$. This implies our assertion about the Newton polygons of p and q .

To prove the second assertion, about the genera of the two curves in question, we use the following two facts about general curves in $\mathbb{P}^3$. These are easily derived by computing the Hilbert series and then reading off the Hilbert polynomial. Recall that, for a curve with the Hilbert polynomial $h(n) = h_1n + h_0$, the degree is h_1 and the arithmetic genus is $1 - h_0$. Moreover, if the curve is smooth, then its geometric genus equals the arithmetic genus.

- A smooth space curve defined by the 2×2 -minors of a 2×3 -matrix with rows of degrees a and b has degree $a^2 + ab + b^2$ and genus $a^3 + \frac{3}{2}a^2b + \frac{3}{2}ab^2 + b^3 - 2a^2 - 2ab - 2b^2 + 1$.
- The complete intersection of two general surfaces of degrees a and b in $\mathbb{P}^3$ is a smooth curve of degree ab and genus $\frac{1}{2}ab(a + b - 4) + 1$.

The genus of the plane curve $V(p)$ is equal to the genus of the space curve C_1 that maps to it, and similarly for $V(q)$ and C_2 . So, it suffices to compute the genera of the affine curves C_1 and C_2 in $\mathbb{C}^3$. We may work with their projective closures $\overline{C}_1$ and $\overline{C}_2$ in $\mathbb{P}^3$. The curve $\overline{C}_1$ has the determinantal representation as in the first bullet, with $a = d_1 - 1$ and $b = d_2 - 1$. Substitution yields the desired formula for $\text{genus}(p)$. The curve $\overline{C}_2$ is the complete intersection of two surfaces in $\mathbb{P}^3$, of degree $a = e$ and $b = d_1 + d_2 + e - 3$. Substituting these expressions into $\frac{1}{2}ab(a + b - 4) + 1$, we obtain the desired formula for $\text{genus}(q)$.

We can regard $V(p)$ and $V(q)$ as curves in the weighted projective plane given by the known Newton polygons. The genus of a general curve of the same degree is the number of interior lattice points on the Newton triangle. That number is equal to

$$\#(\mathbb{Z}^2 \cap \text{int}(\text{conv}\{(0, 0), (0, rd_1), (rd_2, 0)\})) = \frac{1}{2}((rd_1 - 1)(rd_2 - 1) - \gcd(rd_1, rd_2) + 1).$$

Here r is D_p or D_q as before. The number of singular points is the number above minus the genus of the curve. This gives the count in the last assertion of Theorem 1.

We used the computer algebra system Macaulay2 [10] to verify some of the entries in Table 1. Here is the Macaulay2 code we used for a typical computation with $d_1 = d_2 = e = 2$:

```
S = QQ[x, y, u, v, w]; h = u^2+v^2+w^2-1;
f = u*v-u*w+7*v^2+v*w+5*w^2+u+v+w+1;
g = u^2-u*v+u*w-v^2+v*w-w^2+u-v+w-1;
C1 = minors(2, jacobian(ideal(f, g)));
```

```

C2 = minors(3,jacobian(ideal(f,g,h)))+ideal(h);
p = first first entries gens
      eliminate({u,v,w},C1+ideal(x-f,y-g))
Ip = radical(ideal(diff(x,p),diff(y,p),p));
{degree p, # terms p, degree Ip}
q = first first entries gens
      eliminate({u,v,w},C2+ideal(x-f,y-g))
Iq = radical(ideal(diff(x,q),diff(y,q),q));
{degree q, # terms q, degree Iq}

```

The polynomials p and q have degrees 6 and 12 respectively. The command `# terms` verifies that all monomials in the Newton polygons appear with non-zero coefficients. The singular loci of the two curves are given by their radical ideals, I_p and I_q . Applying the command `degree` to these ideals verifies that the number of singular points is 10 and 51 respectively.

3 Topological Complexity

When a soccer ball gets flattened, one generally expects the planar image to be simply connected. However, it is quite possible for $\phi(\mathcal{B})$ to have holes. In other words, the complement $\mathbb{R}^2 \setminus \phi(\mathcal{B})$ can have two or more connected components. In this section we present an explicit construction that makes this happen, with the number of holes being arbitrarily large.

The number of connected components of $\phi(\mathcal{B})$ is at most the number of connected components of $\mathcal{B}$. The number of its holes is counted by the first Betti number of $\phi(\mathcal{B})$. The best upper bounds for Betti numbers of compact semialgebraic sets are due to Basu and Riener [1, Theorem 10] and Basu and Rizzie [2, Theorem 27]. In our setting, the number of holes is bounded by $O(\max(d_1, d_2)^6 e^2)$ if $e \leq \max(d_1, d_2)$ and by $O(\max(d_1, d_2)^8)$ otherwise.

In what follows we assume that $\mathcal{B}$ is the unit ball (3). The image $\phi(\mathcal{B})$ is a compact connected subset of $\mathbb{R}^2$. We are interested in maps ϕ whose image $\phi(\mathcal{B})$ is not simply connected. The construction we shall give furnishes the lower bound $O(d_1 d_2)$ on the number of holes of $\phi(\mathcal{B})$. Based on *Lissajous curves*, it gives rise to some beautiful explicit examples.

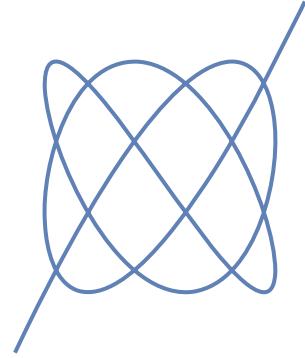
The *Chebyshev polynomials (of the first kind)* are defined recursively by

$$T_0(t) = 1, \quad T_1(t) = t \quad \text{and} \quad T_{d+1}(t) = 2tT_d(t) - T_{d-1}(t) \quad \text{for } d \geq 1.$$

Explicitly, the Chebyshev polynomials are

$$T_2(t) = 2t^2 - 1, \quad T_3(t) = 4t^3 - 3t, \quad T_4(t) = 8t^4 - 8t^2 + 1, \quad T_5(t) = 16t^5 - 20t^3 + 5t, \quad \dots$$

Fig. 3 The Lissajous curve $\mathcal{L}_{5,7}$



They satisfy the trigonometric identity $\cos(d\theta) = T_d(\cos(\theta))$. Fix relatively prime positive integers d_1 and d_2 with $d_1 < d_2$. Let $\mathcal{L}_{d_1, d_2}$ denote the Lissajous curve

$$x = \cos(d_1\theta), \quad y = \cos(d_2\theta). \quad (4)$$

Its Zariski closure is the curve of degree d_2 with polynomial parametrization

$$x = T_{d_1}(t), \quad y = T_{d_2}(t).$$

For instance, Lissajous curve $\mathcal{L}_{2,3}$ is the rational cubic $\{4x^3 - 2y^2 - 3x + 1 = 0\}$. It is singular at $(x, y) = (\frac{1}{2}, 0)$.

Example 4 Figure 3 shows the Lissajous curve $\mathcal{L}_{5,7}$. This curve has 12 singular points. This is the number of bounded regions in the complement of $\mathcal{L}_{5,7}$ in $\mathbb{R}^2$.

Lemma 1 *The curve $\mathcal{L}_{d_1, d_2}$ has precisely $\frac{(d_1-1)(d_2-1)}{2}$ complex singular points. All of these are real and are attained by two distinct values of θ in the trigonometric parametrization (4).*

Proof By the same argument as in the proof of Theorem 1, the Newton polygon of the Lissajous curve $\mathcal{L}_{d_1, d_2}$ is contained in the triangle with vertices $(0, 0)$, $(d_2, 0)$ and $(0, d_1)$. The number of interior lattice points of that triangle is $\frac{(d_1-1)(d_2-1)}{2}$. This is the genus of the generic curve with that Newton polygon. And, it hence is an upper bound on the number of complex singular points of the special curve $\mathcal{L}_{d_1, d_2}$.

We next exhibit $\frac{(d_1-1)(d_2-1)}{2}$ real singular points on $\mathcal{L}_{d_1, d_2}$ that are in the image of (4). Pick any $k \in \{1, \dots, d_1 - 1\}$ and any $l \in \{1, \dots, d_2 - 1\}$. Consider the angles

$$\theta' = \pi\left(\frac{k}{d_1} + \frac{l}{d_2}\right) \quad \text{and} \quad \theta'' = \left|\pi\left(\frac{l}{d_2} - \frac{k}{d_1}\right)\right|. \quad (5)$$

If $\frac{l}{d_2} - \frac{k}{d_1} > 0$, then $\theta' - \theta'' = \frac{2k\pi}{d_1}$ and $\theta' + \theta'' = \frac{2l\pi}{d_2}$; otherwise $\theta' - \theta'' = \frac{2l\pi}{d_2}$ and $\theta' + \theta'' = \frac{2k\pi}{d_1}$. This means that θ' and θ'' map to the same point, and the

Lissajous curve $\mathcal{L}_{d_1, d_2}$ has a node at that point. There are $(d_1 - 1)(d_2 - 1)$ choices of pairs (k, l) . Since the trigonometric parametrization (4) is 2-to-1 on the interval $[0, 2\pi]$, this creates $\frac{(d_1-1)(d_2-1)}{2}$ nodal singularities on $\mathcal{L}_{d_1, d_2}$. This argument is a modification of [4, Section 2.1]. The lower bound we derived matches the upper bound in the previous paragraph, and this completes the proof.

We now apply this to flattening the soccer ball. Consider the map $\phi = (f, g)$ with

$$f(u, v, w) = T_{d_1}(u) + \epsilon \cdot v, \quad g(u, v, w) = T_{d_2}(u) + \epsilon \cdot w, \quad (6)$$

where $T_d(\cdot)$ is the degree- d Chebyshev polynomial, and $\epsilon > 0$ is a small constant. The map ϕ takes the soccer ball $\mathcal{B}$ and creates a two-dimensional image with many holes in $\mathbb{R}^2$.

Example 5 Let $d_1 = 2$ and $d_2 = 3$. The set $\phi(\mathcal{B})$ is the region shown in Fig. 4. It has precisely one hole. This picture was created by the following code in Mathematica, which produces a huge expression:

```
h = 1 - (u^2 + v^2 + w^2);
f = 2*u^2 - 1 + 1/10*v; g = 4*u^3 - 3*u + 1/10*w;
S = Exists[{u, v, w}, h >= 0 && x == f && y == g];
SR = Resolve[S, Reals]
RegionPlot[SR, {x, -1.2, 1.2}, {y, -1.2, 1.2}, PlotPoints -> 50]
```

The command “Resolve” performs quantifier elimination.

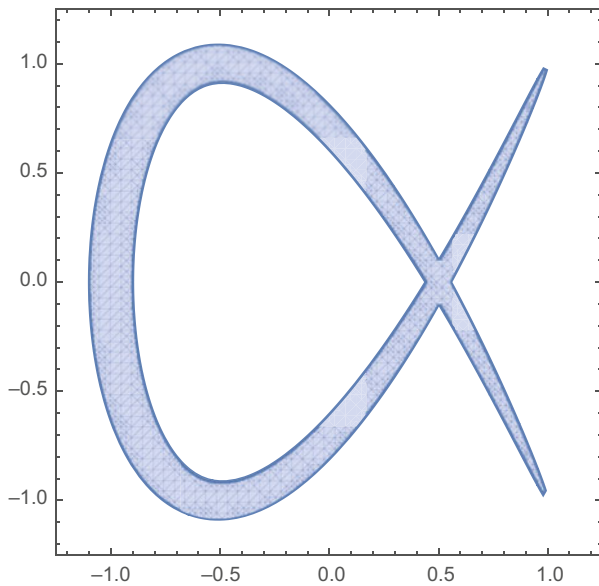


Fig. 4 The modification of the Lissajous curve in Example 5

The following is our main result in this section.

Theorem 2 *Let $d_1 < d_2$ be relatively prime and ϕ as above with $\epsilon > 0$ sufficiently small. Then, the complement of $\phi(\mathcal{B})$ in $\mathbb{R}^2$ has $\frac{(d_1-1)(d_2-1)}{2} + 1$ connected components. The algebraic boundary of $\phi(\mathcal{B})$ is an irreducible curve of degree at most $4d_2 - 2$. It is the branch locus of $\phi_{\{h=0\}}$, so it is defined by the polynomial that was denoted by q in Theorem 1.*

Proof The part of the curve $\mathcal{L}_{d_1, d_2}$ that lies in the square $[-1, 1]^2$ is compact. We regard this as an embedded planar graph, where the vertices are the nodal singularities given in (5) together with the two degree 1 endpoints, and the edges are the pieces of the Lissajous curve that connect the nodes and endpoints. This planar graph is 4-valent, the numbers of vertices, edges and faces satisfy $v - e + f = 2$ and $2e = 4(v - 2) + 2$. This implies $f = v - 1 = \frac{(d_1-1)(d_2-1)}{2} + 1$, i.e. the Lissajous curve has the correct number of holes.

As ϵ increases from 0 to being positive, the curve gets replaced by a two-dimensional region. But the number of holes in the complement does not change.

The algebraic boundary of $\phi(\mathcal{B})$ is given by the polynomials p and q that describe the branch curves of ϕ and $\phi_{\{h=0\}}$ respectively. However, in the present case, the curve $V(p)$ does not exist because the Jacobian of the map ϕ has rank 2 for all $(u, v, w) \in \mathbb{C}^3$. The Jacobian determinant of (f, g, h) with respect to (u, v, w) is the irreducible polynomial

$$\det \begin{pmatrix} \partial T_{d_1}/\partial u & \epsilon & 0 \\ \partial T_{d_2}/\partial u & 0 & \epsilon \\ -2u & -2v & -2w \end{pmatrix} = 2v\epsilon \frac{\partial T_{d_1}}{\partial u} + 2w\epsilon \frac{\partial T_{d_2}}{\partial u} - 2\epsilon^2 u. \quad (7)$$

This is a polynomial of degree d_2 in which v and w occur linearly. The intersection of this surface with the unit sphere is an irreducible curve of degree at most $2d_2$. To compute the image of the curve, we substitute $v = \frac{1}{\epsilon}(x - T_{d_1}(u))$ and $w = \frac{1}{\epsilon}(y - T_{d_2}(u))$ into $h(u, v, w)$ and into (7). This results in two polynomials in u, x, y . Our task is to eliminate u . We do this by taking the determinant of the Sylvester matrix with respect to u . The non-constant entries in the Sylvester matrix have degree one or two in x or y . By examining their pattern in the matrix, we find that the determinant is a polynomial of degree at most $4d_2 - 2$.

Remark 2 We found experimentally that the Newton polygon of q is the triangle with vertices $(0, 0)$, $(4d_2 - 2, 0)$ and $(0, 2d_1 + 2d_2 - 2)$, but we could not prove this.

Example 6 We return to the flattened soccer ball seen in the introduction. To draw this picture from scratch in Mathematica, we run the code in Example 5, modified as follows:

```
f = u*v + v*w + u*w;  g = u*v*w;
```

For this input, the output of the quantifier elimination command `Resolve` equals:

```
(-(1/2) <= x <= 0 && y == 0) ||
(y == -(1/(3*Sqrt[3])) && x == -(1/3)) ||
(-(1/(3*Sqrt[3])) < y < 0 &&
  Root[2*#1^3 + #1^2 - y^2 & , 1] <= x <=
  Root[2*#1^3 + #1^2 - y^2 & , 2]) ||
(y == 0 && Inequality[-(1/2), LessEqual, x, Less,
  0]) || (0 < y < 1/(3*Sqrt[3]) &&
  Root[2*#1^3 + #1^2 - y^2 & , 1] <= x <=
  Root[2*#1^3 + #1^2 - y^2 & , 2]) ||
(y == 1/(3*Sqrt[3]) && x == -(1/3)) ||
(x == -(1/2) && -(1/(3*Sqrt[6])) <= y <=
  1/(3*Sqrt[6])) || (-(1/2) < x < -(1/3) &&
  Root[729*#1^4 + #1^2*(-(92*x^3) + 6*x^2 + 48*x -
    16) + 4*x^6 - 4*x^5 + x^4 & , 1] <= y <=
  Root[729*#1^4 + #1^2*(-(92*x^3) + 6*x^2 + 48*x -
    16) + 4*x^6 - 4*x^5 + x^4 & , 4]) ||
(x == -(1/3) && -(1/(3*Sqrt[3])) <= y <=
  1/(3*Sqrt[3])) ||
(-(1/3) < x < (1/38)*(5*Sqrt[5] - 7) &&
  Root[729*#1^4 + #1^2*(-(92*x^3) + 6*x^2 + 48*x -
    16) + 4*x^6 - 4*x^5 + x^4 & , 1] <= y <=
  Root[729*#1^4 + #1^2*(-(92*x^3) + 6*x^2 + 48*x -
    16) + 4*x^6 - 4*x^5 + x^4 & , 4]) ||
(x == (1/38)*(5*Sqrt[5] - 7) && -Sqrt[2*x^3 + x^2] <=
  y <= Sqrt[2*x^3 + x^2]) ||
((1/38)*(5*Sqrt[5] - 7) < x < 16/43 &&
  Root[729*#1^4 + #1^2*(-(92*x^3) + 6*x^2 + 48*x -
    16) + 4*x^6 - 4*x^5 + x^4 & , 1] <= y <=
  Root[729*#1^4 + #1^2*(-(92*x^3) + 6*x^2 + 48*x -
    16) + 4*x^6 - 4*x^5 + x^4 & , 4]) ||
(16/43 <= x <= 1/2 && -(Sqrt[x^3]/(3*Sqrt[3])) <=
  y <= Sqrt[x^3]/(3*Sqrt[3])) ||
(1/2 < x < 1 && -(Sqrt[x^3]/(3*Sqrt[3])) <= y <=
  Root[729*#1^4 + #1^2*(-(92*x^3) + 6*x^2 + 48*x -
    16) + 4*x^6 - 4*x^5 + x^4 & , 2] ||
  Root[729*#1^4 + #1^2*(-(92*x^3) + 6*x^2 + 48*x -
    16) + 4*x^6 - 4*x^5 + x^4 & , 3] <= y <=
  Sqrt[x^3]/(3*Sqrt[3])) ||
(x == 1 && (y == -(1/(3*Sqrt[3])) ||
  y == 1/(3*Sqrt[3])))
```

This is a quantifier-free formula for the flattened soccer ball in Fig. 2. Most readers will find such an output hard to understand. The next section offers an alternative.

4 Exact Representation of the Image

Quantifier elimination for polynomial systems over $\mathbb{R}$ is usually performed by cylindrical algebraic decomposition [8], abbreviated CAD. Many variants can be found in the recent literature, including *truth table invariant CAD* [5] and *variant quantifier elimination* [11]. CAD represents a semialgebraic set as a union of cells. In dimension one this would be a disjoint union of points and open intervals. Several implementations of CAD are now available, including QEPCAD [7], and the packages RegularChains [14] and ProjectionCAD [9] in Maple. In Example 6, we experimented with the implementation of CAD in Mathematica.

This section is purely expository, aimed at all mathematicians and their students. We show how to obtain a meaningful CAD “by hand” for all instances with parameters $d_1 = 2, d_2 = 3, e = 2$. Experts and CAD developers might find this useful as a family of test cases.

Consider the curve in $\mathbb{R}^2$ defined by the polynomial $p \cdot q$. Our image $\phi(\mathcal{B})$ is the closure of a union of connected components of its complement. We compute a partition of $\mathbb{R}^2$ that refines the partition given by $V(pq)$. A key step is to label each open piece in the finer partition. We then test which pieces lie in $\phi(\mathcal{B})$, and we report the labels of those that do.

Algorithm 1 Quantifier-free representation of the flattened soccer ball $\phi(\mathcal{B})$

Input: Three polynomials $f, g, h \in \mathbb{Q}[u, v, w]$.

Step 1: Compute the polynomial p that defines the branch locus of the map ϕ .

Step 2: Compute the polynomial q that defines the branch locus of the restriction of ϕ to the boundary surface $V(h)$ of $\mathcal{B}$.

Step 3: Compute the sorted list C of all critical x -values of the polynomial $p \cdot q$.

Step 4: Sample points uniformly from $\mathcal{B}$ and compute their images under the map ϕ . Save the result in a list S .

Step 5: For each $(\bar{x}, \bar{y}) \in S$ determine $k \in \mathbb{N}$ such that $\bar{x}$ is between the k th and $(k + 1)$ -st element of C . Compute sorted list R of all real roots of the univariate polynomial $p(\bar{x}, y) \cdot q(\bar{x}, y)$. Determine $l \in \mathbb{N}$ such that $\bar{y}$ is between the l th and $(l + 1)$ st root in R .

Output: The polynomials p, q , the list C , and the set of all pairs (k, l) generated in Step 5.

Algorithm 1 describes what we do. The geometric idea is to project $V(pq)$ onto the x -axis. The critical points of that projection come in four flavors: singular points of $V(p)$, singular points of $V(q)$, points in the intersection $V(p, q)$, and smooth points on $V(pq)$ with vertical tangent lines. The *critical x -values* are the x -coordinates of all real critical points.

Two consecutive critical x -values define a vertical strip. Here, the behavior of the curve segments does not change as x varies. In particular, curve segments do not cross or change direction. Curve segments pass from the left to the right over two consecutive critical x -values, and they divide the vertical strip into open regions. Two of them are unbounded and hence irrelevant. Each bounded region is either contained in $\phi(\mathcal{B})$ or is disjoint from $\phi(\mathcal{B})$.

Our description of $\phi(\mathcal{B})$ has three parts: the polynomials p and q that define the algebraic boundary, the critical x -values, and a set of pairs of positive integers. A pair (k, l) determines a region in $\mathbb{R}^2$ as follows. The x -coordinates are between the k th and $(k + 1)$ st critical x -value, and the y -coordinates lie between l th and $(l + 1)$ st curve segment in the y -direction.

Steps 1 and 2 of Algorithm 1 can be done with `Macaulay2`, as shown at the end of Sect. 2. Step 3 is more delicate because p and q are fairly large, even when f, g, h are small. The task is to compute the critical x -values of the polynomials p and q , and also the x -coordinates of the common zeros of p and q . To find these x -values symbolically, we compute three resultants of two polynomials in (x, y) with respect to y . Namely, we compute

$$\text{Res}_y(p, \partial p / \partial y), \text{Res}_y(q, \partial q / \partial y) \text{ and } \text{Res}_y(p, q). \quad (8)$$

At this stage one might compute the real roots of these three univariate polynomials in x . This can be done numerically using various methods, including the numerical algebraic geometry package in `Macaulay2`. However, we did not do this in our computations. Instead, we identify the real roots of our three resultants purely symbolically, using the command `Solve` with the option `Reals` in `Mathematica`. `Solve` tries to write each solution explicitly in terms of radicals, and if unsuccessful, it creates a representation as a root of a polynomial.

Since $\mathcal{B}$ is compact, we can enclose it inside an appropriate cube in $\mathbb{R}^3$. We sample points (u, v, w) with rational coordinates from that cube, and we throw out points that do not satisfy $h(u, v, w) \geq 0$. For instance, if $\mathcal{B}$ is the unit sphere, then we first uniformly sample points from the cube $[-1, 1]^3$ and keep the points that satisfy $1 - u^2 - v^2 - w^2 \geq 0$.

Step 4 is probabilistic. As the number of samples grows to infinity, every region in the CAD of $\phi(\mathcal{B})$ will be reached and certified eventually. For any finite number of samples, there is a positive probability that some region is missed. Our approach can be turned this into a deterministic method by selecting a sample point in each region and deciding whether it has a preimage in $\mathcal{B}$. This amounts to testing whether a semialgebraic set in $\mathbb{R}^3$ is non-empty. The best known algorithm for deciding non-emptiness of a semialgebraic set is by Renegar [18]. In practice this can be done using implementations of CAD, but we decided not to pursue the deterministic variant in the present study.

In our examples we are also interested in deciding which regions of $\phi(\mathcal{B})$ lie in $\phi(\partial\mathcal{B})$. Note the distinction between the colors in Fig. 2. For instance, to obtain points on the boundary of the unit ball, we uniformly sample points from the square $[-1, 1]^2$, keep the points that satisfy $1 - u^2 - v^2 \geq 0$, and create two boundary points with $w = \pm\sqrt{1 - u^2 - v^2}$.

In Step 5, we use binary search to determine k such that $\bar{x}$ is between the k th and $(k + 1)$ st element of the list C . The symbolic representation derived in *Mathematica* is suitable for doing this. To be precise, we did the binary search with the following commands:

```
Block[{$ContextPath}, Needs["Combinatorica`"]]
k = Combinatorica`BinarySearch[C, x] - 1/2
```

The real roots of $p(\bar{x}, y) \cdot q(\bar{x}, y)$ can again be computed using *Mathematica* command *Solve*. We now illustrate Algorithm 1 by applying it to the instance that launched this project.

Example 7 Fix f, g, h as in Example 1. The first part of the output are the polynomials p and q at the end of Example 1. The second part is the list of critical x -values:

$$-\frac{1}{2}, 0, \frac{16}{43}, \frac{2}{5}, \frac{1}{2}, 1. \quad (9)$$

The third part is a list which represents a partition of $\phi(\mathcal{B})$ into 22 regions:

$$(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (3, 1), (3, 2), (3, 3), \\ (3, 4), (3, 5), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (5, 1), (5, 2), (5, 4), (5, 5).$$

The output above represents a quantifier-free formula for the flattened soccer ball. Using symbolic computation, we can assign one of the 22 labels to any sample point $\phi(u_0, v_0, w_0)$. For instance, $\phi(\mathcal{B}) \setminus \phi(\partial\mathcal{B})$ consists of the six regions $(3, 1), (3, 5), (4, 1), (4, 5), (5, 1), (5, 5)$.

From our output, we drew the picture in Fig. 5. For this, we used the coordinates of all singular points and branch points of $V(pq)$. Branching occurs along the vertical tangent line $y = -\frac{1}{2}$. The real singular points are $(-\frac{1}{2}, -\frac{1}{3\sqrt{6}}), (-\frac{1}{2}, \frac{1}{3\sqrt{6}}), (0, 0), (\frac{16}{43}, -\frac{64}{129\sqrt{129}}), (\frac{16}{43}, \frac{64}{129\sqrt{129}}), (\frac{2}{5}, -\frac{2}{15\sqrt{15}}), (\frac{2}{5}, \frac{2}{15\sqrt{15}}), (\frac{1}{2}, 0), (1, -\frac{1}{3\sqrt{3}}), (1, \frac{1}{3\sqrt{3}})$. Four pairs of critical points have the same x -coordinates, seen in (9). The singular points and the vertical tangent line $y = -\frac{1}{2}$ are the black landmarks in Fig. 5. The 22 regions are shown in different colors. The six regions in $\phi(\mathcal{B}) \setminus \phi(\partial\mathcal{B})$ are colored light.

The output of Algorithm 1 can be interpreted as a semialgebraic formula for $\phi(\mathcal{B})$, as follows: Consider $p \cdot q$ as a univariate polynomial in y . Write down its Sturm sequence. Let n denote the number of sign changes in that sequence evaluated at $-\infty$. The formula for the region (k, l) is a disjunction over all the possible sign assignments of the Sturm sequence with $n - l$ sign changes, in conjunction with x being between the k th and $(k + 1)$ st critical x -value. Such a formula will again be hard to read, just like the *Mathematica* output displayed at the end of Sect. 4. A description like Example 7, accompanied by a picture like Fig. 5, seems to be the most human-friendly way to represent the result of flattening a soccer ball.

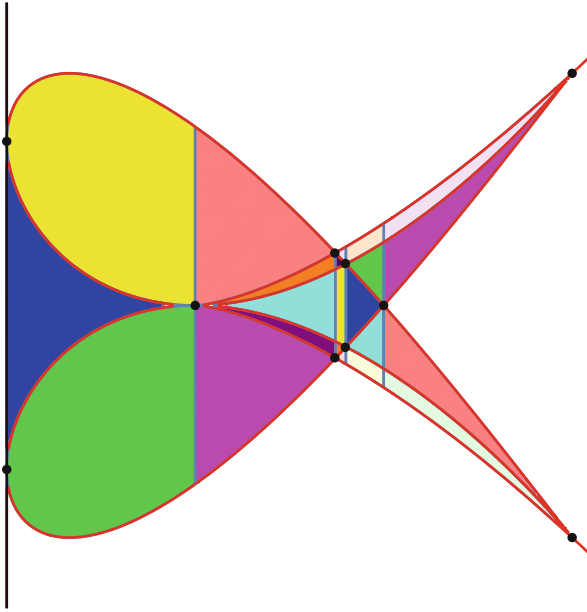


Fig. 5 The flattened soccer ball in Examples 1 and 7 is divided into 22 regions

We next discuss a more serious example, where the `Resolve` command does not terminate. It will demonstrate the pros and cons of the exact symbolic approach.

Example 8 Fix $h = u^2 + v^2 + w^2 - 1$ as before, so $\mathcal{B}$ is the unit ball. We select f and g randomly from polynomials of degree 2 and 3 respectively. This means we are now in the regime covered by Theorem 1. Note the fourth line in Table 1 marked $(d_1, d_2, e) = (2, 3, 2)$. The following instance, picked for us by `Macaulay2`, corresponds to the picture in Fig. 6:

$$\begin{aligned} f &= \frac{3}{5}u^2 + uv + \frac{10}{3}v^2 + \frac{7}{3}uw + \frac{1}{4}vw + \frac{3}{10}w^2 + \frac{7}{4}u + \frac{8}{5}v + \frac{7}{5}w + \frac{10}{9}, \\ g &= \frac{1}{4}u^3 + 3u^2v + uv^2 + \frac{5}{3}v^3 + u^2w + \frac{8}{5}uvw + \frac{4}{7}v^2w + \frac{7}{3}uw^2 + \frac{7}{3}vw^2 + \frac{7}{10}w^3 + \\ &\quad \frac{7}{2}u^2 + 3uv + \frac{5}{9}v^2 + \frac{3}{8}uw + \frac{1}{9}vw + \frac{7}{4}w^2 + \frac{9}{2}u + \frac{3}{4}v + \frac{5}{6}w + \frac{3}{7}. \end{aligned}$$

The polynomial p that describes the branch locus of ϕ has degree 21 and 169 terms. The polynomial q that describes the branch locus of $\phi_{\{h=0\}}$ has degree 24 and 217 terms. In both cases, these numbers count exactly the lattice points of the Newton triangles in Theorem 1.

For Step 3 we compute three univariate polynomials in x , namely the resultants in (8). The resultant of p and $\frac{\partial p}{\partial y}$ has degree 273 with 13 real roots. The resultant of p and $\frac{\partial q}{\partial y}$ has degree 360 with 22 real roots. The resultant of p and q has degree 336 with 16 real roots. All $13 + 22 + 16$ real roots are different, so the total number of critical x -values is 51. In Step 4 we sample points from the cube $[-1, 1]^3$ and from

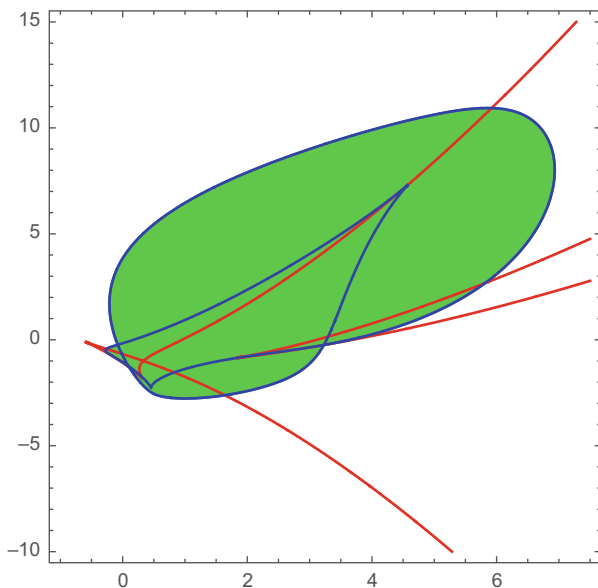


Fig. 6 The flattened soccer ball $\phi(\mathcal{B})$ in Example 8 and its algebraic boundary

its boundary, and we record their images under ϕ . In Step 5, we run over these points in $\mathbb{R}^2$, and we identify the labels (k, l) of the regions that contain these points. Some regions are very small. It takes a long time to identify them. We found that $\phi(\mathcal{B})$ is the union of the following 144 regions:

(12, 2), (13, 2), (13, 3), (14, 2), (14, 3), (14, 5), (15, 1), (15, 2), (15, 3), (15, 5), (16, 1), (16, 2),
 (16, 3), (16, 4), (16, 5), (17, 1), (17, 2), (17, 3), (17, 4), (17, 5), (18, 1), (18, 2), (18, 3), (18, 4),
 (18, 5), (19, 1), (19, 2), (19, 3), (19, 4), (19, 5), (20, 1), (20, 2), (20, 3), (20, 4), (20, 5), (20, 6),
 (20, 7), (21, 1), (21, 2), (21, 3), (21, 4), (21, 5), (21, 6), (21, 7), (22, 1), (22, 2), (22, 3), (22, 4),
 (22, 5), (23, 1), (23, 2), (23, 3), (23, 4), (23, 5), (24, 1), (24, 2), (24, 3), (24, 4), (24, 5), (24, 6),
 (24, 7), (25, 1), (25, 2), (25, 3), (25, 4), (25, 5), (26, 1), (26, 2), (26, 3), (26, 4), (26, 5), (27, 1),
 (27, 2), (27, 3), (27, 4), (27, 5), (28, 2), (28, 3), (28, 4), (28, 5), (29, 2), (29, 3), (29, 4), (29, 5),
 (29, 6), (29, 7), (30, 2), (30, 3), (30, 4), (30, 5), (30, 6), (30, 7), (31, 2), (31, 3), (31, 4), (31, 5),
 (31, 6), (31, 7), (32, 2), (32, 3), (32, 4), (32, 5), (32, 6), (32, 7), (33, 3), (33, 4), (33, 5), (33, 6),
 (33, 7), (34, 3), (34, 4), (34, 5), (34, 6), (34, 7), (35, 3), (35, 4), (35, 5), (35, 6), (35, 7), (36, 3),
 (36, 4), (36, 5), (36, 6), (36, 7), (37, 3), (37, 4), (37, 5), (37, 6), (37, 7), (38, 3), (38, 4), (38, 5),
 (38, 6), (38, 7), (39, 3), (39, 4), (39, 5), (40, 3), (40, 4), (40, 5), (41, 4), (41, 5), (42, 4), (43, 4).

Out of the 50 bounded segments between the critical x -values, precisely 32 arise from $\mathcal{B}$. The 11 left-most segments and the 7 right-most segments do not arise.

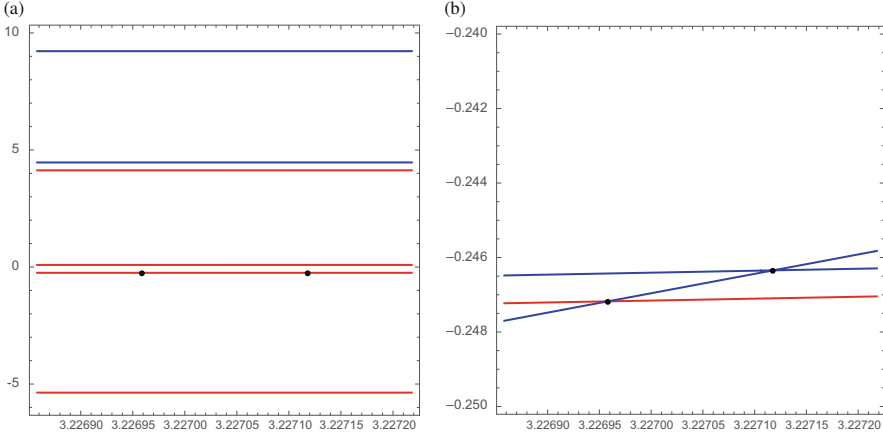


Fig. 7 Magnifications between the 33rd and 34th critical points. (a) Magnification of the x -axis only. (b) Magnification of both coordinate axes

In the list above, the first pair (12, 2) refers to x -coordinates between critical x -values labeled #12 and #13, which are approximately -0.275436 and -0.2599 . The corresponding y -coordinates are between the 2-nd and 3-rd root of $p \cdot q$, regarded as a polynomial in y , with x fixed in segment #12.

Figure 7 shows that the situation is delicate. The regions can be very small, even for curves of degree 21. For example, consider the critical x -values labeled #33 and #34. They are 3.22696 and 3.22712. The unique critical point over #33 is an intersection point of $V(p)$ and $V(q)$. The unique critical point over #34 is a node of $V(q)$. They are shown in Fig. 7.

In Fig. 7a we scaled the x -axis so that it shows the vertical strip #33. The y -axis is left unscaled, so that we can see all curve segments in that strip. It looks like the blue curve $V(p)$ has two segments and the red curve $V(q)$ has four segments. However, two more blue segments are eclipsed by the relevant red segment. The truth becomes visible in Fig. 7b, where we also scale the y -axis. The critical point on the right is a node of the blue curve $V(q)$ and it does not lie on the red curve $V(p)$. The left critical point lies in $V(p, q)$. Our careful analysis also shows that $\phi(\mathcal{B}) \neq \phi(\partial\mathcal{B})$, although this is not visible in Fig. 6. Between critical x -values labeled #15 and #19, the lower boundary is given by the red curve $V(p)$.

5 Convexity and Optimization

The problem of computing images of maps is of considerable interest in polynomial optimization. Magron et al. [16] developed a method for this based on outer approximations. Our study is complementary to theirs, in the sense that we do not

consider approximations but we seek exact descriptions. A related question is how to compute and represent the convex hull of the image $\phi(\mathcal{B})$. This issue will be addressed later in this section.

We start our discussion with a few examples where both $\mathcal{B}$ and $\phi(\mathcal{B})$ are convex. In what follows we retain the assumption that $h = u^2 + v^2 + w^2 = 1$, so $\mathcal{B}$ is the unit ball in $\mathbb{R}^3$. This can be folded into a convex polygon in various interesting ways.

Example 9 It is easy to find quadratic polynomials f and g such that $\phi = (f, g)$ maps the soccer ball onto a triangle or a rectangle. Figure 1 shows a map onto a square. Here are two explicit maps that work. If $f = u^2 + w^2$ and $g = v^2 + w^2$ then $\phi(\mathcal{B})$ is the square with vertices $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$, and $\phi(\partial\mathcal{B})$ is the triangle with vertices $(0, 1)$, $(1, 0)$, $(1, 1)$. The boundary polynomials are $p = xy(x-y)$ and $q = (1-x)(1-y)(x+y-1)$. For a second example let ϕ be defined by $f = u^2 + 2v^2 + w^2 - 1$ and $g = u^2 + v^2 + 2w^2 - 1$. Now the flattened soccer ball $\phi(\mathcal{B})$ is the triangle with vertices $(-1, -1)$, $(0, 1)$, $(1, 0)$, while $\phi(\partial\mathcal{B})$ is the triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$. The difference $\phi(\mathcal{B}) \setminus \phi(\partial\mathcal{B})$ is not a convex set.

If we pass from quadratic maps to cubic maps then we can create other polygons.

Example 10 Use the cubic Chebyshev polynomial $T_3(t) = 4t^3 - 3t$ to define ϕ via

$$f(u, v, w) = \sqrt{3}(T_3(u) - T_3(v)) \quad \text{and} \quad g(u, v, w) = T_3(u) + T_3(v) - 2T_3(w).$$

The image $\phi(\mathcal{B})$ is the regular hexagon with vertices at $(0, \pm 4)$ and $(\pm 2\sqrt{3}, \pm 2)$.

This raises the question whether we can prescribe $\phi(\mathcal{B})$ to be any polyhedral shape. Ueno [21] proves that every unbounded convex polygon in $\mathbb{R}^2$ is the image of $\mathbb{R}^2$ under a polynomial map. It is not known, if his construction extends to polynomial images of the unit ball.

Problem 1 Let P be an arbitrary convex polygon in $\mathbb{R}^2$. Construct explicit polynomials f and g in $\mathbb{R}[u, v, w]$ such that $P = \phi(\mathcal{B})$.

Our next topic is the *flattenings of pancakes*. These arise as special scenarios when we flatten soccer balls. Indeed, suppose that the map ϕ depends only on two of the variables, say

$$\phi : (u, v, w) \mapsto (f(u, v), g(u, v)).$$

Then the image of $\mathcal{B}$ is the same as the image under $\phi' = (f, g)$ of the unit disk $\mathcal{D} = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 \leq 1\}$. In symbols, $\phi(\mathcal{B}) = \phi'(\mathcal{D})$. The unit disk $\mathcal{D}$ serves the role of our *pancake*.

Example 11 Magron et al. [16, Example 1] illustrate their method for the map

$$f = \frac{1}{2}(u + uv) \quad \text{and} \quad g = \frac{1}{2}(v - u^3).$$

It is instructive to compare their output with that of Algorithm 1. The exact description of the flattened pancake begins with the two polynomials that define

the algebraic boundary:

$$\begin{aligned} p &= 2048x^3 + 432y^4 + 864y^3 + 648y^2 + 216y + 27, \\ q &= 64x^6 + 128x^5 + 96x^4 + 128x^3y - 32x^3 + 192x^2y^2 - 44x^2 \\ &\quad + 96xy^3 + 48xy^2 - 24xy - 12x + 16y^4 + 16y^3 - 4y - 1. \end{aligned}$$

The projection of the curve $V(pq)$ onto the x -axis has four critical points:

$$\left(-\frac{3\sqrt{3}}{8}, \frac{-473 + 264\sqrt{3}}{6208 - 3600\sqrt{3}}\right), \left(-\frac{1}{2}, \frac{1}{2}\right), \left(0, -\frac{1}{2}\right), \left(\frac{3\sqrt{3}}{8}, \frac{-473 - 264\sqrt{3}}{6208 + 3600\sqrt{3}}\right).$$

Our quantifier-free description of the planar region $\phi'(\mathcal{D})$ consists of the five pairs:

$$(1, 2), (1, 3), (2, 2), (2, 3), (3, 1).$$

Figure 8 shows $\phi'(\mathcal{D})$ where these five regions are colored in yellow, orange, green, blue and purple. The critical points are black, and the boundary curve $V(pq)$ is red.

Outer approximations of the kind studied in [16] tend to work best when one is interested not in the image itself but in its convex hull $\text{conv}(\phi(\mathcal{B}))$. Its extreme points are the solutions for the parametrized family of optimization problems

$$\gamma(\alpha, \beta) := \max\{\alpha x + \beta y : (x, y) \in \phi(\mathcal{B})\}. \quad (10)$$

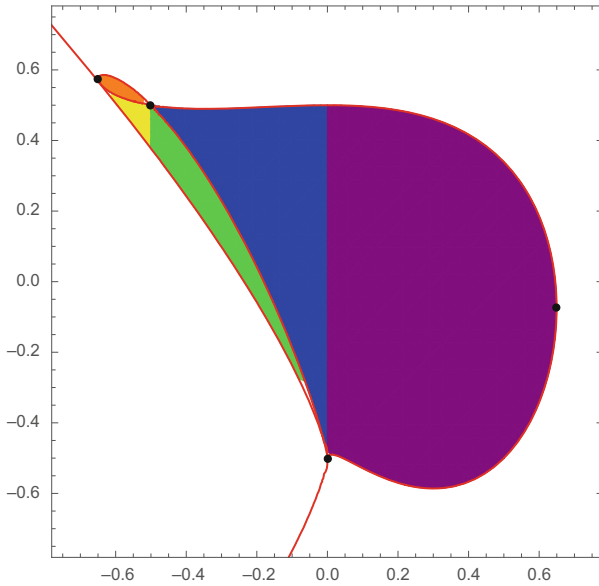


Fig. 8 This flattened pancake was discussed by Magron et al. [16]

The function γ is called the *support function* of the set $\phi(\mathcal{B})$. From the values of this function one obtains a description of the convex hull of $\phi(\mathcal{B})$ as an intersection of closed halfspaces:

$$\text{conv}(\phi(\mathcal{B})) = \bigcap_{(\alpha, \beta) \in \mathbb{R}^2} \{(x, y) : \alpha x + \beta y \leq \gamma(\alpha, \beta)\}.$$

The optimization problem in (10) can be equivalently written as

$$\gamma(\alpha, \beta) = \max_{(u, v, w)} \alpha f(u, v, w) + \beta g(u, v, w) \quad \text{s.t.} \quad h(u, v, w) \geq 0.$$

Both the objective function and the constraint are given by polynomial functions. Although in general this can be difficult to solve, good upper bounds can be obtained using *sum of squares* methods [3]. This method minimizes γ_{SOS} subject to

$$\gamma_{SOS} - (\alpha f(u, v, w) + \beta g(u, v, w)) = s_0(u, v, w) + s_1(u, v, w)h(u, v, w),$$

where s_0 and s_1 are *sums of squares*. The solution satisfies the inequality $\gamma \leq \gamma_{SOS}$. Furthermore, restricting the polynomials s_0, s_1 to have fixed degree, this is a semidefinite optimization problem, so it can be solved efficiently. Since $\gamma(\alpha, \beta) \leq \gamma_{SOS}(\alpha, \beta)$, we obtain an outer approximation of the convex hull:

$$\text{conv}(\phi(\mathcal{B})) \subseteq \bigcap_{(\alpha, \beta) \in \mathbb{R}^2} \{(x, y) : \alpha x + \beta y \leq \gamma_{SOS}(\alpha, \beta)\}. \quad (11)$$

The right hand side is a *spectrahedral shadow*, so it is a desirable set in the context of [3]. If one is lucky then equality holds in (11) and a semidefinite representation of $\text{conv}(\phi(\mathcal{B}))$ has been obtained. From our earlier algebraic perspective, such a representation still involves quantifiers. Any quantifier-free formula has to account for supporting lines that were created when passing from $\phi(\mathcal{B})$ to its convex hull. Those lines are bitangents of our curve $V(pq)$.

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The Smallest Normal Edge Polytopes with No Regular Unimodular Triangulations

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Abstract The edge polytope of a finite graph is the convex hull of the column vectors of its vertex-edge incidence matrix. In this paper, we discuss the existence of a regular unimodular triangulation of normal edge polytopes of finite graphs. For normal edge polytopes of finite graphs with d vertices, n edges, and no regular unimodular triangulations, we determine the polytope that has the following: (1) the smallest number of vertices ($d = 9$), (2) the smallest number of edges ($n = 15$), and (3) the smallest codimension ($n - d = 4$ and $d = 17$).

1 Introduction

Let $\mathcal{P} \subset \mathbb{R}^d$ be an integral convex polytope, i.e., a convex polytope for which all of the vertices are integer points. Let $K[t_1^{\pm 1}, \dots, t_d^{\pm 1}, s]$ denote the Laurent polynomial ring over a field K . We write $K[\mathcal{P}]$ for the subalgebra of $K[t_1^{\pm 1}, \dots, t_d^{\pm 1}, s]$ which is generated by all monomials $t_1^{a_1} \cdots t_d^{a_d} s$, where $(a_1, \dots, a_d) \in \mathcal{P} \cap \mathbb{Z}^d$. An integral convex polytope is called *normal* if the subalgebra $K[\mathcal{P}]$ is normal. We are interested in the combinatorial structures of integral convex polytopes, including which type of triangulation it has. In particular, the following conditions have been discussed in many papers:

- (i) $\mathcal{P}$ has a regular unimodular triangulation;
- (ii) $\mathcal{P}$ has a unimodular triangulation;

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- (iii) $\mathcal{P}$ has a unimodular covering;
- (iv) $\mathcal{P}$ is normal.

In general, (i) $\implies$ (ii) $\implies$ (iii) $\implies$ (iv) holds. However, all of the converse implications are false [2, 10]. A necessary and sufficient condition for the existence of a regular unimodular triangulation is given by the theory of Gröbner bases [6, 16]. An integral convex $\mathcal{P}$ has a regular unimodular triangulation if and only if the defining ideal of $K[\mathcal{P}]$ has a squarefree initial ideal.

In this paper, we consider normal integral convex polytopes with no regular unimodular triangulations; in particular, we consider the class of edge polytopes. Let G be a finite simple graph with vertex set $V = [d] = \{1, \dots, d\}$ and edge set $E = \{e_1, \dots, e_n\}$. Let $\mathbf{e}_i$ be the i th unit coordinate vector of $\mathbb{R}^d$. If $e = \{i, j\}$ is an edge of G , then we set $\rho(e) = \mathbf{e}_i + \mathbf{e}_j \in \mathbb{R}^d$. The edge polytope $\mathcal{P}_G$ of G is the convex hull of $\{\rho(e_1), \dots, \rho(e_n)\} \subset \mathbb{R}^d$ which is discussed in many papers, e.g., [9–11, 14, 15]. Following [9, 11], we study the existence of a regular unimodular triangulation of the normal edge polytopes. Presumably, “most” normal edge polytopes have regular unimodular triangulations; some counterexamples are given in the aforementioned papers.

In Sect. 2, we introduce four examples of normal edge polytopes that have no regular unimodular triangulations. In Sect. 3, we prepare several lemmas that will help to determine minimal examples. In Sect. 4, by performing a computer search, we show that the examples introduced in Sect. 2 are the smallest in terms of one of the following:

- (a) the number of vertices of G ;
- (b) the number of edges of G ;
- (c) the number of vertices under the restriction that $n - d = 4$.

2 Edge Polytopes and Their Regular Unimodular Triangulations

First, we introduce a criterion to determine the normality of the edge polytopes given in [10, 15].

Lemma 2.1 *Let G be a finite connected simple graph. Then, the edge polytope $\mathcal{P}_G$ of G is normal if and only if for every pair of disjoint odd cycles (C, C') in G , C and C' are connected by an edge.*

The condition in Lemma 2.1 for G is called the *odd-cycle condition*. Now, let $K[x_1, \dots, x_n]$ denote a polynomial ring over K with n variables. We define the surjective ring homomorphism $\varphi : K[x_1, \dots, x_n] \rightarrow K[\mathcal{P}_G]$ by setting $\varphi(x_k) = t_i t_j s$ for each $e_k = \{i, j\}$. The toric ideal I_G of G is the kernel of φ . It is known that I_G is generated by those binomials $u - v$ where u and v are monomials of $K[x_1, \dots, x_n]$ with $\deg u = \deg v$ such that $\varphi(u) = \varphi(v)$. The following properties of I_G , $\mathcal{P}_G$, and

$K[\mathcal{P}_G]$ have been further discussed in the literature:

- (i) I_G has a squarefree initial ideal with respect to a monomial order;
- (ii) $\mathcal{P}_G$ has a regular unimodular triangulation;
- (iii) $\mathcal{P}_G$ has a unimodular triangulation;
- (iv) $K[\mathcal{P}_G]$ is normal.

It is known (see, for example [6, Section 5.6]) that (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) is true. What about the converse of each implication? Some counterexamples for (iii) $\Rightarrow$ (ii) are known ([9, 11]).

Example 2.2 The edge polytope of each graph displayed in Fig. 1 has a nonregular unimodular triangulation and hence is normal. However, the edge polytope has no regular unimodular triangulations.

We present two new examples of normal edge polytopes that have no regular unimodular triangulations in Figs. 2 and 3.

Theorem 2.3 *The edge polytopes of the graphs displayed in Figs. 2 and 3 are normal polytopes that have no regular unimodular triangulations.*

Proof The normality of these two polytopes can be determined by using Lemma 2.1.

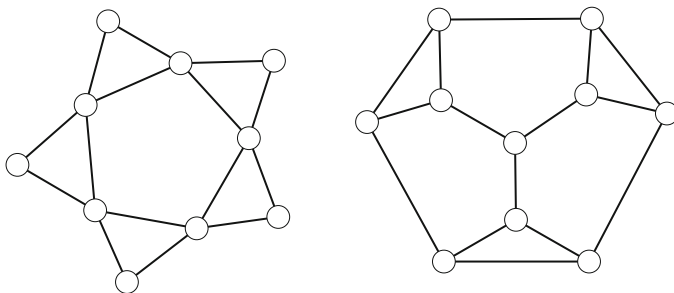


Fig. 1 Graphs with 10 vertices

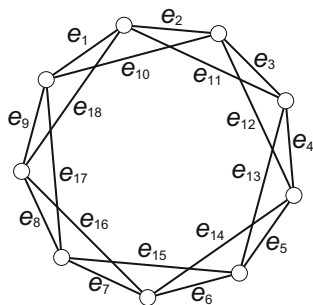
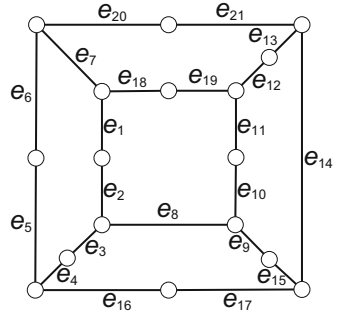


Fig. 2 Graph with 9 vertices

Fig. 3 Graph with 17 vertices and 21 edges



First, we show that there does not exist a monomial order such that the initial ideal of the toric ideal associated with the graph in Fig. 2 is squarefree. Suppose that there exists a monomial order $<$ such that $\text{in}_<(I_G)$ is squarefree. By using [12, Lemma 3.2], the following nine binomials are indispensable, i.e., these binomials appear in the reduced Gröbner basis with respect to any monomial order:

$$\begin{aligned}
 & x_1 x_4 x_{13} x_{18} - x_5 x_9 x_{11}^2 \\
 & x_2 x_5 x_{14} x_{10} - x_6 x_1 x_{12}^2 \\
 & x_3 x_6 x_{15} x_{11} - x_7 x_2 x_{13}^2 \\
 & x_4 x_7 x_{16} x_{12} - x_8 x_3 x_{14}^2 \\
 & x_5 x_8 x_{17} x_{13} - x_9 x_4 x_{15}^2 \\
 & x_6 x_9 x_{18} x_{14} - x_1 x_5 x_{16}^2 \\
 & x_7 x_1 x_{10} x_{15} - x_2 x_6 x_{17}^2 \\
 & x_8 x_2 x_{11} x_{16} - x_3 x_7 x_{18}^2 \\
 & x_9 x_3 x_{12} x_{17} - x_4 x_8 x_{10}^2.
 \end{aligned}$$

Since $\text{in}_<(I_G)$ is squarefree, the initial terms of the above nine binomials with respect to $<$ must be the first terms. On the other hand, by computing the product of all initial terms and the product of all non-initial terms of the above binomials, we obtain

$$x_1^2 x_2^2 \cdots x_{18}^2 < x_1^2 x_2^2 \cdots x_{18}^2.$$

This contradicts the statement that $<$ is a monomial order.

Next, by using a similar argument, we show there does not exist a squarefree initial ideal for the graph in Fig. 3. The following three binomials are indispensable:

$$\begin{aligned} x_2x_3x_5x_7x_9x_{10}x_{12}x_{14} &- x_1x_4x_6x_8^2x_{11}x_{13}x_{15} \\ x_4x_7x_8x_{13}x_{15}x_{17}x_{19}x_{21} &- x_3x_9x_{12}x_{14}^2x_{16}x_{18}x_{20} \\ x_1x_6x_8x_{11}x_{14}x_{16}x_{18}x_{20} &- x_2x_5x_7^2x_{10}x_{17}x_{19}x_{21}. \end{aligned}$$

We may assume that there exists a monomial order $<$ such that for each of the above binomials, the initial term with respect to $<$ is the first term. Then, the product of all initial terms and the product of all non-initial terms are both equal to

$$x_1x_2 \cdots x_{21} \cdot x_7x_8x_{14}.$$

This yields a contradiction. $\square$

3 Theoretical Lower Bounds

In this section, we prepare several lemmas that will help to determine minimal examples. The following lemma shows that we are not interested in any graph that has a vertex of degree 1.

Lemma 3.1 *Let G be a finite connected graph having a vertex v with $\deg v = 1$, and let $G' = G \setminus v$. Then,*

- (a) $\mathcal{P}_G$ is normal if and only if $\mathcal{P}'_G$ is normal;
- (b) $\mathcal{P}_G$ has a regular unimodular triangulation if and only if $\mathcal{P}'_G$ has one.

The following proposition (see, e.g., [13, Theorem 3.3]) shows that we are not interested in any graph with ≤ 5 vertices.

Proposition 3.2 *Let G be a finite connected graph with d vertices. Then, all triangulations of $\mathcal{P}_G$ are unimodular if and only if any two odd cycles of G have a common vertex. In particular, if $d \leq 5$, then all triangulations of $\mathcal{P}_G$ are unimodular.*

Let G be a graph with d vertices and n edges. Then, $n-d$ is called the *codimension* of $K[\mathcal{P}_G]$. The following proposition appeared without proof in [11]; for the sake of completeness, we give its proof here.

Proposition 3.3 *Let G be a finite connected graph with d vertices and n edges. Suppose $n - d \leq 3$. Then, the edge polytope $\mathcal{P}_G$ has a regular unimodular triangulation if and only if $\mathcal{P}_G$ is normal.*

Proof The “only if” part is clear, so we will show the “if” part. By Lemma 3.1, we may assume that the degree of each vertex of G is at least 2. In this proof, we use the following fact:

- If a subgraph G' of G has d' vertices and n' edges, then we have $n' - d' \leq n - d$.

Suppose that $\mathcal{P}_G$ is normal and has no regular unimodular triangulations. Then, by using [8, Theorem 3.5], there exist two pairs of disjoint odd cycles $\Pi = (C_1, C_2)$ and $\Pi' = (C_3, C_4)$ such that there exists an edge e of C_3 that joins C_1 and C_2 . Assume that the sequence of vertices of C_3 is $v_1, v_2, \dots, v_m, v_1$, and assume $e = \{v_1, v_m\}$. Let $G' = C_1 \cup C_2 \cup C_3$. Suppose that C_4 is a cycle in G' . Since C_3 and C_4 have no common vertices, the cycle C_4 is a subgraph of $C_1 \cup C_2$ and does not have vertices v_1 and v_m , which is a contradiction. Thus, C_4 is not contained in G' . We let ν (≥ 2) denote the number of maximal consecutive subsequences of vertices of C_1 or C_2 that appear consecutively in C_3 . Let d' (resp. n') be the number of vertices (resp. edges) of the graph G' . Then, $\nu = n' - d' \leq n - d \leq 3$. Hence, we have $\nu \in \{2, 3\}$.

(Case 1) Suppose $n - d = n' - d'$. Let $E = \{e_1, \dots, e_n\}$ be the edge set of G . If $\{e_{i_1}, \dots, e_{i_\ell}\}$ is the edge set of a cycle C of G , then we set $\tau(C) = \sum_{j=1}^{\ell} \mathbf{e}_{i_j} \in \mathbb{F}_2^n$. Then, the cycle space of G is a subspace of $\mathbb{F}_2^n$ spanned by $\{\tau(C) \mid C \text{ is a cycle of } G\}$. Since both G and G' are connected, the dimension of the cycle space of G and that of G' are both $n - d + 1 = n' - d' + 1$ (see, e.g., [4, Theorem 1.9.6]). Hence, the two cycle spaces are isomorphic, and, in particular, the set of cycles of G coincides with that of G' . Thus, C_4 appears in G' , which is a contradiction. Hence this case is not possible.

(Case 2) Suppose $n - d > n' - d'$. Then, $3 \geq n - d > n' - d' = \nu \geq 2$, and hence $n - d = 3$ and $n' - d' = \nu = 2$. Let V' (resp. E') be the set of vertices (resp. edges) of G' , and let G'' be a subgraph of G with edge set $E'' = E \setminus E'$ and the vertices V'' appearing in E'' . Let $G_1, \dots, G_s$ denote connected components of G'' . Then, we have

$$1 = (n - d) - (n' - d') = \sum_{i=1}^s (|E_i| - |V_i| + |V' \cap V_i|), \quad (1)$$

where V_i is the vertex set of G_i , and E_i is the edge set of G_i . Since G is connected, it follows that $|V' \cap V_i| \geq 1$ for each i . In addition, since G_i is connected, we have $|E_i| - |V_i| \geq -1$. Suppose that $|E_i| - |V_i| + |V' \cap V_i| = 0$. Then, $|E_i| - |V_i| = -1$, and hence G_i is a tree. It is known that any tree has at least 2 vertices of degree 1. Since the degree of each vertex of G is at least 2, we have $|V' \cap V_i| \geq 2$, and hence $|E_i| - |V_i| + |V' \cap V_i| \neq 0$, which is a contradiction. Thus, $|E_i| - |V_i| + |V' \cap V_i| \geq 1$. By Eq. (1), it follows that $s = 1$ and $|E_1| - |V_1| + |V' \cap V_1| = 1$. Therefore, G'' is connected, and either (i) $|E''| - |V''| = 0$ and $|V' \cap V''| = 1$, or (ii) $|E''| - |V''| = -1$ and $|V' \cap V''| = 2$.

Suppose that $V' \cap V'' = \{v'_1, v'_2\}$, and v'_i is a vertex of C_i for $i = 1, 2$. Since $|E''| - |V''| = -1$, G'' is a path. Then, any cycle in $C_1 \cup C_2 \cup G''$ has one of the vertices v_1 and v_m . This contradicts the assumption that C_4 is a subgraph of $C_1 \cup C_2 \cup G''$. Thus, we may assume that $|V' \cap V''| \leq 2$, and $V' \cap V''$ is a subset of

the vertex set of C_1 . Then, C_4 is a subgraph of $C_1 \cup G''$ and has no vertices of C_2 . Since G satisfies the odd-cycle condition, there exists an edge e' of G joining C_4 and C_2 . Since $G'' \cup C_1$ and C_2 have no common vertices, e' is not an edge of $G'' \cup C_1$. Since C_4 and $C_2 \cup C_3$ have no common vertices, e' is not an edge of $C_2 \cup C_3$. Thus, e' is not an edge of $G = G'' \cup C_1 \cup C_2 \cup C_3$, which is a contradiction.

Thus, Cases 1 and 2 are not possible, and hence we have the desired conclusion. $\square$

4 Computational Experiments

We used existing software programs to enumerate all of the finite connected simple graphs G satisfying the conditions listed. We used the following procedure to find the candidates for normal edge polytopes with no regular unimodular triangulations:

- (Step 1) Generate all finite connected simple graphs by using `nauty` [7], and add them to the list of candidates.
- (Step 2) Exclude from the list of candidates every graph G that has a vertex of degree 1.
- (Step 3) Determine the normality of $\mathcal{P}_G$ by using `Normaliz` [3]. If $\mathcal{P}_G$ is not normal, exclude G from the list of candidates.
- (Step 4) Compute $\text{in}_w(I_G)$ with respect to a randomly generated weight vector w by using `Macaulay2` [5] and `4ti2` [1]. If $\text{in}_w(I_G)$ is squarefree, exclude G from the list of candidates.

Repeat Step 4 until a squarefree initial ideal has been found or until the number of iterations reaches 10,000.

4.1 Number of Vertices

First, we search for normal edge polytopes that have no regular unimodular triangulations and that have the minimum number of vertices. The graph shown in Fig. 2 has 9 vertices, a normal edge polytope, and no regular unimodular triangulations. Table 1 shows the number of graphs with 6 through 9 vertices and that also meet the following conditions:

1. they are finite connected simple graphs;
2. they have no vertices of degree 1;
3. their edge polytope is normal;
4. they have no regular unimodular triangulations (noRUT).

Table 1 Numbers of graphs with 6 through 9 vertices

Vertices	(1) #graphs	(2) #deg ≥ 2	(3) #normal	(4) #noRUT
6	112	61	61	0
7	853	507	501	0
8	11,117	7,442	7,281	0
9	261,080	197,772	190,647	1

Table 2 Numbers of graphs with up to 15 edges

Vertices	(1) #graphs	(2) #deg ≥ 2	(3) #normal	(4) #noRUT
10	112, 618	29, 001	24, 511	2
11	313, 670	27, 678	21, 634	0

4.2 Number of Edges

Next, we search for the normal edge polytopes with the minimum number of edges but that have no regular unimodular triangulations. Figure 1 shows graphs with 10 vertices and 15 edges that have a normal edge polytope and no regular unimodular triangulations. By the computational results presented in Sect. 4.1, we consider graphs that have $d > 9$ vertices (and $n \leq 15$ edges). Moreover, by Proposition 3.3, we can obtain the graphs that have $d \leq 11$ vertices, since $d + 4 \leq n \leq 15$. Table 2 shows the number of graphs with 10 or 11 vertices and ≤ 15 edges and that also meet the following conditions:

1. they are finite connected simple graphs;
2. they have no vertices of degree 1;
3. their edge polytope is normal;
4. they have no regular unimodular triangulations (noRUT).

4.3 Codimension

A normal edge polytope with no regular unimodular triangulations and that satisfies $n - d = 4$ was given in [9]; we wish to find the smallest such polytope. Figure 3 shows a graph with 17 vertices and 21 edges that has a normal edge polytope and that has no regular unimodular triangulations. By the results of Sect. 4.2, we may assume that $15 < n = d + 4$, and hence $d \geq 12$. Table 3 shows the number of graphs with $12 \leq d \leq 17$ vertices and that also meet the following conditions:

1. they are finite connected simple graphs;
2. they have no vertices of degree 1;
3. their edge polytope is normal;
4. they have no regular unimodular triangulations (noRUT).

Table 3 Numbers of graphs satisfying $n - d = 4$

Vertices	(1) #graphs	(2) #deg ≥ 2	(3) #normal	(4) #noRUT
12	1,694,642	79,764	58,636	0
13	8,610,789	208,958	145,345	0
14	41,613,518	506,066	334,282	0
15	19,282,712	1,146,003	722,868	0
16	862,572,395	2,451,133	1,481,501	0
17	3,743,001,968	4,987,833	2,901,658	1

4.4 Nonregular Unimodular Triangulations

It is known [9, 11] that the edge polytopes of the graph presented in Fig. 1 have a nonregular unimodular triangulation. In this section, we present a nonregular unimodular triangulation of the edge polytope of the graph in Fig. 3. We used the software *Normaliz* with the following input array:

```

1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0
1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0
0 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0
0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0
0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 1 1 0 0 0
0 0 0 0 1 0 0 0 0 0 0 0 0 1 0 0
0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0
1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 1
0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 1
0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 1;

```

we fortunately obtained an “almost unimodular” regular triangulation Δ of $\mathcal{P}_G$, namely, the normalized volume of the edge polytope is 186, and Δ consists of 185 maximal simplices for which the Stanley–Reisner ideal is generated by

$$\begin{aligned}
& x_3x_5x_7x_{10}x_{19}, \quad x_1x_8x_{11}x_{13}x_{20}, \quad x_1x_3x_5x_{13}x_{19}x_{20}, \quad x_1x_3x_{14}x_{16}x_{20}, \quad x_3x_5x_9x_{14}x_{20}, \\
& x_1x_6x_8x_{11}x_{14}x_{16}x_{18}x_{20}, \quad x_3x_{10}x_{12}x_{14}x_{16}, \quad x_1x_6x_8x_{15}x_{16}, \quad x_1x_3x_{10}x_{15}x_{16}x_{19}, \\
& \underline{x_1x_4x_6x_8x_{11}x_{13}x_{15}}, \quad x_1x_8x_{13}x_{15}x_{19}, \quad x_7x_{10}x_{15}x_{19}x_{21}, \quad x_5x_9x_{11}x_{13}x_{17}x_{20}, \quad x_5x_7x_{13}x_{17}x_{19}, \\
& x_4x_7x_8x_{13}x_{15}x_{17}x_{19}x_{21}.
\end{aligned}$$

We then obtained a nonregular unimodular triangulation Δ' of $\mathcal{P}_G$ by using a geometric bistellar flip supported on a circuit,

$$x_2x_3x_5x_7x_9x_{10}x_{12}x_{14} - x_1x_4x_6x_8^2x_{11}x_{13}x_{15},$$

for which the Stanley–Reisner ideal is generated by

$$\begin{aligned} & x_3x_5x_7x_{10}x_{19}, x_1x_8x_{11}x_{13}x_{20}, x_1x_3x_5x_{13}x_{19}x_{20}, x_1x_3x_{14}x_{16}x_{20}, x_3x_5x_9x_{14}x_{20}, \\ & x_1x_6x_8x_{11}x_{14}x_{16}x_{18}x_{20}, x_3x_{10}x_{12}x_{14}x_{16}, x_1x_6x_8x_{15}x_{16}, x_1x_3x_{10}x_{15}x_{16}x_{19}, \\ & \underline{x_2x_3x_5x_7x_9x_{10}x_{12}x_{14}}, x_1x_8x_{13}x_{15}x_{19}, x_7x_{10}x_{15}x_{19}x_{21}, x_5x_9x_{11}x_{13}x_{17}x_{20}, x_5x_7x_{13}x_{17}x_{19}, \\ & x_4x_7x_8x_{13}x_{15}x_{17}x_{19}x_{21}. \end{aligned}$$

It remains to be determined whether the edge polytope of the graph in Fig. 2 has a nonregular unimodular triangulation.

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Homological Conjectures and Lim Cohen-Macaulay Sequences

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Abstract We discuss the new notion of a lim Cohen-Macaulay sequence of modules over a local ring, and also a somewhat weaker notion, as well as the theory of content for local cohomology modules. We relate both to the problem of proving the direct summand conjecture and other homological conjectures without using almost ring theory and perfectoid space theory, and we also indicate some other open problems whose solution would yield a new proof of the direct summand conjecture.

1 Introduction

This paper is an expanded version of material from a lecture given by the author at the conference *Homological and Computational Methods in Commutative Algebra*, honoring the seventieth birthday of Winfried Bruns, in Cortona, Italy on June 2, 2016.

Throughout, all rings are commutative, associative, with identity, and homomorphisms are assumed to preserve the identity. By a *local ring* $(R, \mathfrak{m}, K)$ we mean a Noetherian ring R with a unique maximal ideal $\mathfrak{m}$ and residue class field K .

Since the talk on which this paper is based was given, the direct summand conjecture has been proved: see [1, 2, 12]. André's paper [2] includes a proof of the existence of big Cohen-Macaulay algebras, and contains the assertion that one can prove a weakly functorial version of the existence of big Cohen-Macaulay algebras for injective local maps of complete local domains, but does not give details. Bhatt's paper [12] gives a short proof of both the direct summand conjecture and a derived form of the conjecture introduced by de Jong. Bhatt's approach uses part of André's work, but is also substantially different: Bhatt utilizes a quantitative form of Scholze's Hebbbarkeitssatz (a perfectoid version of the Riemann extension theorem) instead of André's generalization (the perfectoid Abhyankar lemma) of the almost purity theorem due to Faltings. Moreover, quite recently, Heitmann and Ma [38] have proved a weakly functorial form of the existence of big

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Cohen-Macaulay algebras in mixed characteristic (valid for certain surjections of local rings both of mixed characteristic) that is sufficient to prove the vanishing conjecture for maps of Tor when both rings have mixed characteristic. See also [60]. This yields another proof of the direct summand conjecture that also depends on perfectoid geometry techniques. We discuss some conjectures involving big Cohen-Macaulay algebras further in Sect. 13.

Despite all these exciting events, it would be desirable to have a proof of the direct summand conjecture that does not depend on almost ring theory and perfectoid space theory.

We shall describe here nine conjectures each of which is stronger than the direct summand conjecture. There is a recap of these conjectures in Sect. 13. Despite the breakthroughs made in [1, 2, 12, 38], most of these conjectures remain open, at least in some cases. Among these, we emphasize the conjecture that complete local domains have $\lim$ Cohen-Macaulay sequences, which are introduced in [13] and defined in this manuscript in Sect. 5. In [13] it is shown that $\lim$ Cohen-Macaulay sequences exist in characteristic $p > 0$. It is also shown there that the existence of $\lim$ Cohen-Macaulay sequences in mixed characteristic implies the general case of the positivity of Serre intersection multiplicities, a question that has been open for over 50 years. The new developments using almost ring theory and perfectoid geometry as yet do not seem to help with Serre's multiplicity conjecture.

Although we emphasize here the theory of $\lim$ Cohen-Macaulay sequences, we also describe approaches that involve the theory of content of local cohomology modules and Q -sequences (see Sect. 9) developed in [52, 56], as well as a conjecture that involves tight closure theory in positive characteristic in Sect. 11.

For further background on the direct summand conjecture and related local homological conjectures we refer the reader to [3, 6, 9–11, 22–30, 35–37, 39–41, 44–47, 49, 51, 52, 58, 63–65, 69–73]. Basic references for almost ring theory are [31, 33] and [74] is a basic reference for the theory of perfectoid spaces. For treatments of other notions from commutative algebra not explained explicitly here, we refer the reader to [18, 19, 62], and [75].

2 Big and Small Cohen-Macaulay Modules and Algebras and Homological Conjectures: A First Glance, and Some History

The existence of big Cohen-Macaulay modules (defined below) implies certain local homological conjectures, but not Serre's conjecture on multiplicities. The existence of small Cohen-Macaulay modules also implies Serre's conjecture. Hence, small Cohen-Macaulay modules are better.

But big Cohen-Macaulay modules and algebras are now known to exist in equal and, recently, mixed characteristic [1, 2, 38], while not much is known about existence of small Cohen-Macaulay modules.

We discuss a new notion, introduced in a joint paper [13] with Bhargav Bhatt and Linquan Ma: *lim Cohen-Macaulay sequences*. The existence of these for complete local domains implies both the existence of big Cohen-Macaulay modules and Serre's conjecture on multiplicities. And they do exist in positive characteristic!

We also discuss an alternate notion of *lim Cohen-Macaulay sequences*. The requirements are weaker, but using ideas related to the theory of *content* of local cohomology modules, developed in [52, 56], the existence of the *lim Cohen-Macaulay sequences* in this weakened sense is sufficient to deduce the direct summand conjecture.

If R is a domain, $\text{frac}(R) = \mathcal{F}$ and M is an R -module, $\text{rank}(M) = \dim_{\mathcal{F}} \mathcal{F} \otimes_R M$, the *torsion-free rank* of M . If $(R, \mathfrak{m}, K)$ is local and M is finitely generated, $\nu(M) := \dim_K(K \otimes_R M)$, the least number of generators of M as an R -module. $\ell(H)$ is the length of the finite length module H . If R is a local domain, $\text{rank}(M) \leq \nu(M)$.

A *big Cohen-Macaulay module* over a local ring $(R, \mathfrak{m}, K)$ is a (not necessarily finitely generated) module M such that $\mathfrak{m}M \neq M$ and every system of parameters for R is a regular sequence on M . (In many places in the literature, the term *balanced big Cohen-Macaulay module* is used for this notion. But note that, by the results of [5], if one system of parameters for R is a regular sequence on M , then the $\mathfrak{m}$ -adic completion of M is a balanced big Cohen-Macaulay module.) M is called a *big Cohen-Macaulay algebra* if it also happens to be an R -algebra.

If M is a finitely generated R -module, then M is a big Cohen-Macaulay module for R iff $M \neq 0$ and one system of parameters is a regular sequence on M . In this case, M is called a *small Cohen-Macaulay module* for R .

Big Cohen-Macaulay modules (and algebras) were first shown to exist in equal characteristic and if $\dim(R) \leq 3$. This was shown for modules in [41], and for algebras in [46], and by other methods later in [49, 51]. The existence of big Cohen-Macaulay algebras in dimension 3 in mixed characteristic is shown in [47], based on the results of [37]. The latter results are improved in [36]. These results were obtained without perfectoid geometry. The existence of big Cohen-Macaulay algebras in general [1, 2, 38] was proved recently using perfectoid methods.

Small Cohen-Macaulay modules are only known to exist if $\dim(R) \leq 2$ (this is easy: if R is a complete domain of dimension at most 2, the integral closure is Cohen-Macaulay) or if R is $\mathbb{N}$ -graded over a perfect field of characteristic p and has an isolated non-Cohen-Macaulay point at the origin: this was first observed by R. Hartshorne, and rediscovered independently by Peskine and Szpiro. A proof is given in [42]. While there are a few other special results (see, for example, [55]), there has not been much progress in 40 years!

In the early 1970s I conjectured:

Conjecture 1 Every complete local domain has a small Cohen-Macaulay module.

In the 2000s I conjectured:

Conjecture 2 There are complete local domains that do not have a small Cohen-Macaulay module.

One of these conjectures is bound to be correct.

A result that clarifies what one may hope for is given by Bhatt in [10], where it is shown that small Cohen-Macaulay algebras do not exist in general.

The existence of big Cohen-Macaulay modules suffices to prove many of the local homological conjectures formulated by Maurice Auslander, Hy Bass, and, later by C. Peskine and L. Szpiro. See [3, 6, 63, 64]. Peskine and Szpiro proved a number of these conjectures in characteristic $p > 0$ and in certain cases in equal characteristic 0. I added the direct summand conjecture (regular local rings are direct summands of their module-finite extension rings) and some others [40, 45], and showed that big Cohen-Macaulay modules exist in equal characteristic [41], thus settling a number of these conjectures in the general equicharacteristic case. Paul Roberts proved the new intersection theorem in general (i.e., mixed characteristic) [72, 73], using the theory of local Chern classes in all characteristics developed by Fulton et al. [7], for which there is an excellent exposition in [32]. Ray Heitmann proved the direct summand conjecture in $\dim 3$ [37], and I used his main result to show that big Cohen-Macaulay algebras exist in dimension 3 in mixed characteristic [47]. His results in [36, 37] show that if (R, m) is a complete local domain of dimension 3 and R^+ denotes its *absolute integral closure*, i.e., its integral closure in an algebraic closure of its fraction field, then every element of the maximal of R^+ kills $H_m^2(R^+)$.

The direct summand conjecture implies the canonical element conjecture [45]. The point of view of [45] enables one to show that the direct summand conjecture reduces to the case of a formal power series ring over an unramified mixed characteristic discrete valuation ring. The direct summand conjecture also implies a very powerful form of the new intersection theorem that we refer to here as the *strong intersection theorem* (it has also been called the improved new intersection theorem) that is discussed in the next section, and many other results. The interactions of these various conjectures, all of which are now theorems, are studied in [24, 45].

3 The Strong Intersection Theorem, the Syzygy Problem, and the Generalized Principal Ideal Theorem

In this section and throughout the rest of this paper we often assume that given local rings (R, m, K) are complete local domains, which suffices in all of the applications we have in mind.

For simplicity we also freely assume that K is perfect: while this is not always needed, this case likewise suffices for applications.

The results of [45] show that the direct summand conjecture is equivalent to the canonical element conjecture, and that these imply the following statement, the strong intersection conjecture, stated as Theorem 1 below. All of these are now theorems. Evans and Griffith [30] used the strong intersection conjecture, which they proved in the equicharacteristic case using big Cohen-Macaulay modules,

to prove their syzygy theorem in the equicharacteristic case, which is stated as Theorem 2 below. Sankar Dutta proved that the strong intersection conjecture implies the direct summand conjecture, so that the equivalence of the direct summand conjecture, the canonical element conjecture and the strong intersection conjecture has been known for several decades.

Theorem 1 *If $(R, \mathfrak{m}, K)$ is local and $0 \rightarrow G_s \rightarrow \cdots \rightarrow G_0 \rightarrow 0$ is a complex of finitely generated free modules such that $H_0(G_\bullet)$ has a minimal generator that is killed by a power of $\mathfrak{m}$ and $H_i(G_\bullet)$ has finite length for $i \geq 1$, then $d = \dim(R) \leq s$.*

The strong intersection theorem was proved by Graham Evans and Phil Griffith in equal characteristic in their proof of the syzygy conjecture.

Theorem 2 (Evans-Griffith Syzygy Theorem) *Let $(R, \mathfrak{m})$ be a local catenary domain. Let M be a non-free S_k -module of rank r and finite projective dimension. Then r is greater than or equal to k . In particular, over a regular local ring, every non-free k th module of syzygies has rank at least k .*

It is worth noting that Evans and Griffith comment in [30] that their result, Theorem 2 above, is a sort of converse to one of the beautiful results of Winfried Bruns in [16]. They referred specifically to Bruns's result that if M is a k th syzygy of rank $k + r$ with $r > 0$, then M has an R -free submodule G of rank r such that M/G is a k th syzygy of rank k .

We note that Eisenbud and Evans proved the following result in equal characteristic using the existence of big Cohen-Macaulay modules: the idea used is strongly related to the argument in [30], and only depends on knowing the strong intersection theorem. In Theorem 3 below, with $\kappa_{\mathfrak{p}} := R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$,

$$\text{rank}(M) := \max \{ \dim_{\kappa_{\mathfrak{p}}} \kappa_{\mathfrak{p}} \otimes_R M : \mathfrak{p} \text{ is a minimal prime of } R \}.$$

Theorem 3 (Generalized Principal Ideal Theorem) *Let R be a noetherian ring, M a finitely generated R -module, and $x \in M$. If there is a prime ideal $\mathfrak{p}$ of R with $x \in \mathfrak{p}M_{\mathfrak{p}}$, and I is the ideal of $\{f(x) : f \in \text{Hom}_R(M, R)\}$, then $\text{height}(I) \leq \text{rank}(M)$.*

Bruns gave a beautiful proof in [17] that Theorem 3 holds without the restriction to equal characteristic. Of course, the fact that the direct summand conjecture is now known also gives a proof of the general case, but the argument is much more difficult than the one in [17] when one considers the underlying machinery.

4 Serre Multiplicities

Let $(T, \mathfrak{n}, K)$ be a regular local ring of dimension n and let M, N be nonzero finitely generated modules such that $\ell(M \otimes_T N) < \infty$, i.e., $\text{Supp}(M) \cap \text{Supp}(N) = \{\mathfrak{m}\}$. Here, Serre's intersection multiplicity [75] is defined by the formula

$$\chi^T(M, N) = \chi(M, N) := \sum_{i=0}^n (-1)^i \ell(\text{Tor}_i^T(M, N)).$$

$\chi(M, N)$ is bi-additive in M and N when it is defined on all the pairs occurring. Since M and N have finite filtrations in which all factors are prime cyclic modules T/P , the behavior of χ is determined by its behavior on pairs of such modules T/P , T/Q , where P, Q are prime and $P + Q$ is $\mathfrak{m}$ -primary.

This is a formal situation analogous to studying the intersection of two varieties near an isolated point of intersection.

In equal characteristic, T is regular iff its completion is a formal power series ring over a field. In mixed characteristic, it may be formal power series over a complete DVR (V, pV) (like the p -adic numbers) whose maximal ideal is generated by the characteristic p of the residue class field.

But there is a frequently more difficult ramified case where the ring has the form

$$V[[X_1, \dots, X_d]]/(p - F).$$

Here F is in the square of the maximal ideal. Such a ring, in general, is regular but *not* a formal power series ring over a DVR. For a specific example of this type, consider $V[[X, Y, Z]]/(p - X^3 - Y^5 - Z^7)$.

Serre proved [75] that if T is regular local and its completion is formal power series over a field or a DVR, then the following hold for finitely generated nonzero modules M, N when $\ell(M \otimes_R N) < \infty$ (think of the case where $M = T/P$, $N = T/Q$):

- (a) $\dim(M) + \dim(N) \leq \dim(T)$ (dimension inequality).
- (b) $\dim(M) + \dim(N) < \dim(T) \Rightarrow \chi(M, N) = 0$ (vanishing).
- (c) $\dim(M) + \dim(N) = \dim(T) \Rightarrow \chi(M, N) > 0$ (positivity).

These results are highly non-trivial and very powerful. For example, part (a) implies that the height of a prime ideal in a regular ring cannot increase if it stays proper when you map to another ring, which greatly generalizes the well-known results of Eagon and Northcott on heights of determinantal ideals.

Serre also proved (a) $\dim(M) + \dim(N) \leq \dim(T)$ for any regular local ring T , and, essentially, conjectured (“Il est naturel de conjecturer”) that (b) and (c) hold as well. The remaining case is the ramified case in mixed characteristic. It has been an open question for over 50 years. Serre also proved the case when either M or N is $T/(f_1, \dots, f_h)T$ where $f_1, \dots, f_h$ is a regular sequence, i.e. the case where one of the modules is a complete intersection.

Serre’s results inspired a great deal of additional work on multiplicities defined by similar methods in this and even greater generality. We refer the reader to [22, 23, 25–29, 64] as well as to several other papers cited explicitly just below.

Paul Roberts and, independently, H. Gillet and C. Soulé, proved

- (b) $\dim(M) + \dim(N) < \dim(T) \Rightarrow \chi(M, N) = 0$.

See [71] and [34]. A new, shorter proof of Serre vanishing has been given recently in [15].

O. Gabber, using de Jong’s results on alterations [20], proved that $\chi(M, N) \geq 0$ in the ramified case. Gabber’s argument is given in [8].

All that remains is to show that $\chi(M, N)$ is strictly positive in the ramified case. This remains open. One may assume that T is complete with a perfect (or even algebraically closed) residue field. Also, it suffices to prove the result when $M = T/P$ and $N = T/Q$ are prime cyclic modules.

An important point here is that $\chi(T/P, T/Q)$ is positive when T/P and T/Q both have small Cohen-Macaulay modules. Therefore, the existence of small Cohen-Macaulay modules implies the remaining case of Serre's conjecture. In fact, this idea can be used to settle Serre's conjecture on positivity of multiplicities up through dimension 4.

We give the argument. Keep in mind that $P+Q$ is $\mathfrak{m}$ -primary, and that we have that $\dim(T/P) + \dim(T/Q) = \dim(T)$. Suppose M is a small Cohen-Macaulay module for $R = T/P$ of (torsion-free) rank r and that N is a small Cohen-Macaulay module for $S = T/Q$ of rank s . M has a finite filtration with r factors equal to R and other factors of smaller dimension. N has a finite filtration with s factors equal to S and other factors of smaller dimension. Using the bi-additivity of χ and the fact that the vanishing part (b) of the conjecture is known, one obtains that $\chi(M, N) = rs \chi(R, S)$. But when M, N are Cohen-Macaulay and so $\text{Tor}_i(M, N) = 0$ for $i > 0$ higher Tors vanish. Hence, $\chi(M, N) = \ell(M \otimes_T N) > 0 \Rightarrow \chi(R, S) > 0$. $\square$

5 Lim Cohen-Macaulay Sequences of Modules

One of the main points here is that the idea in the proof given at the end of the preceding section can be made to work with a much weaker assumption than the existence of small Cohen-Macaulay modules for the complete local domains R and S . We want to introduce the notion of a sequence of modules in which the terms are getting "closer and closer" to being Cohen-Macaulay, compared to their torsion-free ranks.

Let $(R, \mathfrak{m}, K)$ be a local ring. If $\underline{x} = x_1, \dots, x_d \in R$ let $H_\bullet(\underline{x}; M)$ denote Koszul homology, and let $h_i(\underline{x}; M)$ denote its length when the length is finite.

The following notations will be convenient. If $f, g : \mathbb{N}_1 \rightarrow (0, \infty)$ where $\mathbb{N}_1 \subseteq \mathbb{N}$ contains all $n \gg 0$, $f(n) = O(g(n))$ if $f(n)/g(n)$ is bounded and $f(n) = o(g(n))$ if $f(n)/g(n) \rightarrow 0$ as $n \rightarrow \infty$.

Let $(R, \mathfrak{m}, K)$ be, in addition, a domain. A sequence of nonzero finitely generated modules $\mathcal{M} = \{M_n\}_n$ of dimension $\dim(R)$ over R is defined to be *lim Cohen-Macaulay* if for some (equivalently, every) system of parameters $\underline{x} = x_1, \dots, x_d$, for $1 \leq i \leq d$ we have

$$h_i(\underline{x}; M_n) = o(v(M_n)).$$

It is not obvious, but it is true that the condition is independent of the choice of parameters $\underline{x}$.

If R is a local domain, and $\{M_n\}_n$ is lim Cohen-Macaulay, then

$$1 \leq \frac{v(M_n)}{\text{rank}(M_n)} \leq e(R) + \epsilon_n$$

where $e(R)$ = the multiplicity of R and $\epsilon_n \rightarrow 0$ as $n \rightarrow \infty$. Hence, $\{M_n\}_n$ is lim Cohen-Macaulay iff for some (equivalently, every) system of parameters $\underline{x}$, $h_i(\underline{x}; M_n) = o(\text{rank}(M_n))$, $1 \leq i \leq d$. Also, one can prove that $h_0(\underline{x}; M_n) = O(\text{rank}(M_n))$.

Note that if R has a small Cohen-Macaulay module M , we may take the sequence to be the constant sequence $M, M, M, \dots, M, \dots$.

We note the following result from [13].

Theorem 4 *If $(R, \mathfrak{m}, K)$ is a complete local domain of characteristic $p > 0$ with K perfect, then $M_n := R^{1/p^n}$ is a lim Cohen-Macaulay sequence for R .*

This follows from standard results in tight closure theory, and also can be deduced from the results of [22]. The same holds if R is F -finite (i.e., the Frobenius map $F : R \rightarrow R$ is module-finite.) This is a vast improvement over what we know about existence of small Cohen-Macaulay modules.

We note the following two facts:

If $R \rightarrow S$ is local and module-finite, a sequence of nonzero S -modules is lim Cohen-Macaulay over S if and only if it is lim Cohen-Macaulay over R .

If R is a regular local, a sequence $\{M_n\}_n$ of R -modules is lim Cohen-Macaulay if and only if all higher Betti numbers of M_n are quite small, in the sense of $o(_)$, compared to the 0th Betti number, which is $v(M_n)$.

Conjecture 3 Every complete local domain of mixed characteristic with algebraically closed residue class field has a lim Cohen-Macaulay sequence of modules.

We shall see in the next section, Theorem 5 and Corollary 1, that this conjecture is sufficient to prove the Serre conjecture on positivity of multiplicities, and that it also implies the existence of big Cohen-Macaulay modules (hence, also, the direct summand conjecture) in mixed characteristic: see Theorem 9.

We conclude this section by mentioning a recent application of the lim Cohen-Macaulay sequence idea in a different context: to settle Lech's conjecture in equal characteristic in dimension 3. We first recall that over 50 years ago, Lech made the following:

Conjecture 4 (Lech) If $(R, \mathfrak{m}) \rightarrow (S, \mathfrak{n})$ is a flat local extension of local rings, then $e(R) \leq e(S)$.

The case where $\dim(R) = \dim(S)$ and R, S are complete implies the general case. Lech proved the result in dimension at most 2. Recently, Linquan Ma [59] used the fact that R^{1/p^n} is a lim Cohen-Macaulay sequence as well as other ideas to prove the case where R is of dimension 3 and characteristic $p > 0$, and this implies the case where R has dimension 3 and contains a field, by reduction to characteristic $p > 0$.

6 Lim Cohen-Macaulay Sequences and the Positivity of Serre Multiplicities

We note the following result from [13]:

Theorem 5 *Let P and Q be primes of a complete regular local ring $(T, \mathfrak{n}, K)$, let $R := T/P$ and $S := T/Q$, and suppose that $P + Q$ is $\mathfrak{n}$ -primary and that*

$$\dim(R) + \dim(S) = \dim(T).$$

If R and S have lim Cohen-Macaulay sequences, say $\{M_n\}_n$ and $\{N_n\}_n$, respectively, then $\chi(R, S) > 0$. In fact, we have

$$\chi(R, S) = \lim_{n \rightarrow \infty} \frac{\ell(M_n \otimes_T N_n)}{\text{rank}(M_n)\text{rank}(N_n)} \geq \frac{v(M_n)}{\text{rank}(M_n)} \frac{v(N_n)}{\text{rank}(N_n)} \geq 1.$$

Corollary 1 *If all complete local mixed characteristic domains with perfect (or even algebraically closed) residue class fields have lim Cohen-Macaulay sequences of modules, then Serre's conjecture on multiplicities is true in general.*

This case gives the general result by using standard reductions in the problem.

To give some idea of the proof of Theorem 5, we need to define multiple Tor over a ring T , i.e., Tor applied to a sequence of inputs, not just two. The inputs are left complexes of flat T -modules or simply T -modules.

If the input is a module, replace it by a projective (or flat) resolution: the choice won't matter. Multiple Tor is the homology of the total tensor product of these complexes. This agrees with the usual definition of $\text{Tor}_\bullet^T(M, N)$ for the case of two modules. By viewing the total complex as a double complex in various ways, one gets a number of spectral sequences.

Theorem 5 follows if $\ell(\text{Tor}_i^T(M_n, N_n)) = o(\text{rank}(M_n)\text{rank}(N_n))$ for $i \geq 1$. One can choose a system of parameters $\underline{x}, \underline{y}$ for T such that the $x_i \in Q$ and their images form a system of parameters in $R = T/P$ and the y_j are in P and their images form a system of parameters in $S = T/Q$. One uses a spectral sequence argument to show that $\Theta_n := \text{Tor}_{i+d}^T(T/(\underline{x}, \underline{y}), M_n, N_n)$ maps onto $\text{Tor}_i^T(M_n, N_n)$. $\Theta_n \cong \text{Tor}_{i+d}^T(T/(\underline{x}), T/(\underline{y}), M_n, N_n)$ (quadruple Tor) $\cong \text{Tor}_{i+d}^T(T/(\underline{x}), M_n, T/(\underline{y}), N_n)$.

By grouping together the first two inputs and the last two, one gets, using two spectral sequences, that the length of Θ_n is bounded by the sum of the lengths of the $\text{Tor}_r^T(H_s(\underline{x}; M_n), H_t(\underline{y}; N_n))$ for $r + s + t = i + d$. If $r > d$ the Tor vanishes. If $r \leq d$ then $s + t \geq i > 0$. The respective inputs to Tor_r^T are at worst $O(\text{rank}(M_n))$ and $O(\text{rank}(N_n))$ and at least one of s, t is positive, so at least one of these can be replaced by o . It follows that Θ_n is $o(\text{rank}(M_n)\text{rank}(N_n))$. $\square$

7 Lim Cohen-Macaulay Sequences Yield Closure Operations

We shall use lim Cohen-Macaulay sequences over a local ring $(R, \mathfrak{m}, K)$ to define a closure operation $\natural$ on submodules of finitely generated modules. The good properties of this closure operation enable one to show that if a local ring has a lim Cohen-Macaulay sequence, then it is a big Cohen-Macaulay module. Moreover, one can get many other closure operations from sequences of finitely generated modules (or even nets of modules: the index set is a directed set, not necessarily the positive integers) over any local ring.

Let $\mathcal{M}$ denote $\{M_n\}_n$, a sequence of nonzero modules over a local ring $(R, \mathfrak{m}, K)$. Let α be a function from the set of modules $\{M_n : n \in \mathbb{N}\}$ to $\mathbb{N}$, the nonnegative integers. Our main interest is in the cases where α is ν (number of generators) or torsion-free rank. For each such α we define a closure operation on submodules of finitely generated R -modules, which we refer to as $\mathcal{M}$ -closure with respect to α . If $A \subseteq B$ are R -modules, we use the notation $A_B^\natural$ for the $\mathcal{M}$ -closure of A in B with respect to α .

These closure operations on submodules A of finitely generated R -modules B are defined as follows. If B/A has finite length, we define the closure $A_B^\natural$ to be the largest submodule A' of B containing A such that

$$\begin{aligned} (\dagger_\alpha) \quad \ell(\operatorname{Im}(M_n \otimes_R A'/A \rightarrow M_n \otimes_R B/A)) \\ = o(\alpha(M_n)). \end{aligned}$$

In general, we define the $\mathcal{M}$ -closure of A in B with respect to α as

$$\bigcap_{i=1}^{\infty} (A + \mathfrak{m}^i B)_B^\natural.$$

This gives the same result as the original definition if B/A has finite length.

The subscript B is often omitted if B is clear from context. In discussing ideals of R , $I^\natural$ is always $I_R^\natural$.

Example If R is a local Noetherian domain, consider the net of all nonzero ideals, where $I \leq J$ means that $J = II'$ for some nonzero ideal I' . The closure with respect to rank is integral closure.

Theorem 6 *Let $(R, \mathfrak{m}, K)$ be a local domain and let $\mathcal{M} = \{M_n\}_n$ be a sequence of finitely generated R -modules that are not torsion. Suppose that $\widehat{R}$ is equidimensional, which is true if R is excellent. Then for every ideal I of R , the closure of I with respect to $\mathcal{M}$ and rank is contained in $\overline{I}$, the integral closure of I .*

Equivalently, integrally closed ideals of I are $\mathcal{M}$ -closed with respect to rank. Hence, radical ideals are $\mathcal{M}$ -closed with respect to rank, and so prime ideals are

$\mathcal{M}$ -closed with respect to rank. In particular, 0 and $\mathfrak{m}$ are $\mathcal{M}$ -closed in R with respect to rank.

From now on we discuss only closures coming from lim Cohen-Macaulay sequences with respect to ν . If the ring is a domain, this is the same as closure with respect to rank. Here is one rather trivial example.

Example If R is Cohen-Macaulay, the closure coming from the lim Cohen-Macaulay sequence $R, R, R, \dots$ and ν (or rank) is the identity closure. (Every submodule of every module is closed.)

The lim Cohen-Macaulay sequences discussed in Theorem 4 give tight closure:

Theorem 7 *If R is a local domain of characteristic $p > 0$, then the closure operation given by the lim Cohen-Macaulay sequence $\{R^{1/p^n}\}_n$ (i.e., $\{F_*^n(R)\}_n$) is tight closure.*

This can be deduced from length characterizations of tight closure in Sect. 8 of [48].

All lim Cohen-Macaulay closures $^\natural$ over a complete local domain R satisfy colon-capturing: if $x_1, \dots, x_d$ is a system of parameters then if $I = (x_1, \dots, x_k)R$, $I :_R x_{k+1} \subseteq I^\natural$. Better: $I_R^\natural :_R x_{k+1} \subseteq I^\natural$.

Also, if $s \geq 1$, $(x_1, \dots, x_{k-1}, x_k^{s+1})^\natural :_R x_k \subseteq (x_1, x_2, \dots, x_{k-1}, x_k^s)^\natural$.

Moreover, lim Cohen-Macaulay closures satisfy the somewhat mysterious Dietz colon-capturing axiom from [21], which is axiom (7) on the list of Dietz axioms at the beginning of the next section.

8 The Dietz Axioms

Let $(R, \mathfrak{m})$ be a fixed complete local domain. Let $^\natural$ denote a closure operation over R chart assigns to every R -submodule A of a finitely generated R -module B a submodule $A_B^\natural$ of B . Let A, B , and C be arbitrary finitely generated R -modules with $A \subseteq B$. By the *Dietz axioms* we mean the following seven conditions on $^\natural$.

- (1) $A_B^\natural$ is a submodule of B containing A .
- (2) $(A_B^\natural)_B^\natural = A_B^\natural$.
- (3) If $A \subseteq B \subseteq C$, then $A_C^\natural \subseteq B_C^\natural$.
- (4) Let $f : B \rightarrow C$ be an R -linear homomorphism. Then $f(A_B^\natural) \subseteq f(A)_C^\natural$.
- (5) If $A_B^\natural = A$, then $0_{B/A}^\natural = 0$.
- (6) The ideals $\mathfrak{m}$ and 0 are $^\natural$ -closed in R .
- (7) Let $x_1, \dots, x_{k+1}$ be part of a system of parameters for R , and let $J = (x_1, \dots, x_k)R$. Suppose that one has a surjective map $f : B \twoheadrightarrow R/J$ and $v \in B$ such that $f(v) = x_{k+1} + J \in R/J$. Then $(Rv)_B^\natural \cap \text{Ker}(f) \subseteq (Jv)_B^\natural$.

Theorem 8 (Dietz) *A complete local domain R has a big Cohen-Macaulay module if and only if it has a closure operation on submodules of finitely generated modules satisfying axioms (1)–(7) above.*

In [13] it is proved that the closure coming from a lim Cohen-Macaulay sequence of modules is a Dietz closure, i.e., it satisfies the Dietz axioms. Hence:

Theorem 9 *If a local ring R has a lim Cohen-Macaulay sequence of modules, then R has a big Cohen-Macaulay module.*

Of course, this also means that proving the existence of lim Cohen-Macaulay sequences of modules in mixed characteristic would give a new proof of the direct summand conjecture.

9 The Direct Summand Conjecture and Q-Sequences

Let M be a finitely generated R -module killed by a power of an ideal I of a ring R . We define the *quasileneth* $\mathcal{L}_I(M)$ of M with respect I to be the length of a shortest finite filtration of M in which each factor is a homomorphic image of R/I , i.e., a cyclic R -module annihilated by I . This notion is introduced and studied in [52], and the theory is developed further in [56].

If I is an ideal of R generated by d elements, say $\underline{x} = x_1, \dots, x_d$, let $I_t = (x_1^t, \dots, x_d^t)R$. The *content* of a d th local cohomology module $H_I^d(M)$ with respect to a specific sequence of d generators $\underline{x} = x_1, \dots, x_d$ of I is defined as

$$\lim_{t \rightarrow \infty} \frac{\mathcal{L}_I(M/I_t M)}{t^d},$$

if it exists. Evidently, the order of the elements in the sequence does not matter.

We are primarily interested here in the case where $M = R$. Since R/I_t always has a filtration by t^d monomial ideals in which the factors are homomorphic images of R/I , the content of $H_I^d(R)$, if it exists, is always in the unit interval $[0, 1]$. It is shown in [52] that whether the content of the local cohomology $H_I^d(R)$ is 1 is independent of the choice of d generators for I . In this case, the local cohomology $H_I^d(R)$ is called *robust*, and the elements $x_1, \dots, x_d$ are called a *Q-sequence* in R . Alternatively, we may simply say that $\underline{x}$ is a *Q-sequence* if for all positive integers t , every finite filtration of R/I_t with factors that are cyclic modules that are homomorphic images of R/I has at least t^d terms.

A regular sequence in a Noetherian ring is a Q-sequence. Surprisingly, it is not known whether a regular sequence in a ring that is not Noetherian must be a Q-sequence, even in the case where $d = 2$. In [52], it is shown that a system of parameters for an equicharacteristic local ring is a Q-sequence. This is not known in mixed characteristic, even in dimension 3, although it is known in mixed characteristic that the content of the d th local cohomology of a local ring of dimension d with respect to a system of parameters exists and is positive.

Conjecture 5 Every system of parameters of every local ring is a Q-sequence.

It is shown in [52] that this conjecture implies the direct summand conjecture. Conjecture 5 remains open.

In [56], the notion of a *robust* algebra over a local ring R is studied: S is defined to be *robust* if every system of parameters for R is a Q-sequence in S . This leads to a notion of closure for ideals: J is closed in the sense of robust closure if whenever an element $u \in R$ is such that $u \in JS$ for some robust R -algebra S , the element u is in J . The robust closure of an ideal is defined to be the smallest ideal containing it that is closed in the sense of robust closure. It is shown in [52] that this closure agrees with tight closure for complete local domains of characteristic p . It remains an open question whether this notion gives a valuable generalization of tight closure in the mixed characteristic case, since we do not even know that a complete local domain is robust as an algebra over itself: this would follow from Conjecture 5.

Note that a number of other closures that might play, in mixed characteristic, a role similar to the one played by tight closure in equal characteristic are studied in [43, 45] and [54].

10 A Variant Notion of Lim Cohen-Macaulay Sequences

Let $e_I(M)$ denote the multiplicity of M with respect to the $\mathfrak{m}$ -primary ideal I . A finitely generated R -module of dimension $d = \dim(R)$ is Cohen-Macaulay if and only if for one (equivalently, every) system of parameters $\underline{x} = x_1, \dots, x_d$ or R we have

$$\ell(M/(\underline{x})M) = e_{(\underline{x})}(M).$$

The definition we have given for a lim Cohen-Macaulay sequence implies that for any given system of parameters $\underline{x} = x_1, \dots, x_d$ for the local ring R ,

$$(\dagger) \quad \lim_{n \rightarrow \infty} \frac{\ell(M_n/(\underline{x})M_n)}{e_{(\underline{x})}(M_n)} = 1.$$

When one has a sequence of modules of dimension equal to that of R that satisfies this weaker condition for a system of parameters $\underline{x}$, one can show that the monomial conjecture holds for these parameters. That is, for all positive integers t , one has

$$(*) \quad (x_1 \cdots x_d)^{t-1} \notin (x_1^t, \dots, x_d^t)R.$$

To see why, let $I_s = (x_1^s, \dots, x_d^s)R$. Note that contradicting $(*)$ gives a filtration of R/I_t , constructed using ideals generated by monomials, that has length $t^d - 1$ instead of t^d . The fact that

$$(x_1 \cdots x_d)^{t-1} \in I_t$$

enables one to drop one factor from the obvious filtration. Each factor is a homomorphic image of R/I , where $I = I_1 = (x_1, \dots, x_d)R$.

This implies that for every s , R/I_s has a filtration by $(t^d - 1)s^d$ monomial ideals, where each factor is a homomorphic image of R/I .

One first filters with s^d homomorphic images of R/I_t , and then filters each of these as above. Expanding the ideals to M_n , we obtain a filtration of $M_n/I_s M_n$ by $(t^d - 1)s^d$ homomorphic images of M_n/IM_n .

This shows that

$$\ell(M_n/I_s M_n) \leq (t^d - 1)s^d \ell(M_n/IM_n)$$

or

$$\frac{\ell(M_n/I_s M_n)/(st)^d}{\ell(M_n/IM_n)} \leq \frac{(t^d - 1)s^d}{t^d s^d} = 1 - \frac{1}{t^d}.$$

As $n \rightarrow \infty$ the numerator on the left has limit $e_{(\underline{x})}(M_n)$. If we now let $n \rightarrow \infty$, we obtain a contradiction. $\square$

This argument is a variant of one given in [52] proving that the content of local cohomology of a local ring with respect to a system of parameters is positive: see Sect. 9.

Even if $R = K[[x, y]]$, one can let $M_n := R^n \oplus K$, where $K = R/(x, y)$, which gives a lim Cohen-Macaulay sequence in the original (i.e., strong) sense, or one can let $M_n := R^n + R/m^n$, which gives a lim Cohen-Macaulay sequence in the weak sense (the multiplicity is n , the length of $M_n/(x, y)M_n$ is $n + 1$, but $H_2(x, y; M_n) \cong m^{n-1}/m^n$ is enormous compared to $v(M_n) = n + 1$).

The following is, evidently, a weakening of Conjecture 3

Conjecture 6 A complete local domain with algebraically closed residue class field has a sequence of nonzero finitely generated modules such that

$$\lim_{n \rightarrow \infty} \frac{\ell(M_n/(\underline{x})M_n)}{e_{(\underline{x})}(M_n)} = 1.$$

11 A Local Cohomology Conjecture and a Tight Closure Conjecture

We recall that the *absolute integral closure* R^+ of a domain R is the integral closure of R in an algebraic closure of its fraction field. If R is a complete local domain of characteristic $p > 0$, then R^+ is a big Cohen-Macaulay algebra [49]. In mixed characteristic p , Heitmann [37] first proved that if $(R, \mathfrak{m})$ is a mixed characteristic p complete local domain of dimension 3, then for all n , $p^{1/n}$ kills $H_{\mathfrak{m}}^2(R^+)$, although the result is not stated in precisely this way in [49], and that this implies the direct

summand conjecture. Heitmann's result was used in [46] to prove the existence of big Cohen-Macaulay algebras in dimension 3. In [36], Heitmann proved that the entire maximal ideal of R^+ kills $H_{\mathfrak{m}}^2(R^+)$ for mixed characteristic complete local domains $(R, \mathfrak{m})$ of dimension 3.

The following two conjectures are open: a proof of either would give a new proof of the direct summand conjecture.

Conjecture 7 Let $(R, \mathfrak{m})$ be a complete local domain of mixed characteristic p and Krull dimension d . Then the maximal ideal of R^+ kills $H_{\mathfrak{m}}^{d-1}(R^+)$.

In [56] it is shown that Conjecture 7 implies:

Conjecture 8 Let R be a complete local domain of mixed characteristic p . Suppose that $x_1, \dots, x_{d-1}, x_d = p$ is a system of parameters for R , and let t be a positive integer. Then the image of $(x_1, \dots, x_{d-1})R :_R p^t$ in $\bar{R} = R/pR$ is contained in the tight closure of $(x_1, \dots, x_{d-1})\bar{R}$.

This results of [36, 37] and [56] show that both these conjectures are true if $d = 3$ (or is smaller).

For background on tight closure theory, we refer the reader to [18, 48, 50], and [57].

The initial motivation for Conjecture 8 came from the much stronger conjecture that there is a notion of tight closure for all local rings that agrees with the usual notion for local rings of positive characteristic, has the colon-capturing property in complete domains, and is persistent.

12 Ranganathan's Strong Direct Summand Conjecture and the Vanishing Conjecture for Maps of Tor

While the direct summand conjecture has now been settled, then strong form, stated as Conjecture 9 below, remains open, and the vanishing conjecture for maps of Tor, stated below as Conjecture 10, remains open as well, although many of the remaining cases are settled in [38]. There is a discussion of the vanishing conjecture for maps of Tor in [51, Sect. 4], where it is shown to follow from the weakly functorial existence of big Cohen-Macaulay algebras. In equal characteristic, it can be proved using tight closure theory. The vanishing conjecture for maps of Tor implies that if a Noetherian ring R is a direct summand, as an R -module, of a regular Noetherian ring S , then R is Cohen-Macaulay. Cf. [14, 53]. In [14] it is shown in the affine case that in characteristic 0 R has rational singularities. It appears that the recent results of André may be used to prove that direct summands of regular rings are Cohen-Macaulay in general. Whether that is the case or not, the issue is resolved in [38], where it is proved that direct summands of regular rings are Cohen-Macaulay and, in fact, pseudo-rational in mixed characteristic.

Conjecture 9 (Ranganathan's Strong Direct Summand Conjecture) For $A \subseteq R$, suppose R is a module-finite domain over A , where A is regular local. Let x be a

regular parameter (i.e., that $x \in \mathfrak{m}_A - \mathfrak{m}_A^2$) and let q be a height 1 prime in R lying over xA . Then xA is a direct summand of q as an A -module.

It is shown in [65] that this is equivalent to the vanishing conjecture for maps of Tor:

Conjecture 10 (vanishing Conjecture for Maps of Tor) Let $A \rightarrow R \rightarrow S$ be Noetherian rings, where A is a regular domain, S is module-finite and torsion-free over A , and S is regular. Then for every A -module M and integer $i \geq 1$, the map $\mathrm{Tor}_i^A(M, R) \rightarrow \mathrm{Tor}_i^A(M, S)$ is 0.

Since [65] is not published, we give the proof of the equivalence of Conjectures 9 and 10 here. We make use of several preliminary results, given below. First note that because both conjectures imply the direct summand conjecture, we do not need to use that the direct summand conjecture is known in establishing the equivalence. For Conjecture 10, this is proved in [51, (4.4), p. 81]. For the strong direct summand conjecture, if $A \rightarrow S$ is a module-finite extension of a local ring, we may kill a minimal prime of S to reduce to the case where S is a domain, choose a regular parameter $x \in \mathfrak{m}_A - \mathfrak{m}_A^2$, and choose a minimal prime Q of S lying over xA . Since xA is a direct summand of Q , it is a direct summand of $xS \subseteq Q$, and this shows A is a direct summand of S .

From [4] we have:

Theorem 10 *If $A \rightarrow T$ is a local homomorphism of complete Noetherian local rings, and A is regular, then there is a complete regular local ring A' and a factorization $A \rightarrow A' \twoheadrightarrow T$, where the maps are local homomorphisms, A' is faithfully flat over A , and $A' \twoheadrightarrow T$ is surjective.*

We use the following fact [43] which holds under a mild condition (approximately Gorenstein) on the local ring A , and, in particular, holds when A is regular.

Theorem 11 *If A is regular local and W is a finitely generated A -module, then an injection $A \rightarrow W$ sending $1 \mapsto w$ splits if and only if for every ideal I of A , $IW \cap Aw = Iw$.*

Remark 1 In fact, the results of [43] show that when A is regular local or even Gorenstein local the map splits if for one system of parameters $x_1, \dots, x_d$ for A and every $I_t = (x'_1, \dots, x'_d)$ with $t \geq 1$, one has $I_t W \cap Aw = I_t w$.

We now state and prove the third preliminary result.

Lemma 1 *Let $A \hookrightarrow R \twoheadrightarrow T$ be maps of Noetherian rings, where A is a local domain, R is a domain, $A \hookrightarrow R$ is a module-finite extension, T is regular, and $Q = \mathrm{Ker}(R \twoheadrightarrow T)$ is a prime ideal of R . Assume that A is a direct summand of R . Assume also that $Q \cap A = P$ is the prime ideal of A generated by part of a regular system of parameters $x_1, \dots, x_h$ (i.e., part of a minimal set of generators of the maximal ideal $\mathfrak{m}_A$), and $A/xA \cong R/Q \cong T$. Then for every ideal I of A , the map $\theta_I : \mathrm{Tor}_1^A(A/I, R) \rightarrow \mathrm{Tor}_1^A(A/I, T)$ is 0 if and only if $IQ \cap P = IP$.*

Hence, in the case where $h = 1$ and $x = x_1$, xA is a direct summand of Q if and only if all of the maps θ_I are 0.

Proof We may tensor the short exact sequence $0 \rightarrow Q \rightarrow R \rightarrow T \rightarrow 0$ with A/I over A to obtain

$$\cdots \rightarrow \operatorname{Tor}_1^A(A/I, R) \xrightarrow{\theta_I} \operatorname{Tor}_1^A(A/I, T) \rightarrow Q/IQ \rightarrow \cdots$$

and so $\theta_I = 0$ if and only if

$$\operatorname{Tor}_1(A/I, T) \cong \operatorname{Tor}_1(A/I, A/P) \cong (I \cap P)/IP$$

injects into Q/IQ , i.e., if and only if $(I \cap P) \cap IQ \subseteq PI$. (Note that IQ is not necessarily contained in I here, because Q is an ideal of R .) Now $IQ \cap P \subseteq IR \cap A = I$, since $A \rightarrow R$ splits, so that I may be omitted from the triple intersection, and, hence, the vanishing of θ_I is equivalent to the condition that $P \cap IQ \subseteq IP$. Since the opposite inclusion is automatic, we have proved the equivalence stated in the first paragraph.

The final statement follows at once from Theorem 11.

The final statement in Lemma 1 yields at once that the vanishing conjecture for maps of Tor implies the strong direct summand conjecture: in fact, we only need the vanishing conjecture under the specific hypotheses of Lemma 1 in the case where $h = 1$, and we can assume the usual direct summand conjecture (so that we know A is a direct summand of R).

From Lemma 1 in the case where $h = 1$ we immediately have a global version:

Corollary 2 *Let $A \hookrightarrow R \twoheadrightarrow T$ be maps of Noetherian rings, where A, R are domains, $A \hookrightarrow R$ is a module-finite extension, T is regular, and $Q = \operatorname{Ker}(R \twoheadrightarrow T)$ is a height one prime of R . Assume that A is a direct summand of R . Assume also that $Q \cap A$ is a principal prime $P = xA$ such that A/xA is regular. Then xA splits from Q if and only for all ideals I of A , $xA \cap IQ \subseteq Ix$.*

Proof Both statements are local on the maximal ideals of A . If x becomes a unit after localization, xA becomes A , Q becomes R , and the result is clear. If x does not become a unit, A is local and the conclusion follows from the final statement in Lemma 1.

The final preliminary result is:

Lemma 2 *Let A be a regular ring, let P be a prime of A such that A/P is regular, let R be a module-finite extension ring of A , and let Q be a prime ideal of R lying over P . Then for every positive integer $n \in \mathbb{N}$, $P^n Q \cap A = P^{n+1}$.*

Proof The issue is local on the maximal ideals of A . Hence, we may assume that A is local and then we know that P is generated by part of a regular system of parameters for A , and that P^{n+1} is primary to P . Suppose that $P^n Q \cap A$ contains an element b that is not P^{n+1} . Then this is preserved when we local at $A - P$, and so we may assume that P is the maximal ideal of the regular local ring A . Moreover, the counterexample is preserved when we complete with respect to P , and also when we kill a minimal prime of R lying over (0) in A . Thus, we may assume that $A \subseteq R$ are complete

domains, and, hence, that R is local. The P -adic valuation on A extends to a discrete rank one valuation v on R (with values in a cyclic subgroup of $\mathbb{Q}$) whose values on elements of Q are positive. This shows that the order of b with respect to v is greater than n , and since $b \in A$, $v(a)$ must be at least $n + 1$.

It remains only to show that the strong direct summand conjecture implies the vanishing conjecture for maps of Tor. Henceforth, we assume that the strong direct summand conjecture is true, and prove the vanishing conjecture for maps of Tor. We make several reductions in the problem.

Consider $A \subseteq R \rightarrow T$, where R is module-finite and torsion-free over the regular ring A , and let M be any A -module. We must show that for $i \geq 1$, the map $\mathrm{Tor}_i^A(M, R) \rightarrow \mathrm{Tor}_i^A(M, T)$ is 0. Since M is a direct limit of finitely generated A -modules, we may assume that M is finitely generated. By replacing M by a first module of syzygies of M repeatedly, we may assume that $i = 1$.

There is a presentation of M as the quotient of a free A -module:

$$M = \frac{Au_1 + \cdots + Au_s}{\mathrm{Span}\{\sum_{j=1}^s a_{ij}u_j : 1 \leq i \leq s\}}$$

By replacing A by $A[u] := A[u_1, \dots, u_s]$, R by $R[u]$, T by $T[u]$, and M by the symmetric algebra of M over A , $\mathrm{Sym}_A(M) \cong A[u]/(\sum_{j=1}^s a_{ij}u_j : 1 \leq i \leq s)$, we may assume that M is a cyclic A -module A/I . Because a free resolution $G_\bullet$ of $\mathrm{Sym}_A(M)$ by free $A[u]$ -modules is also a free resolution over A and $G_\bullet \otimes_{A[u]} R[u] \cong G_\bullet \otimes_A R$ (we may think of $R[u]$ as $A[u] \otimes_A R$), we may identify

$$\mathrm{Tor}_1^{A[u]}(\mathrm{Sym}_A(M), R[u]) \cong \mathrm{Tor}_1^A(\mathrm{Sym}_A(M), R)$$

and, similarly,

$$\mathrm{Tor}_1^{A[u]}(\mathrm{Sym}_A(M), T[u]) \cong \mathrm{Tor}_1^A(\mathrm{Sym}_A(M), T).$$

If the map between the Tor modules involving $\mathrm{Sym}_A(M)$ is 0, this is also true for M , because M is a direct summand of $\mathrm{Sym}_A(M)$ as A -modules.

In this way, we reduce to the case where M is a cyclic A -module, A/I . The issue of whether the map is 0 is local on T . We may therefore replace T by a localization, and complete T . We may likewise localize and complete A at the contraction of the maximal ideal of T to A . Since R is torsion-free over A , it is embeddable in a free A -module, and so remains torsion-free when we complete. The map $R \rightarrow T$ has a prime ideal as its kernel, and this prime contains a minimal prime of R lying over 0 in A . We may replace R by its quotient by this minimal prime. Hence, we may also assume that R is a complete local domain.

Now let A' be as in Theorem 10. By replacing A , M , and R by A' , $M \otimes_A A'$ and $R \times_A A'$ respectively, we may assume that the map from A to T is a surjection, with kernel a prime ideal P . Note also that P must be generated by part of a regular system of parameters for A , since $A/P \cong T$ is regular. Let $P = (y_1, \dots, y_h)$, where the y_i

are part of a regular system of parameters for A and h is the height of P . We may again reduce to the case where R is a domain. We are still in the situation where M is a cyclic A -module, say A/I . Since A maps onto T , R also maps onto T . Let Q be the kernel of this map. Note that $Q \cap A = P$.

We are now in the situation of Lemma 1, with $M = A/I$. Hence, to complete the proof, it suffices to show that for all ideals I of A , $IQ \cap P = IP$. We shall achieve this by using an extended Rees ring to reduce to the case where $h = 1$.

Let t be a new indeterminate over R , let $v := 1/t$, and let $\tilde{A}$ be the extended Rees ring of P , which is $A[Pt, v] = A[y_1t, \dots, y_h t, v] \subseteq \tilde{A}[t, 1/t]$. Note that $\tilde{A}$ is regular, since localizing at v yields $A[t, 1/t]$, while killing $v\tilde{A}$ yields the associated graded ring of A with respect to $(y_1, \dots, y_h)A$, which is a polynomial ring in h variables z_i over $T \cong A/P$. This, $v\tilde{A}$ is a principal prime ideal of $\tilde{A}$ such that the quotient is regular.

Let $\tilde{R} := R[Pt, v]$. Evidently, this domain is module-finite over the regular domain $\tilde{A}$. We want to show that there is a homogeneous prime ideal $\tilde{Q}$ of $\tilde{R}$ that contains Q and contracts to $v\tilde{A}$. Let $J = Q\tilde{R} + v\tilde{R}$. Then

$$J = \dots + Qv^n + \dots + Qv + Q + QPt + QP^2t^2 + \dots + QP^st^s + \dots.$$

By Lemma 2, we have that $QP^s \cap P^s = P^{s+1}$. Hence, $J \cap \tilde{A} = v\tilde{A}$. It follows that there is a prime $\tilde{Q}$ of $\tilde{R}$ containing J and lying over $v\tilde{A}$, and since it may be taken to be a minimal prime of J , it is homogeneous. Because we are assuming the strong direct summand conjecture, we have that $v\tilde{A} \hookrightarrow \tilde{Q}$ splits. For every ideal I of A , we then have that

$$(\dagger) \quad I\tilde{Q} \cap v\tilde{A} = Iv\tilde{A}.$$

The degree 0 part of $\tilde{Q}$ must be Q , since it contains Q and contracts to P in A . Comparing degree 0 components in $(\dagger)$, we obtain that $IQ \cap P = IP$, as required. Hence, the strong direct summand conjecture implies the vanishing conjecture for maps of Tor. $\square$

13 Big Cohen-Macaulay Algebras, Tight Closure in all Characteristics, and a Recap of All the Conjectures

In this section we collect all of the conjectures made so far, and also an additional conjecture on a very strong form of existence of big Cohen-Macaulay algebras. The conjectures below, most of which have been described earlier, are all open, and each of them implies the direct summand theorem. Three of the statements below are now theorems, and that is indicated. A diagram at the end shows what is known about the implications among them. Note that in this section the number of conjectures has been changed from their earlier occurrences: it is now 13.0 through 13.9.

Conjecture 13.0 (Positivity of Serre Intersection Multiplicities) If $M, N \neq 0$ are finitely generated modules over a regular local ring T such that $\ell(M \otimes_T N)$ is finite and $\dim(M) + \dim(N) = \dim(T)$, then the Serre intersection multiplicity $\chi(M, N) > 0$.

The following three statements are now known, and we have used letters for them instead of numbers.

Theorem DS (Direct Summand Theorem) *A regular ring is a direct summand as a module over itself of every module-finite extension.*

Theorem CMA (Existence of Big Cohen-Macaulay Algebras) *Every local ring has a big Cohen-Macaulay algebra.*

Theorem CMM (Existence of Big Cohen-Macaulay Modules) *Every local ring has a big Cohen-Macaulay module.*

The remaining statements here are open. They all imply Theorem DS.

Conjecture 13.1 (Existence of Lim Cohen-Macaulay Sequences in a Weak Sense) If R is a complete local domain of mixed characteristic with algebraically closed residue class field, then R has a sequence of modules M_n such that

$$\lim_{n \rightarrow \infty} \frac{\ell(M_n / (\underline{x})M_n)}{e_{(\underline{x})}(M_n)} = 1.$$

See Sect. 10.

Conjecture 13.2 (Existence of Lim Cohen-Macaulay Sequences) If R is a complete local domain of mixed characteristic with algebraically closed residue class field, then R has a lim Cohen-Macaulay sequence of modules.

See Sect. 5.

Conjecture 13.3 (Systems of Parameters are Q-Sequences) Every system of parameters of every local ring is a Q-sequence.

See Sect. 9. For the next two conjectures, see Sect. 11.

Conjecture 13.4 (Capturing Mixed Characteristic Parameter Colon Ideals with Tight Closure) Let R be a complete local domain of mixed characteristic p . Suppose that $x_1, \dots, x_{d-1}, x_d = p$ is a system of parameters for R , and let t be a positive integer. Then the image of $(x_1, \dots, x_{d-1})R :_R p^t$ in $\bar{R} = R/pR$ is contained in the tight closure of $(x_1, \dots, x_{d-1})\bar{R}$.

Conjecture 13.5 (Next to Top Local Cohomology is Almost 0) If $(R, \mathfrak{m})$ is a complete local domain of mixed characteristic of dimension d with algebraically closed residue class field, then $H_{\mathfrak{m}}^{d-1}(R^+)$ is killed by the maximal ideal of R^+ .

For the next two conjectures, see Sect. 12. The two are equivalent, and are known in equal characteristic and in many cases in mixed characteristic by the results of [38]. We note that is shown in [61] that Conjecture 13.5 implies Conjectures 13.6 and 13.7.

Conjecture 13.6 (Ranganathan's Strong Direct Summand Conjecture) For $A \subseteq R$, suppose R is a module-finite domain over A , where A is regular local. Let x be a regular parameter (i.e., let x be in $\mathfrak{m}_A - \mathfrak{m}_A^2$) and let q be a height 1 prime in R lying over xA . Then xA is a direct summand of q as an A -module.

Conjecture 13.7 (Vanishing Conjecture for Maps of Tor) Let $A \rightarrow R \rightarrow S$ be Noetherian rings, where A is a regular domain, S is module-finite and torsion-free over A , and S is regular. Then for every A -module M and integer $i \geq 1$, the map $\mathrm{Tor}_i^A(M, R) \rightarrow \mathrm{Tor}_i^A(M, S)$ is 0.

The last two conjectures have not been discussed separately. The first is known in equal characteristic [51], and is now known in many cases in mixed characteristic [38].

Conjecture 13.8 (Existence of Weakly Functorial Big Cohen-Macaulay Algebras) If $R \rightarrow S$ is a local map of complete local domains, there exists a map from a big Cohen-Macaulay algebra B over R to a big Cohen-Macaulay algebra C over S such that the diagram

$$\begin{array}{ccc} B & \longrightarrow & C \\ \uparrow & & \uparrow \\ R & \longrightarrow & S \end{array}$$

commutes.

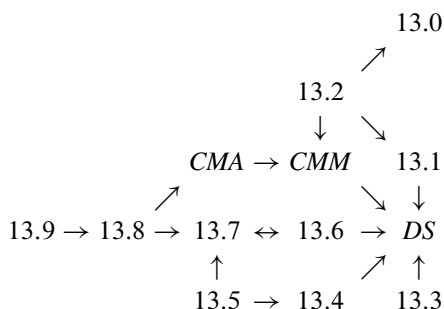
Conjecture 13.9 (Existence of Weakly Functorial Big Cohen-Macaulay Algebras, Strong Form) One can assign a big Cohen-Macaulay algebra B_R to every complete local domain R such that B_R contains R^+ and given any local map $R \rightarrow S$ of complete local domains, there is a commutative diagram:

$$\begin{array}{ccc} B_R & \longrightarrow & C_R \\ \uparrow & & \uparrow \\ R & \longrightarrow & S \end{array}$$

Conjecture 13.9, which is true in positive characteristic, would prove the existence of an analogue of tight closure in mixed characteristic with properties similar to those of tight closure in equal characteristic. If $N \subseteq M$ are finitely generated R -modules, one defines the closure $\mathrm{cl}_M(N)$ of N in M to be the kernel of the map $M \rightarrow B_R \otimes_R (M/N)$ such that $u \mapsto 1 \otimes (u + N)$. The characteristic p notions of tight closure (and plus closure) can be obtained in this way. For more details about closure operations induced by big Cohen-Macaulay modules and algebras, see [66–68].

With the recent breakthroughs, Conjecture 13.9 may be in reach.

The following diagram, in which the arrows indicate implications, shows the relationships among these conjectures.



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Algebras with the Weak Lefschetz Property

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Abstract This is a survey on some works in which the Weak Lefschetz Property (WLP) for Artinian standard graded algebras is investigated, see for instance (Ragusa and Zappalà, arXiv:1112.1498. To appear in Rend Circ Mat Palermo; Colloq Math 64:73–83, 2013; Favacchio et al, J Pure Appl Algebra 217:1955–1966, 2013). In particular, it is shown that the Hilbert function of an almost complete intersection Artinian standard graded algebra of codimension 3 is a Weak Lefschetz sequence, i.e. it is the Hilbert function of some Artinian algebra with WLP or equivalently it is unimodal and the positive part of their first differences is a O -sequence. Moreover we give both some numerical condition on the Hilbert function and other conditions on the graded Betti numbers in order to force Artinian Gorenstein standard graded algebras of codimension 3 to enjoy the WLP. For Artinian standard graded algebras with the WLP we study the behavior of their linear quotients both with respect to the Hilbert function and to the graded Betti numbers. From this we produce a new property denominated Betti Weak Lefschetz Property (β -WLP) which permits a good behavior of the grade Betti numbers for the linear quotients of Artinian standard graded algebras with the WLP. We find conditions on the generators' degrees of a complete intersection Artinian graded algebra with the WLP which force the algebra to have the β -WLP.

1 Motivations

The origin of this investigation goes back to the following result.

Proposition 1.1 *Let $R = k[x_1, \dots, x_r]$, k a field of characteristic zero and let I be an Artinian monomial complete intersection ideal of R , i.e. $I = (x_1^{a_1}, \dots, x_r^{a_r})$. Let $h \in R$ be a generic linear form. Then for every positive integers d and i , the multiplication map by h^d , $\times h^d : (R/I)_i \rightarrow (R/I)_{i+d}$ has maximal rank. (In particular, this is true for $d = 1$).*

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The above result was proved by different approaches by R. Stanley in [24] by algebraic topology, by J. Watanabe in [25] using representation theory, by Reid, Roberts and Roitman in [22] by algebraic methods, by Herzog and Popescu (see [8]) by linear algebra and by Ikeda [10] using combinatorics.

So, it seems of great interest to study Artinian standard graded algebras with such a property. Moreover, recently, other relationships were found for these algebras with ordinary projective curves by Mirò Roig in [16] and with the projections of the Veronese variety which satisfy at least a Laplace equation of order $d - 1$ (see Ottaviani, Mezzetti, Mirò Roig in [11]). Of course, in the literature there are very nice papers which give survey on many aspects of this very important property. For instance one can give a look at the paper by Harima et al. [7], and to the paper of Migliore and Nagel [14]. Most of the results of this survey can be found in [5, 19, 21].

2 Definitions and Basic Facts

Definition 2.1 Let $R = k[x_1, \dots, x_n]$ be the polynomial ring on a field k and let $A = R/I$ be an Artinian standard graded algebra of codimension n , we say that A has the Weak Lefschetz Property (WLP) if there exists a linear form $l \in R_1$ such that, for every i , the map $\times l : A_i \rightarrow A_{i+1}$ obtained by multiplication by l , has maximal rank.

In the last few years the Weak Lefschetz Property for Artinian standard graded algebras has been investigated very deeply. It is trivial to see that all Artinian standard graded algebras of codimension 2 have the Weak Lefschetz Property, but already in codimension 3 many questions are still open. One of the most important result on this direction is the proof of the Weak Lefschetz Property for all Artinian complete intersections of codimension 3, due to T. Harima, J.C. Migliore, U. Nagel, J. Watanabe, in [6]. Other results on Weak Lefschetz algebras and the corresponding Hilbert functions can be found, for instance, in [12, 15]. Nevertheless, again in codimension 3, it is enough to consider almost complete intersection ideals, i.e. perfect ideals of height 3 and minimally generated by 4 elements, to produce examples of Artinian standard graded algebras not enjoying the Weak Lefschetz Property (see [2]). On the other hand, in the same paper [6] there exists a characterization of the sequences which occur as Hilbert functions of Weak Lefschetz algebras. These sequences are exactly those which are unimodal and such that the positive part of their first difference is an O -sequence. So we give the following definition.

Definition 2.2 Let H be the Hilbert function of an Artinian standard graded k -algebra. We will say that H is a *Weak Lefschetz sequence* if it is unimodal and $(\Delta H)^+$ is an O -sequence.

A first result on this direction states.

Theorem 2.3 *Let H be a Hilbert function of an almost complete intersection Artinian standard graded k -algebra of codimension 3. Then H is a Weak Lefschetz sequence.*

Proof (Sketch) Let R/I_Q be a quotient of $R = k[x_1, x_2, x_3]$ with I_Q an almost complete intersection ideal of codimension 3, such that $H_Q = H$. Then there exist ideals $I_Z \subseteq I_G$ such that $I_Q = I_Z : I_G$ with I_Z a complete intersection ideal of height 3 with generators' degrees $a_1 \leq a_2 \leq a_3$ and I_G a Gorenstein ideal of height 3 with generators' degrees $d_1 \leq \dots \leq d_{2m+1}$, $\vartheta_Z - \vartheta_G \leq a_1$ (here ϑ_Z and ϑ_G are the degrees of the last syzygy for R/I_Z and R/I_G , i.e. $\vartheta_Z = \sum a_i$ and $\vartheta_G = \sum d_i/m$) and for every $3 \leq i \leq m+1$ $\vartheta_G > d_i + d_{2m+4-i}$. In such a situation we set $t_{G|Z} := \min\{n \in \mathbb{Z}_0^+ \mid \Delta H_G(n) > \Delta H_Z(n)\}$.

Thus, in order to show that H_Q is unimodal we need to find an integer u such that $H_Q(i-1) < H_Q(i)$ for $0 < i \leq u$ and $H_Q(i-1) \geq H_Q(i)$ for $i > u$, or, equivalently, $\Delta H_Q(i) > 0$ for $0 < i \leq u$ and $\Delta H_Q(i) \leq 0$ for $i > u$. Now, by the equality $\Delta H_Q(i) = \Delta H_G(\vartheta_Z - 2 - i) - \Delta H_Z(\vartheta_Z - 2 - i)$, obtained by liaison, the previous inequalities become $\Delta H_G(j) > \Delta H_Z(j)$ for $\vartheta_Z - 2 - u \leq j \leq \vartheta_Z - 2$ and $\Delta H_G(j) \leq \Delta H_Z(j)$ for $0 \leq j < \vartheta_Z - 2 - u$.

We will show that the required u is $u := \vartheta_Z - 2 - t_{G|Z}$. One proves first that $t_{G|Z} > \vartheta_G - d_2 - 1$. Thus, for every $0 \leq i < t_{G|Z}$, by definition of $t_{G|Z}$, we have $\Delta H_G(i) \leq \Delta H_Z(i)$, i.e. $\Delta H_G(i) \leq \Delta H_Z(i)$ for $0 \leq i < \vartheta_Z - 2 - u$. Now, we want to show that for every i , $\vartheta_Z - 2 - u \leq i \leq \vartheta_Z - 2$ we have $\Delta H_G(i) > \Delta H_Z(i)$. If not, let $j = \min\{i \mid \vartheta_Z - 2 - u < i \leq \vartheta_Z - 2, \Delta H_G(i) \leq \Delta H_Z(i)\}$, then $\Delta H_G(j-1) > \Delta H_Z(j-1)$. Now, since $j > \vartheta_Z - 2 - u = t_{G|Z}$, by the stated result on $t_{G|Z}$, $j > \vartheta_G - d_2 - 1$, thus $\Delta^2 H_G(j) \geq 0$. If $j \leq \vartheta_Z - a_1 - 1$ then $\Delta^2 H_Z(j) \leq 0$. From this, we get

$$\Delta H_G(j) = \Delta H_G(j-1) + \Delta^2 H_G(j) > \Delta H_Z(j-1) + \Delta^2 H_Z(j) = \Delta H_Z(j)$$

a contradiction. If $\vartheta_Z - a_1 - 1 < j \leq \vartheta_G - 2$ then $\Delta^2 H_G(j) = \Delta^2 H_Z(j) = 1$, so again

$$\Delta H_G(j) = \Delta H_G(j-1) + 1 > \Delta H_Z(j-1) + 1 = \Delta H_Z(j).$$

Finally, if $\vartheta_G - 2 < j \leq \vartheta_Z - 2$, trivially $0 = \Delta H_G(j) > \Delta H_Z(j)$. In this way we can conclude that $H = H_Q$ is unimodal.

In order to conclude the proof we need to show that the sequence $(\Delta H)^+$ is an O -sequence. In our situation, if the generators' degrees of I_Q are $e_1 \leq e_2 \leq e_3 \leq e_4$ one shows that there exists an almost complete intersection ideal $I_{Q'}$ with the same generators' degrees and Hilbert function as I_Q such that $I_{Q'} = I_{Z'} : I_{G'}$ where $I_{Z'}$ is a complete intersection ideal of type (e_2, e_3, e_4) , $I_{G'}$ is a Gorenstein ideal with $I_{Z'} \subset I_{G'}$, $\vartheta_{Z'} - \vartheta_{G'} = e_1$ and for every $3 \leq i \leq m+1$ $\vartheta_{G'} > d'_i + d'_{2m+4-i}$ ($d'_1, \dots, d'_{2m+1}$ the generators' degrees of $I_{G'}$). Therefore it is enough to show that, for $i \geq \vartheta_{Z'} - \vartheta_{G'}$, $(\Delta H)^+(i-1) \geq (\Delta H)^+(i)$. This is equivalent to prove that, for $t_{G'|Z'} + 1 \leq j \leq \vartheta_{G'} - 1$, $\Delta^2 H_{G'}(j) \geq \Delta^2 H_{Z'}(j)$. Since $\Delta^2 H_{G'}(j) = 1$ for

$j > \vartheta_{G'} - 1 - d'_1$ the inequality trivially holds in this range. Now, take $t_{G'|Z'} + 1 \leq j \leq \vartheta_{G'} - 1 - d'_1$. Since $\vartheta_{G'} < d'_1 + e_3 + a_4$, as $\text{mci}(d_1, \dots, d_{2m+1}) \leq (d'_1, e_3, e_4)$, we have that $j \leq \vartheta'_Z - e_2 - 1$, therefore $\Delta^2 H'_G(j) = 0$ and $\Delta^2 H'_Z(j) \leq 0$ so we get again the required inequality. $\square$

3 Weak Lefschetz Property for Gorenstein Algebras

In order to generalize the result in [6] for complete intersections the first step is to study the WLP for Artinian standard graded Gorenstein algebras. By a result of Ikeda in [10] we know that there are Artinian standard graded Gorenstein algebras of codimension 4 which do not have the WLP, but in codimension 3 the problem is still wide open.

Let H be the Hilbert function of an Artinian algebra A of codimension 3 and let $\vartheta - 3$ be the socle degree of A . Then H is a Gorenstein sequence of codimension 3 (i.e. the Hilbert function of some Artinian standard graded Gorenstein algebra) iff $H(i) = H(\vartheta - 3 - i)$, for every i and $(\Delta H)^+$ is an O -sequence (see Theorem 4.2 in [23]).

For every Gorenstein sequence H we set

$$\vartheta_H := \max\{i \mid H(i) > 0\} + 3$$

$$u_H := \min\{i \mid \Delta H(i) < 0\} - \max\{i \mid \Delta H(i) > 0\} - 1;$$

$$s_H := \min\{i \mid \Delta H(i) < i + 1\};$$

$$t_H := \max\{i < \vartheta_H - 1 \mid \Delta H(i) \geq 0\}.$$

Since in the sequel the most interesting situations will happen when $u_H = 0$ or $u_H = 1$ we emphasize that in these cases $t_H + 1 = \frac{\vartheta_H - 1}{2}$ or $t_H + 1 = \frac{\vartheta_H}{2}$ respectively.

We will show that given a Gorenstein sequence H (or a Gorenstein Betti sequence β) of codimension 3 the WLP holds for the generic Artinian standard graded Gorenstein algebra with Hilbert function H (respectively with Betti sequence β).

We start with this observation.

Proposition 3.1 *Let $A = R/I$ be an Artinian standard graded Gorenstein algebra, with socle degree $\vartheta - 3$ and Hilbert function H . Then A enjoys the WLP iff there exists a linear form $l \in R$ such that the multiplication map $l : A_{t-1} \rightarrow A_t$ is injective, where $t = t_H$.*

Proof By hypothesis and Proposition 2.1, (b) in [13], we get that the multiplication map $l : A_{i-1} \rightarrow A_i$ is injective for $i \leq t$. Now since $t \geq \frac{\vartheta-3}{2}$, by Gorenstein duality we deduce that the multiplication map $l : A_{i-1} \rightarrow A_i$ is surjective for $i > t$, so it has maximal rank in any degree. $\square$

The following lemma, whose proof is quite easy, is a key point for our purpose.

Lemma 3.2 *Let $A = R/I$ be an Artinian standard graded algebra. Let us suppose that there exists an integer t , a perfect ideal $J \subset R$ of height ≤ 2 and a form $f \in R$ (possibly a constant) such that $I_i = (fJ)_i$ for $i = t-1, t$. Then there exists a linear form $l \in R$ such that the multiplication map $l : A_{t-1} \rightarrow A_t$ is injective.*

Now we collect in the next proposition some cases in which the Hilbert function H forces the WLP for every Artinian standard graded Gorenstein algebras having H as Hilbert function.

Proposition 3.3 *Let H be a Gorenstein sequence of codimension 3 and let $u = u_H$ and $t = t_H$. If H satisfies one of the following conditions*

- 1) $\Delta H(t-1) = \Delta H(t)$;
- 2) $\Delta H(t-1) = 1$ and $\Delta H(t) = 0$;
- 3) $u = 0$ and $\Delta H(t) \geq t-1$;

then every Artinian standard graded Gorenstein algebra with Hilbert function H has the WLP.

Proof

- 1) Note that by Theorem 3.1 and Corollary 3.4 in [18], every Artinian standard graded Gorenstein algebra $A = R/I$ with Hilbert function H is such that $I_{\leq t} = fJ$, where f is a form of degree $\Delta H(t)$ and J is perfect ideal of height ≤ 2 . So, applying Lemma 3.2, we are done.
- 2) In this case $\sum_{i=0}^t \Delta^3 H(i) = \Delta^2 H(t) = -1$. Using the same arguments of proof of Theorem 3.1 in [18], if we set p the number of the minimal generators of I of degree $\leq t$ and q the number of the minimal first syzygies of I of degree $\leq t+1$, we get that $p - q = 1$. Taking a look at the Buchsbaum-Eisenbud matrix (see [3]) of A , again as in the proof of Theorem 3.1 in [18], we get that $I_{\leq t}$ is a perfect ideal of height 2, so by Lemma 3.2 and Proposition 3.1, A enjoys the WLP.
- 3) Since $(\Delta H)^+$ is an O -sequence, in this case the ideal $I_{\leq t}$ is minimally generated by at most two elements. Therefore $I_{\leq t} = 0$ or $I_{\leq t}$ is principal or $I_{\leq t} = fJ$ where J is generated by a regular sequence of length 2. By Lemma 3.2 the conclusion follows. $\square$

Remark 3.4 Note that if $u_H \geq 2$, then the hypothesis 1) of Proposition 3.3 is satisfied. Therefore, since when $u_H \geq 2$ the WLP holds, in the sequel we will treat only the cases $u_H = 0, 1$.

Now we concentrate on the relationship between the WLP and the Betti sequences of the Artinian standard graded Gorenstein algebras.

We recall the following results from [18].

Proposition 3.5 *Let $A = R/I$ be an Artinian Gorenstein algebra with Hilbert function H . Let us denote by ϑ the degree of the only second syzygy of A , $s = \min\{i \in \mathbb{N} \mid I_i \neq 0\}$. Then*

- 1) $\vartheta = \max\{i \mid \Delta^3 H(i) \neq 0\}$.
- 2) $v_i(A) \geq -\Delta^3 H(i)$ for $i < \vartheta$; $\sigma_i(A) \geq \Delta^3 H(i)$ for all i .

- 3) $v_s(A) = -\Delta^3 H(s)$, $v_i(A) \leq -\Delta^2 H(i)$ for $s+1 \leq i \leq \vartheta - s - 1$; and $v_i(A) = 0$ for $i \geq \vartheta - s$; consequently, $\sigma_{\vartheta-s}(A) = \Delta^3 H(\vartheta - s)$ and $\sigma_i(A) \leq -\Delta^2 H(i-1)$ for $s+1 \leq i \leq \vartheta - s - 1$

(here $v_i(A)$ means the minimal number of generators of degree i for I and $\sigma_i(A)$ means the minimal number of first syzygies of degree i for I).

Proposition 3.6 *Let H be a Gorenstein sequence of codimension 3 and let BG_H be the poset of all Gorenstein Betti sequences compatible with H .*

- 1) BG_H has only one minimal and only one maximal element, respectively $\beta_{\min}(H)$ and $\beta_{\max}(H)$.
- 2) If $d_1 \leq \dots \leq d_{2m+1}$ are the degrees of the minimal generators associated to $\beta_{\max}(H)$, then $m = d_1$ and $d_i + d_{2m+3-i} + 1 = \vartheta_H$, for every $2 \leq i \leq m+1$.
- 3) Each element in BG_H can be obtained by $\beta_{\max}(H)$ by performing couples of cancellations of the type $i, \vartheta_H - i$. All these cancellations are allowed.

One key result for our purpose is the following theorem.

Theorem 3.7 *Let $A = R/I$ be an Artinian standard graded Gorenstein algebra. Let H be the Hilbert function of A and let $t = t_H$. If $v_{t+1}(A) = v_{t+1}^{\max}(H)$ then A enjoys the WLP.*

Proof By Remark 3.4 we can suppose that $u := u_H \leq 1$. Let $s := s_H$ and $\vartheta := \vartheta_H$. Note that $s-1 \leq t \leq \vartheta - s - 2$ and when $t = s-1$ we have that $A_i = R_i$ for $i \leq t$, so the WLP is trivial by Proposition 3.1. So we can assume $s+1 \leq t+1 \leq \vartheta - s - 1$. If $v_{t+1}(A) > v_{t+1}^{\min}(H)$, applying Theorem 4.4 in [20], we get that $I_{\leq t} = fJ$ where f is a form and J is perfect of height ≤ 2 . So the hypotheses of Lemma 3.2 are satisfied and by Proposition 3.1 A has the WLP. Let us suppose now that $v_{t+1}(A) = v_{t+1}^{\min}(H)$. If $u = 0$ then

$$v_{t+1}^{\min}(H) = \max\{0, -\Delta^3 H(t+1)\} = \max\{0, -\Delta^2 H(t+1)\} = -\Delta^2 H(t+1) > 0$$

i.e. $\Delta^3 H(t+1) = \Delta^2 H(t+1)$; this implies that $\Delta H(t-1) = \Delta H(t)$. So by Proposition 3.3 1) we get that A has the WLP.

If $u = 1$ the assumption $v_{t+1}(A) = v_{t+1}^{\min}(H)$ can happen only if $\Delta H(t-1) = 1$, so the conclusion follows by Proposition 3.3 item 2. $\square$

Corollary 3.8 *Let $A = R/I$ be an Artinian standard graded Gorenstein algebra. Let β be the Betti sequence of A and H be the Hilbert function of A . If $\beta = \beta_{\max}(H)$ then A enjoys the WLP. In particular, for every Gorenstein sequence H of codimension 3, there is an Artinian standard graded Gorenstein algebra whose Hilbert function is H and having the WLP.*

Proof This is just a particular case of Theorem 3.7. $\square$

Corollary 3.9 *Let H be a Gorenstein sequence of codimension 3. Then there exists a Gorenstein Betti sequence β_0 compatible with H such that for every Gorenstein*

Betti sequence $\beta \in \text{BG}_H$, $\beta \geq \beta_0$ all Artinian standard graded Gorenstein algebras having Betti sequence β enjoy the WLP.

Proof Let us consider the sequence v_0 defined as follows

$$(v_0)_i := \begin{cases} v_i^{\max}(H) & \text{for } i = t_H + 1, \vartheta_H - t_H - 1 \\ v_i^{\min}(H) & \text{otherwise} \end{cases}.$$

Let β_0 be the Betti sequence corresponding to v_0 . Note that β_0 is admissible for an Artinian standard graded Gorenstein algebra by Proposition 3.6, item 3. So if A is an Artinian standard graded Gorenstein algebra such that $\beta(A) \geq \beta_0$ then it has the WLP by Theorem 3.7. $\square$

Example 3.10 Let us consider the following Gorenstein sequence

$$H = (1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 76, 84, 89, 90, \\ 89, 84, 76, 66, 55, 45, 36, 28, 21, 15, 10, 6, 3, 1).$$

Then $\beta^{\max}(H)$ is defined by the sequence

$$v_{11} = 2, v_{12} = 2, v_{13} = 3, v_{14} = 4, v_{15} = 2, v_{16} = 4, \\ v_{17} = 3, v_{18} = 2, v_{19} = 1,$$

and β_0 is defined by the sequence

$$v_{11} = 2, v_{12} = 1, v_{13} = 1, v_{14} = 1, v_{15} = 2, v_{16} = 4.$$

Remark 3.11 Note that if k is algebraically closed by the irreducibility of $\text{Gor } H$, the previous results imply that the generic element in $\text{Gor } H$ has the WLP.

Theorem 3.12 *Let $A = R/I$ be an Artinian standard graded Gorenstein algebra, enjoying the WLP, whose Hilbert function is H and Betti sequence is β . Let $\beta' \in \text{BG}_H$, $\beta' < \beta$ such that $]\beta', \beta[= \emptyset$. Then there exists a 1-dimensional family of Artinian standard graded Gorenstein algebras with Betti sequence β' and the WLP.*

Proof Since $]\beta', \beta[= \emptyset$ we get β' from β by performing a couple of cancellations of degrees r and $\vartheta_H - r$ among the degrees of minimal generators and the degrees of first syzygies (see Remark 3.8 in [18]). Let M be an alternating matrix whose submaximal pfaffians minimally generate I . Let $M_{\hat{i}}$ be the submatrix obtained by $M = (m_{ij})$ by deleting the i -th row and the i -th column. Let $p_i = \text{pf } M_{\hat{i}}$. Then I is minimally generated by $p_1, \dots, p_{2m+1}$. So we have a minimal generator p_a of degree r and a minimal generator p_b of degree $\vartheta_H - r$ and two minimal first syzygies of the same degrees. So, by minimality, $m_{ab} = m_{ba} = 0$. Let $\lambda \in k$ and let $M_{(\lambda)} = (m'_{ij})$ be the alternating matrix such that $m'_{ab} = \lambda$, $m'_{ba} = -\lambda$ and $m'_{ij} = m_{ij}$ otherwise. Let

$I_{(\lambda)}$ be the ideal generated by the submaximal pfaffians of $M_{(\lambda)}$. So $A_{(\lambda)} = R/I_{(\lambda)}$ is an Artinian standard graded Gorenstein algebra with Betti sequence β for $\lambda = 0$ and Betti sequence β' for $\lambda \neq 0$. Since A has the WLP there exists $l \in R_1$ such that the multiplication map $l : A_{i-1} \rightarrow A_i$ has maximal rank for every i . Now let us consider the multiplication map $l : (A_{(\lambda)})_{t-1} \rightarrow (A_{(\lambda)})_t$, where $t = t_H$. Of course the rank of the latter map depends on λ and it is maximal for $\lambda = 0$, since $A_{(0)} = A$. So this rank is maximal for the generic $\lambda \in k$, therefore the generic $A_{(\lambda)}$ has the WLP. $\square$

Corollary 3.13 *Let β be any Gorenstein Betti sequence of codimension 3. Then there are infinitely many Artinian standard graded Gorenstein algebra with Betti sequence β and enjoying the WLP. In particular, if k is algebraically closed, the generic element of $\text{Gor } \beta$ has the WLP.*

Proof Let H be the Hilbert function corresponding to β . By Corollary 3.8 every Artinian standard graded Gorenstein algebra with Betti sequence $\beta_{\max}(H)$ has the WLP. Now applying iteratively Theorem 3.12 we can reach β from $\beta_{\max}(H)$, obtaining families of Artinian standard graded Gorenstein algebra with Betti sequence β and enjoying the WLP. The last assertion follows from the irreducibility of the stratum of $\text{Gor } H$ defined by β (see [4, Sect. 3]). $\square$

4 Linear Quotients of Weak Lefschetz Algebras

When we study an Artinian algebra A with the WLP we see that, in some sense, it has a “generic” linear quotient $A/\ell A$. So, it seems totally natural to study those algebras which arise as generic linear quotient of WL Artinian graded algebras and try to understand what they inherit from the starting algebra. Such an investigation is similar to what one does in Algebraic Geometry when one studies the generic hyperplane section of a projective variety. It is well known that if one starts from an arithmetically Cohen-Macaulay variety of dimension >0 then the generic hyperplane section, since is done by a regular element, has the same graded Betti numbers of the starting variety and consequently its Hilbert function is just the first difference of the Hilbert function of the variety. In the case of the generic linear quotient of a WL Artinian graded algebra A the question is not so simple, since the form ℓ is no more a regular element. So, while it is easy to see that its Hilbert function is just the positive part of the first difference of the Hilbert function of A , the question is more tricky for the graded Betti numbers.

For this we will use the following terminology.

Let k be an infinite field and $R = k[x_1, \dots, x_c]$ the polynomial ring in c indeterminates and I an ideal of R of height c .

Let $\ell \in R_1$ be a linear form, we will set: $I_{[\ell]} := I + (\ell)$, $A_{[\ell]} := R/I_{[\ell]}$, $\bar{R}_{[\ell]} := R/(\ell)$, $\bar{I}_{[\ell]} := I_{[\ell]}/(\ell)$ and $\bar{A}_{[\ell]} := \bar{R}_{[\ell]}/\bar{I}_{[\ell]}$ (of course, $\bar{A}_{[\ell]} \cong A_{[\ell]}$ as rings), when there is no ambiguity, we will write $\bar{R}$, $\bar{I}$ and $\bar{A}$, instead of $\bar{R}_{[\ell]}$, $\bar{I}_{[\ell]}$ and $\bar{A}_{[\ell]}$. Let i be an integer, we will denote by $\varphi_{\ell,i} : A_i \rightarrow A_{i+1}$ the linear map (as k -vector spaces)

obtained by multiplication by ℓ . With this terminology we have, for every integer i , the following exact sequence

$$0 \rightarrow \text{Ker } \varphi_{\ell,i} \rightarrow A_i \xrightarrow{\varphi_{\ell,i}} A_{i+1} \rightarrow (A_{[\ell]})_{i+1} \rightarrow 0.$$

From this sequence we get immediately

$$H_{A_{[\ell]}}(i+1) = \Delta H_A(i+1) + \dim_k \text{Ker } \varphi_{\ell,i}. \quad (1)$$

We will denote by $H_{[\ell]}$ the Hilbert function of $A_{[\ell]}$ and

$$\mathcal{H}_A = \{H_{[\ell]} \mid \ell \in R_1\};$$

as a poset with the natural order of the functions induced by the usual ordering on $\mathbb{Z}$. Moreover we define

$$S_{A,H} := \{[\ell] \in \mathbb{P}_k(R_1) \mid H_{[\ell]} = H\}.$$

So we have that set-theoretically $\mathbb{P}_k(R_1) = \bigcup_{H \in \mathcal{H}_A} S_{A,H}$. Observe that $\{S_{A,H}\}_H$ is a partition of $\mathbb{P}_k(R_1)$. Since $\mathcal{H}_A$ is a finite set there exists $\bar{\ell} \in R_1$ such that $S_{A,H_{[\bar{\ell}]}}$ contains a non empty open subset $U \subseteq \mathbb{P}_k(R_1)$ and there is only one element in $\mathcal{H}_A$ with such a property.

Definition 4.1 With the above notation we say that $A_{[\ell]}$ has the generic Hilbert function with respect to A iff $[\ell] \in S_{A,H_{[\bar{\ell}]}}$. In this case $H_{[\bar{\ell}]}$ will be called the Hilbert function of the generic linear section of A and will be denoted by H_A^{gen} .

Proposition 4.2 Let $A = R/I$ be an Artinian standard graded R -algebra. The poset $\mathcal{H}_A$ has only one minimal element, precisely H_A^{gen} .

Proof Let i be an integer, $\ell \in R_1$ and let $\varphi_{\ell,i} : A_i \rightarrow A_{i+1}$ be the multiplication map by ℓ (as k -vector spaces). Let $r_i := \max\{\text{rank } \varphi_{\ell,i} \mid \ell \in R_1\}$. Of course, the set $T_i := \{[\ell] \in \mathbb{P}_k(R_1) \mid \text{rank } \varphi_{\ell,i} = r_i\}$ contains a non empty open subset of $\mathbb{P}_k(R_1)$. Since A is Artinian only a finite number of T_i is different from $\mathbb{P}_k(R_1)$ so the set $\bigcap_i T_i$ contains a non empty open subset of $\mathbb{P}_k(R_1)$. Let H_A^{min} the function defined by $H_A^{\text{min}}(i) := H_A(i) - r_{i-1}$. According to (1) for every $[\ell] \in \mathbb{P}_k(R_1)$ $H_{A_{[\ell]}} \geq H_A^{\text{min}}$. Moreover for every $[\ell] \in \bigcap_i T_i$, $H_{A_{[\ell]}} = H_A^{\text{min}}$ and $H_A^{\text{min}} = H_A^{\text{gen}}$. $\square$
In the sequel we denote $S^{\text{gen}} = \{[\ell] \in \mathbb{P}_k(R_1) \mid H_{A_{[\ell]}} = H_A^{\text{gen}}\}$. If we denote, as usual, $\Delta H_A^+(n) = \max\{\Delta H_A(n), 0\}$, we can deduce the following simple characterizations of WL algebras.

Proposition 4.3 An Artinian standard graded R -algebra A has the WLP iff if there is a linear form $\ell \in R_1$ such that $H_{A_{[\ell]}} = \Delta H_A^+$. Or, equivalently, A has the WLP iff $H_A^{\text{gen}} = \Delta H_A^+$.

Now, we can do similar discussions with respect to the Betti sequences of linear quotients of an Artinian standard graded algebra. So, if $\ell \in R_1$ we define $\beta_\ell := \beta_{A_{[\ell]}}$ the Betti sequence of $A_{[\ell]}$ and

$$\mathcal{B}_A = \{\beta_\ell \mid [\ell] \in S^{gen}\}.$$

Note that $\mathcal{B}_A \subseteq \mathcal{B}_{H^{gen}}$.

Since every element in $\mathcal{B}_A$ is associated to the same Hilbert function H_A^{gen} , by a well known result by Bigatti, Hulett and Pardue (see [1, 9, 17]) the set $\mathcal{B}_A$ is finite. Now we set

$$Z_\beta := \{[\ell] \in S^{gen} \mid \beta_\ell = \beta\}.$$

Obviously $\{Z_\beta\}_{\beta \in \mathcal{B}_A}$ is a finite partition of S^{gen} . Consequently, as before, there exists $\bar{\ell} \in R_1$ such that $Z_{\beta_{\bar{\ell}}}$ contains a nonempty open subset of S^{gen} .

Definition 4.4 With the above notation we say that $A_{[\ell]}$ has the generic Betti sequence with respect to A iff $[\ell] \in Z_{\beta_{\bar{\ell}}}$. In this case $\beta_{\bar{\ell}}$ will be called the Betti sequence of the generic linear section of A and will be denoted by β^{gen} .

Proposition 4.5 Let $A = R/I$ be an Artinian standard graded R -algebra. The poset $\mathcal{B}_A$ has only one minimal element, precisely β^{gen} .

Proof Let $[\ell] \in S^{gen}$; we want to show that $\beta^{gen} \leq \beta_\ell$. Since $[\ell]$ is in the closure of $Z_{\beta^{gen}}$ and both Betti sequences are associated to the same Hilbert function H^{gen} , we can apply Lemma 1.2 in [18] to get our assertion. $\square$

Example 4.6 Let $R = k[x, y, z]$ and $I = (x^4 + x^2z(y+z), x^3y - xyz(y+z), x^2y^2 + y^2z^2, z^5, y^5, xy^4) \subset R$. Then

$$H_{R/I} = (1, 3, 6, 10, 12, 9, 2, 1, 0)$$

and

$$\beta_{R/I} = ((4^3, 5^3), (6, 7^{10}, 9), (8^6, 10)).$$

The linear forms $\ell_0 = x + y + z$, $\ell_1 = y + z$ and $\ell_2 = z$ are WL forms for R/I ; we have that

$$H_{R/I_{\ell_0}} = H_{R/I_{\ell_1}} = H_{R/I_{\ell_2}} = \Delta H_{R/I}^+ = H^{gen} = (1, 2, 3, 4, 2, 0)$$

and

$$\beta_{A_{[\ell_0]}} = ((4^3), (6^2));$$

$$\beta_{A_{[\ell_1]}} = ((4^3, 5), (5, 6^2));$$

$$\beta_{A_{[\ell_2]}} = ((4^3, 5^2), (5^2, 6^2)).$$

Consequently $\beta^{gen} = ((4^3), (6^2))$. Moreover one can see that $\mathcal{B}_A$ contains only these three Betti sequences.

Now we apply the previous general results to study the graded Betti numbers of linear quotients of Artinian standard graded algebras which enjoy the WLP. From now on, in this section, $A = R/I$ will be a Weak Lefschetz Artinian standard graded R -algebra and $\ell \in R_1$ a WL form for A .

We can choose a change of variables such that $\ell = x_c$ and we still denote $R = k[x_1, \dots, x_{c-1}, \ell]$. Let us consider on R the graded reverse lexicographic monomial order, with $x_1 > x_2 > \dots > \ell$. We write $\text{lt}(f)$ for the leading term of a polynomial f and $\text{LT}(I)$ for the ideal generated by the leading terms of the polynomials in I . We first recall some results on the algebra $R/\text{LT}(I)$.

Remark 4.7 Notice that using the above monomial order, if $f \in R$ and $\text{lt}(f)$ is divisible by ℓ , then f is divisible by ℓ too.

Let $\mathcal{G} = \{g_1, \dots, g_N\}$ be a minimal set of generators of I , contained in a Gröbner basis.

Let H_A be the Hilbert function of A ; since A has the Weak Lefschetz property, H_A is unimodal and ΔH_A^+ is an O -sequence. We set

$$t := \max\{j \mid \Delta H_A(j) > 0\}.$$

We have the following facts.

Remark 4.8 If $m \in \text{LT}(I)_{\leq t}$ is a monomial and m is divisible by ℓ , then $\frac{m}{\ell} \in \text{LT}(I)$. Namely, let $f \in I$ such that $m = \text{lt}(f)$. Since ℓ divides m , by Remark 4.7, ℓ divides f ; by the injectivity of $\varphi_{\ell,i}$ for $i < t$ we have that $\frac{f}{\ell} \in I$ so $\frac{m}{\ell} \in \text{LT}(I)$.

Remark 4.9 $k[x_1, \dots, x_{c-1}]_{t+1} \subseteq \text{LT}(I)$. Indeed, if there is

$$m \in k[x_1, \dots, x_{c-1}]_{t+1} \setminus \text{LT}(I),$$

by the surjectivity of $\varphi_{\ell,t}$, there exists $u \in R_t$ such that $m - \ell u \in I$, so $m = \text{lt}(m - \ell u) \in \text{LT}(I)$ and this leads a contradiction.

Moreover we can prove the following result.

Proposition 4.10 *$R/\text{LT}(I)$ is a Weak Lefschetz algebra and ℓ is a Weak Lefschetz form for it.*

Proof Let $L := \text{LT}(I)$ and assume that there exists an integer j such that the multiplication map by ℓ , $\mu : (R/L)_j \rightarrow (R/L)_{j+1}$, has not maximal rank. We split the proof in two parts.

If $\dim(R/L)_j \leq \dim(R/L)_{j+1}$, since μ is not injective, there is a monomial $m \in R_j \setminus L$ such that $\ell m \in L$. This implies that there is a form $f \in I$ such that $\text{lt}(f) = \ell m$; by Remark 4.7 ℓ should divide f . Since L and I have the same Hilbert function $\varphi_{\ell,j}$ is injective by the WLP, therefore $\frac{f}{\ell} \in I$. This is a contradiction since $m = \text{lt}(\frac{f}{\ell}) \in L$.

If $\dim(R/L)_j > \dim(R/L)_{j+1}$, since μ is not surjective, we can find a monomial $m \in R_{j+1} \setminus L$ which is not divisible by ℓ . Of course $m \notin I$ and, with the same

argument as before we see that $\varphi_{\ell,j}$ is surjective, hence there exists $f \in R_j$ such that $m - \ell f \in I$. Again we get a contradiction since $m = \text{lt}(m - \ell f) \in L$. $\square$

Now we concentrate on studying the graded Betti numbers, $\bar{\beta}_{ij}(\bar{A})$, of the algebra

$$\bar{A} := A/\ell A \cong \bar{R}/\bar{I} \cong R/(I + (\ell))$$

as an $\bar{R}$ -algebra.

We denote by $\mathcal{G}_{\leq j}$ the subset of $\mathcal{G}$ consisting of the elements which have degree less than or equal to j and by $\bar{\mathcal{G}}$ the set $\{\bar{g} \in \bar{I} \mid g \in \mathcal{G}\}$. Moreover we denote by $I_{\leq j}$ the ideal generated by $\mathcal{G}_{\leq j}$.

Remark 4.11 Since $\varphi_{\ell,i}$ is injective for $i < t$, we see that $\{\ell\} \cup \mathcal{G}_{\leq t}$ is a minimal set of generators for $(I_{[\ell]})_{\leq t}$. Hence $\bar{\mathcal{G}}_{\leq t}$ is a minimal set of generators for $\bar{I}_{\leq t}$.

The Hilbert function of $\bar{A}$ vanishes in degrees $\geq t + 1$, therefore every minimal set of generators of $\bar{I}$ is contained in $\bar{I}_{\leq t+1}$. So, to determinate $\bar{\beta}_0(\bar{A})$, it is enough to compute $\bar{\beta}_{0,t+1}(\bar{A})$ i.e. the number of elements of $\mathcal{G}$ in degree $t + 1$ which are not divisible by ℓ (this follows from the fact that $\mathcal{G}$ is contained in a Gröbner basis).

Let $\{g_1, \dots, g_r\}$ be the elements of $\mathcal{G}$ of degree $\leq t$ and $\{g_{r+1}, \dots, g_s\}$ the elements of $\mathcal{G}$ of degree $t + 1$. From now on we set $J := I_{\leq t}$. We consider the following commutative diagram:

$$\begin{array}{ccc} (R/I)_t & \xrightarrow{\psi} & (R/J)_{t+1} \\ & \searrow \varphi & \downarrow \pi \\ & & (R/I)_{t+1} \end{array}$$

where the maps ψ and φ are both the multiplication by ℓ (hence $\varphi = \varphi_{\ell,t}$ and π is the natural map).

Proposition 4.12 $\bar{\beta}_{0,t+1}(\bar{A}) = \dim_k(\text{Coker } \psi)$.

Proof Since φ is surjective, we can choose a k -basis of $(R/I)_{t+1}$, $\{f_1 + I, \dots, f_d + I\}$, in such a way that ℓ divides each f_i . On the other hand a k -basis of $(R/J)_{t+1}$ consists of

$$\{g_{r+1} + J, \dots, g_s + J\} \cup \{f_1 + J, \dots, f_d + J\}.$$

Now we can assume (just by re-ordering) that $g_{r+1}, \dots, g_{r'}$ are not divisible by ℓ and $g_{r'+1}, \dots, g_s$ are divisible by ℓ (with $r' \geq r$). To conclude the proof it will be enough to show that the set $\tilde{g}_{r+1}, \dots, \tilde{g}_{r'}$ is a k -basis of $\text{Coker } \psi$ where $\tilde{g}_i = (g_i + J) + \text{im } \psi$.

Since $\{\tilde{g}_{r+1}, \dots, \tilde{g}_s, \tilde{f}_1, \dots, \tilde{f}_d\}$ generate $\text{Coker } \psi$ and $g_{r'+1} + J, \dots, g_s + J, f_1 + J, \dots, f_d + J \in \text{im } \psi$, it follows that $\tilde{g}_{r+1}, \dots, \tilde{g}_{r'}$ is a set of generators for $\text{Coker } \psi$. Now let us suppose that $\lambda_{r+1}\tilde{g}_{r+1} + \dots + \lambda_{r'}\tilde{g}_{r'} = 0$, with each $\lambda_i \in k$. This implies that

$$\lambda_{r+1}(g_{r+1} + J) + \dots + \lambda_{r'}(g_{r'} + J) \in \text{im } \psi$$

i.e. $\lambda_{r+1}g_{r+1} + \cdots + \lambda_{r'}g_{r'} + \ell h \in J$ for some $h \in R_t$. If $\lambda_{r+1}g_{r+1} + \cdots + \lambda_{r'}g_{r'} \neq 0$, then $\text{lt}(\lambda_{r+1}g_{r+1} + \cdots + \lambda_{r'}g_{r'} + \ell h) = \text{lt}(g_e) \in \text{LT}(J) = \text{LT}(g_1, \dots, g_r)$, for some e , $r+1 \leq e \leq r'$. This is a contradiction since $\mathcal{G}$ is contained in a Gröbner basis. $\square$

Theorem 4.13 *With the above notation*

$$\beta_{0 \ t+1}(A) - \bar{\beta}_{0 \ t+1}(\bar{A}) = \dim_k(\text{Ker } \varphi) - \dim_k(\text{Ker } \psi).$$

Proof From the following exact sequence

$$0 \longrightarrow \text{Ker } \psi \longrightarrow (R/I)_t \xrightarrow{\psi} (R/J)_{t+1} \longrightarrow \text{Coker } \psi \longrightarrow 0.$$

we get

$$\dim_k(\text{Ker } \psi) - \dim_k(R/I)_t + \dim_k(R/J)_{t+1} - \dim_k(\text{Coker } \psi) = 0.$$

By Proposition 4.12 we have

$$\dim_k(\text{Ker } \psi) - H_A(t) + \dim_k(R/J)_{t+1} - \bar{\beta}_{0 \ t+1}(\bar{A}) = 0;$$

since $J = I_{\leq t}$ we get

$$\begin{aligned} \beta_{0 \ t+1}(A) - \bar{\beta}_{0 \ t+1}(\bar{A}) &= H_A(t) - H_A(t+1) - \dim_k(\text{Ker } \psi) = \\ &= \dim_k(\text{Ker } \varphi) - \dim_k(\text{Ker } \psi), \end{aligned}$$

using the surjectivity of the map φ . $\square$

Corollary 4.14 *With the above notation*

- i) ψ is surjective iff $\bar{\beta}_{0 \ t+1}(\bar{A}) = 0$;
- ii) ψ is injective iff $\bar{\beta}_{0 \ t+1}(\bar{A}) = \beta_{0 \ t+1}(A) + \Delta H_A(t+1)$.

Proof Just using Proposition 4.12 and Theorem 4.13. $\square$

In order to study all the other graded Betti numbers of $\bar{A}$ we consider a graded minimal free resolution of R -modules of A

$$F_{\bullet} : 0 \rightarrow F_{c-1} \rightarrow \cdots \rightarrow F_i \xrightarrow{d_i} F_{i-1} \rightarrow \cdots \rightarrow F_0 \rightarrow R \rightarrow A \rightarrow 0 \quad (2)$$

and a graded minimal free resolution of $\bar{R}$ -modules of $\bar{A}$

$$G_{\bullet} : 0 \rightarrow G_{c-2} \rightarrow \cdots \rightarrow G_i \xrightarrow{d'_i} G_{i-1} \rightarrow \cdots \rightarrow G_0 \rightarrow \bar{R} \rightarrow \bar{A} \rightarrow 0. \quad (3)$$

Let $\pi_{\bullet} : F_{\bullet} \rightarrow G_{\bullet}$ a lifting of the natural map of R -modules $\pi : A \rightarrow \bar{A}$.

To show the results in Theorem 4.17 we make use of the following technical lemmas.

Lemma 4.15 *Let $d : F \rightarrow M$ be a map of graded R -modules, with F free R -module. Let $\{y_1, \dots, y_r\}$ be a minimal set of homogeneous generators for $\text{im } d$, $\deg y_i \leq \deg y_{i+1}$ for $1 \leq i \leq r-1$. Let us suppose that there is a free basis $\{e_1, \dots, e_r\}$ of F such that $d(e_i) = y_i$ for $1 \leq i \leq r$. Let $z_1, \dots, z_s \in F$ be homogeneous elements such that $d(z_i) = y_i$ for $1 \leq i \leq s$.*

Then $\{z_1, \dots, z_s\}$ is a part of a free basis of F .

One can check easily that the set $\{z_1, \dots, z_s, e_{s+1}, \dots, e_r\}$ is a free basis of F , precisely, it can be obtained by $\{e_1, \dots, e_r\}$ with a transformation whose matrix is a triangular matrix with 1's in the diagonal.

Lemma 4.16 *Let R and $\bar{R} = R/(\ell)$ be as above and consider the following commutative diagram*

$$\begin{array}{ccc} F & \xrightarrow{d} & M \\ \downarrow \pi & & \downarrow \tau \\ G & \xrightarrow{d'} & N \end{array}$$

where F and M are R -modules, F R -free, G and N are $\bar{R}$ -modules, G $\bar{R}$ -free, d, π, τ are R -morphisms and d' is a $\bar{R}$ -morphism.

Let $\{e_1, \dots, e_r\}$ be a free basis of F such that $\{d(e_1), \dots, d(e_r)\}$ is a m.s.o.g. for $\text{im } d$ and $\{\tau d(e_1), \dots, \tau d(e_r)\}$ is part of a m.s.o.g. for $\text{im } d'$.

If $z \in \ker \pi$ then $z = \ell y$ for some $y \in F$.

It is enough to use the commutativity of the diagram and to apply the previous lemma.

Theorem 4.17 *With the above notation, for every $i \geq 0$, let $\{\gamma_{i1}, \dots, \gamma_{i\beta_i}\}$, $\deg \gamma_{i1} \leq \dots \leq \deg \gamma_{i\beta_i}$, be a minimal set of generators for $\text{im } d_i$, and $u_i := |\{j \mid \deg \gamma_{ij} \leq t + i\}|$. If $u_i > 0$ then $\{\pi_{i-1}(\gamma_{i1}), \dots, \pi_{i-1}(\gamma_{iu_i})\}$ can be completed to a minimal set of generators for $\text{im } d'_i$ with elements of degree $\geq t + i$.*

Proof For the case $i = 0$ see Remark 4.11.

Let $\gamma_{i1}, \dots, \gamma_{iu_i}$ be as in the hypotheses and assume that there is γ_{ij} , $1 \leq j \leq u_i$ such that $\pi_{i-1}(\gamma_{ij}) = \sum_{h \neq j} \bar{\mu}_h \pi_{i-1}(\gamma_{ih})$, $\bar{\mu}_h \in \bar{R}$. Then $\gamma := \gamma_{ij} - \sum_{h \neq j} \mu_h \gamma_{ih} \in \text{Ker } \pi_{i-1}$. Now let $(e_1, \dots, e_{\beta_{i-1}})$ be a basis of F_{i-1} such that $d_{i-1}(e_h) = \gamma_{i-1h}$, for $1 \leq h \leq \beta_{i-1}$. Since $\gamma \in F_{i-1}$ it can be written as $\gamma = \sum_{h=1}^{\beta_{i-1}} a_h e_h$. Note that since $u_i > 0$ then $u_h > 0$ for $0 \leq h \leq i$. On the other hand $\gamma \in \text{Ker } d_{i-1}$ and $\deg \gamma \leq t + i$, therefore $a_h = 0$ when $\deg e_h \geq t + i$, so $\gamma = \sum_{h=1}^{u_{i-1}} a_h e_h$, (notice that the number of e_h of degree $\leq t + i - 1$ is u_{i-1}). By induction $\{\pi_{i-2}(\gamma_{i-11}), \dots, \pi_{i-2}(\gamma_{i-1u_{i-1}})\}$ can be completed to a minimal set of generators for $\text{im } d'_{i-1}$ with elements of degree $\geq t + i - 1$. Now

$$d'_{i-1} \pi_{i-1}(e_h) = \pi_{i-2} d_{i-1}(e_h) = \pi_{i-2}(\gamma_{i-1h}), \text{ for } 1 \leq h \leq \beta_{i-1};$$

so we can apply Lemma 4.16 to get $\gamma = \ell\gamma'$, $\gamma' \in \text{Ker } d_{i-1}$. Hence $\gamma_{ij} = \sum_{h \neq j} \mu_h \gamma_{ih} + \ell\gamma'$; this contradicts the minimality of the set $\{\gamma_{i1}, \dots, \gamma_{i\beta_i}\}$.

In order to conclude the proof it is enough to show that each element in $\text{im } d'_i$ of degree $\leq t + i - 1$ is in the submodule generated by

$$\pi_{i-1}(\gamma_{i1}), \dots, \pi_{i-1}(\gamma_{iu_i}).$$

Now, let $\delta \in \text{im } d'_i$ be an element of degree $\leq t + i - 1$. By Lemma 4.15, the elements $\pi_{i-1}(e_1), \dots, \pi_{i-1}(e_{u_{i-1}})$ can be completed to a basis of G_{i-1} with elements of degree $\geq t + i$. Therefore $\delta = \sum_{h=1}^{u_{i-1}} b_h \pi_{i-1}(e_h)$, so $\delta = \pi_{i-1}(\xi)$, for some ξ . Consequently $d_{i-1}(\xi) \in \text{Ker } \pi_{i-2}$; using again Lemma 4.16 one finds $d_{i-1}(\xi) = \ell\xi'$, with $\xi' \in \text{Ker } d_{i-2}$ and $\deg \xi' \leq t + i - 2$, therefore $d_{i-1}(\xi) = \ell d_{i-1}(\eta)$, for some η , so $\xi - \ell\eta \in \text{im } d_i$, hence $\xi - \ell\eta = \sum_{h=1}^{u_i} c_h \gamma_{ih}$; applying π_{i-1} to both sides of the previous equality we get $\delta = \sum_{h=1}^{u_i} c_h \pi_{i-1}(\gamma_{ih})$. $\square$

We can use Theorem 4.17 to give a partial structure to the minimal free resolution $G_\bullet$ of $\bar{A}$. To do that we decompose each F_i and each G_i in (2) and (3) in this way

$$F_i = F'_i \oplus F''_i \text{ and } G_i = G'_i \oplus G''_i$$

where $F'_i := (e_1, \dots, e_{u_i})$ (therefore F''_i is generated in degree $> t + i$); $G'_i := (\pi_i(e_1), \dots, \pi_i(e_{u_i}))$ and G''_i is generated only in degree $t + i$ and $t + i + 1$ (this is due to the fact that the degree of the last syzygy is $t + c - 1$ and $G_\bullet$ is a minimal graded free resolution of a Cohen-Macaulay ring).

Corollary 4.18 *Let us consider the commutative diagram*

$$\begin{array}{ccc} F'_i \oplus F''_i & \xrightarrow{d_i} & F'_{i-1} \oplus F''_{i-1} \\ \downarrow \pi_i & & \downarrow \pi_{i-1} \\ G'_i \oplus G''_i & \xrightarrow{d'_i} & G'_{i-1} \oplus G''_{i-1}. \end{array}$$

With a suitable choice of the free bases, if

$$\begin{pmatrix} M & N_1 \\ 0 & N_2 \end{pmatrix}$$

is a matrix representing d_i , where $M = (m_{hk})$ is a matrix of size $u_{i-1} \times u_i$, then

$$\begin{pmatrix} \pi(M) & P_1 \\ 0 & P_2 \end{pmatrix}$$

is a matrix associated to d'_i , where $\pi(M) = (\pi(m_{hk}))$.

Proof This is a direct consequence of Theorem 4.17. $\square$

Corollary 4.19

$$\bar{\beta}_{ih}(\bar{A}) = \beta_{ih}(A), \text{ for } i \geq 0 \text{ and } h \leq t + i - 1.$$

Moreover $\bar{\beta}_{i+t+1}(\bar{A}) \geq \beta_{i+t+1}(A)$.

Proof It follows immediately by Theorem 4.17. $\square$

By Theorem 4.17 we can easily deduce also a property on the last graded Betti numbers of the Weak Lefschetz algebras.

Corollary 4.20 $\beta_{c-1+j}(A) = 0$ per ogni $j \leq t + c - 1$.

Proof Just apply Corollary 4.19, to the case $i = c - 1$. $\square$

The result in Corollary 4.19 can be clarified in the case $i = 1$.

Corollary 4.21 $\beta_{1+t+1}(A) + \dim_k(\text{Ker } \psi) = \bar{\beta}_{1+t+1}(\bar{A})$.

Proof Since $\Delta H_A(i) = H_{\bar{A}}(i)$, for $i \leq t$ we have

$$\Delta^c H_A(t+1) - \Delta^{c-1} H_{\bar{A}}(t+1) = \Delta H_A(t+1).$$

Then

$$\sum_{i=0}^{c-1} (-1)^i \beta_{i+t+1}(A) = \sum_{i=0}^{c-2} (-1)^i \bar{\beta}_{i+t+1}(\bar{A}) - \Delta H_A(t+1).$$

By Corollary 4.19 we know that $\beta_{i+t+1}(A) = \bar{\beta}_{i+t+1}(\bar{A})$ for every $i \geq 2$; now applying Proposition 4.13 and using the fact that φ is surjective, we get our conclusion. $\square$

In Corollary 4.19 we gave an inequality for the i -th graded Betti numbers of A and $\bar{A}$ in degree $t + i$. Now we will give a characterization when the equality happens.

Proposition 4.22 $\beta_{i+t+1}(A) = \bar{\beta}_{i+t+1}(\bar{A})$ for every $i \geq 0$ iff ψ is injective.

Proof If $\beta_{i+t+1}(A) = \bar{\beta}_{i+t+1}(\bar{A})$ for every i then $\beta_{1+t+1}(A) = \bar{\beta}_{1+t+1}(\bar{A})$, hence by Corollary 4.21 we get the injectivity of ψ .

Now let us suppose that ψ is injective. For $i = 0$ see Remark 4.11. For $i = 1$ see Corollary 4.21. Let $i > 1$; it is enough to show that for every $\sigma \in \text{Ker } d'_{i-1}$, with $\deg \sigma \leq t + i$, there exists $\rho \in \text{Ker } d_{i-1}$ such that $\pi_{i-1}(\rho) = \sigma$. Since $\sigma \in \text{Ker } d'_{i-1}$, it belongs to the submodule of G_{i-1} generated by the elements of degree $\leq t + i - 1$. So, by the inductive hypothesis and by Lemma 4.15, called $D = \{j \mid \deg e_j < t + i\}$, we can write

$$\sigma = \sum_{j \in D} a_j \pi_{i-1}(e_j).$$

Applying d'_{i-1} we have

$$0 = d'_{i-1}(\sum_{j \in D} a_j \pi_{i-1}(e_j)) = \pi_{i-2} d_{i-1}(\sum_{j \in D} a_j e_j) = \pi_{i-2}(\sum_{j \in D} a_j \gamma_{i-1j})$$

consequently

$$\sum_{j \in D} a_j \gamma_{i-1j} \in \ker \pi_{i-2} \cap \ker d_{i-2}$$

so by Lemma 4.16

$$\sum_{j \in D} a_j \gamma_{i-1j} = \ell \eta,$$

$\ell \eta \in \ker d_{i-2} = \text{im } d_{i-1}$. Since $\ell \eta \in \ker d_{i-2}$ we have $\eta \in \ker d_{i-2} = \text{im } d_{i-1}$ i.e. $\eta = d_{i-1}(\gamma)$, for some γ . Now we set $\rho := \sum_{j \in D} a_j e_j - \ell \gamma \in F_{i-1}$ hence $\pi_{i-1}(\rho) = \sigma$ and $d_{i-1}(\rho) = 0$. $\square$

5 Betti Weak Lefschetz Property

Collecting some results of the previous section we can give a description of the graded Betti numbers of $\bar{A}$.

$$\bar{\beta}_{ij}(\bar{A}) = \begin{cases} \beta_{ij}(A) & \text{if } j \leq t + i - 1 \\ \beta_{ij}(A) + m_i & \text{if } j = t + i \\ \sum_{h \geq t+1} (-1)^{h+i+1} \beta_{hj}(A) + (-1)^{i+1} \Delta^{c-1} \Delta H_A^+(j) + m_{i+1} & \text{if } j = t + i + 1 \\ 0 & \text{if } j > t + i + 1 \end{cases} \quad (4)$$

where $m_i \geq 0$ and in particular $m_0 = 0$ and $m_1 = \dim_k \ker \psi$.

Indeed, this follows by Corollary 4.19 and by the following computation.

$$-\Delta^{c-1} H_{\bar{A}}(t + i + 1) = \sum_{h \geq 0} (-1)^h \bar{\beta}_{h \ t+i+1}(\bar{A})$$

then

$$\begin{aligned} & -\Delta^{c-1} H_{\bar{A}}(t + i + 1) = \\ & = (-1)^i \bar{\beta}_{i \ t+i+1}(\bar{A}) + (-1)^{i+1} \bar{\beta}_{i+1 \ t+i+1}(\bar{A}) + \sum_{h \geq i+2} (-1)^h \bar{\beta}_{h \ t+i+1}(\bar{A}) = \\ & = (-1)^i \bar{\beta}_{i \ t+i+1}(\bar{A}) + (-1)^{i+1} (\beta_{i+1 \ t+i+1}(A) + m_{i+1}) + \sum_{h \geq i+2} (-1)^h \beta_{h \ t+i+1}(A), \end{aligned}$$

so by multiplying by $(-1)^i$ we get $\bar{\beta}_{i+t+1}(\bar{A}) =$

$$= m_{i+1} + (-1)^{i+1} \Delta^{c-1} H_{\bar{A}}(t+i+1) + \sum_{h \geq i+1} (-1)^{h+i+1} \beta_{h+t+1}(A).$$

Remark 5.1 If $c = 3$ the graded Betti numbers of $\bar{A}$ are determined by $\dim_k \text{Ker } \psi$.

The Weak Lefschetz property for an Artinian standard graded algebra A induces a natural relationship between its Hilbert function and the Hilbert function of the generic linear quotient of A . The previous results suggest to study a property which preserves a well behavior also for the graded Betti numbers.

In the sequel, if I is an ideal of R and $\ell \in R_1$ is a Weak Lefschetz form for R/I , we will denote by ψ_ℓ the map $\psi_\ell : (R/I)_t \rightarrow (R/(I_{\leq t}))_{t+1}$ defined by multiplication by ℓ (t as defined after Remark 4.7).

Definition 5.2 We say that $A = R/I$ has the *Betti Weak Lefschetz Property*, briefly β -WLP, if there exists $\ell \in R_1$ such that

- (1) ℓ is a Weak Lefschetz form for A ;
- (2) ψ_ℓ is injective.

A linear form ℓ satisfying the conditions as above will be said a β -WL form. In the Example 4.6 the linear form ℓ_0 is a β -WL form (hence R/I is a β -WL algebra) and the linear forms ℓ_1 and ℓ_2 are WL forms but they are not β -WL forms.

Proposition 5.3 *Let A be a standard graded R -algebra enjoying the β -WLP and ℓ a β -WL form and $\bar{A} = A/(\ell)$. Then the graded Betti numbers of $\bar{A}$ are determined by the graded Betti numbers of A . Precisely, we can compute them by (4), with $m_i = 0$ for every i .*

Proof Using Proposition 4.22 and (4) we get $m_i = 0$ for every i . □

Note that there are algebras with the WLP enjoying trivially β -WLP. For instance, if $A = R/I$ has the WLP and I is generated in degree $\geq t+1$, then A has the β -WLP. (It is enough to observe that $I_{\leq t} = (0)$).

Next results put into relation the β -WLP with some particular WL algebras.

It is known from [6] that if H is a Weak Lefschetz sequence then the set

$$\mathcal{B}_H^{\text{WL}} = \{\beta_A \mid H_A = H \text{ and } A \text{ has the WLP}\}$$

admits exactly one maximal element, say, β^H .

Proposition 5.4 *Let H be a Weak Lefschetz sequence and let $A = R/I$ be an Artinian algebra with $H_A = H$ such that A has the WLP. If $\beta_{0+t+1}(A) = \beta_{0+t+1}^H$ then A has the β -WLP.*

Proof Let ℓ be a WL form for A , from Theorem 3.20 in [6] it follows that

$$\beta_{0\ t+1}^H = \hat{\beta}_{0\ t+1} - \Delta H_A(t+1),$$

where $\hat{\beta}$ is the Betti sequence of the lex-segment ideal $L \subset \bar{R} = R/(\ell)$ such that $H_{\bar{R}/L} = \Delta H^+$. By Theorem 4.13 we have

$$\beta_{0\ t+1}(A) = \bar{\beta}_{0\ t+1}(\bar{A}) - \Delta H_A(t+1) - \dim_k \text{Ker } \psi_\ell.$$

So

$$\bar{\beta}_{0\ t+1}(\bar{A}) - \dim_k \text{Ker } \psi_\ell = \hat{\beta}_{0\ t+1},$$

and, using the maximality of $\hat{\beta}$, we get $\text{Ker } \psi_\ell = 0$. □

Corollary 5.5 *Let $A = R/I$ be an Artinian algebra with the WLP such that $\beta_A = \beta^{H_A}$ then A has the β -WLP.*

It is known that if H is the Hilbert function of an Artinian Gorenstein standard graded R -algebra of codimension 3 and $\vartheta - 3$ is its socle degree then the set of the Gorenstein Betti sequences compatible with H

$$\mathcal{G}_H = \{\beta_A \mid H_A = H \text{ and } A \text{ is a Gorenstein Algebra}\}$$

has only one maximal element $\beta^{\max}$ and only one minimal element $\beta^{\min}$. According to the paper [19] there exists a Gorenstein Betti sequence $\gamma^H \in \mathcal{G}_H$, such that every Artinian Gorenstein standard graded R -algebra with Betti sequence $\geq \gamma^H$ has the WLP (see Corollary 2.7 in [19]). We recall that

$$\gamma_{0i}^H = \begin{cases} \beta_{0i}^{\max} & \text{for } i = t+1, \vartheta - t - 1 \\ \beta_{0i}^{\min} & \text{otherwise} \end{cases}.$$

Actually in the next proposition we can improve that result.

Proposition 5.6 *Let H be the Hilbert function of an Artinian Gorenstein standard graded R -algebra of codimension 3. Then every R -algebra A with Betti sequence $\beta_A \in \mathcal{G}_H$ and $\beta_A \geq \gamma^H$ has the β -WLP.*

Proof Let $A = R/I$ be an Artinian Gorenstein standard graded R -algebra of codimension 3 such that $\beta_A \geq \gamma^H$. Repeating the same arguments in Theorem 2.5 in [19] $I_{\leq t} = fI'$ where I' is a perfect ideal of height ≤ 2 , and f some form in R . So if we take ℓ to be a linear regular form in R/I' , such that ℓ does not divide f , then ℓ is a β -WL form. □

6 Examples

Let $A = R/I$ be a Weak Lefschetz Artinian graded standard complete intersection R -algebra with $I = (g_1, \dots, g_c)$, $\deg g_i \leq \deg g_{i+1}$ for $1 \leq i \leq c-1$. For such an algebra it is easy to study the behavior with respect to the β -WLP defined in the previous section.

Proposition 6.1 *Let A be as above then*

- 1) *If $\deg g_c > t$ then A has the β -WLP.*
- 2) *If $\deg g_c \leq t$ and $\Delta H_A(t+1) = 0$ then A has the β -WLP.*
- 3) *If $\deg g_c \leq t$ and $\Delta H_A(t+1) \neq 0$ then A has not the β -WLP.*

Proof If $\deg g_c > t$ then $I_{\leq t}$ is generated by a regular sequence of length $< c$, hence $\text{depth } R/(I_{\leq t}) > 0$, so there is a linear form ℓ which is regular for $R/(I_{\leq t})$, therefore ψ_ℓ is injective.

If $\deg g_c \leq t$ then $I_{\leq t} = I$, therefore $\psi_\ell = \varphi_{\ell,t}$ for every $\ell \in R_1$, so the conclusions of items 2 and 3 follow by the Weak Lefschetz property of A . $\square$

The item 3 of the previous proposition in particular says that ψ_ℓ is not injective but still it has maximal rank. This suggests to give a weaker form of the Definition 5.2.

Definition 6.2 We say that $A = R/I$ has the *generators Weak Lefschetz Property*, briefly β_0 -WLP, if there exists $\ell \in R_1$ such that

- (1) ℓ is a Weak Lefschetz form for A ;
- (2) ψ_ℓ has maximal rank.

Note that for complete intersections the WLP and the β_0 -WLP are equivalent.

Of course, not all β_0 -WL Artinian algebras are β -WL algebras. Just take a WL complete intersection whose generators have degree $\leq t$ and $\Delta H_A(t+1) \neq 0$ (as in the item 3 in Proposition 6.1).

Remark 6.3 Note that the graded Betti numbers of a β -WL algebra A determine the graded Betti numbers of $A/(\ell)$, for a generic $\ell \in R_1$. Analogously when A is a β_0 -WL algebra, $\beta_0(A)$ determines $\overline{\beta}_0(A/(\ell))$, for a generic $\ell \in R_1$. Precisely $\overline{\beta}_{0j}(A/\ell) = \beta_{0j}(A)$ for $j \leq t$ and $\overline{\beta}_{0j}(A/\ell) = 0$, otherwise (see Corollary 4.14).

Next two examples will show that there are Weak Lefschetz Artinian algebras which have not β_0 -WLP.

Example 6.4 $R = k[x, y, z]$, $I = (x^4, x^2y^2, xy^3, xz^3, y^5, z^5)$

$$H_{R/I} = (1, 3, 6, 10, 11, 8, 4, 2, 1, 0)$$

In this case $t = 4$ and $J := I_{\leq 4} = (x^4, x^2y^2, xy^3, xz^3)$. One can check that $\ell = x + y + z$ is a WL form for R/I . On the other hand the Hilbert function of R/J is

$$H_{R/J} = (1, 3, 6, 10, 11, 10, 8, 8, 9, \dots)$$

and $H_{R/J}(n) = n + 1$ for $n \geq 7$. If $\psi_\ell : (R/J)_4 \rightarrow (R/J)_5$ were surjective then the multiplication by ℓ should be surjective also in the successive degrees. This is clearly impossible in degrees ≥ 7 .

Example 6.5 $R = k[x, y, z]$, $I = (x^5, x^3y^2, x^4z, y^6, z^6, y^3z^3)$

$$H_{R/I} = (1, 3, 6, 10, 15, 18, 17, 14, 8, 3, 0, \dots).$$

In this case $t = 5$ and $J := I_{\leq 5} = (x^5, x^3y^2, x^4z)$. One can check that $\ell = x + y + z$ is a WL form for R/I . On the other hand

$$H_{R/J}(5) = 18 < 20 = H_{R/J}(6).$$

For every linear form ℓ we have $\ell x^4y \in J$ and $x^4y \notin J$. So ψ_ℓ cannot be injective.

In the next example we have a β_0 -WL algebra A without the β -WLP for which not all WL forms are β_0 -WL forms.

Example 6.6 $R = k[x, y, z]$, $I = (x^4, x^3y, x^2y^2 - xz^3, xy^3 + yz^3, y^5 + z^5)$

$$H_{R/I} = (1, 3, 6, 10, 11, 9, 5, 2, 0, \dots)$$

$$\beta_A = ((4^4, 5), (5, 6, 7^3, 8, 9^2), (8, 9, 10^2))$$

$\ell_0 = z$ and $\ell_1 = x + y + z$ are both WL forms since one can check that

$$H_{\bar{A}_{[\ell_0]}} = H_{\bar{A}_{[\ell_1]}} = (1, 2, 3, 4, 1, 0, \dots).$$

Computing the Betti sequences, we have

$$\beta_{\bar{A}_{[\ell_0]}} = ((4^4, 5), (5^3, 6)),$$

$$\beta_{\bar{A}_{[\ell_1]}} = ((4^4), (5^2, 6)).$$

Thus ℓ_0 is not a β_0 -WL form since $\bar{\beta}_{05}(\bar{A}_{[\ell_0]}) \neq 0$. ℓ_1 is a β_0 -WL form for A and A is not β -WL algebra since $\dim_k(R/J)_4 = 11 > \dim_k(R/J)_5 = 10$.

Next example shows as the β_0 -WLP does not determine the graded Betti numbers of the its generic linear quotient.

Example 6.7 $R = k[x, y, z, w]$

$$I = (x, y)^5 + (x, z)^5 + (y, z)^5 + (x^2y^2z + w^5),$$

$$H_{R/I} = (1, 4, 10, 20, 35, 40, 38, 32, 22, 7, 1, 0, \dots)$$

$\ell_0 = w$ and $\ell_1 = x + y + z + w$ are both β_0 -WL forms, in fact

$$H_{\overline{A}[\ell_0]} = H_{\overline{A}[\ell_1]} = (1, 3, 6, 10, 15, 5, 0, \dots).$$

Computing the Betti sequences, we have that

$$\beta_{\overline{A}[\ell_0]} = ((5^{16}), (6^{20}, 7^4), (7^4, 8^5))$$

whereas

$$\beta_{\overline{A}[\ell_1]} = ((5^{16}), (6^{20}, 7^3), (7^3, 8^5)).$$

Next example shows as we can have a β -WL algebra R/I even if $R/I_{\leq t}$ is already Artinian.

Example 6.8 $R = k[x, y, z]$,

$$I = (x^5, y^5, z^5, x^3y^3, y^3z^3)$$

$$H_{R/I} = (1, 3, 6, 10, 15, 18, 17, 12, 7, 3, 1, 0)$$

So $t = 5$, $I_{\leq 5} = J := (x^5, y^5, z^5)$ and $\dim_k(R/J)_5 = 18$, $\dim_k(R/J)_6 = 19$.

Now $x + y + z$ is a WL-form for J so ψ_{x+y+z} is injective.

Next example shows that although for a β -WL algebra A the graded Betti numbers of its generic linear quotient $\overline{A}$ are determined, we can find in such a sequence ghost terms which were not in β_A .

Example 6.9 $R = k[x, y, z, w]$,

$$I = (x^3, x^2y, x^2z, w^3, (x + y + z + w)^4, (x - y + z + w)^4, (x + y - z + w)^4, \\ (x - y - z + w)^4, (x + y + 2z + w)^4, (x + 2y + z + w)^4);$$

the Hilbert function of $A = R/I$ is

$$H_A = (1, 4, 10, 16, 16, 3, 0, \dots)$$

observe that $t = 3$. If we take the weak Lefschetz form $\ell = x + 3y + z + 2w$, we have

$$H_{\overline{A}} = (1, 3, 6, 6, 0, \dots).$$

Looking at the Betti sequences of both algebras

$$\beta_{R/I} = ((3^4, 4^6), (4^3, 6^{30}), (5, 7^{30}, 8^2), (8^6, 9^3))$$

$$\beta_{\overline{R}/\overline{I}} = ((3^4, 4^6), (4^3, 5^{13}), (5, 6^6))$$

we find a ghost term for $\overline{A}$ which is not so for A . From $\beta_{14}(A) = \overline{\beta}_{14}(\overline{A})$, it follows that ℓ is a β -WL form for A .

In next example we have a β_0 -WL algebra A without the β -WLP, and not all WL forms are β_0 -WL forms.

Example 5.10 $R = k[x, y, z]$

$$I = (x^4, x^3y, x^2y^2 - xz^3, xy^3 + yz^3, y^5 + z^5)$$

$$H_{R/I} = (1, 3, 6, 10, 11, 9, 5, 2, 0, \dots)$$

$$\beta_A = ((4^4, 5), (5, 6, 7^3, 8, 9^2), (8, 9, 10^2))$$

$\ell_0 = z$ and $\ell_1 = x + y + z$ are both WL forms, in fact

$$H_{A[\ell_0]} = H_{A[\ell_1]} = (1, 2, 3, 4, 1, 0, \dots).$$

Computing the Betti sequences, we have that

$$\beta_{A[\ell_0]} = ((4^4, 5), (5^3, 6))$$

and ℓ_1 is a β_0 -WL form for A

$$\beta_{A[\ell_1]} = ((4^4), (5^2, 6)).$$

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About Multiplicities and Applications to Bezout Numbers

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Abstract Let $(A, \mathfrak{m}, \mathbb{k})$ denote a local Noetherian ring and $\mathfrak{q}$ an ideal such that $\ell_A(M/\mathfrak{q}M) < \infty$ for a finitely generated A -module M . Let $\underline{a} = a_1, \dots, a_d$ denote a system of parameters of M such that $a_i \in \mathfrak{q}^{c_i} \setminus \mathfrak{q}^{c_i+1}$ for $i = 1, \dots, d$. It follows that $\chi := e_0(\underline{a}; M) - c \cdot e_0(\mathfrak{q}; M) \geq 0$, where $c = c_1 \cdot \dots \cdot c_d$. The main results of the report are a discussion when $\chi = 0$ resp. to describe the value of χ in some particular cases. Applications concern results on the multiplicity $e_0(\underline{a}; M)$ and applications to Bezout numbers.

1 Introduction

Let $(A, \mathfrak{m}, \mathbb{k})$ denote a local ring. Let $\mathfrak{q} \subset A$ be an $\mathfrak{m}$ -primary ideal and $\underline{a} = a_1, \dots, a_d$ be a system of parameters in A such that $a_i \in \mathfrak{q}^{c_i}$, $i = 1, \dots, d$, with $c_i > 0$. The main interest in the present report is a comparison of the multiplicities $e_0(\underline{a}; A)$ and $e_0(\mathfrak{q}; A)$.

Let M be a finitely generated A -module. Note that for an ideal $\mathfrak{q} \subset A$ such that the length $\ell_A(M/\mathfrak{q}M)$ is finite, the multiplicity $e_0(\mathfrak{q}; M)$ is defined as the leading term of the Hilbert-Samuel polynomial

$$\ell_A(M/\mathfrak{q}^{n+1}M) = \sum_{i=0}^d e_i(\mathfrak{q}; M) \binom{n+d-i}{d-i} \text{ for } n \gg 0, \text{ with } d = \dim_A M$$

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(see for instance [9, 15] for all the details or more general [12] for generalizations to filtered modules). With the previous assumption, clearly $e_0(\underline{a}; A) \geq e_0(\mathfrak{q}; A)$. We will discuss this relation in more detail. First let us recall some known results:

- (a) For $\mathfrak{q} = \underline{a}A$ we get $e_0(\underline{a}^n; A) = n \cdot e_0(\underline{a}; A)$, where $\underline{a}^n = a_1^{n_1}, \dots, a_d^{n_d}$ and $n = n_1 \cdot \dots \cdot n_d$ for some $(n_1, \dots, n_d) \in \mathbb{N}^d$.
- (b) If $\underline{a}$ is a minimal reduction of $\mathfrak{q}$, then $e_0(\underline{a}; A) = e_0(\mathfrak{q}; A)$. The converse is true (see [11]) provided A is formally equidimensional.
- (c) It follows that $e_0(\underline{a}; A) \geq c \cdot e_0(\mathfrak{q}; A)$, where $c = c_1 \cdot \dots \cdot c_d$ with the above notation.

For the proof of (a) we refer to [1]. The first part of (b) is well-known, while the converse is an outstanding result of Rees (see [11]). The claim in (c) is easy to prove (see [2] for details). The main goal of the present report is a discussion of the difference $\chi := e_0(\underline{a}; A) - c \cdot e_0(\mathfrak{q}; A) \geq 0$ of (c) in various situations, its vanishing resp. a simplified proof of some known results.

The importance of the understanding of χ has to do with Bezout's Theorem in the plane. Let $C = V(F), D = V(G) \subset \mathbb{P}_{\mathbb{k}}^2$, $\mathbb{k}$ an algebraically closed field, be two curves in the projective plane without a common component. Then

$$\sum_{P \in C \cap D} \mu(P; C, D) = \deg C \cdot \deg D,$$

where $\mu(P; C, D)$ denotes the local intersection multiplicity of P in $C \cap D$. In the case of P is the origin, it follows that $\mu(P; C, D) = e_0(f, g; A)$, where $A = \mathbb{k}[x, y]_{(x, y)}$ and f, g denote the equations in A . Since C, D have no component in common, $\{f, g\}$ forms a system of parameters in A . Then

$$e_0(f, g; A) \geq c \cdot d \cdot e_0(\mathfrak{m}; A) = c \cdot d,$$

since $e_0(\mathfrak{m}; A) = 1$. Here c, d denote the initial degree of f, g respectively. This estimate is well-known (see for instance [3] or [5]) proved by resultants or Puiseux expansions. Moreover equality holds if and only if C, D intersect transversally in the origin. In other words, f^*, g^* , the initial forms of f, g in the form ring $G_A(\mathfrak{m}) \cong \mathbb{k}[X, Y]$ are a homogeneous system of parameters.

Here we shall provide another argument with extensions to arbitrary local rings. Let A denote a local ring with $\underline{a} = a_1, \dots, a_d \in A$ a system of parameters. Put $c = c_1 \cdot \dots \cdot c_d$ for $a_i \in \mathfrak{m}^{c_i} \setminus \mathfrak{m}^{c_i+1}$. In his paper (see [10]) the author claimed that $e_0(\underline{a}; A) = c \cdot e_0(\mathfrak{m}; A)$ if and only if the sequence of initial elements $a_1^*, \dots, a_d^* \in G_A(\mathfrak{m})$ forms a regular sequence. This is not true as the following example shows.

Example 1.1 Let $\mathbb{k}$ denote a field and $A = \mathbb{k}[t^4, t^5, t^{11}] \subset \mathbb{k}[[t]]$, where t is an indeterminate over $\mathbb{k}$. Then A is a one-dimensional domain and therefore a Cohen-Macaulay ring with $A \simeq \mathbb{k}[[X, Y, Z]]/(X^4 - YZ, Y^3 - XZ, Z^2 - X^3Y^2)$. Clearly, the residue class $a = x$ of X is a parameter with $a \in \mathfrak{m} \setminus \mathfrak{m}^2$, so that $c = 1$.

Furthermore, by easy calculations it follows that $e_0(a, A) = \ell_A(A/aA) = 4$ and $e_0(\mathfrak{m}, A) = 4$. So, the equation $e_0(a, A) = c \cdot e_0(\mathfrak{m}, A)$ holds, while $G_A(\mathfrak{m}) = \mathbb{K}[X, Y, Z]/(XZ, YZ, Y^4, Z^2)$ is not a Cohen-Macaulay ring (see [2, Sect. 3] for the details).

In Sect. 2 we start with some preliminaries and Koszul complexes. In Sect. 3 we derive some new complexes from certain Koszul complexes important for the study of multiplicities. Euler characteristics are the feature of Sect. 4. As an application we derive a short argument for computing certain multiplicities as Euler characteristics of Koszul complexes (originally done by Auslander and Buchsbaum (see [1]) and Serre (see [13]) by spectral sequence arguments). In Sect. 5 we study the equality $e_0(\underline{a}; M) = c_1 \cdot \dots \cdot c_d \cdot e_0(\mathfrak{q}; M)$. Under some additional regularity condition on the sequence of initial forms $a_1^*, \dots, a_{d-1}^*$ in $G_A(\mathfrak{q})$ we estimate the difference $\ell_A(M/\underline{a}M) - c_1 \cdot \dots \cdot c_d \cdot e_0(\mathfrak{q}; M)$. As an application we get a bound of the local Bezout intersection numbers of two curves in the projective plane without common component.

2 Preliminaries

First let us fix the notations we will use in the following. For the basics on $\mathbb{N}$ -graded structures we refer e.g. to [6].

Notation 2.1

- (A) We denote by A a commutative Noetherian ring with $0 \neq 1$. For an ideal we write $\mathfrak{q} \subset A$. An A -module is denoted by M . Mostly we consider M as finitely generated.
- (B) We consider the Rees and form rings of A with respect to $\mathfrak{q}$ by

$$R_A(\mathfrak{q}) = \bigoplus_{n \geq 0} \mathfrak{q}^n T^n \subseteq A[T] \text{ and } G_A(\mathfrak{q}) = \bigoplus_{n \geq 0} \mathfrak{q}^n / \mathfrak{q}^{n+1}.$$

Here T denotes an indeterminate over A . Both rings are naturally $\mathbb{N}$ -graded. For an A -module M we define the Rees and form modules in the corresponding way by

$$R_M(\mathfrak{q}) = \bigoplus_{n \geq 0} \mathfrak{q}^n M T^n \subseteq M[T] \text{ and } G_M(\mathfrak{q}) = \bigoplus_{n \geq 0} \mathfrak{q}^n M / \mathfrak{q}^{n+1} M.$$

Note that $R_M(\mathfrak{q})$ is a graded $R_A(\mathfrak{q})$ -module and $G_M(\mathfrak{q})$ is a graded $G_A(\mathfrak{q})$ -module. Note that $R_A(\mathfrak{q})$ and $G_A(\mathfrak{q})$ are both Noetherian rings. In case M is a finitely generated A -module then $R_M(\mathfrak{q})$ resp. $G_M(\mathfrak{q})$ is finitely generated over $R_A(\mathfrak{q})$ resp. $G_A(\mathfrak{q})$.

(C) There are the following two short exact sequences of graded modules

$$\begin{aligned} 0 \rightarrow R_M(\mathfrak{q})_+[1] \rightarrow R_M(\mathfrak{q}) \rightarrow G_M(\mathfrak{q}) \rightarrow 0 \text{ and} \\ 0 \rightarrow R_M(\mathfrak{q})_+ \rightarrow R_M(\mathfrak{q}) \rightarrow M \rightarrow 0, \end{aligned}$$

where $R_M(\mathfrak{q})_+ = \bigoplus_{n>0} \mathfrak{q}^n M T^n$.

(D) Let $m \in M$ and $m \in \mathfrak{q}^c M \setminus \mathfrak{q}^{c+1} M$. Then we define $m^* := m + \mathfrak{q}^{c+1} M \in [G_M(\mathfrak{q})]_c$. If $m \in \bigcap_{n \geq 1} \mathfrak{q}^n M$, then we write $m^* = 0$. m^* is called the initial element of m in $G_M(\mathfrak{q})$ and c is called the initial degree of m . Here $[X]_n, n \in \mathbb{Z}$, denotes the n -th graded component of an $\mathbb{N}$ -graded module X .

For these and related results we refer to [6] and [14]. Another feature for the investigations will be the use of Koszul complexes.

Remark 2.2 (Koszul Complex)

(A) Let $\underline{a} = a_1, \dots, a_t$ denote a system of elements of the ring A . The Koszul complex $K_\bullet(\underline{a}; A)$ is defined as follows: Let F denote a free A -module with basis $e_1, \dots, e_t$. Then $K_i(\underline{a}; A) = \bigwedge^i F$ for $i = 1, \dots, t$. A basis of $K_i(\underline{a}; A)$ is given by the wedge products $e_{j_1} \wedge \dots \wedge e_{j_i}$ for $1 \leq j_1 < \dots < j_i \leq t$. The boundary homomorphism $K_i(\underline{a}; A) \rightarrow K_{i-1}(\underline{a}; A)$ is defined by

$$d_{j_1 \dots j_i} : e_{j_1} \wedge \dots \wedge e_{j_i} \mapsto \sum_{k=1}^i (-1)^{k+1} a_{j_k} e_{j_1} \wedge \dots \wedge \widehat{e_{j_k}} \wedge \dots \wedge e_{j_i}$$

on the free generators $e_{j_1} \wedge \dots \wedge e_{j_i}$.

(B) Another way of the construction of $K_\bullet(\underline{a}; A)$ is inductively by the mapping cone. To this end let X denote a complex of A -modules. Let $a \in A$ denote an element of A . The multiplication by a on each A -module $X_i, i \in \mathbb{Z}$, induces a morphism of complexes $m_a : X \rightarrow X$. We define $K_\bullet(a; X)$ as the mapping cone $\text{Mc}(m_a)$. Then we define inductively

$$K_\bullet(a_1, \dots, a_t; A) = K_\bullet(a_t, K_\bullet(a_1, \dots, a_{t-1}; A)).$$

It is easily seen that

$$K_\bullet(\underline{a}; A) \cong K_\bullet(a_1; A) \otimes_A \dots \otimes_A K_\bullet(a_t; A).$$

Therefore it follows that $K_\bullet(\underline{a}; A) \cong K_\bullet(\underline{a}_\sigma; A)$, where $\underline{a}_\sigma = a_{\sigma(1)}, \dots, a_{\sigma(t)}$ with a permutation σ on t letters. For an A -complex X we define $K_\bullet(\underline{a}; X) = K_\bullet(\underline{a}; A) \otimes_A X$. We write $H_i(\underline{a}; X), i \in \mathbb{Z}$, for the i -th homology of $K_\bullet(\underline{a}; X)$. A short exact sequence of A -complexes $0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$ induces a long exact homology sequence for the Koszul homology

$$\dots \rightarrow H_i(\underline{a}; X') \rightarrow H_i(\underline{a}; X) \rightarrow H_i(\underline{a}; X'') \rightarrow H_{i-1}(\underline{a}; X') \rightarrow \dots$$

Let $\underline{a}$ as above a system of t elements in A and $b \in A$. Then the mapping cone construction provides a long exact homology sequence

$$\dots \rightarrow H_i(\underline{a}; X) \rightarrow H_i(\underline{a}; X) \rightarrow H_i(\underline{a}, b; X) \rightarrow H_{i-1}(\underline{a}; X) \rightarrow H_{i-1}(\underline{a}; X) \rightarrow \dots,$$

where the homomorphism $H_i(\underline{a}; X) \rightarrow H_i(\underline{a}; X)$ is multiplication by $(-1)^i b$. Moreover, $\underline{a}H_i(\underline{a}; X) = 0$ for all $i \in \mathbb{Z}$.

For the proof of the last statement, we recall the following well-known argument.

Lemma 2.3 *Let X denote a complex of A -modules. Let $a \in A$ denote an element. Then $aH_i(a, X) = 0$ for all $i \in \mathbb{Z}$.*

3 The Construction of Complexes

First we fix notations for this section. As above let A denote a commutative Noetherian ring and $\mathfrak{q} \subseteq A$. Let $\underline{a} = a_1, \dots, a_t$ denote a system of elements of A . Suppose that $a_i \in \mathfrak{q}^{c_i}$ for some integers $c_i \in \mathbb{N}$ for $i = 1, \dots, t$. Let M denote a finitely generated A -module. We define two complexes here, see also [8] for more detail.

Notation 3.1 Let n denote an integer. We define a complex $K_\bullet(\underline{a}, \mathfrak{q}, M; n)$ in the following way:

- (a) For $0 \leq i \leq t$ put $K_i(\underline{a}, \mathfrak{q}, M; n) = \bigoplus_{1 \leq j_1 < \dots < j_i \leq t} \mathfrak{q}^{n-c_{j_1}-\dots-c_{j_i}} M$ and $K_i(\underline{a}, \mathfrak{q}, M; n) = 0$ for $i > t$ or $i < 0$.
- (b) The boundary map $K_i(\underline{a}, \mathfrak{q}, M; n) \rightarrow K_{i-1}(\underline{a}, \mathfrak{q}, M; n)$ is defined by maps on each of the direct summands $\mathfrak{q}^{n-c_{j_1}-\dots-c_{j_i}} M$. On $\mathfrak{q}^{n-c_{j_1}-\dots-c_{j_i}} M$ it is the map given by $d_{j_1 \dots j_i} \otimes 1_M$ restricted to $\mathfrak{q}^{n-c_{j_1}-\dots-c_{j_i}} M$, where $d_{j_1 \dots j_i}$ denotes the homomorphism as defined in Remark 2.2.

It is clear that the image of the map is contained in $\bigoplus_{1 \leq j_1 < \dots < j_{i-1} \leq t} \mathfrak{q}^{n-c_{j_1}-\dots-c_{j_{i-1}}} M$. Clearly it is a boundary homomorphism. By the construction it follows that $K_\bullet(\underline{a}, \mathfrak{q}, M; n)$ is a sub complex of the Koszul complex $K_\bullet(\underline{a}; M)$ for each $n \in \mathbb{N}$.

Another way for the construction is the following.

Remark 3.2 Let $R_A(\mathfrak{q})$ and $R_M(\mathfrak{q})$ denote the Rees ring and the Rees module. For $a_i, i = 1, \dots, t$, we consider $a_i T^{c_i} \in [R_A(\mathfrak{q})]_{c_i}$. Then we have the system $\underline{aT^c} = a_1 T^{c_1}, \dots, a_t T^{c_t}$ of elements of $R_A(\mathfrak{q})$. Note that $\deg a_i T^{c_i} = c_i, i = 1, \dots, t$. Then we may consider the Koszul complex $K_\bullet(\underline{aT^c}; R_M(\mathfrak{q}))$. This is a complex of graded $R_A(\mathfrak{q})$ -modules. It is easily seen that the degree n -component $[K_\bullet(\underline{aT^c}; R_M(\mathfrak{q}))]_n$ of $K_\bullet(\underline{aT^c}; R_M(\mathfrak{q}))$ is the complex $K_\bullet(\underline{a}, \mathfrak{q}, M; n)$ as introduced in Notation 3.1. We write $H_i(\underline{a}, \mathfrak{q}, M; n)$ for the i -th homology of $K_\bullet(\underline{a}, \mathfrak{q}, M; n)$ for $i \in \mathbb{Z}$.

We come now to the definition of a second complex.

Definition 3.3 With the previous notation we define $\mathcal{L}_\bullet(\underline{a}, \mathfrak{q}, M; n)$ the quotient of the embedding $K_\bullet(\underline{a}, \mathfrak{q}, M; n) \rightarrow K_\bullet(\underline{a}; M)$. That is there is a short exact sequence of complexes

$$0 \rightarrow K_\bullet(\underline{a}, \mathfrak{q}, M; n) \rightarrow K_\bullet(\underline{a}; M) \rightarrow \mathcal{L}_\bullet(\underline{a}, \mathfrak{q}, M; n) \rightarrow 0.$$

Note that $\mathcal{L}_i(\underline{a}, \mathfrak{q}, M; n) \cong \oplus_{1 \leq j_1 < \dots < j_i \leq t} M / \mathfrak{q}^{n-c_{j_1}-\dots-c_{j_i}} M$. The boundary maps are those induced by the Koszul complex. We write $L_i(\underline{a}, \mathfrak{q}, M; n)$ for the i -th homology of $\mathcal{L}_\bullet(\underline{a}, \mathfrak{q}, M; n)$ for any $i \in \mathbb{Z}$.

For a construction by mapping cones we need the following technical result. For a morphism $f : X \rightarrow Y$ we write $C(f)$ for the mapping cone of f .

Lemma 3.4 *With the previous notation let $b \in \mathfrak{q}^d$ an element. The multiplication map by b induces the following morphisms*

$$m_b(K) : K_\bullet(\underline{a}, \mathfrak{q}, M; n-d) \rightarrow K_\bullet(\underline{a}, \mathfrak{q}, M; n) \text{ and } m_b(\mathcal{L}) : \mathcal{L}_\bullet(\underline{a}, \mathfrak{q}, M; n-d) \rightarrow \mathcal{L}_\bullet(\underline{a}, \mathfrak{q}, M; n)$$

of complexes. They induce isomorphism of complexes

$$C(m_b(K)) \cong K_\bullet(\underline{a}, b, \mathfrak{q}, M; n) \text{ and } C(m_b(\mathcal{L})) \cong \mathcal{L}_\bullet(\underline{a}, b, \mathfrak{q}, M; n).$$

Proof The proof follows easily by the structure of the complexes and the mapping cone construction. $\square$

We begin with a few properties of the previous complexes.

Theorem 3.5 *Let $\underline{a} = a_1, \dots, a_t$ denote a system of elements of A , $\mathfrak{q} \subset A$ an ideal and M a finitely generated A -module. Let $n \in \mathbb{N}$ denote an integer.*

- (a) $H_i(\underline{a}, \mathfrak{q}, M; n) \cong H_i(\underline{a}_\sigma, \mathfrak{q}, M; n)$ and $L_i(\underline{a}, \mathfrak{q}, M; n) \cong L_i(\underline{a}_\sigma, \mathfrak{q}, M; n)$ for all $i \in \mathbb{Z}$ and any σ , a permutation on t letters.
- (b) $\underline{a}H_i(\underline{a}, \mathfrak{q}, M; n) = 0$ and $\underline{a}L_i(\underline{a}, \mathfrak{q}, M; n) = 0$ for all $i \in \mathbb{Z}$.
- (c) $H_i(\underline{a}, \mathfrak{q}, M; n)$ and $L_i(\underline{a}, \mathfrak{q}, M; n)$ are finitely generated $A/\underline{a}A$ -modules for all $i \in \mathbb{Z}$.

Proof The statement in (a) follows by virtue of the short exact sequence of complexes in Definition 3.3 and the long exact homology sequence. Note that the homology of Koszul complexes is isomorphic under permutations.

The claim in (c) is a consequence of (b) since the homology modules $H_i(\underline{a}, \mathfrak{q}, M; n)$ and $L_i(\underline{a}, \mathfrak{q}, M; n)$ are finitely generated A -modules.

For the proof of (b) we follow the mapping cone construction of Lemma 3.4 with the arguments of Lemma 2.3. To this end let $K_n = K_\bullet(\underline{a}, \mathfrak{q}, M; n)$ and $C = C(m_b(K)_n) = K_\bullet(\underline{a}, b, \mathfrak{q}, M; n)$. Then there is a short exact sequence of complexes

$$0 \rightarrow K_n \rightarrow C \rightarrow K_{n-d}[-1] \rightarrow 0.$$

The differential ∂_i on $(x, y) \in C_i = (K_{n-d})_{i-1} \oplus (K_n)_i$ is given by

$$\partial_i(x, y) = (d_{i-1}(x), d_i(y) + (-1)^{i-1}bx).$$

Suppose that $\partial_i(x, y) = 0$, i.e., $d_{i-1}(x) = 0$ and $d_i(y) = (-1)^{i-1}bx$. Then

$$(y, 0) \in (K_{n-d})_i \oplus (K_n)_{i+1} = C_{i+1} \text{ because } (K_n)_i \subseteq (K_{n-d})_i$$

and therefore $\partial_{i+1}((-1)^i y, 0) = b(x, y)$. That is $bH_i(C) = 0$ for all $i \in \mathbb{Z}$.

In order to show the claim in (b) we use the previous argument. So let us consider $K_\bullet(\underline{a}, q, M; n) = C(m_{a_i}(K_\bullet(\underline{a}', q, M; n)))$, where $\underline{a}' = a_1, \dots, a_{t-1}$. The previous argument shows $a_t H_i(\underline{a}, q, M; n) = 0$ for all $i \in \mathbb{Z}$. By view of (a) this finishes the proof in the case of $H_i(\underline{a}, q, M; n)$.

For the proof of $\underline{a}L_i(\underline{a}, q, M; n) = 0$ we follow the same arguments. Instead of the injection $(K_n)_i \subseteq (K_{n-d})_i$ we use the surjection $(\mathcal{L}_n)_i \twoheadrightarrow (\mathcal{L}_{n-d})_i$, where $\mathcal{L}_n = \mathcal{L}_\bullet(\underline{a}, q, M; n)$. We skip the details here. $\square$

4 Euler Characteristics

Let A denote a commutative ring. Let X denote a complex of A -modules.

Definition 4.1 Let $X : 0 \rightarrow X_n \rightarrow \dots \rightarrow X_1 \rightarrow X_0 \rightarrow 0$ denote a bounded complex of A -modules. Suppose that $H_i(X)$, $i = 0, 1, \dots, n$, is an A -module of finite length. Then

$$\chi_A(X) = \sum_{i=0}^n (-1)^i \ell_A(H_i(X))$$

is called the Euler characteristic of X .

We collect a few well known facts about Euler characteristics.

Lemma 4.2 Let A denote a Noetherian commutative ring.

- (a) Let $0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$ denote a short exact sequence of complexes such that all the homology modules are of finite length. Then $\chi_A(X) = \chi_A(X') + \chi_A(X'')$.
- (b) Suppose $X : 0 \rightarrow X_n \rightarrow \dots \rightarrow X_1 \rightarrow X_0 \rightarrow 0$ is a bounded complex such that X_i , $i = 0, \dots, n$, is of finite length. Then $\chi_A(X) = \sum_{i=0}^n (-1)^i \ell_A(X_i)$

Proof The statement in (a) follows by the long exact homology sequence derived by $0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$. The second statement might be proved by induction on n , the length of the complex X . $\square$

As an application we get the following result about multiplicities, originally shown by Auslander and Buchsbaum [1] and Serre [13].

Proposition 4.3 *Let $(A, \mathfrak{m})$ be a local ring and $a_1, \dots, a_d \in \mathfrak{m}$ a system of parameters for M , a finitely generated A -module. Then*

$$\chi_A(\underline{a}; M) = e_0(\underline{a}; M),$$

where $\chi_A(\underline{a}; M) = \chi_A(K_\bullet(\underline{a}; M))$ and $e_0(\underline{a}; M)$ denotes the Hilbert-Samuel multiplicity.

Proof Let $\underline{a} = a_1, \dots, a_d$ be the system of parameters and $\underline{a}A = \mathfrak{q}$. We choose $c_i = 1, i = 1, \dots, d$. Then the short exact sequence of Definition 3.3 has the following form

$$0 \rightarrow K_\bullet(\underline{a}, \underline{a}, M; n) \rightarrow K_\bullet(\underline{a}; M) \rightarrow \mathcal{L}_\bullet(\underline{a}, \underline{a}, M; n) \rightarrow 0.$$

All of the three complexes have homology modules of finite length and therefore $\chi_A(\underline{a}; M) = \chi_A(K_\bullet(\underline{a}, \underline{a}, M; n)) + \chi_A(\mathcal{L}_\bullet(\underline{a}, \underline{a}, M; n))$ for all $n \in \mathbb{N}$.

First we show that $\chi_A(K_\bullet(\underline{a}, \underline{a}, M; n)) = 0$ for all $n \gg 0$. To this end recall that $H_i(\underline{a}, \underline{a}, M; n) = [H_i(\underline{aT}; R_M(\underline{a}))]_n$. We know that $\underline{aT}(H_i(\underline{aT}; R_M(\underline{a}))) = 0$ for all $i = 0, \dots, d$. Therefore $H_i(\underline{aT}; R_M(\underline{a}))$ is a finitely generated module over $R_A(\underline{a})/\underline{aT}R_A(\underline{a}) = A/\underline{a}A$. This implies that $[H_i(\underline{aT}; R_M(\underline{a}))]_n = H_i(\underline{a}, \underline{a}, M; n) = 0$ for all $n \gg 0$. That is $\chi_A(K_\bullet(\underline{a}, \underline{a}, M; n)) = 0$ for all $n \gg 0$.

By view of Lemma 4.2 (b) we get

$$\chi_A(\mathcal{L}_\bullet(\underline{a}, \underline{a}, M; n)) = \sum_{i=0}^d (-1)^i \binom{d}{i} \ell_A(M/\underline{a}^{n-i}M).$$

For $n \gg 0$ the length $\ell_A(M/\underline{a}^n M)$ is given by the Hilbert polynomial $e_0(\underline{a}; M) \binom{d+n}{d} + \dots + e_d(\underline{a}; M)$. Therefore, it follows that $\chi_A(\mathcal{L}_\bullet(\underline{a}, \underline{a}, M; n)) = e_0(\underline{a}; M)$. This completes the argument. $\square$

The more general situation of a system of parameters $\underline{a} = a_1, \dots, a_d$ of a finitely generated A -module M of a local ring $(A, \mathfrak{m})$ and an ideal $\mathfrak{q} \supset \underline{a}$ with $a_i \in \mathfrak{q}^{c_i}, i = 1, \dots, d$ is investigated in the following.

Proposition 4.4 *With the previous notation we have the equality*

$$e_0(\underline{a}; M) = c_1 \cdot \dots \cdot c_d \cdot e_0(\mathfrak{q}; M) + \chi_A(K_\bullet(\underline{a}, \mathfrak{q}, M; n))$$

for all $n \gg 0$. In particular, for all $n \gg 0$ the Euler characteristic $\chi_A(K_\bullet(\underline{a}, \mathfrak{q}, M; n))$ is a constant.

Proof The proof follows by the inspection of the short exact sequence of complexes

$$0 \rightarrow K_\bullet(\underline{a}, \mathfrak{q}, M; n) \rightarrow K_\bullet(\underline{a}; M) \rightarrow \mathcal{L}_\bullet(\underline{a}, \mathfrak{q}, M; n) \rightarrow 0$$

of Definition 3.3. By view of Proposition 4.3 we have $e_0(\underline{a}; M)$ for the Euler characteristic of the complex in the middle. For the Euler characteristic on the right

we get (see Lemma 4.2 (b))

$$\chi_A(\mathcal{L}_\bullet(\underline{a}, \mathfrak{q}, M; n)) = \sum_{i=0}^d (-1)^i \sum_{1 \leq j_1 < \dots < j_i \leq d} \ell_A(M/\mathfrak{q}^{n-c_{j_1}} \dots \mathfrak{q}^{n-c_{j_i}} M),$$

which gives the first summand in the above formula (see also [2] for the details in the case of $M = A$). This finally proves the claim. $\square$

For several reasons it would be interesting to have an answer to the following problem.

Problem 4.5 With the notation of Proposition 4.4 it would be of some interest to give an interpretation of $\chi_A(\underline{a}, \mathfrak{q}, M) := \chi_A(K_\bullet(\underline{a}, \mathfrak{q}, M; n))$ for large $n \gg 0$ independently of n . By a slight modification of an argument given in [2] it follows that $\chi_A(\underline{a}, \mathfrak{q}, M) \geq 0$.

First we shall use the previous results in order to prove a few formulas for the multiplicity. The following Theorem provides a simplified proof of some of the main results of Auslander and Buchsbaum (see [1]), originally proved by the use of spectral sequences.

Theorem 4.6 Let $(A, \mathfrak{m})$ denote a local ring with $\underline{a} = a_1, \dots, a_d, d = \dim_A M$, a system of parameters for a finitely generated A -module M .

- (a) $e_0(\underline{a}; M) = e_0(\underline{a}'; M/aM) - e_0(\underline{a}'; 0 :_M a)$, where $a = a_1$ and $\underline{a}' = a_2, \dots, a_d$.
- (b) $e_0(\underline{a}; M) = e_0(a, \underline{a}'; M) + e_0(b, \underline{a}'; M)$, where $a_1 = a \cdot b$ and $\underline{a}' = a_2, \dots, a_d$.
- (c) $e_0(\underline{a}^{\underline{n}}; M) = n \cdot e_0(\underline{a}; M)$, where $\underline{a}^{\underline{n}} = a_1^{n_1}, \dots, a_d^{n_d}$ for $\underline{n} = (n_1, \dots, n_d) \in \mathbb{N}^d$ and $n = n_1 \cdot \dots \cdot n_d$.

Proof We start with the comparison of the Koszul complexes $K_\bullet(b; M)$ and $K_\bullet(ab; M)$. This leads to the following commutative diagram with exact rows

$$\begin{array}{ccccccc} & & 0 & & 0 & & \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & M & \xlongequal{\quad} & M & \longrightarrow & 0 \\ & & \downarrow b & & \downarrow ab & & \\ 0 & \longrightarrow & 0 :_M a & \longrightarrow & M & \xrightarrow{a} & M \longrightarrow M/aM \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \\ & & 0 & & 0 & & \end{array}$$

where the columns are the Koszul complexes. That is, we have an exact sequence of complexes

$$0 \rightarrow 0 :_M a \rightarrow K_\bullet(b; M) \rightarrow K_\bullet(ab; M) \rightarrow M/aM \rightarrow 0.$$

We tensor now by $K_\bullet(\underline{a}'; A)$, a bounded complex of free A -modules. By the definition it induces an exact sequence of complexes

$$0 \rightarrow K_\bullet(\underline{a}'; 0 :_M a) \rightarrow K_\bullet(b, \underline{a}'; M) \rightarrow K_\bullet(ab, \underline{a}'; M) \rightarrow K_\bullet(\underline{a}'; M/aM) \rightarrow 0.$$

Now we inspect the previous sequence in the case $b = 1$. Then it follows that $K_\bullet(1, \underline{a}'; M)$ is exact. By virtue of the Euler characteristic and by Proposition 4.3, the statement in (a) follows.

Next we look at the case of a general b . With the previous result (a) it implies that

$$e_0(ab, \underline{a}'; M) = e_0(a, \underline{a}'; M) + e_0(b, \underline{a}'; M)$$

which proves (b).

By induction and permutability of the sequence this yields the statement in (c). $\square$

It is noteworthy to say that $e_0(\underline{a}'; 0 :_M a) \neq 0$ if and only if $\dim_A 0 :_M a = \dim_A M - 1$.

5 On An Equality

As before let $(A, \mathfrak{m})$ denote a local ring with M a finitely generated A -module. Let $\mathfrak{q}$ denote an ideal of A such that $\ell_A(M/\mathfrak{q}M) < \infty$. For a system of parameters $\underline{a} = a_1, \dots, a_d, d = \dim_A M$, of M suppose that $a_i \in \mathfrak{q}^{c_i} \setminus \mathfrak{q}^{c_i+1}, i = 1, \dots, d$. Besides of the Rees module $R_M(\mathfrak{q})$ we need the following.

Definition 5.1 With the previous notation we investigate the ring $\mathfrak{R} = A[a_1 T^{c_1}, \dots, a_d T^{c_d}] \subset A[T]$. Note that $\mathfrak{R}$ is a graded A -algebra. Because of $a_i \mathfrak{q}^n \subseteq \mathfrak{q}^{n+c_i}, i = 1, \dots, d$ it follows that the Rees module $R_M(\mathfrak{q})$ is an $\mathfrak{R}$ -module.

In the following we shall explore when $R_M(\mathfrak{q})$ is a finitely generated $\mathfrak{R}$ -module.

Lemma 5.2 *With the previous notation the following conditions are equivalent:*

- (i) $R_M(\mathfrak{q})$ is a finitely generated $\mathfrak{R}$ -module.
- (ii) There is a positive integer k such that $\mathfrak{q}^n M = \sum_{i=1}^d a_i \mathfrak{q}^{n-c_i} M$ for all $n > k$.
- (iii) The initial forms $a_1^*, \dots, a_d^*$ are a system of parameters of $G_M(\mathfrak{q})$.

Proof First we prove (i) $\implies$ (ii). If $R_M(\mathfrak{q})$ is finitely generated over $\mathfrak{R}$, then

$$R_M(\mathfrak{q}) = \mathfrak{R}(m_1 T^{d_1}, \dots, m_r T^{d_r}) \text{ for } m_i T^{d_i} \in R_M(\mathfrak{q})_{d_i}.$$

Let $k = \max\{d_1, \dots, d_r\}$ and $n > k$. Let $m \in \mathfrak{q}^n M$ and therefore

$$m T^n = \sum_{j=1}^r \left(\sum_{\underline{a}_j} r_{\underline{a}_j}^{(j)} (a_1 T^{c_1})^{\alpha_{j,1}} \dots (a_d T^{c_d})^{\alpha_{j,d}} \right) m_j T^{d_j} \text{ with } r_{\underline{a}_j}^{(j)} \in A$$

and $d_j + \sum c_i \alpha_{j,i} \geq n$. The last inequality implies

$$\sum c_i \alpha_{j,i} \geq n - d_j \geq n - k > 0.$$

Therefore for each j there is an i such that $\alpha_{j,i} > 0$ and $mT^n \in \sum_{i=1}^d a_i q^{n-c_i} MT^n$, which proves the statement in (ii).

For the converse we claim $R_M(q) = \Re(M, qMT, \dots, q^k MT^k)$. Let mT^n with $m \in q^n M$. If $n \leq k$ then clearly $mT^n \in \Re(M, qMT, \dots, q^k MT^k)$. If $n > k$, then $m = \sum_{i=1}^d a_i m_i$, $m_i \in q^{n-c_i} M$. By induction $m_i T^{n-c_i} \in \Re(M, qMT, \dots, q^k MT^k)$ and therefore $mT^n = \sum_{i=1}^d a_i T^{c_i} m_i T^{n-c_i} \in \Re(M, qMT, \dots, q^k MT^k)$, which finishes the argument.

For the equivalence of (ii) and (iii) recall that $a_1^*, \dots, a_d^*$ is a homogeneous system of parameters if and only if $[G_M(q)/(a_1^*, \dots, a_d^*)G_M(q)]_n = 0$ for all $n > k$. This is equivalent to

$$q^n M = \sum_{i=1}^d a_i q^{n-c_i} M + q^{n+1} M \text{ for all } n > k.$$

By Nakayama Lemma this is equivalent to the condition in (ii). $\square$

For the following we define $c = c_1 \cdot \dots \cdot c_d$ and $e_i = c/c_i$, $i = 1, \dots, d$. Then $\underline{a}^c = a_1^{e_1}, \dots, a_d^{e_d}$ is a system of parameters of M and $a_i^{e_i} \in q^c$. With these notation we get the following commutative diagram

$$\begin{array}{ccc} R_M(q^c) & \subset & R_M(q) \\ \cup & & \cup \\ \mathfrak{S} & \subset & \Re, \end{array}$$

where $\mathfrak{S} = A[a_1^{e_1} T^c, \dots, a_d^{e_d} T^c]$. It is easily seen that $\mathfrak{S} \subset \Re$ is a finitely generated extension since it is integral. Note that $a_i^{e_i} T^c = (a_i T^{c_i})^{e_i}$, $i = 1, \dots, d$.

Corollary 5.3 *With the previous notation the following conditions are equivalent:*

- (i) *The initial forms $a_1^*, \dots, a_d^*$ are a system of parameters of $G_M(q)$.*
- (ii) *There is an integer $k > 0$ such that $q^{nc} M = \sum_{i=1}^d a_i^{e_i} q^{nc-c} M$ for all $n > k$.*
- (iii) *The initial forms $(a_1^{e_1})^*, \dots, (a_d^{e_d})^*$ in $G_A(q^c)$ are a system of parameters in $G_M(q^c)$.*

Proof We have the isomorphism $R_M(q^c) \cong M[q^c T^c] =: \mathfrak{M}$ and

$$R_M(q) \cong \mathfrak{M} \oplus qT\mathfrak{M} \oplus \dots \oplus q^{c-1} T^{c-1} \mathfrak{M}.$$

Whence $\mathfrak{S} \subset R_M(q^c)$ is a finitely generated extension if and only if $\mathfrak{S} \subset R_M(q)$ is a finitely generated extension. Then Artin-Rees lemma yields the equivalence of (i) and (ii) by view of Lemma 5.2.

The equivalence of (ii) and (iii) follows by Lemma 5.2. $\square$

Corollary 5.4 *With the previous notation suppose that $a_1^*, \dots, a_d^*$ forms a system of parameters in $G_M(\mathfrak{q})$. Then $e_0(\underline{a}; M) = c_1 \cdot \dots \cdot c_d \cdot e_0(\mathfrak{q}; M)$ and therefore $\chi_A(K_\bullet(\underline{a}, \mathfrak{q}, M; n)) = 0$ for $n \gg 0$.*

Proof By view of Corollary 5.3 we get $(\mathfrak{q}^c)^n M = \sum_{i=1}^d a_i^{e_i} (\mathfrak{q}^c)^{n-1} M$ and $(\mathfrak{q}^c)^l M \subset \underline{a}^{\underline{e}} M \subset \mathfrak{q}^c M$ for some $l \in \mathbb{N}$. Moreover it follows that $(\mathfrak{q}^c)^{n+k} M \subseteq (\underline{a}^{\underline{e}})^k M \subseteq (\mathfrak{q}^c)^k M$ for all $k \geq l$. Then

$$\ell_A(M/(\mathfrak{q}^c)^{n+k} M) \geq \ell_A(M/(\underline{a}^{\underline{e}})^k M) \geq \ell_A(M/(\mathfrak{q}^c)^k M),$$

which implies that $e_0(\mathfrak{q}^c; M) = e_0(\underline{a}^{\underline{e}}; M)$. Because of $e_0(\mathfrak{q}^c; M) = c^d \cdot e_0(\mathfrak{q}; M)$ as easily seen and $e_0(\underline{a}^{\underline{e}}; M) = e_1 \cdot \dots \cdot e_d \cdot e_0(\underline{a}; M)$ (see Theorem 4.6). This finishes the proof. $\square$

The previous result is a generalization of [2, Theorem 5.1] to the situation of finitely generated A -modules. In the case of a formally equidimensional ring the converse is also true (see [2, Theorem 5.2]).

6 The Subregular Case

As a consequence of the definition of $\mathcal{L}_\bullet(\underline{a}, \mathfrak{q}, M; n)$ we get the following equality

$$c_1 \cdot \dots \cdot c_d \cdot e_0(\mathfrak{q}; M) = \sum_{i=1}^d (-1)^i \ell_A(L_i(\underline{a}, \mathfrak{q}, M; n)) \text{ for all } n \gg 0.$$

We have $L_0(\underline{a}, \mathfrak{q}, M; n) \cong M/(\underline{a}, \mathfrak{q}^n)M$. The other homology modules are difficult to describe. For the vanishing of some of them in relation to the existence of $G_M(\mathfrak{q})$ -regular sequences we refer to [7].

In the previous section we have shown that $e_0(\underline{a}; M) = c \cdot e_0(\mathfrak{q}; M)$ provided $a_1^*, \dots, a_d^*$ forms a system of parameters of $G_M(\mathfrak{q})$. In the next, we consider the case that $(a_1^*, \dots, a_d^*)G_A(\mathfrak{q})$ contains a $G_M(\mathfrak{q})$ -regular sequence of length $d - 1$.

We start with the behavior of $\chi(\underline{a}, \mathfrak{q}, M; n)$ by passing to a certain element. To this end, let $\underline{a}' = a_2, \dots, a_d, a = a_1$ and $c_1 = f$.

Lemma 6.1 *Suppose that a^* is $G_M(\mathfrak{q})$ regular. Then*

$$\chi(\underline{a}, \mathfrak{q}, M; n) = \chi(\underline{a}', \mathfrak{q}, M/aM; n) \text{ for all } n \in \mathbb{Z}.$$

Proof If a^* is $G_M(\mathfrak{q})$ -regular, then $0 :_M a = 0$ and $\mathfrak{q}^n M :_M a = \mathfrak{q}^{n-f} M$ for all $n \geq 0$. So it follows that $0 :_{R_M(\mathfrak{q})} aT^f = 0$ and $R_M(\mathfrak{q})/(aT^f)R_M(\mathfrak{q}) \cong R_{M/aM}(\mathfrak{q})$.

Moreover, there is the following diagram with exact rows

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & R_M(\mathfrak{q})(-f) & \xlongequal{\quad} & R_M(\mathfrak{q})(-f) & \longrightarrow & 0 \\
 & & \parallel & & \downarrow aT^f & & \\
 0 & \longrightarrow & R_M(\mathfrak{q})(-f) & \xrightarrow{aT^f} & R_M(\mathfrak{q}) & \longrightarrow & R_{M/aM}(\mathfrak{q}) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & &
 \end{array}$$

where the columns are the Koszul complexes $K_\bullet(1; R_M(\mathfrak{q}))(-f)$ and $K_\bullet(aT^f; R_M(\mathfrak{q}))$ resp. That is, we have a short exact sequence of complexes

$$0 \rightarrow K_\bullet(1; R_M(\mathfrak{q}))(-f) \xrightarrow{aT^f} K_\bullet(aT^f; R_M(\mathfrak{q})) \rightarrow R_{M/aM}(\mathfrak{q}) \rightarrow 0.$$

By tensoring with $K_\bullet(a'T^{c'}, R_A(\mathfrak{q}))$ it provides a short exact sequence of complexes

$$0 \rightarrow K_\bullet(1, \underline{a'T^{c'}}, R_M(\mathfrak{q}))(-f) \xrightarrow{aT^f} K_\bullet(\underline{aT^c}; R_M(\mathfrak{q})) \rightarrow K_\bullet(\underline{a'T^{c'}}; R_{M/aM}(\mathfrak{q})) \rightarrow 0.$$

Since the first Koszul complex in the previous sequence is exact (see Lemma 2.3), the claim follows by the definition. $\square$

The following result is a particular case of [15, VIII, Lemma 3] resp. to [12, Lemma 1.6], a generalization to filtered modules.

Lemma 6.2 *Let $\mathfrak{q} \subset A$ denote an ideal such that $\ell_A(M/\mathfrak{q}M) < \infty$.*

- (a) *Suppose that a^* is $G_M(\mathfrak{q})$ regular. Then $f \cdot e_0(\mathfrak{q}; M) = e_0(\mathfrak{q}; M/aM)$.*
- (b) *Let $\dim_A M = 1$ and $a \in \mathfrak{q}^f$ be a parameter of M . Then*

$$f \cdot e_0(\mathfrak{q}; M) = \ell_A(M/aM) - \ell_A(\mathfrak{q}^n M :_M a/\mathfrak{q}^{n-f} M)$$

for all $n \gg 0$. In particular $\ell_A(\mathfrak{q}^n M :_M a/\mathfrak{q}^{n-f} M)$ is a constant for all $n \gg 0$.

Proof The statements follow by counting the lengths in the exact sequence

$$0 \rightarrow \mathfrak{q}^n M :_M a/\mathfrak{q}^{n-f} M \rightarrow M/\mathfrak{q}^{n-f} M \xrightarrow{a} M/\mathfrak{q}^n M \rightarrow M/(\mathfrak{q}, \mathfrak{q}^n)M \rightarrow 0$$

for all $n \in \mathbb{N}$. If a^* is $G_M(q)$ regular, then $q^n M :_M a = q^{n-f} M$ for all $n \in \mathbb{N}$. If $\dim_A M = 1$ and $a \in q^f$ is a parameter of M , then $\ell_A(M/q^n M) - \ell_A(M/q^{n-f} M) = f \cdot e_0(q; M)$ for all $n \gg 0$. Moreover $q^n M \subset aM$ for all $n \gg 0$. For details see [15, VIII, Lemma 3] resp. [12, Lemma 1.6]. $\square$

Now we are prepared for the main statement of this section.

Corollary 6.3 *With the previous notation assume that $a_1^*, \dots, a_{d-1}^*$ is a $G_M(q)$ -regular sequence. Then*

$$c_1 \cdot \dots \cdot c_d \cdot e_0(q; M) = \ell_A(M/\underline{a}M) - \ell_A((\underline{a}', q^n)M :_M a_d/(\underline{a}', q^{n-c_d})M)$$

for all $n \gg 0$ where $\underline{a}' = a_1, \dots, a_{d-1}$.

Proof By an iterative application of Lemma 6.2 (a) it follows

$$c_1 \cdot \dots \cdot c_{d-1} \cdot e_0(q; M) = e_0(q; M/\underline{a}'M).$$

By view of Lemma 6.2 (b) it implies $c_d \cdot e_0(q; M/\underline{a}'M) = \ell_A(M/\underline{a}M) - \ell_A((\underline{a}', q^n)M :_M a_d/(\underline{a}', q^{n-c_d})M)$. Putting both together yields the claim. $\square$

It is a problem to give an interpretation of the constant

$$\ell_A((\underline{a}', q^n)M :_M a_d/(\underline{a}', q^{n-c_d})M)$$

in intrinsic data of the module M . As a partial result in this direction we get the following.

Lemma 6.4 *With the notation and assumption of Corollary 6.3 we have the following inequality*

$$\ell_A(M/\underline{a}M) \geq c \cdot e_0(q; M) + \mathfrak{x},$$

where $\mathfrak{x} = \ell_A([\text{Ext}_{G_A(q)}^{d-1}(G_A(q)/(\underline{a}^*), G_M(q))]_{n-s-1})$ is a constant for $n \gg 0$ and s denotes $s = c_1 + \dots + c_d$.

Proof There is an injection of

$$\mathfrak{X} := [(\underline{a}'^*)G_M(q) : a_d^*/(\underline{a}'^*)G_M(q)]_{n-c_d-1} \cong ((\underline{a}', q^n)M :_M a_d) \cap (\underline{a}', q^{n-c_d-1})M / (\underline{a}', q^{n-c_d})M$$

into $(\underline{a}', q^n)M :_M a_d/(\underline{a}', q^{n-c_d})M$. Moreover, there is an isomorphism

$$\mathfrak{X} \cong [\text{Hom}_{G_A(q)}(G_A(q)/(\underline{a}^*), G_M(q)/(\underline{a}'^*)G_M(q))]_{n-c_d-1}.$$

Since $\underline{a}'^*$ is a $G_M(q)$ -regular sequence of length $d-1$, it follows that

$$\mathfrak{X} \cong [\text{Ext}_{G_A(q)}^{d-1}(G_A(q)/(\underline{a}^*), G_M(q))]_{n-s-1},$$

which proves the claim. $\square$

We conclude with a geometric application on the local Bezout numbers.

Example 6.5 Let $C = V(F), D = V(G) \subset \mathbb{P}_{\mathbb{k}}^2$ be two curves in the projective plane without a common component. Let $\mu(P; C, D)$ denote the local intersection multiplicity of P in $C \cap D$. In the case of P is the origin, it follows that $\mu(P; C, D) = \ell_A(A/(f, g))$, where $A = \mathbb{k}[x, y]_{(x, y)}$ and f, g denote the equations in A . Since C, D have no component in common, $\{f, g\}$ forms a system of parameters in A . Let $\mathfrak{m}$ denote the maximal ideal of A . Then $B := \mathbb{k}[X, Y] \cong G_A(\mathfrak{m})$ and $1 = e_0(\mathfrak{m}; A)$. We distinguish two cases:

1. C and D intersect transversally in the origin. Then f^*, g^* form a homogeneous system of parameters in B and therefore

$$\ell_A(A/(f, g)) = c \cdot d,$$

where c, d denote the initial degree of f, g resp.

2. Suppose that C and D do not intersect transversally. Then

$$\ell_A(A/(f, g)) \geq c \cdot d + t,$$

where t denotes the number of common tangents of f and g at the origin counted with multiplicities.

Proof First note that A is a Cohen-Macaulay ring and therefore $e_0(f, g; A) = \ell_A(A/(f, g))$. Then the equality in the first case is a consequence of Corollary 5.4. To this end note that f^*, g^* forms a system of parameters in B provided C and D intersect transversally in the origin.

For the second case we use Lemma 6.4. To this end we have to show that $\mathfrak{r} = t$. We put $\mathfrak{Y} = f^*B :_B g^*/f^*B$. Since f^*, g^* are not relatively prime, we write $f^* = h \cdot f', g^* = h \cdot g'$ with homogeneous polynomials $f', g', h \in B$, where f', g' are relatively prime. Then

$$\mathfrak{Y} = f'hB :_B g'h/f'hB \cong f'B/f'hB \cong B/hB[-\deg f']$$

and $\dim_{\mathbb{k}} \mathfrak{Y}_n = \deg h$ for all $n \gg 0$. Since $\deg h$ is the number of common tangents counted with multiplicities, this confirms the second case. $\square$

The second case was also proved by Bydžovský (see [4]). Note that this is an improvement of the corresponding result in [5] where it is shown that $\ell_A(A/(f, g)) \geq c \cdot d + 1$ in case there is a common tangent.

A further discussion of the difference $\ell_A(A/(f, g)) - c \cdot d - t \geq 0$ is given in [2]. There is also another approach by blowing-ups.

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A Polynomial Identity via Differential Operators

Anurag K. Singh

Abstract We give a new proof of a polynomial identity involving the minors of a matrix, that originated in the study of integer torsion in a local cohomology module.

1 Introduction

Our study of integer torsion in local cohomology modules began in the paper [12], where we constructed a local cohomology module that has p -torsion for each prime integer p , and also studied the determinantal example $H_{I_2}^3(\mathbb{Z}[X])$ where X is a 2×3 matrix of indeterminates, and I_2 the ideal generated by its size 2 minors. In that paper, we constructed a polynomial identity that shows that the local cohomology module $H_{I_2}^3(\mathbb{Z}[X])$ has no integer torsion; it then follows that this module is a rational vector space. Subsequently, in joint work with Lyubeznik and Walther, we showed that the same holds for all local cohomology modules of the form $H_{I_t}^k(\mathbb{Z}[X])$, where X is a matrix of indeterminates, I_t the ideal generated by its size t minors, and k an integer with $k > \text{height } I_t$, [9, Theorem 1.2]. In a related direction, in joint work with Bhatt, Blickle, Lyubeznik, and Zhang, we proved that the local cohomology of a polynomial ring over $\mathbb{Z}$ can have p -torsion for at most finitely many p ; we record a special case of [2, Theorem 3.1]:

Theorem 1 *Let R be a polynomial ring over the ring of integers, and let $f_1, \dots, f_m$ be elements of R . Let n be a nonnegative integer. Then each prime integer that is a nonzerodivisor on the Koszul cohomology module $H^n(f_1, \dots, f_m; R)$ is also a nonzerodivisor on the local cohomology module $H_{(f_1, \dots, f_m)}^n(R)$.*

These more general results notwithstanding, a satisfactory proof or conceptual understanding of the polynomial identity from [12] had previously eluded us; extensive calculations with *Macaulay2* had led us to a conjectured identity, which

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we were then able to prove using the hypergeometric series algorithms of Petkovšek et al. [11], as implemented in *Maple*. The purpose of this note is to demonstrate how techniques using differential operators underlying the papers [2] and [9] provide the “right” proof of the identity, and, indeed, provide a rich source of similar identities.

We remark that there is considerable motivation for studying local cohomology of rings of polynomials with integer coefficients such as $H_{I_i}^k(\mathbb{Z}[X])$: a matrix of indeterminates X specializes to a given matrix of that size over an arbitrary commutative noetherian ring (this is where $\mathbb{Z}$ is crucial), which turns out to be useful in proving vanishing theorems for local cohomology supported at ideals of minors of arbitrary matrices. See [9, Theorem 1.1] for these vanishing results, that build upon the work of Bruns and Schwänzl [5].

2 Preliminary Remarks

We summarize some notation and facts. As a reference for Koszul cohomology and local cohomology, we mention [4]; for more on local cohomology as a $\mathcal{D}$ -module, we point the reader towards [7] and [2].

2.1 Koszul and Čech Cohomology

For an element f in a commutative ring R , the Koszul complex $K^\bullet(f; R)$ has a natural map to the Čech complex $C^\bullet(f; R)$ as follows:

$$\begin{array}{ccccccc} K^\bullet(f; R) & := & 0 & \longrightarrow & R & \xrightarrow{f} & R \longrightarrow 0 \\ & & \downarrow & & \parallel & & \downarrow \frac{1}{f} \\ C^\bullet(f; R) & := & 0 & \longrightarrow & R & \longrightarrow & R_f \longrightarrow 0. \end{array}$$

For a sequence of elements $\mathbf{f} := f_1, \dots, f_m$ in R , one similarly obtains

$$K^\bullet(\mathbf{f}; R) := \bigotimes_i K^\bullet(f_i; R) \longrightarrow \bigotimes_i C^\bullet(f_i; R) =: C^\bullet(\mathbf{f}; R),$$

and hence, for each $n \geq 0$, an induced map on cohomology modules

$$H^n(\mathbf{f}; R) \longrightarrow H^n_{(\mathbf{f})}(R). \quad (1)$$

Now suppose R is a polynomial ring over a field $\mathbb{F}$ of characteristic $p > 0$. The Frobenius endomorphism φ of R induces an additive map

$$H^n_{(f)}(R) \longrightarrow H^n_{(f^p)}(R) = H^n_{(f)}(R),$$

where $f^p = f_1^p, \dots, f_m^p$. Set $R\{\varphi\}$ to be the extension ring of R obtained by adjoining the Frobenius operator, i.e., adjoining a generator φ subject to the relations $\varphi r = r^p \varphi$ for each $r \in R$; see [8, Sect. 4]. By an $R\{\varphi\}$ -module we will mean a left $R\{\varphi\}$ -module. The map displayed above gives $H^n_{(f)}(R)$ an $R\{\varphi\}$ -module structure. It is not hard to see that the image of $H^n(f; R)$ in $H^n_{(f)}(R)$ generates the latter as an $R\{\varphi\}$ -module; what is much more surprising is a result of Álvarez et al. [1, Corollary 4.4], by which the image of $H^n(f; R)$ in $H^n_{(f)}(R)$ generates the latter as a $\mathcal{D}(R, \mathbb{F})$ -module; see below for the definition. The result is already notable in the case $m = 1 = n$, where the map (1) takes the form

$$\begin{aligned} H^1(f; R) = R/fR &\longrightarrow R_f/R = H^1_{(f)}(R) \\ [1] &\longmapsto [1/f]. \end{aligned}$$

By [1], the element $1/f$ generates R_f as a $\mathcal{D}(R, \mathbb{F})$ -module. It is of course evident that $1/f$ generates R_f as an $R\{\varphi\}$ -module since the elements $\varphi^e(1/f) = 1/f^{p^e}$ with $e \geq 0$ serve as R -module generators for R_f . See [3] for an algorithm to explicitly construct a differential operator δ with $\delta(1/f) = 1/f^{p^e}$, along with a *Macaulay2* implementation.

2.2 Differential Operators

Let A be a commutative ring, and x an indeterminate; set $R = A[x]$. The divided power partial differential operator

$$\frac{1}{k!} \frac{\partial^k}{\partial x^k}$$

is the A -linear endomorphism of R with

$$\frac{1}{k!} \frac{\partial^k}{\partial x^k}(x^m) = \binom{m}{k} x^{m-k} \quad \text{for } m \geq 0,$$

where we use the convention that the binomial coefficient $\binom{m}{k}$ vanishes if $m < k$. Note that

$$\frac{1}{r!} \frac{\partial^r}{\partial x^r} \cdot \frac{1}{s!} \frac{\partial^s}{\partial x^s} = \binom{r+s}{r} \frac{1}{(r+s)!} \frac{\partial^{r+s}}{\partial x^{r+s}}.$$

For the purposes of this paper, if R is a polynomial ring over A in the indeterminates $x_1, \dots, x_d$, we define the ring of A -linear differential operators on R , denoted $\mathcal{D}(R, A)$, to be the free R -module with basis

$$\frac{1}{k_1!} \frac{\partial^{k_1}}{\partial x_1^{k_1}} \cdot \dots \cdot \frac{1}{k_d!} \frac{\partial^{k_d}}{\partial x_d^{k_d}} \quad \text{for } k_i \geq 0,$$

with the ring structure coming from composition. This is consistent with more general definitions; see [6, 16.11]. By a $\mathcal{D}(R, A)$ -module, we will mean a *left* $\mathcal{D}(R, A)$ -module; the ring R has a natural $\mathcal{D}(R, A)$ -module structure, as do localizations of R . For a sequence of elements f in R , the Čech complex $C^\bullet(f; R)$ is a complex of $\mathcal{D}(R, A)$ -modules, and hence so are its cohomology modules $H_{(f)}^n(R)$. Note that for $m \geq 1$, one has

$$\frac{1}{k!} \frac{\partial^k}{\partial x^k} \left(\frac{1}{x^m} \right) = (-1)^k \binom{m+k-1}{k} \frac{1}{x^{m+k}}.$$

We also recall the Leibniz rule, which states that

$$\frac{1}{k!} \frac{\partial^k}{\partial x^k} (fg) = \sum_{i+j=k} \frac{1}{i!} \frac{\partial^i}{\partial x^i} (f) \frac{1}{j!} \frac{\partial^j}{\partial x^j} (g).$$

3 The Identity

Let R be the ring of polynomials with integer coefficients in the indeterminates

$$\begin{pmatrix} u & v & w \\ x & y & z \end{pmatrix}.$$

The ideal I generated by the size 2 minors of the above matrix has height 2; our interest is in proving that the local cohomology module $H_I^3(R)$ is a rational vector space. We label the minors as $\Delta_1 = vz - wy$, $\Delta_2 = wx - uz$, and $\Delta_3 = uy - vx$. Fix a prime integer p , and consider the exact sequence

$$0 \longrightarrow R \xrightarrow{p} R \longrightarrow \bar{R} \longrightarrow 0,$$

where $\bar{R} = R/pR$. This induces an exact sequence of local cohomology modules

$$\longrightarrow H_I^2(R) \xrightarrow{\pi} H_I^2(\bar{R}) \longrightarrow H_I^3(R) \xrightarrow{p} H_I^3(R) \longrightarrow H_I^3(\bar{R}) \longrightarrow 0.$$

The ring $\overline{R}/I\overline{R}$ is Cohen-Macaulay of dimension 4, so [10, Proposition III.4.1] implies that $H_I^3(\overline{R}) = 0$. As p is arbitrary, it follows that $H_I^3(R)$ is a divisible abelian group. To prove that it is a rational vector space, one needs to show that multiplication by p on $H_I^3(R)$ is injective, equivalently that π is surjective. We first prove this using the identity (2) below, and then proceed with the proof of the identity.

For each $k \geq 0$, one has

$$\begin{aligned} \sum_{i,j \geq 0} \binom{k}{i+j} \binom{k+i}{k} \binom{k+j}{k} \frac{(-wx)^i (vx)^j u^{k+1}}{\Delta_2^{k+1+i} \Delta_3^{k+1+j}} \\ + \sum_{i,j \geq 0} \binom{k}{i+j} \binom{k+i}{k} \binom{k+j}{k} \frac{(-uy)^i (wy)^j v^{k+1}}{\Delta_3^{k+1+i} \Delta_1^{k+1+j}} \\ + \sum_{i,j \geq 0} \binom{k}{i+j} \binom{k+i}{k} \binom{k+j}{k} \frac{(-vz)^i (uz)^j w^{k+1}}{\Delta_1^{k+1+i} \Delta_2^{k+1+j}} = 0. \quad (2) \end{aligned}$$

Since the binomial coefficient $\binom{k}{i+j}$ vanishes if i or j exceeds k , this equation may be rewritten as an identity in the polynomial ring $\mathbb{Z}[u, v, w, x, y, z]$ after multiplying by the element $(\Delta_1 \Delta_2 \Delta_3)^{2k+1}$.

Computing $H_I^2(R)$ as the cohomology of the Čech complex $C^\bullet(\Delta_1, \Delta_2, \Delta_3; R)$, Eq. (2) gives a 2-cocycle in

$$C^2(\Delta_1, \Delta_2, \Delta_3; R) = R_{\Delta_1 \Delta_2} \oplus R_{\Delta_1 \Delta_3} \oplus R_{\Delta_2 \Delta_3};$$

we denote the cohomology class of this cocycle in $H_I^2(R)$ by η_k . When $k = p^e - 1$, one has

$$\binom{k}{i+j} \binom{k+i}{k} \binom{k+j}{k} \equiv 0 \pmod{p} \quad \text{for } (i, j) \neq (0, 0),$$

so (2) reduces modulo p to

$$\frac{u^{p^e}}{\Delta_2^{p^e} \Delta_3^{p^e}} + \frac{v^{p^e}}{\Delta_3^{p^e} \Delta_1^{p^e}} + \frac{w^{p^e}}{\Delta_1^{p^e} \Delta_2^{p^e}} \equiv 0 \pmod{p},$$

and the cohomology class η_{p^e-1} has image

$$\pi(\eta_{p^e-1}) = \left[\left(\frac{w^{p^e}}{\Delta_1^{p^e} \Delta_2^{p^e}}, \frac{-v^{p^e}}{\Delta_1^{p^e} \Delta_3^{p^e}}, \frac{u^{p^e}}{\Delta_2^{p^e} \Delta_3^{p^e}} \right) \right] \quad \text{in } H_I^2(\overline{R}).$$

Since $\bar{R}$ is a regular ring of positive characteristic, $H_I^2(\bar{R})$ is generated as an $\bar{R}\{\varphi\}$ -module by the image of

$$H^2(\Delta_1, \Delta_2, \Delta_3; \bar{R}) \longrightarrow H_I^2(\bar{R}).$$

The Koszul cohomology module $H^2(\Delta_1, \Delta_2, \Delta_3; \bar{R})$ is readily seen to be generated, as an $\bar{R}$ -module, by elements corresponding to the relations

$$u\Delta_1 + v\Delta_2 + w\Delta_3 = 0 \quad \text{and} \quad x\Delta_1 + y\Delta_2 + z\Delta_3 = 0.$$

These two generators of $H^2(\Delta_1, \Delta_2, \Delta_3; \bar{R})$ map, respectively, to

$$\alpha := \left[\left(\frac{w}{\Delta_1\Delta_2}, \frac{-v}{\Delta_1\Delta_3}, \frac{u}{\Delta_2\Delta_3} \right) \right] \quad \text{and} \quad \beta := \left[\left(\frac{z}{\Delta_1\Delta_2}, \frac{-y}{\Delta_1\Delta_3}, \frac{x}{\Delta_2\Delta_3} \right) \right]$$

in $H_I^2(\bar{R})$. Thus, $H_I^2(\bar{R})$ is generated over $\bar{R}$ by $\varphi^e(\alpha)$ and $\varphi^e(\beta)$ for $e \geq 0$. But

$$\varphi^e(\alpha) = \pi(\eta_{p^e-1})$$

is in the image of π , and hence so is $\varphi^e(\beta)$ by symmetry. Thus, π is surjective.

3.1 The Proof of the Identity

We start by observing that $C^2(\Delta_1, \Delta_2, \Delta_3; R)$ is a $\mathcal{D}(R, \mathbb{Z})$ -module. The element

$$\left(\frac{w}{\Delta_1\Delta_2}, \frac{-v}{\Delta_1\Delta_3}, \frac{u}{\Delta_2\Delta_3} \right)$$

is a 2-cocycle in $C^2(\Delta_1, \Delta_2, \Delta_3; R)$ since

$$\frac{w}{\Delta_1\Delta_2} + \frac{v}{\Delta_1\Delta_3} + \frac{u}{\Delta_2\Delta_3} = 0. \quad (3)$$

We claim that the identity (2) is simply the differential operator

$$D = \frac{1}{k!} \frac{\partial^k}{\partial u^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial y^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial z^k}$$

applied termwise to (3); we first explain the choice of this operator: set $k = p^e - 1$, and consider $\bar{D} = D \bmod p$ as an element of

$$\mathcal{D}(R, \mathbb{Z})/p\mathcal{D}(R, \mathbb{Z}) = \mathcal{D}(R/pR, \mathbb{Z}/p\mathbb{Z}).$$

It is an elementary verification that

$$\begin{aligned}\overline{D}(u\Delta_2^{p^e-1}\Delta_3^{p^e-1}) &\equiv u^{p^e} \\ \overline{D}(v\Delta_3^{p^e-1}\Delta_1^{p^e-1}) &\equiv v^{p^e} \pmod{p} \\ \overline{D}(w\Delta_1^{p^e-1}\Delta_2^{p^e-1}) &\equiv w^{p^e}\end{aligned}$$

Since $k < p^e$, the differential operator $\overline{D}$ is $\overline{R}^{p^e}$ -linear; dividing the above equations by $\Delta_2^{p^e}\Delta_3^{p^e}$, $\Delta_3^{p^e}\Delta_1^{p^e}$, and $\Delta_1^{p^e}\Delta_2^{p^e}$ respectively, we obtain

$$\overline{D}\left(\frac{w}{\Delta_1\Delta_2}, \frac{-v}{\Delta_1\Delta_3}, \frac{u}{\Delta_2\Delta_3}\right) \equiv \left(\frac{w^{p^e}}{\Delta_1^{p^e}\Delta_2^{p^e}}, \frac{-v^{p^e}}{\Delta_1^{p^e}\Delta_3^{p^e}}, \frac{u^{p^e}}{\Delta_2^{p^e}\Delta_3^{p^e}}\right) \pmod{p},$$

which maps to the desired cohomology class $\varphi^e(\alpha)$ in $H_I^2(\overline{R})$. Of course, the operator D is not unique in this regard.

Using elementary properties of differential operators recorded in Sect. 2, we have

$$\begin{aligned}D\left(\frac{v}{\Delta_3\Delta_1}\right) &= \frac{1}{k!} \frac{\partial^k}{\partial u^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial y^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial z^k} \left[\frac{v}{(uy - vx)(vz - wy)} \right] \\ &= \frac{1}{k!} \frac{\partial^k}{\partial u^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial y^k} \left[\frac{v(-v)^k}{(uy - vx)(vz - wy)^{k+1}} \right] \\ &= \frac{1}{k!} \frac{\partial^k}{\partial y^k} \left[\frac{v(-v)^k(-y)^k}{(uy - vx)^{k+1}(vz - wy)^{k+1}} \right] \\ &= v^{k+1} \frac{1}{k!} \frac{\partial^k}{\partial y^k} \left[\frac{y^k}{(uy - vx)^{k+1}(vz - wy)^{k+1}} \right] \\ &= v^{k+1} \sum_{i,j} \left[\frac{1}{i!} \frac{\partial^i}{\partial y^i} \frac{1}{(uy - vx)^{k+1}} \right] \left[\frac{1}{j!} \frac{\partial^j}{\partial y^j} \frac{1}{(vz - wy)^{k+1}} \right] \left[\frac{1}{(k-i-j)!} \frac{\partial^{k-i-j}}{\partial y^{k-i-j}} y^k \right] \\ &= v^{k+1} \sum_{i,j} \binom{k+i}{i} \frac{(-u)^i}{(uy - vx)^{k+1+i}} \binom{k+j}{j} \frac{w^j}{(vz - wy)^{k+1+j}} \binom{k}{i+j} y^{i+j} \\ &= v^{k+1} \sum_{i,j} \binom{k+i}{i} \binom{k+j}{j} \binom{k}{i+j} \frac{(-uy)^i (wy)^j}{\Delta_3^{k+1+i} \Delta_1^{k+1+j}}.\end{aligned}$$

A similar calculation shows that

$$D\left(\frac{w}{\Delta_1\Delta_2}\right) = w^{k+1} \sum_{i,j} \binom{k+i}{i} \binom{k+j}{j} \binom{k}{i+j} \frac{(-vz)^i (uz)^j}{\Delta_1^{k+1+i} \Delta_2^{k+1+j}}.$$

It remains to evaluate $D\left(\frac{u}{\Delta_2\Delta_3}\right)$; we reduce this to the previous calculation as follows. First note that the differential operators $\frac{\partial}{\partial u} \cdot \frac{\partial}{\partial y}$ and $\frac{\partial}{\partial v} \cdot \frac{\partial}{\partial x}$ commute; it is readily checked that they agree on $\frac{u}{\Delta_2\Delta_3}$. Consequently the operators

$$\frac{1}{k!} \frac{\partial^k}{\partial u^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial y^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial z^k} \quad \text{and} \quad \frac{1}{k!} \frac{\partial^k}{\partial v^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial z^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial x^k}$$

agree on $\frac{u}{\Delta_2\Delta_3}$ as well. But then

$$D\left(\frac{u}{\Delta_2\Delta_3}\right) = \frac{1}{k!} \frac{\partial^k}{\partial v^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial z^k} \cdot \frac{1}{k!} \frac{\partial^k}{\partial x^k} \left[\frac{u}{(wx - uz)(uy - vx)} \right]$$

which, using the previous calculation and symmetry, equals

$$u^{k+1} \sum_{i,j} \binom{k+i}{i} \binom{k+j}{j} \binom{k}{i+j} \frac{(-wx)^i (vx)^j}{\Delta_2^{k+1+i} \Delta_3^{k+1+j}}.$$

3.2 Identities in General

Suppose $\mathbf{f} = f_1, \dots, f_m$ are elements of a polynomial ring R over $\mathbb{Z}$, and $g_1, \dots, g_m$ are elements of R such that

$$g_1 f_1 + \dots + g_m f_m = 0.$$

Then, for each prime integer p and $e \geq 0$, the Frobenius map on $\overline{R} = R/pR$ gives

$$g_1^{p^e} f_1^{p^e} + \dots + g_m^{p^e} f_m^{p^e} \equiv 0 \pmod{p}. \quad (4)$$

Now suppose p is a nonzerodivisor on the Koszul cohomology module $H^m(\mathbf{f}; R)$. Then Theorem 1 implies that (4) lifts to an equation

$$G_1 f_1^N + \dots + G_m f_m^N = 0 \quad (5)$$

in R in the sense that the cohomology class corresponding to (5) in $H_{(\mathbf{f})}^{m-1}(R)$ maps to the cohomology class corresponding to (4) in $H_{(\mathbf{f})}^{m-1}(\overline{R})$.

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F-Thresholds, Integral Closure and Convexity

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Abstract The purpose of this note is to revisit the results of the paper of Henriques and Varbaro from a slightly different perspective, outlining how, if the integral closures of a finite set of prime ideals abide the expected convexity patterns, then the existence of a peculiar polynomial f allows one to compute the F -jumping numbers of all the ideals formed by taking sums of products of the original ones. The note concludes with the suggestion of a possible source of examples falling in such a framework.

1 Properties A, A+ and B for a Finite Set of Prime Ideals

Let S be a standard graded polynomial ring over a field $\mathbb{k}$ and let m be a positive integer. Fix homogeneous prime ideals of S :

$$\mathfrak{p}_1, \mathfrak{p}_2, \dots, \mathfrak{p}_m.$$

For any $\sigma = (\sigma_1, \dots, \sigma_m) \in \mathbb{N}^m$ and $k = 1, \dots, m$, denote by

$$I^\sigma := \mathfrak{p}_1^{\sigma_1} \cdots \mathfrak{p}_m^{\sigma_m} \quad \text{and} \quad e_k(\sigma) := \max\{\ell : I^\sigma \subseteq \mathfrak{p}_k^{(\ell)}\}.$$

Obviously we have $I^\sigma \subseteq \bigcap_{k=1}^m \mathfrak{p}_k^{(e_k(\sigma))}$. Since $S_{\mathfrak{p}_k}$ is a regular local ring with maximal ideal $(\mathfrak{p}_k)_{\mathfrak{p}_k}$, we have that $(\mathfrak{p}_k)_{\mathfrak{p}_k}^\ell$ is integrally closed in $S_{\mathfrak{p}_k}$ for any $\ell \in \mathbb{N}$. Therefore $\mathfrak{p}_k^{(\ell)} = (\mathfrak{p}_k)_{\mathfrak{p}_k}^\ell \cap S$ is integrally closed in S for any $\ell \in \mathbb{N}$. Eventually we conclude that $\bigcap_{k=1}^m \mathfrak{p}_k^{(e_k(\sigma))}$ is integrally closed in S , so:

$$\overline{I^\sigma} \subseteq \bigcap_{k=1}^m \mathfrak{p}_k^{(e_k(\sigma))}. \quad (1)$$

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Definition 1 We say that $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy condition **A** if

$$\overline{I^\sigma} = \bigcap_{k=1}^m \mathfrak{p}_k^{(e_k(\sigma))} \quad \forall \sigma \in \mathbb{N}^m.$$

If $\Sigma \subseteq \mathbb{N}^m$, denote by $I(\Sigma) := \sum_{\sigma \in \Sigma} I^\sigma$ and by $\overline{\Sigma} \subseteq \mathbb{Q}^m$ the convex hull of $\Sigma \subseteq \mathbb{Q}^m$.

Lemma 1 For any $\Sigma \subseteq \mathbb{N}^m$, $\overline{I(\Sigma)} \supseteq \sum_{\mathbf{v} \in \overline{\Sigma}} I^{[\mathbf{v}]}$, where $[\mathbf{v}] := ([v_1], \dots, [v_m])$ for $\mathbf{v} = (v_1, \dots, v_m) \in \mathbb{Q}^m$.

Proof Since S is Noetherian, we can assume that $\Sigma = \{\sigma^1, \dots, \sigma^N\}$ is a finite set. Take $\mathbf{v} \in \overline{\Sigma}$. Then there exist nonnegative rational numbers $q_1, \dots, q_N$ such that

$$\mathbf{v} = \sum_{i=1}^N q_i \sigma^i \quad \text{and} \quad \sum_{i=1}^N q_i = 1.$$

Let d be the product of the denominators of the q_i 's and $\sigma = d \cdot \mathbf{v} \in \mathbb{N}^m$. Clearly:

$$(I^{[\mathbf{v}]})^d = I^{d \cdot [\mathbf{v}]} \subseteq I^\sigma.$$

Setting $a_i = dq_i$, notice that $\sigma = \sum_{i=1}^N a_i \sigma^i$ and $\sum_{i=1}^N a_i = d$. Therefore

$$(I^{[\mathbf{v}]})^d \subseteq I^\sigma \subseteq I(\Sigma)^d.$$

This implies that $I^{[\mathbf{v}]}$ is contained in the integral closure of $I(\Sigma)$.

From the above lemma, $\overline{I(\Sigma)} \supseteq \sum_{\mathbf{v} \in \overline{\Sigma}} \overline{I^{[\mathbf{v}]}}$. In particular:

$$\mathfrak{p}_1, \dots, \mathfrak{p}_m \text{ satisfy condition } \mathbf{A} \Rightarrow \overline{I(\Sigma)} \supseteq \sum_{\mathbf{v} \in \overline{\Sigma}} \left(\bigcap_{k=1}^m \mathfrak{p}_k^{(e_k([\mathbf{v}]))} \right). \quad (2)$$

Definition 2 We say that $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy condition **A+** if

$$\overline{I(\Sigma)} = \sum_{\mathbf{v} \in \overline{\Sigma}} \left(\bigcap_{k=1}^m \mathfrak{p}_k^{(e_k([\mathbf{v}]))} \right) \quad \forall \Sigma \subseteq \mathbb{N}^m$$

Remark 1 If $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy condition **A+**, then they satisfy **A** as well (for $\sigma \in \mathbb{N}^m$, just consider the singleton $\Sigma = \{\sigma\}$).

Lemma 2 Let $\sigma^1, \dots, \sigma^N$ be vectors in $\mathbb{N}^m$, and $a_1, \dots, a_N \in \mathbb{N}$. Then

$$e_k \left(\sum_{i=1}^N a_i \sigma^i \right) = \sum_{i=1}^N a_i e_k(\sigma^i) \quad \forall k = 1, \dots, m.$$

Proof Set $\sigma = \sum_{i=1}^N a_i \sigma^i$, and notice that

$$I^\sigma = \prod_{i=1}^N \left(I^{\sigma^i} \right)^{a_i} \subseteq \prod_{i=1}^N \left(\mathfrak{p}_k^{(e_k(\sigma^i))} \right)^{a_i} \subseteq \prod_{i=1}^N \mathfrak{p}_k^{(a_i e_k(\sigma^i))} \subseteq \mathfrak{p}_k^{(\sum_{i=1}^N a_i e_k(\sigma^i))},$$

so the inequality $e_k(\sigma) \geq \sum_{i=1}^N a_i e_k(\sigma^i)$ follows directly from the definition.

For the other inequality, for each $i = 1, \dots, N$ choose $f_i \in I^{\sigma^i}$ such that its image in $S_{\mathfrak{p}_k}$ is not in $(\mathfrak{p}_k)_{\mathfrak{p}_k}^{e_k(\sigma^i)+1}$. Then the class $\bar{f}_i$ is a nonzero element of degree $e_k(\sigma^i)$ in the associated graded ring G of $S_{\mathfrak{p}_k}$. Since G is a polynomial ring (in particular a domain), the element $\prod_{i=1}^N \bar{f}_i^{a_i}$ is a nonzero element of degree $\sum_{i=1}^N a_i e_k(\sigma^i)$ in G . Therefore

$$\prod_{i=1}^N f_i^{a_i} \in I_{\mathfrak{p}_k}^\sigma \setminus \mathfrak{p}_k^{(\sum_{i=1}^N a_i e_k(\sigma^i)+1)}.$$

This means that $e_k(\sigma) \leq \sum_{i=1}^N a_i e_k(\sigma^i)$.

Consider the function $e : \mathbb{N}^m \rightarrow \mathbb{N}^m$ defined by

$$\sigma \mapsto e(\sigma) := (e_1(\sigma), \dots, e_m(\sigma)).$$

From the above lemma we can extend it to a $\mathbb{Q}$ -linear map $e : \mathbb{Q}^m \rightarrow \mathbb{Q}^m$.

Given $\Sigma \subseteq \mathbb{N}^m$, the above map sends $\overline{\Sigma}$ to the convex hull $P_\Sigma \subseteq \mathbb{Q}^m$ of the set $\{e(\sigma) : \sigma \in \Sigma\} \subseteq \mathbb{Q}^m$. In particular we have the following:

Proposition 1 *The prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy condition A+ if and only if*

$$\overline{I(\Sigma)} = \sum_{(v_1, \dots, v_m) \in P_\Sigma} \left(\bigcap_{k=1}^m \mathfrak{p}_k^{(\lceil v_k \rceil)} \right) \quad \forall \Sigma \subseteq \mathbb{N}^m$$

If $\Sigma \subseteq \mathbb{N}^m$ and $s \in \mathbb{N}$, define $\Sigma^s := \{\sigma^{i_1} + \dots + \sigma^{i_s} : \sigma^{i_k} \in \Sigma\}$. Then

$$I(\Sigma)^s = I(\Sigma^s).$$

Furthermore $\overline{\Sigma^s} = s \cdot \overline{\Sigma}$, i.e. $P_{\Sigma^s} = s \cdot P_\Sigma$. So:

Proposition 2 *If $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy condition A+, then*

$$\overline{I(\Sigma)^s} = \sum_{(v_1, \dots, v_m) \in P_\Sigma} \left(\bigcap_{k=1}^m \mathfrak{p}_k^{(\lceil s v_k \rceil)} \right) \quad \forall \Sigma \subseteq \mathbb{N}^m, s \in \mathbb{N}.$$

We conclude this section by stating the following definition:

Definition 3 We say that $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy condition **B** if there exists a polynomial $f \in \bigcap_{k=1}^m \mathfrak{p}_k^{\text{ht}(\mathfrak{p}_k)}$ such that $\text{in}_{\prec}(f)$ is a square-free monomial for some term order $\prec$ on S .

Example 1 Let $S = \mathbb{k}[x, y]$, $m = 2$ and $\mathfrak{p}_1 = (x)$ and $\mathfrak{p}_2 = (y)$. Of course these ideals satisfy condition **B** by considering $f = xy$.

If $\sigma = (\sigma_1, \sigma_2)$, then $I^\sigma = (x^{\sigma_1}y^{\sigma_2})$ and $e_k(\sigma) = \sigma_k$, therefore they trivially satisfy condition **A** so.

Though less trivial, it is a well-known fact that (x) and (y) satisfy condition **A+** as well: for example, $\overline{(x^3, y^3)} = (x^3, x^2y, xy^2, y^3)$.

2 Generalized Test Ideals and F -Thresholds

Let $p > 0$ be the characteristic of $\mathbb{k}$, I be an ideal of S and $\mathfrak{m}$ be the homogeneous maximal ideal of S . For all $e \in \mathbb{N}$, denoting by $\mathfrak{m}^{[p^e]} = (x_1^{p^e}, \dots, x_n^{p^e})$, define

$$v_e(I) := \max\{r \in \mathbb{N} : I^r \not\subseteq \mathfrak{m}^{[p^e]}\}.$$

The *F-pure threshold* of I is then

$$\text{fpt}(I) := \lim_{e \rightarrow \infty} \frac{v_e(I)}{p^e}.$$

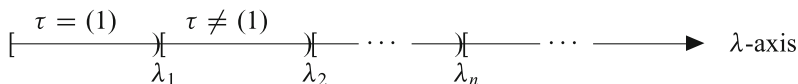
The p^e -th root of I , denoted by $I^{[1/p^e]}$, is the smallest ideal $J \subseteq S$ such that $I \subseteq J^{[p^e]}$. By the flatness of the Frobenius over S the q -th root is well defined. If λ is a positive real number, then it is easy to see that

$$\left(I^{[\lambda p^e]}\right)^{[1/p^e]} \subseteq \left(I^{[\lambda p^{e+1}]}\right)^{[1/p^{e+1}]}$$

The *generalized test ideal* of I with coefficient λ is defined as:

$$\tau(\lambda \cdot I) := \left(I^{[\lambda p^e]}\right)^{[1/p^e]}, \quad e \gg 0.$$

Note that $\tau(\lambda \cdot I) \supseteq \tau(\mu \cdot I)$ whenever $\lambda \leq \mu$. By Blickle et al. [1, Corollary 2.16], $\forall \lambda \in \mathbb{R}_{>0}$, $\exists \epsilon \in \mathbb{R}_{>0}$ such that $\tau(\lambda \cdot I) = \tau(\mu \cdot I) \quad \forall \mu \in [\lambda, \lambda + \epsilon)$. A $\lambda \in \mathbb{R}_{>0}$ is called an *F-jumping number* for I if $\tau((\lambda - \epsilon) \cdot I) \subsetneq \tau(\lambda \cdot I) \quad \forall \epsilon \in \mathbb{R}_{>0}$.



$$(1) \supsetneq \tau(\lambda_1 \cdot I) \supsetneq \tau(\lambda_2 \cdot I) \supsetneq \dots \supsetneq \tau(\lambda_n \cdot I) \supsetneq \dots$$

The λ_i above are the F -jumping numbers. Notice that $\lambda_1 = \text{fpt}(I)$.

Theorem 1 If $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy conditions **A** and **B**, then $\forall \lambda \in \mathbb{R}_{>0}$ we have

$$\tau(\lambda \cdot I^\sigma) = \bigcap_{k=1}^m \mathfrak{p}_k^{(\lfloor \lambda e_k(\sigma) \rfloor + 1 - \text{ht}(\mathfrak{p}_k))} \quad \forall \sigma \in \mathbb{N}^m.$$

If $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy conditions **A+** and **B**, then $\forall \lambda \in \mathbb{R}_{>0}$ we have

$$\tau(\lambda \cdot I(\Sigma)) = \sum_{(v_1, \dots, v_m) \in P_\Sigma} \left(\bigcap_{k=1}^m \mathfrak{p}_k^{(\lfloor \lambda v_k \rfloor + 1 - \text{ht}(\mathfrak{p}_k))} \right) \quad \forall \Sigma \subseteq \mathbb{N}^m.$$

Proof The first part immediately follows from [3, Theorem 3.14], for if $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy conditions **A** and **B**, then I^σ obviously enjoys condition $(\diamond +)$ of [3] $\forall \sigma \in \mathbb{N}^m$.

Concerning the second part, Proposition 2 implies that $I(\Sigma)$ enjoys condition $(*)$ of [3] $\forall \Sigma \subseteq \mathbb{N}^m$ whenever $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfy conditions **A+** and **B**. Therefore the conclusion follows once again by Henriques and Varbaro [3, Theorem 4.3].

3 Where to Fish?

Let $\mathbb{k}$ be of characteristic $p > 0$. So far we have seen that, if we have graded primes $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ of S enjoying **A** and **B**, then we can compute lots of generalized test ideals. If they enjoy **A+** and **B**, we get even more.

That looks nice, but how can we produce $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ like these? Before trying to answer this question, let us notice that, as explained in [3], the ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ of the following examples satisfy conditions **A+** and **B**:

1. $S = \mathbb{k}[x_1, \dots, x_m]$ and $\mathfrak{p}_k = (x_k)$ for all $k = 1, \dots, m$.
2. $S = \mathbb{k}[X]$, where X is an $m \times n$ generic matrix (with $m \leq n$) and $\mathfrak{p}_k = I_k(X)$ is the ideal generated by the k -minors of X for all $k = 1, \dots, m$.
3. $S = \mathbb{k}[Y]$, where Y is an $m \times m$ generic symmetric matrix and $\mathfrak{p}_k = I_k(Y)$ is the ideal generated by the k -minors of Y for all $k = 1, \dots, m$.
4. $S = \mathbb{k}[Z]$, where Z is a $(2m+1) \times (2m+1)$ generic skew-symmetric matrix and $\mathfrak{p}_k = P_{2k}(Z)$ is the ideal generated by the $2k$ -Pfaffians of Z for all $k = 1, \dots, m$.

Even for a simple example like 1., Theorem 1 is interesting: it gives a description of the generalized test ideals of any monomial ideal.

In my opinion, a class to look at to find new examples might be the following: **fix $f \in S$ a homogeneous polynomial such that $\text{in}_{\prec}(f)$ is a square-free monomial for some term order $\prec$** (better if lexicographical) **on S** , and let $\mathcal{C}_f$ be the set of ideals of S defined, recursively, like follows:

- $(f) \in \mathcal{C}_f$;
- If $I \in \mathcal{C}_f$, then $I : J \in \mathcal{C}_f$ for all $J \subseteq S$;
- If $I, J \in \mathcal{C}_f$, then both $I + J$ and $I \cap J$ belong to $\mathcal{C}_f$.

If f is an irreducible polynomial, $\mathcal{C}_f$ consists of only the principal ideal generated by f , but otherwise things can get interesting. Let us give two guiding examples:

1. If $u := x_1 \cdots x_m$, then the associated primes of (u) are $(x_1), \dots, (x_m)$. Furthermore all the ideals of $S = \mathbb{k}[x_1, \dots, x_m]$ generated by variables are sums of the principal ideals above, and all square-free monomial ideals can be obtained by intersecting ideals generated by variables. Therefore, any square-free monomial ideal belongs to $\mathcal{C}_u$, and one can check that indeed:

$$\mathcal{C}_u = \{\text{square-free monomial ideals of } S\}.$$

2. Let $X = (x_{ij})$ be an $m \times n$ matrix of variables, with $m \leq n$. For positive integers $a_1 < \dots < a_k \leq m$ and $b_1 < \dots < b_k \leq n$, recall the standard notation for the corresponding k -minor:

$$[a_1, \dots, a_k | b_1, \dots, b_k] := \det \begin{pmatrix} x_{a_1 b_1} & x_{a_1 b_2} & \cdots & x_{a_1 b_k} \\ \vdots & \vdots & \ddots & \vdots \\ x_{a_k b_1} & x_{a_k b_2} & \cdots & x_{a_k b_k} \end{pmatrix}.$$

For $i = 0, \dots, n-m$, let $\delta_i := [1, \dots, m | i+1, \dots, m+i]$. Also, for $j = 1, \dots, m-1$ set $g_j := [j+1, \dots, m | 1, \dots, m-j]$ and $h_j := [1, \dots, m-j | n-m+j+1, \dots, n]$.

Let Δ be the product of the δ_i 's, the g_j 's and the h_j 's:

$$\Delta := \prod_{i=0}^{n-m} \delta_i \cdot \prod_{j=1}^{m-1} g_j h_j.$$

By considering the lexicographical term order $\prec$ extending the linear order

$$x_{11} > x_{12} > \cdots > x_{1n} > x_{21} > \cdots > x_{2n} > \cdots > x_{m1} > \cdots > x_{mn},$$

we have that

$$\text{in}(\Delta) = \prod_{i=0}^{n-m} \text{in}(\delta_i) \cdot \prod_{j=1}^{m-1} \text{in}(g_j) \text{in}(h_j) = \prod_{(i,j)} x_{ij}$$

is a square-free monomial. Since each (δ_i) belongs to $\mathcal{C}_\Delta$, the height- $(n - m + 1)$ complete intersection

$$J := (\delta_0, \dots, \delta_{n-m})$$

is an ideal of $\mathcal{C}_\Delta$ too. Notice that the ideal $I_m(X)$ generated by all the maximal minors of X is a height- $(n - m + 1)$ prime ideal containing J . So $I_m(X)$ is an associated prime of J , and thus an ideal of $\mathcal{C}_\Delta$ by definition. With more effort, one should be able to show that the ideals of minors $I_k(X)$ stay in $\mathcal{C}_\Delta$ for any size k .

The ideals of $\mathcal{C}_f$ have quite strong properties. First of all, $\mathcal{C}_f$ is a finite set by Schwede [5]. Then, all the ideals in $\mathcal{C}_f$ are radical. Even more, Knutson proved in [4] that they have a square-free initial ideal!

In order to produce graded prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ satisfying conditions **A** (or even **A+**) and **B**, it seems natural to seek for them among the prime ideals in $\mathcal{C}_f$. This is because, at least, f is a good candidate for the polynomial needed for condition **B**: if $f = f_1 \cdots f_r$ is the factorization of f in irreducible polynomials, then for each $A \subseteq \{1, \dots, r\}$ the ideal

$$J_A := (f_i : i \in A) \subseteq S$$

is a complete intersection of height $|A|$. If $\mathfrak{p}$ is an associated prime ideal of J_A , then f obviously belongs to $\mathfrak{p}^{|A|} \subseteq \mathfrak{p}^{(|A|)}$. So such a $\mathfrak{p}$ satisfies **B**.

Question 1 Does the ideal $\mathfrak{p}$ above satisfy condition **A**? Even more, is it true that for prime ideals $\mathfrak{p}$ as above $\mathfrak{p}^s = \mathfrak{p}^{(s)}$ for all $s \in \mathbb{N}$?

If the above question admitted a positive answer, Theorem 1 would provide the generalized test ideals of $\mathfrak{p}$. A typical example, is when $J_A = (\delta_0, \dots, \delta_{n-m})$ and $\mathfrak{p} = I_m(X)$ (see 2. above), in which case it is well-known that $I_m(X)^s = I_m(X)^{(s)}$ for all $s \in \mathbb{N}$ (e.g. see [2, Corollary 9.18]).

Remark 2 Unfortunately, it is not true that $\mathfrak{p}$ satisfies **B** for all prime ideal $\mathfrak{p} \in \mathcal{C}_f$: for example, consider $f = \Delta$ in the case $m = n = 2$, that is $\Delta = x_{21}(x_{11}x_{22} - x_{12}x_{21})x_{12}$. Notice that $(x_{21}, x_{11}x_{22} - x_{12}x_{21}) = (x_{21}, x_{11}x_{22}) = (x_{21}, x_{11}) \cap (x_{21}, x_{22})$, so

$$\mathfrak{p} = (x_{21}, x_{11}) + (x_{21}, x_{22}) = (x_{21}, x_{11}, x_{22}) \in \mathcal{C}_\Delta.$$

However $\Delta \notin \mathfrak{p}^{(3)}$.

Problem 1 Find a large class of prime ideals in $\mathcal{C}_f$ (or even characterize them) satisfying condition **B**.

If $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ are prime ideals satisfying **A+**, then (by definition)

$$\overline{\sum_{i \in A} \mathfrak{p}_i} = \sum_{i \in A} \mathfrak{p}_i \quad \forall A \subseteq \{1, \dots, m\}.$$

If $\mathfrak{p}_1, \dots, \mathfrak{p}_m$ are in $\mathcal{C}_f$, then the above equality holds true because $\sum_{i \in A} \mathfrak{p}_i$, belonging to $\mathcal{C}_f$, is a radical ideal.

Problem 2 Let $\mathcal{P}_f$ be the set of prime ideals in $\mathcal{C}_f$. Is it true that $\mathcal{P}_f$ satisfies condition **A+**? If not, find a large subset of $\mathcal{P}_f$ satisfying condition **A+**.

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